Algebraic Number Theory

— ***EYNTKA*** **Series** —

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1

Relating Algebra to Number Theory

A central object of study in mathematics is, naturally, the natural numbers $\mathbb{N}$ and, if the negatives are included, the integers $\mathbb{Z}$. So far, it is arguable to say that these objects have played a relatively minor algebraic role in this book: we know $\mathbb{N}$ (resp. $\mathbb{Z}$) is the initial abelian monoid (resp. ring) in their respective categories, that they are free objects, and that from $\mathbb{N}$ we may construct $\mathbb{Z}$, then $\mathbb{Q}$, and from there is any other field of characteristic 0. One aspect of $\mathbb{N}$ and $\mathbb{Z}$ we have barely touched on are their *primes* (recall the Fundamental Theorem of Arithmetic, [3, $\mathbb{Z}$ is a UFD (Fundamental Theorem of Arithmetics)]). There are infinitely many primes, since for any primes $p_1, \ldots, p_k$ a prime factor of $p_1 \cdots p_k + 1$ is none of them. Since $\mathbb{Z}$ is a PID, being irreducible is equivalent to being prime. However, we have explored little the nature of primes, and more generally have explored little of functions with integer solutions (elements of $f \in \mathbb{Z}[x_1, ..., x_n]$ are called *Diophantine equations*).

Unlike fields, it is in general much harder to ascertain whether an algebraic equation has solutions over the integers (or even integral extensions of the integers). Perhaps the most famous number theory problem demonstrating this difficulty is the following:

Show that for $n \geq 3$, the equation $x^n + y^n = z^n$ has no integer solutions

This problem is known as *Fermat's Last Theorem*. This is trivial if $n = 1$ (ex. $1 + 2 = 3$). It is more interesting to see the case of $n = 2$, that is we want to find all $(a, b, c) \in \mathbb{Z}^3$ ($\gcd(a, b, c) = 1$) such that $x^2 + y^2 = z^2$. Such a triple is called a *Pythagorean triple* and there are infinitely many. However, for $n \geq 3$, it is incredibly more difficult to find whether solutions exist or not, partly due to the fact that many rings that naturally arise in solving this case are *not* unique factorization domains, and partly because there turn out to be some "bad number" that make many of the number theory strategies fail. This particular problem has spurred a lot of the development of modern algebraic and analytic number theory, and shall be touched on later due to some interesting general theory

that came from it. In this section we shall cover the mathematics that build built-up in solving Diophantine equations, including some special cases of Fermat's last theorem, and prepare the reader for Class Field theory which shall definitively answer the question of factoring integer-polynomials mod primes[1].

One final intuition that shall become more evident is that the theory of prime factorization in field extensions tie into solving irreducible polynomials mod some prime ideals. This theory extends results from the classical results of solutions or lack thereof for an irreducible polynomial over $\mathbb{Q}$ when considering it mod p for all primes $p \in \mathbb{Z}$. Recall that factorization and irreducibility of monic polynomials over $\mathbb{Q}$, more generally a finite extension $K/\mathbb{Q}$ is exactly reflected to the factorization and irreducibility of monic polynomials over $\mathbb{Z}$ (which will have a ring $\mathcal{O}_K$ that shall be the equivalent[2]) by Gauss's lemma ([3, Gauss's Lemma]). This was the historical origin of number theory, as it is much easier to work with polynomials finite fields (recall [3, Reduction Test] and [3, Limitation of reduction method]).

Material Overview

In this chapter, we shall start by exploring when do $\mathbb{Z}$-polynomials solutions over a field k. The space of solutions shall naturally be a subset of the field k. It shall be labeled as $\mathcal{O}_K$ and called *rings of integers*, and will have many important properties to determine. Similarly to how we extended fields to find solutions to polynomials over fields, rings of integers shall be extended (in parallel to a field extension) to find more solutions. Some key functionals such as the trace, norm, and discriminant shall become important tools to find properties of elements of $\mathcal{O}_K$ and the structure of $\mathcal{O}_K$ itself. As we classify $\mathcal{O}_K$, we shall see that we may consider $\mathcal{O}_K$ as a *lattice* when embedded in $\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ for appropriate $r_1, r_1 \in \mathbb{N}$. This geometric interpretation of $\mathcal{O}_K$ shall add some important new perspectives onto the elements of $\mathcal{O}_K$[3], and will allow us to define some important inequalities.

After having done a sweep over the structure of $\mathcal{O}_K$, we shall tackle some of it's algebraic structure. Namely, we shall see that they form a new type of ring called a *Dedekind domain*. Dedekind domains are very important types of rings. For those who are keen on algebraic geometry, Dedekind domains are the algebraic counter-part to 1-dimensional smooth curves (or schemes). In fact, we have seen Dedekind domains before in the previous chapter (for example, $\mathbb{Q}[x, y]/(x - y^2)$). Hence, Dedekind domains will be studied in their own right. They shall have a very rich structure: for example, many $\mathcal{O}_K$ will *not* be unique factorization domain (a fact that led to many wrong proofs of Fermat's last theorem), however we shall be able to define an abelian group structure on the ideals of $\mathcal{O}_K$ which *will* have a unique factorization (in the case where $\mathcal{O}_K$ is a PID, we shall see the ideal group I_K is isomorphic to $\mathbb{Q}^\times$, exactly mirroring the structure of Frac($\mathbb{N}$), however in general it may be much more complicated[4]). The discussion on Dedekind domains shall finish with a discussion of the unit's in Dedekind rings, which shall be important for finding the factorization of ideals.

After we realise we ought to be focusing on the *prime ideals* of $\mathcal{O}_K$ rather than the irreducible elements of $\mathcal{O}_K$ (for they are not necessarily prime), we shall start exploring how prime may further decompose as we extend to larger rings of integers. This helps in finding solutions to $\mathbb{Z}$-polynomials for the prime ideals in larger rings will have very rigid structure that can be taken advantage of. We have already seen this before: It shall turn out that $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$ (the Gaussian integers). In the

[1] In particular, the *Artin Reciprocity Law* shall give the over-arching result showing how to factorize Diophantine equations

[2] See definition 50.1.3

[3] and of the norm

[4] which is measured by the *class group*, as we shall see

Gaussian integers, we saw that 2 is not prime, for

$$2 = (1+i)(1-i) = 1^2 - i^2 = 1 + 1$$

or in terms of ideals, we shall have $(2) = (1+i)(1-i)$. Hence, 2 has further factored into new primes. In [3, Application: Sums of Primes and Gaussian Integers], this was used to prove Fermat's Theorem for sums of squares. More generally, we shall see that any prime ideal $\mathfrak{p} \subseteq \mathcal{O}_K$ may factor:

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_k^{e_k} \tag{1.1}$$

Though it may currently seem unrelated, the study of the factorization of primes *almost* mirrors that of the factorization of irreducible polynomials mod $\mathfrak{p}$ after extending fields. In fact, it shall turn out to be the case that *all but finitely many primes* shall factor exactly like how a polynomial factors, the "bad primes" essentially being primes for which an exponent in equation (50.1) is greater than 1, ≥ 2. This sort of exploration leads to a lot of very interesting theory that have applications beyond classical number theory, for example we shall be able to fully classify all abelian extensions of $\mathbb{Q}$.

For those who get motivation through algebraic geometry, algebraic number theory naturally ties into it. It may help to think of the main object of study of this chapter (rings of integers) as further expanding the dictionary between commutative algebra and geometry, namely they shall be important for *normal schemes* which links to the notion of a smooth curve[5]

At the end of this chapter, we shall briefly touch on some analytic number theory, for it's study is central to modern number theory and the author of this books considers that anyone that has reached this stage of the book would find it enlightening to see how the Riemann-zeta function play an important role in number theory.

1.0.1 Warning Some elementary analysis will be assumed in this section, including some properties of the real numbers.

1.1 Ring of Integers

In this chapter, whenever the notation $L/K/\mathbb{Q}$ is used for some letter's L, K, it is implied that L/K are both finite extensions of $\mathbb{Q}$, and L is an extension of K. Unless otherwise specified, the characteristic of any field is implicitly assumed to be 0. We start by defining our terms.

Definition 1.1.1: Algebraic Number Fields

Let K be a field. If $K/\mathbb{Q}$ is a finite extension of $\mathbb{Q}$, then K is called an *algebraic number field* or *number field* if unambiguous.

Example 1.1.2: Number Fields

1. Let $\zeta_n = e^{2\pi i/n}$ be a root of unity. Then $\mathbb{Q}(\zeta_n)$ is a number field, called the *Cyclotomic field*.
2. Let $m \in \mathbb{Z}$ be not-a-square. Then Fields of the form $\mathbb{Q}(\sqrt{m})$ is called a *quadratic field*. Such fields always have degree 2. Notice that we can consider any $m \in \mathbb{Z}$, however we get many

[5] See a textbook in algebraic geometry for further details

equality of fields, for example $\mathbb{Q}(\sqrt{12}) = \mathbb{Q}(\sqrt{3})$. The field $\mathbb{Q}(i)$ is both a quadratic field and a Cyclotomic field since $i = \zeta_4$. As an exercise, show that $\mathbb{Q}(\sqrt{-3})$ is a Cyclotomic field.

3. The field $\mathbb{Q}(\sqrt{n}, \sqrt{m})$ is called a *biquadratic* field where n, m are both not-a-square.
4. The field $\mathbb{R}$ and $\mathbb{C}$ are *not* algebraic number fields, are they are not algebraic extensions of $\mathbb{Q}$ (they both contain transcendental elements).
5. Similarly, $k(x)$ for any field k is not an algebraic number field (it is an [algebraic] function field). However, in many ways this field is very similar to algebraic number fields, and shall be given some more attention in section 50.2.

Of key interest to this chapter the study of integer polynomials with solutions fields:

Definition 1.1.3: Ring of Integers (of Number Fields)

Let K be a field of characteristic 0 and identify $\mathbb{Z} \subseteq K$ via the (unique) map $\mathbb{Z} \hookrightarrow K$. Then the *ring of integers* of K is the integral closure of $\mathbb{Z}$ over K ([2, Integral Elements]):

$$\mathcal{O}_K := \overline{\mathbb{Z}}^K = \{a \in K \mid f(a) = 0 \text{ for some monic } f(x) \in \mathbb{Z}[x]\}$$

As all fields in consideration shall be number fields $K/\mathbb{Q}$, there is already $\mathbb{Z} \subseteq K$ identified. The generality in the definition is to show that the ring of integers is defined over any field of characteristic 0. Note that there is an equivalent definition for fields of positive characteristic, however as $\text{im}(\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ any integral closure of $\mathbb{Z}/p\mathbb{Z}$, $\overline{\mathbb{Z}/p\mathbb{Z}}^K$, is a field extension by [2, Integral Extension over Fields] and so there is no notion of prime ideals which essentially erases the key extra information that is studied. Instead, $\mathbb{F}_p[t]$ becomes the new $\mathbb{Z}$ and a parallel theory of finite extension $K/\mathbb{F}_p(t)$ is studied. These are called *algebraic function fields* and the corresponding $\overline{\mathbb{F}_p[t]}^K$ is the *ring of integers of function fields.*. The two theories have many parallels, but exploring these is beyond the scope of this book, see [8] for more details.

Although, once this chapter becomes it's own book, it may be a good idea to dedicate a section or chapter to this.

By [2, Integral Elements Form Rings], $\mathcal{O}_K \subseteq K$ is a ring. Since we will often work over $\mathbb{Z}$, it is useful to know that instead of taking all roots of polynomials, it is equivalent to take all the roots of *irreducible polynomials*. By using Gauss's lemma, if $f = gh$, $f \in \mathbb{Z}[x]$ and $g, h \in \mathbb{Q}[x]$, then in fact $g, h \in \mathbb{Z}[x]$. Hence, the smallest degree polynomial for which α is a root will be irreducible. An important ring of integer that is worth isolating is the following:

Definition 1.1.4: Algebraic Integers [of Number Fields]

The collection of elements

$$\mathcal{O}_{\overline{\mathbb{Q}}} = \mathcal{O}_{\mathbb{C}} := \overline{\mathbb{Z}}^{\mathbb{C}}$$

are called the *algebraic integers*, and is sometimes denoted $\mathbb{A}$[a].

[a] this notation comes from the theory of *idele rings* that is an important tool in class field theory, see [8]

Just like for fields, there may be multiple "integral closures", and they are all isomorphic which motivates saying they are *the* algebraic integers.

Lemma 1.1.5: Equivalence Definition

Let $K/\mathbb{Q}$ be a finite extension and let $\mathcal{O}_{\mathbb{C}}$ be the algebraic integers. Then:

$$\mathcal{O}_{\mathbb{C}} \cap K = \mathcal{O}_K$$

***Proof*:**
The statement identifies K with a subfield of $\mathbb{C}$. By definition 50.1.3 and definition 50.1.4, both $\mathcal{O}_K$ and $\mathcal{O}_{\mathbb{C}}$ consist of the roots, in K and in $\mathbb{C}$ respectively, of monic polynomials in $\mathbb{Z}[x]$. The defining condition mentions only the element and $\mathbb{Z}[x]$, never the ambient field. Hence if $a \in \mathcal{O}_K$, then a is a root of a monic polynomial in $\mathbb{Z}[x]$ and $a \in K \subseteq \mathbb{C}$, so $a \in \mathcal{O}_{\mathbb{C}} \cap K$. Conversely, if $a \in \mathcal{O}_{\mathbb{C}} \cap K$, then $a \in K$ and a is a root of a monic polynomial in $\mathbb{Z}[x]$, so $a \in \mathcal{O}_K$.

Example 1.1.6: Ring of Integers

1. $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$. For this reason, the integers in algebraic number theory are sometimes called the rational integers

2. $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$. To see this, let $\alpha = a + bi$, $a, b \in \mathbb{Q}$. Then for $\alpha \in \mathcal{O}_{\mathbb{Q}(i)}$, we need to show that the minimal polynomial of α:

   $$\alpha^2 - 2a\alpha + (a^2 + b^2) = 0$$

   has integer coefficients, that is $-2a$ and $a^2 + b^2$ are integers. First, if $c, d \in \mathbb{Z}$, $c + di \in \mathcal{O}_{\mathbb{Q}(i)}$. Since $\mathcal{O}_{\mathbb{Q}(i)}$ is a ring, if $a + bi \in \mathcal{O}_{\mathbb{Q}(i)}$, then so is $a + bi - c - di = a' + b'i$ for $c, d \in \mathbb{Z}$ where $a', b' \leq 1$. But then from $(a')^2 + (b')^2 = k \in \mathbb{Z}$, we get $a', b' \in \{-1, 0, 1\}$. Then that implies $a, b \in \mathbb{Z}$, showing that $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$.

3. In general, it is difficult to find the ring of integers. The difficulty of finding the ring of integers can even show up for quadratic extensions. Consider $F = \mathbb{Q}(\sqrt{-3})$. Then we might be tempted to say that $\mathcal{O}_F = \mathbb{Z}[\sqrt{-3}]$. However, this is not the case, take:

   $$\omega = \frac{-1 + \sqrt{-3}}{2} \in \mathbb{Q}(\sqrt{-3})$$

   Then $\omega^2 + \omega + 1 = 0$, however $\omega \notin \mathbb{Z}[\sqrt{-3}]$.

4. In general, more tools are needed to find $\mathcal{O}_k$, but in the quadratic case the polynomials are simple enough to find all rings of integers. Let $k = \mathbb{Q}(\sqrt{m})$ with $m \neq 0, 1$ square-free. Then:

   $$\mathcal{O}_k = \mathbb{Z}[\sqrt{m}] \qquad m \equiv 2, 3 \bmod 4$$
   $$\mathcal{O}_k = \mathbb{Z}\left[\frac{1 + \sqrt{m}}{2}\right] \qquad m \equiv 1 \bmod 4$$

   The computation uses the trace and norm of $r + s\sqrt{m}$, and is carried out in example 50.1.17 once these are available.

The ring of integers $\mathcal{O}_K/\mathbb{Z}$ hold the core information of $K/\mathbb{Q}$. In particular, if $\mathbf{Fld}_0^{\text{fin}}$ is the category

whose objects are finite extensions of $\mathbb{Q}$ and morphisms are field morphisms and $\mathbf{AlgInt}_0^{\text{fin}}$ is the category whose objects are the rings of integers $\mathcal{O}_K$ of finite extensions $K/\mathbb{Q}$ and whose morphisms are ring homomorphisms, these two categories are *equivalent*, and the theorem below proves it.

First, if $\varphi : F \to L$ be a field homomorphism (not necessarily a k-homomorphism), then $\varphi|_{\mathcal{O}_F} : \mathcal{O}_F \to \mathcal{O}_L$ is certainly well defined since if $\alpha \in \mathcal{O}_F$, then:

$$a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1} + \alpha^n = 0 \qquad a_i \in \mathbb{Z}$$

Then $\varphi(a_i\alpha^i) = a_i\varphi(\alpha)^i$, since $a_i \in \mathbb{Z}$. Hence $\varphi(\alpha) \in \mathcal{O}_L$. From this, define $\mathcal{O}_- : \mathbf{Fld}_0^{\text{fin}} \to \mathbf{AlgInt}_0^{\text{fin}}$ which gives the commutative diagram:

$$\begin{array}{ccc} F & \xhookrightarrow{\varphi} & L \\ \downarrow{\scriptstyle \mathcal{O}_-} & & \downarrow{\scriptstyle \mathcal{O}_-} \\ \mathcal{O}_F & \xrightarrow{\varphi|_{\mathcal{O}_F}} & \mathcal{O}_L \end{array}$$

Now given a map $\psi : \mathcal{O}_F \to \mathcal{O}_K$, can we find a map $\Psi : F \to K$ which is furthermore an inverse to the above functor? Yes. To find this functor, we first find how to retrieve the field from the ring of integers:

Proposition 1.1.7: Field of Fraction of Integral Closures of Extensions

Let R be an integral domain, $K = K(R)$ its field of fraction, and $S = \overline{R}^L$ where L/K is an algebraic extension. Then for every $\alpha \in L$ there is a nonzero $d \in R$ with $d\alpha \in S$. Consequently $L = K(S)$. Equivalently, $L = \langle S \rangle_{K(R)}$.

If $R = \mathbb{Z}$ and $K(R) = \mathbb{Q}$ so that $S = \overline{\mathbb{Z}}^L = \mathcal{O}_L$, then $L = K(\mathcal{O}_L)$.

***Proof*:**
Let R, K, S and L be defined as in the question. We shall first show that if $\alpha \in L$ is algebraic over K, then there exists a $d \in R$ such that $d\alpha$ is integral over R. Since α is algebraic over K,

$$\alpha^m + a_1\alpha^{m-1} + \cdots + a_m = 0 \qquad a_i \in K$$

Let $d \neq 0$ be the product of denominators of the a_i, so that $da_i \in R$ for all i. Then multiplying the above equation by d^m, we get:

$$d^m\alpha^m + d^m a_1\alpha^{m-1} + \cdots + d^m a_m = 0 \qquad a_i \in K$$

re-writing:

$$(d\alpha)^m + da_1(d\alpha)^{m-1} + \cdots + d^m a_m = 0 \qquad a_i \in K$$

where each $a_i d, ..., a_m d^m \in R$, hence $d\alpha$ is integral over R. Note that this immediately gives the equivalence $L = \langle S \rangle_{K(S)}$.

With that, to see that $L = K(S)$, notice that every $\alpha \in L$ can be written as $\alpha = b/d$ for $d \in R$, hence $L \subseteq K(S)$. Conversely, if $a/b \in K(S)$, where $a, b \in S$ and each satisfy an R-polynomial, then they are certainly satisfy a K-polynomial, and furthermore S is all elements of L that satisfy an R-polynomial, hence $a/b \in L$, giving equality.

Corollary 1.1.8: Basis For Number Field

Let $F/\mathbb{Q}$ be a finite extension of degree n and $(\alpha_1, \alpha_2, ..., \alpha_n)$ an $\mathbb{Q}$-basis for F. Then there exists an $m_i \in \mathbb{Z}$ such that $m_1\alpha_1, m_2\alpha_2, ..., m_n\alpha_n \in \mathcal{O}_F$

***Proof*:**
let $R = \mathbb{Z}$ in proposition 50.1.7.

Theorem 1.1.9: Equivalent Categories: Fields and Rings of Integers

Let $K/\mathbb{Q}$ be a number field and $\mathcal{O}_K$ the ring of integers of K. Then:

1. If L/K, then $\mathcal{O}_L \cap K = \mathcal{O}_K$. In particular $\mathcal{O}_K \cap \mathbb{Q} = \mathbb{Z}$.
2. $\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Q} \cong K$, by $\beta \otimes q \mapsto q\beta$.
3. For number fields F and L, restriction $\varphi \mapsto \varphi|_{\mathcal{O}_F}$ is a bijection from the field homomorphisms $F \to L$ to the ring homomorphisms $\mathcal{O}_F \to \mathcal{O}_L$.
4. For subfields K, K' of a number field L, $\mathcal{O}_{K\cap K'} = \mathcal{O}_K \cap \mathcal{O}_{K'}$.

Hence $F \mapsto \mathcal{O}_F$, $\varphi \mapsto \varphi|_{\mathcal{O}_F}$ is an equivalence of categories $\mathbf{Fld}_0^{\text{fin}} \to \mathbf{AlgInt}_0^{\text{fin}}$ ([3, Equivalence of Categories]), with quasi-inverse $(-) \otimes_{\mathbb{Z}} \mathbb{Q}$.

***Proof*:**
Step 1: proof of (1). An element of $\mathcal{O}_L \cap K$ is an element of K that is a root of a monic polynomial in $\mathbb{Z}[x]$, which is the defining property of $\mathcal{O}_K$ (definition 50.1.3), so $\mathcal{O}_L \cap K = \mathcal{O}_K$. Taking $K = \mathbb{Q}$ and $L = K$ gives $\mathcal{O}_K \cap \mathbb{Q} = \mathcal{O}_{\mathbb{Q}}$. Each $a \in \mathbb{Z}$ is a root of $x - a$, so $\mathbb{Z} \subseteq \mathcal{O}_{\mathbb{Q}}$. Conversely, an element of $\mathcal{O}_{\mathbb{Q}}$ is a rational number integral over $\mathbb{Z}$. The ring $\mathbb{Z}$ is a UFD ([3, $\mathbb{Z}$ is a UFD (Fundamental Theorem of Arithmetics)]), hence integrally closed in its fraction field $\mathbb{Q}$ ([2, A Unique Factorization Domain Is Integrally Closed]), so the element lies in $\mathbb{Z}$. Hence $\mathcal{O}_K \cap \mathbb{Q} = \mathbb{Z}$.

Step 2: proof of (2). With $D = \mathbb{Z} \setminus \{0\}$, [2, Characterization of Localization] gives $\mathcal{O}_K[D^{-1}] \cong D^{-1}\mathbb{Z} \otimes_{\mathbb{Z}} \mathcal{O}_K = \mathbb{Q} \otimes_{\mathbb{Z}} \mathcal{O}_K$. As $\mathcal{O}_K \subseteq K$ has no nonzero element killed by a nonzero integer, its localization is the subring $\{\beta/d \mid \beta \in \mathcal{O}_K, d \in D\}$ of K, and this is all of K because every $\alpha \in K$ has $d\alpha \in \mathcal{O}_K$ for some $d \in D$ (proposition 50.1.7 with $R = \mathbb{Z}$). Composing gives $\beta \otimes q \mapsto q\beta$.

Step 3: proof of (3). A field homomorphism $\varphi\colon F \to L$ fixes $\mathbb{Z}$, so it carries a monic integer equation of $\alpha \in \mathcal{O}_F$ to one of $\varphi(\alpha)$, and $\varphi(\mathcal{O}_F) \subseteq \mathcal{O}_L$. Restriction is injective: if $\varphi|_{\mathcal{O}_F} = \varphi'|_{\mathcal{O}_F}$ and $\alpha \in F$, choose $0 \neq d \in \mathbb{Z}$ with $d\alpha \in \mathcal{O}_F$ (proposition 50.1.7). Then $d\varphi(\alpha) = \varphi(d\alpha) = \varphi'(d\alpha) = d\varphi'(\alpha)$. For surjectivity, let $\psi\colon \mathcal{O}_F \to \mathcal{O}_L$ be a ring homomorphism, so $\psi(n) = n$ for $n \in \mathbb{Z}$. Let $0 \neq \beta \in \mathcal{O}_F$ and let $x^m + a_{m-1}x^{m-1} + \cdots + a_0 \in \mathbb{Z}[x]$ be monic of least degree with root β. Then $a_0 \neq 0$, since otherwise dividing by x gives a smaller degree, and $a_0 = -\beta\gamma$ with $\gamma = \beta^{m-1} + \cdots + a_1 \in \mathcal{O}_F$ ([2, Integral Elements Form Rings]). So $a_0 = \psi(a_0) = -\psi(\beta)\psi(\gamma)$ is nonzero in L, and $\psi(\beta) \neq 0$. Thus ψ sends $\mathcal{O}_F \setminus \{0\}$ into $L^\times$, and as F is the fraction field of $\mathcal{O}_F$ (proposition 50.1.7), [2, Universal Property of Localization] extends ψ to a ring homomorphism

$\Psi\colon F \to L$ with $\Psi|_{\mathcal{O}_F} = \psi$.

Step 4: proof of (4). By (1) applied to $L/K \cap K'$, L/K and L/K', $\mathcal{O}_{K\cap K'} = \mathcal{O}_L \cap K \cap K' = (\mathcal{O}_L \cap K) \cap (\mathcal{O}_L \cap K') = \mathcal{O}_K \cap \mathcal{O}_{K'}$.

Step 5: the equivalence. By (3), $\mathcal{O}_{(-)}$ is a functor, and so is $G = (-) \otimes_{\mathbb{Z}} \mathbb{Q}$, with $G(\psi) = \psi \otimes \mathrm{id}_{\mathbb{Q}}$. The isomorphisms $\epsilon_F\colon \mathcal{O}_F \otimes_{\mathbb{Z}} \mathbb{Q} \to F$ of (2) are natural: for $\varphi\colon F \to L$, both $\varphi \circ \epsilon_F$ and $\epsilon_L \circ (\varphi|_{\mathcal{O}_F} \otimes \mathrm{id}_{\mathbb{Q}})$ send $\beta \otimes q$ to $q\varphi(\beta)$. So $G \circ \mathcal{O}_{(-)} \cong \mathrm{id}$. For an object $B = \mathcal{O}_F$, the field isomorphism ϵ_F restricts by Step 3 to a ring isomorphism $\mathcal{O}_{G(B)} \to \mathcal{O}_F = B$, natural in B by the naturality of ϵ, so $\mathcal{O}_{(-)} \circ G \cong \mathrm{id}$, completing the proof.

Note that finiteness is important: it shall be shown that $\mathcal{O}_K$ for $K/\mathbb{Q}$ a finite extension is a domain with factorization (or atomic domain, recall [3, Atomic Domains]), however $\mathcal{O}_{\mathbb{C}}$ is *not* an atomic domain as shown in [3, Domain with no Factorization] which is certainly desirable as we are generalizing the factorization of primes.

This shows how to relate $\mathcal{O}_K$ with K and a bit about how to move down the lattice, but we have not said much about the structure of $\mathcal{O}_K$. We know it is a ring, but is it a PID? A UFD? Every ring is a $\mathbb{Z}$-module; it is a finitely generated $\mathbb{Z}$-module? Noetherian $\mathbb{Z}$-module? Projective? The strategy in example 50.1.6 to find the structure of $\mathcal{O}_K$ gets much harder for more complicated roots of irreducible, even once we have degree 3, and especially beyond degree 4. Our first goal is to establish the structure of $\mathcal{O}_K$. To find the structure, we will require some tools for detecting elements of $\mathcal{O}_K$: we shall find numerical values for the elements of K to determine whether they are elements of $\mathcal{O}_K$, i.e. we shall find functionals. We shall see that $\mathcal{O}_K$ will be a free $\mathbb{Z}$-module of finite rank, in fact it shall be a particular $\mathbb{Z}$-module known as a *complete* lattice, as shall be demonstrated in section 50.1.4.

1.1.1 Functionals: Field Trace and Field Norms

(this link is useful: here)

Let M/k be a finite extension, and take $\alpha \in M$. We shall define two invariances on α that will detect whether α is algebraic over k or not, detect when it is an algebraic integer, and eventually help us find the structure of $\mathcal{O}_M$ as well as the factorization of elements of $\mathcal{O}_M$.

Define $L_\alpha : M \to M$ by $m \mapsto \alpha m$. This is a k-linear transformation, since:

$$L_\alpha(a + kb) = \alpha(a + kb) = \alpha a + k\alpha b = L_\alpha(a) + kL_\alpha(b)$$

Thus, we can represent this linear transformation in a matrix form (usually with basis given by α). Since the trace and determinant are basis independent calculations, it is meaningful to ponder what the trace/determinant of this transformation and what information it may provide us:

Definition 1.1.10: Trace And Norm

Let M/k be a finite extension, $L_\alpha : M \to M$ be a linear transformation defined by multiplication Then we call the trace of this linear transformation the *trace* of α and the determinant of this linear transformation the *norm* of α.

Example 1.1.11: Trace And Norm

1. Let $\mathbb{C}/\mathbb{R}$ be the extension and $\alpha = x + iy$. Pick basis $\{1, i\}$. Then multiplication is:

$$\begin{pmatrix} x & y \\ -y & x \end{pmatrix}$$

and so the trace is $2x = 2\text{Re}\alpha$ and the norm is $x^2 + y^2 = \|\alpha\|^2$. If you know some complex analysis, then we see that the norm does indeed match with our usual understanding of the norm, and the trace can be thought of as a generalization of taking the real part of the complex number (up to a factor).

2. Let $\alpha = a + b \cdot \left(\frac{1+\sqrt{-7}}{2}\right) \in \mathbb{Z}\left[\frac{1+\sqrt{-7}}{2}\right]$. Then:

$$\text{Nm}_{\mathbb{Q}(\sqrt{-7})/\mathbb{Q}}(\alpha) = a^2 + ab + 2b^2$$

3. Let M/k be a finite extension and $\alpha \in M$, with $d = [k(\alpha) : k]$ and $s = [M : k(\alpha)]$, and let

$$m_{\alpha,k}(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$$

be the minimal polynomial. If $M = k(\alpha)$, it equals the characteristic polynomial χ of L_α: by [3, Cayley-Hamilton Theorem] $\chi(L_\alpha) = 0$, applying this map to 1 gives $\chi(\alpha) = 0$, so $m_{\alpha,k}$ divides χ, and both are monic of degree d. Then [3, Properties Of The Characteristic Polynomial] shows that the trace of α is $-a_{d-1}$ and the norm is $(-1)^d a_0$. In general, a basis $\beta_1, \ldots, \beta_s$ of M over $k(\alpha)$ splits $M = \bigoplus_j k(\alpha)\beta_j$ into s subspaces stable under L_α, on each of which L_α acts as on $k(\alpha)$. So the matrix of L_α is block diagonal with s equal blocks, $\chi = m_{\alpha,k}^s$ by [3, Determinant of a Block Diagonal Matrix], and

$$\text{tr}_{M/k}(\alpha) = -sa_{d-1} \qquad \text{Nm}_{M/k}(\alpha) = \left((-1)^d a_0\right)^s$$

Naturally, if F/k, $a_1, a_2 \in F$ and $b \in k$:

$$\text{tr}_{F/k}(a_1 + a_2) = \text{tr}_{F/k}(a_1) + \text{tr}_{F/k}(a_2) \quad \text{tr}_{F/k}(ba_1) = b\text{tr}_{F/k}(a_1) \quad \text{tr}_{F/k}(a) = na$$

and:

$$\text{Nm}_{F/k}(a_1a_2) = \text{Nm}_{F/k}(a_1)\text{Nm}_{F/k}(a_2) \quad \text{Nm}_{F/k}(ba_1) = b^n\text{Nm}_{F/k}(a_1) \quad \text{Nm}_{F/k}(b) = b^n$$

Most of the properties of the trace and norm come from linear algebra, and furthermore since every finite separable extension has a primitive element generating the entire field, the norm and trace always show up in the minimal polynomial. In [3, Symmetric Functions], we saw another way of characterizing the trace and norm, which will be emphasized here due to its importance:

Proposition 1.1.12: Trace and Norm via Galois Conjugates

Let M/k be a finite extension where $[M:k]=n$ and let $\alpha \in M$ be algebraic over k. Then, listing each root of $m_{\alpha,k}$ in $\overline{k}$ as often as its multiplicity:

$$\mathrm{tr}_{M/k}(\alpha) = [M:k(\alpha)] \cdot \sum_{\substack{r \\ \text{root of } m_{\alpha,k}}} r \qquad \mathrm{Nm}_{M/k}(\alpha) = \left(\prod_{\substack{r \\ \text{root of } m_{\alpha,k}}} r \right)^{[M:k(\alpha)]}$$

***Proof*:**

Let $d=[k(\alpha):k]$, $s=[M:k(\alpha)]$ and $m_{\alpha,k}(x)=x^d+a_{d-1}x^{d-1}+\cdots+a_1x+a_0$. Over $\overline{k}$ write $m_{\alpha,k}(x)=\prod_{j=1}^{d}(x-r_j)$, each root repeated according to its multiplicity. Expanding the product and comparing coefficients gives

$$a_{d-1} = -\sum_{j=1}^{d} r_j \qquad a_0 = (-1)^d \prod_{j=1}^{d} r_j$$

Step 1: the case $M=k(\alpha)$. The elements $1,\alpha,\ldots,\alpha^{d-1}$ form a k-basis of $k(\alpha)$. Multiplication by α sends α^i to α^{i+1} for $i<d-1$, and sends α^{d-1} to $\alpha^d=-a_0-a_1\alpha-\cdots-a_{d-1}\alpha^{d-1}$. So its matrix in this basis is the companion matrix C of $m_{\alpha,k}$ ([3, Companion Matrix]): 1's on the subdiagonal, last column $(-a_0,\ldots,-a_{d-1})^t$, zeros elsewhere. Its diagonal entries are 0 except the last one, $-a_{d-1}$, so $\mathrm{tr}(C)=-a_{d-1}$. In the expansion of $\det(C)$ ([3, Determinant]), every permutation other than the one sending row 1 to column d and row i to column $i-1$ for $i\geq 2$ has a factor equal to 0. That permutation is a d-cycle, of sign $(-1)^{d-1}$, so $\det(C)=(-1)^{d-1}(-a_0)=(-1)^d a_0$. By definition 50.1.10 and the display above,

$$\mathrm{tr}_{k(\alpha)/k}(\alpha) = \sum_{j=1}^{d} r_j \qquad \mathrm{Nm}_{k(\alpha)/k}(\alpha) = \prod_{j=1}^{d} r_j$$

Step 2: the general case. Let $\beta_1,\ldots,\beta_s$ be a $k(\alpha)$-basis of M. The ds elements $\alpha^i\beta_j$, for $0\leq i<d$ and $1\leq j\leq s$, span M over k, because each coefficient in $k(\alpha)$ is a k-combination of $1,\alpha,\ldots,\alpha^{d-1}$. Since $ds=[M:k]$ ([3, Multiplicity of Finite Extensions (Tower Law)]), they form a k-basis. For each j, multiplication by α maps the span of $\beta_j,\alpha\beta_j,\ldots,\alpha^{d-1}\beta_j$ into itself and acts on it by the matrix C of Step 1. Ordering the basis by j, the matrix of multiplication by α on M is block diagonal with s blocks equal to C. Hence $\mathrm{tr}_{M/k}(\alpha)=s\,\mathrm{tr}(C)$, and $\mathrm{Nm}_{M/k}(\alpha)=\det(C)^s$ by [3, Determinant of a Block Diagonal Matrix]. With Step 1 this gives the two formulas of the statement. No separability was used. Under the standing convention of chapter 50 that fields have characteristic 0, the irreducible polynomial $m_{\alpha,k}$ is separable ([3, Separability Over Characteristic 0]), so its roots are distinct and each is listed once, as we sought to show.

Corollary 1.1.13: Trace and Norm Via Embeddings

Let L/k be a separable extension of degree n, an $\sigma_1, \sigma_2, ..., \sigma_n$ the embedding of L into it's Galois closure or $\overline{k}$. Then for $\alpha \in L$:

$$\mathrm{tr}_{L/k}(\alpha) = \sum_i^n \sigma_i(\alpha) \qquad \mathrm{Nm}_{L/k}(\alpha) = \prod_i^n \sigma_i(\alpha)$$

***Proof*:**

Fix an algebraic closure $\overline{k}$ of k containing L: an algebraically closed field containing L exists by [3, Union Of The Root-Adjoining Chain Is Algebraically Closed] and the chain construction before it, and its elements algebraic over k form such an algebraic closure by [3, Existence Of Algebraic Closure]. Fix the Galois closure of L/k inside $\overline{k}$. A k-embedding $L \to \overline{k}$ sends each $\beta \in L$ to a root of the minimal polynomial of β over k, which splits in the Galois closure, so the k-embeddings of L into the Galois closure and into $\overline{k}$ are the same maps. There are n of them by [3, Separable Degree] and [3, Finite Separable Extension Equivalence], since L/k is separable.

Let $\alpha \in L$, $d = [k(\alpha) : k]$ and $s = [L : k(\alpha)]$. The polynomial $m_{\alpha,k}$ is separable, as L/k is, so it has d distinct roots $r_1, \ldots, r_d$ in $\overline{k}$, and by [3, Separable Degree of Simple Extensions] the k-embeddings of $k(\alpha)$ into $\overline{k}$ are the maps $\rho_1, \ldots, \rho_d$ with $\rho_j(\alpha) = r_j$. Fix j, and regard $\overline{k}$ as an extension of $k(\alpha)$ through ρ_j. It is then an algebraic closure of $k(\alpha)$, and the k-embeddings $L \to \overline{k}$ restricting to ρ_j are exactly the $k(\alpha)$-embeddings of L into it. By [3, Separable Degree], whose count does not depend on the algebraic closure, there are $[L : k(\alpha)]_s$ of them. The extension $L/k(\alpha)$ is separable ([3, Tower Law For Separable Extensions]), so $[L : k(\alpha)]_s = s$ by [3, Finite Separable Extension Equivalence]. Each σ_i restricts to exactly one ρ_j, so the list $\sigma_1(\alpha), \ldots, \sigma_n(\alpha)$ consists of each r_j repeated s times, and

$$\sum_{i=1}^n \sigma_i(\alpha) = s\sum_{j=1}^d r_j \qquad \prod_{i=1}^n \sigma_i(\alpha) = \left(\prod_{j=1}^d r_j\right)^s$$

By proposition 50.1.12, the right-hand sides are $\mathrm{tr}_{L/k}(\alpha)$ and $\mathrm{Nm}_{L/k}(\alpha)$, as claimed.

In the notation of example 50.1.11(3), $\mathrm{tr}_{M/k}(\alpha) = -sa_{d-1}$ and $\mathrm{Nm}_{M/k}(\alpha) = ((-1)^d a_0)^s$, so both lie in k. If $\alpha \neq 0$, then $a_0 \neq 0$: a minimal polynomial with constant term 0 is divisible by x, so being irreducible it equals x, which forces $\alpha = 0$. Hence $\mathrm{Nm}_{M/k}(\alpha) \neq 0$ for $\alpha \neq 0$. The trace and norm also give a first test for whether $\alpha \in M$ belongs to $\mathcal{O}_M$. For $k = \mathbb{Q}$, integrality forces $\mathrm{tr}_{M/\mathbb{Q}}(\alpha), \mathrm{Nm}_{M/\mathbb{Q}}(\alpha) \in \mathbb{Z}$, and the same holds over any integrally closed base, which matters for intermediate extensions:

Corollary 1.1.14: Trace and Norm on Rings of Integers

Let R be an integral domain that is integrally closed in its field of fractions $K(R)$, and let $F/K(R)$ be a finite field extension. Then if $\alpha \in F$ is integral over R,

$$\mathrm{tr}_{F/K(R)}(\alpha) \in R \qquad \mathrm{Nm}_{F/K(R)}(\alpha) \in R$$

In particular, $\mathrm{tr}_{L/k}(\mathcal{O}_L) \subseteq \mathcal{O}_k$ and $\mathrm{Nm}_{L/k}(\mathcal{O}_L) \subseteq \mathcal{O}_k$ for every finite extension L/k of number fields.

***Proof*:**
Step 1: the roots of the minimal polynomial are integral. Put $K = K(R)$, let $f \in R[x]$ be monic with $f(\alpha) = 0$, and let $m_{\alpha,K}(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_0$. Since $m_{\alpha,K}$ divides f in $K[x]$, every root r of $m_{\alpha,K}$ in the algebraic closure $\overline{K}$ of proposition 50.1.12 satisfies $f(r) = 0$, so r is integral over R.

Step 2: trace and norm. List the roots of $m_{\alpha,K}$ as $r_1, \ldots, r_d$, with multiplicity, and put $s = [F : K(\alpha)]$. By proposition 50.1.12, $\mathrm{tr}_{F/K}(\alpha) = s\sum_j r_j$ and $\mathrm{Nm}_{F/K}(\alpha) = (\prod_j r_j)^s$. Sums and products of elements of $\overline{K}$ integral over R are integral over R ([2, Integral Elements Form Rings]), so both values are integral over R. Comparing coefficients in $m_{\alpha,K} = \prod_j (x - r_j)$ gives $\sum_j r_j = -a_{d-1}$ and $\prod_j r_j = (-1)^d a_0$, so both values lie in K. As R is integrally closed in K, both lie in R.

Step 3: number fields. Let L/k be a finite extension of number fields and take $R = \mathcal{O}_k$. Its field of fractions is k (proposition 50.1.7 with $R = \mathbb{Z}$). It is integrally closed in k: if $x \in k$ is integral over $\mathcal{O}_k$, then every element of $\mathcal{O}_k[x]$ is integral over $\mathcal{O}_k$ ([2, Integral Elements Form Rings]), and $\mathcal{O}_k$ is integral over $\mathbb{Z}$, so x is integral over $\mathbb{Z}$ ([2, Transitivity of Integrality]) and $x \in \mathcal{O}_k$ by definition 50.1.3. Every $\alpha \in \mathcal{O}_L$ is integral over $\mathbb{Z} \subseteq \mathcal{O}_k$, so Step 2 with $F = L$ gives $\mathrm{tr}_{L/k}(\alpha), \mathrm{Nm}_{L/k}(\alpha) \in \mathcal{O}_k$, completing the proof.

If $R = \mathbb{Z}$ and $F/\mathbb{Q}$ is a finite extension, the trace and norm thus give a necessary condition for integrality. Since we may be working with intermediate fields $M/L/k$, it is useful to know how to compute the trace and norm given information about the trace and norm of the intermediate fields:

Proposition 1.1.15: Trace and Norm w.r.t. Towers

Let $M/L/K$ be finite extensions with M/L and L/K separable. Then for $\alpha \in M$:

$$\mathrm{tr}_{M/K}(\alpha) = \mathrm{tr}_{L/K}(\mathrm{tr}_{M/L}(\alpha)) \qquad \mathrm{Nm}_{M/K}(\alpha) = \mathrm{Nm}_{L/K}(\mathrm{Nm}_{M/L}(\alpha))$$

***Proof*:**
The extension M/K is separable ([3, Tower Law For Separable Extensions]). Let Ω be an algebraically closed field containing M, which exists by [3, Union Of The Root-Adjoining Chain Is Algebraically Closed] and the chain construction before it, and replace Ω by the algebraic closure of K inside it given by [3, Existence Of Algebraic Closure]. It contains M, as M/K is algebraic, and it is an algebraic closure of L and of M as well. Let $\sigma_1, \ldots, \sigma_n$ be the K-

embeddings $L \to \Omega$ and $\tau_1, \ldots, \tau_m$ the L-embeddings $M \to \Omega$. By [3, Separable Degree] and [3, Finite Separable Extension Equivalence], $n = [L : K]$, $m = [M : L]$, and M has exactly $[M : K] = nm$ embeddings into Ω over K. Regarding Ω as an extension of L through σ_i, [3, Embedding Into Algebraic Extension] extends σ_i to a field homomorphism $\tilde{\sigma}_i \colon \Omega \to \Omega$.

By corollary 50.1.13, $\mathrm{tr}_{M/L}(\alpha) = \sum_j \tau_j(\alpha)$, an element of L, and $\mathrm{tr}_{L/K}(\beta) = \sum_i \sigma_i(\beta)$ for $\beta \in L$. Since $\tilde{\sigma}_i$ extends σ_i and is additive,

$$\mathrm{tr}_{L/K}(\mathrm{tr}_{M/L}(\alpha)) = \sum_{i=1}^{n} \tilde{\sigma}_i \left(\sum_{j=1}^{m} \tau_j(\alpha) \right) = \sum_{i,j} (\tilde{\sigma}_i \circ \tau_j)(\alpha)$$

and in the same way $\mathrm{Nm}_{L/K}(\mathrm{Nm}_{M/L}(\alpha)) = \prod_{i,j} (\tilde{\sigma}_i \circ \tau_j)(\alpha)$. Each $\tilde{\sigma}_i \circ \tau_j$ is a homomorphism $M \to \Omega$ fixing K. They are distinct: if $\tilde{\sigma}_i \tau_j = \tilde{\sigma}_{i'} \tau_{j'}$ on M, then restricting to L, where every τ_j is the identity, gives $\sigma_i = \sigma_{i'}$, so $i = i'$, and then $\tau_j = \tau_{j'}$ because $\tilde{\sigma}_i$ is injective. So the nm maps $\tilde{\sigma}_i \circ \tau_j$ are all the K-embeddings of M into Ω, and corollary 50.1.13 identifies the two right-hand sides with $\mathrm{tr}_{M/K}(\alpha)$ and $\mathrm{Nm}_{M/K}(\alpha)$, completing the proof.

The condition is not sufficient:

Example 1.1.16: Trace And Norm Insufficient To Detect Integral Elements

The polynomial $y^3 + 6y + 24$ is irreducible over $\mathbb{Q}$ by [3, Eisenstein's Criterion For $\mathbb{Z}[x]$] at the prime 3. Let γ be a root, $\alpha = \gamma/2$ and $L = \mathbb{Q}(\alpha) = \mathbb{Q}(\gamma)$, a field of degree 3. Dividing $\gamma^3 + 6\gamma + 24 = 0$ by 8 shows that α is a root of the monic polynomial $x^3 + \frac{3}{2}x + 3$ of degree $[L : \mathbb{Q}]$, so

$$m_{\alpha,\mathbb{Q}}(x) = x^3 + \frac{3}{2}x + 3$$

and example 50.1.11(3) gives

$$\mathrm{tr}_{L/\mathbb{Q}}(\alpha) = 0 \qquad \mathrm{Nm}_{L/\mathbb{Q}}(\alpha) = (-1)^3 \cdot 3 = -3$$

Both are integers, but $\alpha \notin \mathcal{O}_L$. Suppose $f(\alpha) = 0$ for a monic $f \in \mathbb{Z}[x]$. Then $f = m_{\alpha,\mathbb{Q}}\, g$ for a monic $g \in \mathbb{Q}[x]$. If g is constant, then $f = m_{\alpha,\mathbb{Q}} \notin \mathbb{Z}[x]$. Otherwise [3, Gauss's Lemma] gives nonzero $r, t \in \mathbb{Q}$ with $r m_{\alpha,\mathbb{Q}}, tg \in \mathbb{Z}[x]$ and $f = (r m_{\alpha,\mathbb{Q}})(tg)$. Comparing leading coefficients, r and t are integers with $rt = 1$, so $r = \pm 1$ and $m_{\alpha,\mathbb{Q}} \in \mathbb{Z}[x]$, which is false.

For quadratic fields the condition is sufficient, since an element of degree 2 has the trace and the norm as the coefficients of its minimal polynomial:

Example 1.1.17: Rings of Integers of Quadratic Extensions

Let $n \neq 0, 1$ be a square-free integer, $K = \mathbb{Q}(\sqrt{n})$, and $\alpha = a + b\sqrt{n}$ with $a, b \in \mathbb{Q}$. In the basis $1, \sqrt{n}$, multiplication by α has matrix $\begin{pmatrix} a & nb \\ b & a \end{pmatrix}$, so

$$\mathrm{tr}_{K/\mathbb{Q}}(\alpha) = 2a \qquad \mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = a^2 - nb^2$$

If $\alpha \in \mathcal{O}_K$, both are integers by corollary 50.1.14, $\mathbb{Z}$ being integrally closed ([2, A Unique Factorization Domain Is Integrally Closed]). Conversely, if both are integers, then α is a root of the monic polynomial $x^2 - 2ax + (a^2 - nb^2) \in \mathbb{Z}[x]$, as $(\alpha - a)^2 = nb^2$, so $\alpha \in \mathcal{O}_K$. Hence $\alpha \in \mathcal{O}_K$

if and only if $2a \in \mathbb{Z}$ and $a^2 - nb^2 \in \mathbb{Z}$.

Suppose both hold, and put $u = 2a \in \mathbb{Z}$ and $v = 2b$. Then $u^2 - nv^2 = 4(a^2 - nb^2) \in 4\mathbb{Z}$, so $nv^2 \in \mathbb{Z}$. Writing $v = r/t$ in lowest terms with $t > 0$, t^2 divides nr^2 and is coprime to r^2, so $t^2 | n$, and $t = 1$ as n is square-free. So $u, v \in \mathbb{Z}$ and $u^2 \equiv nv^2 \bmod 4$. Squares are 0 or 1 modulo 4, according as the number is even or odd.

- If $n \equiv 2, 3 \bmod 4$ and v is odd, then $nv^2 \equiv 2$ or $3 \bmod 4$, which is not a square modulo 4. So v is even, then $u^2 \equiv 0 \bmod 4$ and u is even, that is, $a, b \in \mathbb{Z}$.
- If $n \equiv 1 \bmod 4$, then $u^2 \equiv v^2 \bmod 4$, so u and v have the same parity, and $\alpha = \frac{u-v}{2} + v \cdot \frac{1+\sqrt{n}}{2}$ with $\frac{u-v}{2}, v \in \mathbb{Z}$. Conversely $\frac{1+\sqrt{n}}{2}$ has trace 1 and norm $\frac{1-n}{4} \in \mathbb{Z}$, so it lies in $\mathcal{O}_K$.

As n is square-free, $n \not\equiv 0 \bmod 4$, and the two cases give

$$\mathcal{O}_{\mathbb{Q}(\sqrt{n})} = \begin{cases} \mathbb{Z}[\sqrt{n}] & n \equiv 2, 3 \bmod 4 \\ \mathbb{Z}\left[\frac{1+\sqrt{n}}{2}\right] & n \equiv 1 \bmod 4 \end{cases}$$

where in the second case $\mathbb{Z}[\frac{1+\sqrt{n}}{2}] = \mathbb{Z} + \mathbb{Z}\frac{1+\sqrt{n}}{2}$, because $\omega = \frac{1+\sqrt{n}}{2}$ satisfies $\omega^2 = \omega + \frac{n-1}{4}$.

A final result in this section is to find the group of units of $\mathcal{O}_K$. Note that in general, if $\alpha \in \mathcal{O}_K$, then $1/\alpha \notin \mathcal{O}_K$. For example $2 \in \mathcal{O}_\mathbb{Q}$, but certainly $1/2 \notin \mathcal{O}_\mathbb{Q}$. In fact, the only units in $\mathcal{O}_\mathbb{Q}$ are ± 1. If $\mathcal{O}_K$ contains roots of unity, then certainly their inverse is in $\mathcal{O}_K$ (for if $\alpha = \zeta_n$ $1/\zeta = \zeta_n^{n-1}$). There are in general more units in $\mathcal{O}_K$, and the group of units of a ring of integers shall be fully characterized in section 50.2.4. With the norm, we have a way of detecting units:

Proposition 1.1.18: Norm And Units

Let L/K be a finite field extension. Then

$$\alpha \in \mathcal{O}_L^* \qquad \Leftrightarrow \qquad \mathrm{Nm}_{L/K}(\alpha) \in \mathcal{O}_K^*$$

In the space case where $K = \mathbb{Q}$, then $\alpha \in \mathcal{O}_L^*$ if and only if $\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) = \pm 1$.

***Proof*:**
Let $\alpha \in \mathcal{O}_L^*$ so that $\mathrm{Nm}_{L/K}(\alpha) \in \mathcal{O}_K$. If α is a unit in $\mathcal{O}_L$, then $1/\alpha \in \mathcal{O}_L$ exists. Then:

$$1 = \mathrm{Nm}_{L/K}(1) = \mathrm{Nm}_{L/K}(\alpha \cdot 1/\alpha) = \mathrm{Nm}_{L/K}(\alpha)\mathrm{Nm}_{L/K}(1/\alpha)$$

hence, $\mathrm{Nm}_{L/K}(\alpha), \mathrm{Nm}_{L/K}(1/\alpha) \in \mathcal{O}_K^\times$. Conversely, if $\mathrm{Nm}_{L/K}(\alpha) \in \mathcal{O}_K$, then there exists a β such that

$$\beta \mathrm{Nm}_{L/K}(\alpha) = 1$$

Then by definition of the norm:

$$1 = \beta \mathrm{Nm}_{L/K}(\alpha) = \beta \prod_\sigma \sigma(\alpha) = \alpha \left(\beta \prod_{\sigma \neq \mathrm{id}} \sigma(\alpha) \right) = \alpha y$$

where $y \in \mathcal{O}_L$ since each $\sigma(\alpha) \in \mathcal{O}_L$, which gives the result.

Proposition 1.1.19: Norm And Irreducible

Let $L/\mathbb{Q}$ be a finite extension and $\alpha \in \mathcal{O}_L$. Then if p is prime:

$$\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) = p \implies \alpha \text{ is irreducible}$$

***Proof*:**
Suppose α where reducible so that $\alpha = \beta\gamma$ where β, γ or not units, hence have norm greater than 1. Then:

$$\begin{aligned}\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) &= \mathrm{Nm}_{L/\mathbb{Q}}(\beta\gamma) \\ &= \mathrm{Nm}_{L/\mathbb{Q}}(\beta)\mathrm{Nm}_{L/\mathbb{Q}}(\gamma) \\ &= p\end{aligned}$$

a contradiction.

Note that the converse is not true

Example 1.1.20: Prime, But Norm Not Prime

Let $K = \mathbb{Q}(\sqrt{2})$ so that $\mathcal{O}_K = \mathbb{Z}[\sqrt{2}]$. Then $\mathrm{Nm}_{K/\mathbb{Q}}(5) = 5^2 = 25$, which is certainly not prime, however 5 is prime in $\mathbb{Z}[\sqrt{2}]$. This is at this stage a tedious calculation: assume that it factors $5 = \alpha\beta$ where $\alpha, \beta \in \mathbb{Z}[\sqrt{2}]$, $\alpha = a + bi$, $\beta = c + di$, and:

$$\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = a - 2b = 5 = c - 2d = \mathrm{Nm}_{K/\mathbb{Q}}(\beta)$$

but no such a, b, c, d exist. Later, we shall be able to deduce this almost immediately using Dedekind-Kummer's Theorem (see theorem 50.4.7).

In lemma 50.2.29, we shall give a partial converse which shall as a corollary imply:

$$\alpha \text{ prime} \implies \left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| = p^f$$

for appropriate f. Note how α must be prime, which is *stronger* than irreducible. However, we will often be working over rings where we have few prime elements but many irreducible elements. In section 50.2 we shall show how to "fix" this problem to make the norm more useful.

1.1.2 Functionals: Discriminant

As we saw, if $\alpha \in F/\mathbb{Q}$ is integral, then $\mathrm{tr}(\alpha), \mathrm{Nm}(\alpha) \in \mathbb{Z}$. However, the converse is not necessarily true as we saw in example 50.1.16. We will require a more powerful result, one that will verify that an element is an element of a ring of integers. For this, we shall bring in some multilinear algebra and define a new tool that shall become invaluable finding a *basis* for $\mathcal{O}_K$ (recall that it was mentioned that $\mathcal{O}_K$ shall be proven to be a finitely generated free $\mathbb{Z}$-module). Given a basis can be found, then an element can be determined to be integral using the basis. The tool that shall be used to detect a basis for $\mathcal{O}_K$ is called the *field discriminant*. The norm and the trace shall both be used in the definition of the field discriminant.

Some Multi-Linear Algebra

Recall from [3, Multilinear Forms] that multi-linear forms have matrix representations and so the determinant of multi-linear maps is well-defined:

Definition 1.1.21: [General] Discriminant

Let $\varphi : V \times V \to k$ be a symmetric k-bilinear map ($\varphi(x, y) = \varphi(y, x)$). Then given a basis $v_1, v_2, ..., v_n$, the *discriminant* of φ is:

$$\operatorname{disc}(\varphi) = \det \begin{pmatrix} \varphi(v_1, v_1) & \varphi(v_1, v_2) & \cdots & \varphi(v_1, v_n) \\ \varphi(v_2, v_1) & \varphi(v_2, v_2) & \cdots & \varphi(v_2, v_n) \\ \vdots & \ddots & \vdots & \\ \varphi(v_n, v_1) & \varphi(v_n, v_2) & \cdots & \varphi(v_n, v_n) \end{pmatrix}$$

and is denoted $\operatorname{disc}(\varphi)$.

Let $\{\beta_1, \beta_2, ..., \beta_n\}$ be a basis for V and $\{v_1, v_2, ..., v_n\}$ be some collection of elements so that

$$v_i = \sum_i a_i \beta_i$$

Then:

$$\varphi(v_k, v_l) = \sum_{i,j} \varphi(a_{ki}\beta_i, a_{lj}\beta_j) = a_{ki}\varphi(\beta_i, \beta_j)a_{kl}$$

hence, as matrices, we have the (component-wise) equality:

$$(\varphi(v_k, v_l)) = A \cdot (\varphi(e_i, e_j)) \cdot A^{\mathrm{tr}}$$

where $A = (a_{ij})$. Then if we have a change of basis, we have the determinant of the matrix is:

$$\det((\varphi(f_i, f_j)) = \det(A)^2 \cdot \det((\varphi(\beta_i, \beta_j))) \tag{1.2}$$

Since $A \in \mathrm{GL}(k)$, we see that the discriminant is defined up to the square of a unit. Hence, if $\operatorname{disc}(\varphi) \neq 0$, $\operatorname{disc}(\varphi)$ is well-defined in $k^\times/(k^\times)^2$ If $K = \mathbb{Q}$, then $(\mathbb{Q}^\times)^2 = 1$, hence the discriminant is well-defined in $\mathbb{Q}^\times$.

Invertible matrices allowed us to define an isomorphism between V and $V^\vee$ when V is finite dimensional. We shall require this same property for bilinear forms. For this, the notion of non-degenerate form needs to be upgraded for bilinear forms:

Definition 1.1.22: Non-degenerate Form

let φ be a bilinear form. Then φ is *non-degenerate* if φ has a non-zero discriminant

Proposition 1.1.23: Non-degenerate Form Equivalences

let φ be a symmetric bilinear form on a finite-dimensional k-vector space V. Then φ is *non-degenerate* if any of the following equivalent conditions hold:

1. φ has a non-zero discriminant
2. the left kernel is zero: $\{v \in V \mid \varphi(v,x) = 0,\ \forall x \in V\}$
3. the right kernel is zero: $\{v \in V \mid \varphi(x,v) = 0,\ \forall x \in V\}$

***Proof*:**
By definition 50.1.21, φ is symmetric, so $\varphi(v,x) = \varphi(x,v)$ for all $v, x \in V$ and the sets in (2) and (3) coincide. It remains to prove that (1) is equivalent to (3).

Fix a basis $\beta_1, \dots, \beta_n$ of V and let $B = (\varphi(\beta_i, \beta_j))$, so that $\operatorname{disc}(\varphi) = \det(B)$ computed in this basis. By (50.2), a change of basis multiplies $\det(B)$ by $\det(A)^2$ with A invertible, so whether $\det(B) = 0$ does not depend on the chosen basis. Write $v = \sum_i x_i\beta_i$. By bilinearity, $\varphi(x,v) = 0$ for all $x \in V$ exactly when $\varphi(\beta_j, v) = \sum_i \varphi(\beta_j, \beta_i)x_i = 0$ for every j, that is, exactly when $Bx = 0$ for the coordinate vector $x = (x_1, \dots, x_n)$.

If $\det(B) \neq 0$, then $\det(B)$ is a unit of k and B is invertible by [3, Invertibility And Determinants], so $Bx = 0$ forces $x = B^{-1}Bx = 0$ and the right kernel is zero. If $\det(B) = 0$, then B is not invertible, so by [3, Nonsingular Square Matrix] there exists $x \neq 0$ with $Bx = 0$, and then $v = \sum_i x_i\beta_i \neq 0$ lies in the right kernel. Hence (1) holds if and only if (3) holds, as we sought to show.

If φ is non-degenerate, it always induces an isomorphism

$$V \mapsto V^\vee \qquad v \mapsto (x \mapsto \varphi(v,x))$$

Showing that non-degenerate maps φ give a "representation" of the hom-dual. The particular symmetric k-bilinear form that will be useful in finding the rank of the ring of integers is the one built from the trace, [3, Trace of a Finite Field Extension]. In this chapter we call it the *trace pairing* and write it with the lower-case tr used throughout,

$$\operatorname{tr}_{F/k} : F \times F \to k \qquad (\alpha, \beta) \mapsto \operatorname{tr}_{F/k}(\alpha\beta)$$

Its nondegeneracy is a statement about *separability*, not about F alone: if F/k is separable then the trace pairing is nondegenerate ([3, The Trace Form of a Separable Extension Is Nondegenerate]). The hypothesis is not decorative. Over $k = \mathbb{F}_p(t)$ the extension $F = k(t^{1/p})$ has $\operatorname{tr}_{F/k} \equiv 0$, so the pairing vanishes identically. Nor can separability be dodged by arguing that $\operatorname{tr}_{F/k}(\alpha\alpha^{-1}) = \operatorname{tr}_{F/k}(1) = [F : k] \neq 0$ for $\alpha \neq 0$: that degree is computed *in* k, and it dies whenever $\operatorname{char} k \mid [F : k]$, already for $\mathbb{F}_4/\mathbb{F}_2$, which is separable and whose trace pairing is nondegenerate all the same.

Every extension we pair below is separable, so by proposition 50.1.23 the left kernel is trivial and we have an isomorphism

$$F \xrightarrow{\cong} \operatorname{Hom}_k(F,k) = F^* \qquad x \mapsto (y \mapsto \operatorname{tr}_{F/k}(xy))$$

The submodule of F that this isomorphism carries onto a prescribed integrality condition is the *integral trace dual*, defined in definition 50.1.35 once we have the ring of integers in hand to state it for.

Field Discriminant

With a general overview of discriminants, we return to detecting algebraic integers more precisely using the discriminant. First the determinant, det, is an alternating n-form, meaning $(\det)^2$ is a symmetric n-form. Note that this is the same discriminant introduced in [3, Important Functional: Resultant and Discriminant]. The equivalence shall soon be shown.

Definition 1.1.24: [Field] Discriminant

Let L/k be a finite extension of degree n, and let $\sigma_1, \sigma_2, ..., \sigma_n \in \mathrm{Emb}_k(L)$ be the k-embeddings of L in $\mathbb{C}$. Then for any n-tuple of elements $\alpha_1, \alpha_2, ..., \alpha_n \in L$, define the *discriminant* of $\alpha_1, \alpha_2, ..., \alpha_n$ to be:

$$\mathrm{disc}_{L/k}(\alpha_1, \alpha_2, ..., \alpha_n) = \det \begin{pmatrix} \sigma_1(\alpha_1) & \cdots & \sigma_1(\alpha_n) \\ \vdots & \ddots & \vdots \\ \sigma_n(\alpha_1) & \cdots & \sigma_n(\alpha_n) \end{pmatrix}^2 = \det((\sigma_i(\alpha_j))_{ij})^2$$

Notice that the square makes the discriminant independent of the ordering of the σ_i and α_j. If we chose a different set of elements $(\beta_1, \beta_2, ..., \beta_n)$ that differ from $(\alpha_1, \alpha_2, ..., \alpha_n)$ by a change of basis, that is $\vec{\alpha} = A\vec{\beta}$, then as we observed in equation 50.2, we get

$$\mathrm{disc}_{L/k}(\beta_1, \beta_2, ..., \beta_n) = \det(A)^2 \cdot \mathrm{disc}_{L/k}(\alpha_1, \alpha_2, ..., \alpha_n)$$

meaning the discriminant is not quite basis independent, meaning it is properly defined for operators in $L^\times/L^{\times 2}$ (where $L^\times$ are the units of L, since det maps invertible matrices to units, see [3, Invertibility And Determinants]).

Example 1.1.25: Field Discriminant

1. Let $K = \mathbb{Q}(\sqrt{m})$ where m is square free. Then K has a basis $A = \{1, \sqrt{m}\}$ and $B = \left\{1, \frac{1+\sqrt{m}}{2}\right\}$. If $m \equiv 2, 3 \bmod 4$, then the former is a basis for $\mathcal{O}_K$ and if $m \equiv 1 \bmod 4$, then then the latter is a basis for $\mathcal{O}_K$. Computing both, we get:

$$\mathrm{disc}_{K/\mathbb{Q}}(A) = \det \begin{pmatrix} 1 & \sqrt{m} \\ 1 & -\sqrt{m} \end{pmatrix}^2 = (-\sqrt{m} - \sqrt{m})^2 = 4m$$

$$\mathrm{disc}_{K/\mathbb{Q}}(B) = \det \begin{pmatrix} 1 & \frac{1+\sqrt{m}}{2} \\ 1 & \frac{1-\sqrt{m}}{2} \end{pmatrix}^2 = \left(\frac{-2\sqrt{m}}{2}\right)^2 = m$$

Yet another basis for $\mathbb{Q}(\sqrt{m})$ is $C = \{1, \sqrt{m}/10\}$. In that case:

$$\mathrm{disc}_{K/\mathbb{Q}}(C) = \det \begin{pmatrix} 1 & \frac{\sqrt{m}}{10} \\ 1 & -\frac{\sqrt{m}}{10} \end{pmatrix}^2 = \left(\frac{-\sqrt{m}}{5}\right)^2 = \frac{m}{25}$$

However, C is not a basis for $\mathcal{O}_K$, namely since the denominator is too big. Notice that the first two discriminant integers, while the third one is not.

2. Let $K = \mathbb{Q}(\sqrt[3]{2})$. Then $A = \{1, \sqrt[3]{2}, \sqrt[3]{4}\}$ is a basis for K, and:

$$\mathrm{disc}_{K/\mathbb{Q}}(A) = \det \begin{pmatrix} 1 & \sqrt[3]{2} & \sqrt[3]{4} \\ 1 & \sqrt[3]{2}\zeta_3 & \sqrt[3]{4}\zeta_3^2 \\ 1 & \sqrt[3]{2}\zeta_3^2 & \sqrt[3]{4}\zeta_3 \end{pmatrix}^2 = -108 = -2^2 \cdot 3^3$$

showing that the discriminant can be negative. We shall come back to seeing how this is useful for finding whether $\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{2}]$.

The following alternative form is useful for an alternate way of computing the discriminant as well as important for deducing some theoretical properties of the elements inputted into the discriminant:

Proposition 1.1.26: Discriminant Via Trace-Pairing

Let L/k be a finite separable extension of degree n, with k-embeddings $\sigma_1, \ldots, \sigma_n$ of L into $\mathbb{C}$ as in definition 50.1.24, and let $\alpha_1, \ldots, \alpha_n \in L$. Then

$$\mathrm{disc}_{L/k}(\alpha_1, \alpha_2, ..., \alpha_n) = \det(\mathrm{tr}_{L/k}(\alpha_i \alpha_j))$$

***Proof*:**
Let $S = (\sigma_i(\alpha_j))_{ij}$. The (j, l) entry of $S^t S$ is

$$\sum_{i=1}^{n} \sigma_i(\alpha_j)\sigma_i(\alpha_l) = \sum_{i=1}^{n} \sigma_i(\alpha_j \alpha_l)$$

because each σ_i is a ring homomorphism. The σ_i take values in the elements of $\mathbb{C}$ algebraic over k, which form an algebraic closure of k ([3, Existence Of Algebraic Closure]), so they are the embeddings of corollary 50.1.13, and the sum equals $\mathrm{tr}_{L/k}(\alpha_j \alpha_l)$. Hence $(\mathrm{tr}_{L/k}(\alpha_j \alpha_l))_{jl} = S^t S$, and by [3, Multiplicativity of the Determinant] and [3, Properties Of The Determinant],

$$\det(\mathrm{tr}_{L/k}(\alpha_j \alpha_l)) = \det(S^t)\det(S) = \det(S)^2 = \mathrm{disc}_{L/k}(\alpha_1, \ldots, \alpha_n)$$

completing the proof.

In particular, the above form allows for the detection of sets of algebraic integers:

Corollary 1.1.27: Detecting Algebraic Integers using Discriminant

Let K be a number field and $\alpha_1, \alpha_2, ..., \alpha_n$ a $\mathbb{Q}$-basis. Then $\mathrm{disc}_{K/\mathbb{Q}}(\alpha_1, ..., \alpha_n) \in \mathbb{Q}$, and if all $\alpha_1, \alpha_2, ..., \alpha_n$ are algebraic integers, then $\mathrm{disc}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Z}$.

***Proof*:**
we know that $\mathrm{tr}_{K/\mathbb{Q}}(\alpha_i \alpha_j) \in \mathbb{Q}$ by proposition 50.1.12 and are integers via corollary 50.1.14, hence $\mathrm{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Q}$, and if they are all algebraic integers then $\mathrm{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Z}$.

Corollary 1.1.28: Discriminant And Linear Independence

Let K be a number field of degree n and $\alpha_1, \alpha_2, ..., \alpha_n \in K$. Then $\mathrm{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) = 0$ if and only if $\alpha_1, \alpha_2, ..., \alpha_n$ are $\mathbb{Q}$-linearly dependent.

***Proof*:**

Step 1: dependence gives 0. Suppose $\sum_j c_j \alpha_j = 0$ with $c_j \in \mathbb{Q}$ and some $c_k \neq 0$. Every embedding σ_i fixes $\mathbb{Q}$, so $\sum_j c_j \sigma_i(\alpha_j) = 0$ for every i, that is, $\sum_j c_j C_j = 0$ for the columns C_j of $(\sigma_i(\alpha_j))_{ij}$. Replacing the kth column by $\sum_j c_j C_j = 0$ multiplies the determinant by c_k, by multilinearity and the alternating property ([3, Properties Of The Determinant]), and produces a matrix with a zero column, of determinant 0. So $c_k \det((\sigma_i(\alpha_j))_{ij}) = 0$, and $\mathrm{disc}_{K/\mathbb{Q}}(\alpha_1, \ldots, \alpha_n) = 0$.

Step 2: independence gives a nonzero value. Suppose the α_i are $\mathbb{Q}$-linearly independent, hence a $\mathbb{Q}$-basis of K, and that $\mathrm{disc}_{K/\mathbb{Q}}(\alpha_1, \ldots, \alpha_n) = 0$. By proposition 50.1.26, the rational matrix $T = (\mathrm{tr}_{K/\mathbb{Q}}(\alpha_i \alpha_j))_{ij}$ has determinant 0. Its rows are linearly dependent over $\mathbb{Q}$: otherwise they form a basis of $\mathbb{Q}^n$, the linear map $x \mapsto xT$ is invertible, T has an inverse matrix S, and $\det(S)\det(T) = \det(I) = 1$ ([3, Multiplicativity of the Determinant]). So there are $a_i \in \mathbb{Q}$, not all zero, with $\sum_i a_i \mathrm{tr}_{K/\mathbb{Q}}(\alpha_i \alpha_j) = 0$ for every j. Put $\alpha = \sum_i a_i \alpha_i$, which is nonzero by independence. The trace is $\mathbb{Q}$-linear, so $\mathrm{tr}_{K/\mathbb{Q}}(\alpha \alpha_j) = 0$ for every j, and then $\mathrm{tr}_{K/\mathbb{Q}}(\alpha t) = 0$ for every $t \in K$, as the α_j span K. For $t = \alpha^{-1}$ this gives $\mathrm{tr}_{K/\mathbb{Q}}(1) = n = 0$, which is false. So the discriminant is nonzero, completing the proof.

The corollary can be thought of as motivation for the name discriminant: If $K/\mathbb{Q}$ is a number field of dimension n, the discriminant will "discriminate" sets that are bases for $K/\mathbb{Q}$.

Since number fields are finite separable extensions, we always find a primitive element for a separable algebraic extension, the following theorem gives another important interpretation of the discriminant:

Proposition 1.1.29: Discriminant via Primitive Element

Let L/K be a finite separable extension of degree n with K-embeddings $\sigma_1, \ldots, \sigma_n$ of L as in definition 50.1.24, let α be a primitive element, and put $\alpha_i = \sigma_i(\alpha)$. Then $\alpha_1, \ldots, \alpha_n$ are the roots of the minimal polynomial $m_{\alpha,K}$, and

$$\mathrm{disc}_{L/K}(1, \alpha, \ldots, \alpha^{n-1}) = \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2 = (-1)^{n(n-1)/2} \mathrm{Nm}_{L/K}(m'_{\alpha,K}(\alpha))$$

where $m'_{\alpha,K}$ is the derivative of the minimal polynomial. This number is denoted $\mathrm{disc}_{L/K}(\alpha)$, the *discriminant* of α[a].

[a] By [3, Discriminant As A Product], it is the discriminant of the polynomial $m_{\alpha,K}$ in the sense of [3, Galois Groups of Polynomials]

***Proof*:**

Step 1: the roots. Each σ_i fixes the coefficients of $m_{\alpha,K}$, so $m_{\alpha,K}(\alpha_i) = \sigma_i(m_{\alpha,K}(\alpha)) = 0$. A K-embedding of $L = K(\alpha)$ is determined by the image of α, so the n distinct embeddings give n distinct roots α_i, which are all the roots of $m_{\alpha,K}$, of degree n. So $m_{\alpha,K} = \prod_i (x - \alpha_i)$.

Step 2: a Vandermonde determinant. For elements $\gamma_1, \ldots, \gamma_N$ of a field, let $W(\gamma_1, \ldots, \gamma_N)$ be the $N \times N$ matrix with uth row $(1, \gamma_u, \ldots, \gamma_u^{N-1})$. We show

$$\det W(\gamma_1, \ldots, \gamma_N) = \prod_{1 \leq u < v \leq N} (\gamma_v - \gamma_u)$$

by induction on N, the case $N = 1$ being 1. For $c = N-1, N-2, \ldots, 1$ in this order, subtract γ_1 times column c from column $c+1$. This changes no determinant ([3, Properties Of The Determinant]), and turns the entry in row u and column $c+1$ into $\gamma_u^c - \gamma_1\gamma_u^{c-1} = \gamma_u^{c-1}(\gamma_u - \gamma_1)$. The first row becomes $(1, 0, \ldots, 0)$. Subtracting the first row from every other row then makes the first column $(1, 0, \ldots, 0)^t$ and changes nothing else, leaving a block diagonal matrix with blocks (1) and the matrix with entries $\gamma_u^{c-2}(\gamma_u - \gamma_1)$ for $2 \leq u, c \leq N$. By [3, Determinant of a Block Diagonal Matrix] and multilinearity in rows,

$$\det W(\gamma_1, \ldots, \gamma_N) = \prod_{v=2}^{N} (\gamma_v - \gamma_1) \det W(\gamma_2, \ldots, \gamma_N)$$

and the induction hypothesis gives the claim.

Step 3: the product formula. By definition 50.1.24, $\mathrm{disc}_{L/K}(1, \alpha, \ldots, \alpha^{n-1})$ is the square of the determinant of $(\sigma_i(\alpha^{j-1}))_{ij} = (\alpha_i^{j-1})_{ij} = W(\alpha_1, \ldots, \alpha_n)$. Step 2 gives $\mathrm{disc}_{L/K}(1, \alpha, \ldots, \alpha^{n-1}) = \prod_{i<j}(\alpha_j - \alpha_i)^2 = \prod_{i<j}(\alpha_i - \alpha_j)^2$.

Step 4: the norm formula. By corollary 50.1.13, and since each σ_r fixes the coefficients of $m'_{\alpha,K}$,

$$\mathrm{Nm}_{L/K}(m'_{\alpha,K}(\alpha)) = \prod_{r=1}^{n} \sigma_r(m'_{\alpha,K}(\alpha)) = \prod_{r=1}^{n} m'_{\alpha,K}(\alpha_r)$$

Writing $m_{\alpha,K} = (x - \alpha_r)h_r$ with $h_r = \prod_{s \neq r}(x - \alpha_s)$, the Leibniz rule gives $m'_{\alpha,K} = h_r + (x - \alpha_r)h'_r$, so $m'_{\alpha,K}(\alpha_r) = \prod_{s \neq r}(\alpha_r - \alpha_s)$. Multiplying over r and pairing the factors $\alpha_r - \alpha_s$ and $\alpha_s - \alpha_r = -(\alpha_r - \alpha_s)$ for each of the $n(n-1)/2$ pairs $r < s$,

$$\prod_{r=1}^{n} m'_{\alpha,K}(\alpha_r) = (-1)^{n(n-1)/2} \prod_{r<s} (\alpha_r - \alpha_s)^2$$

and as $((-1)^{n(n-1)/2})^2 = 1$ this is the second equality, completing the proof.

If $L = K[\alpha]$, then we shall often write $\mathrm{disc}_{L/K}(\alpha)$ instead of $\mathrm{disc}_{L/K}(\alpha, \alpha_2, .., \alpha_n)$. The discriminant detects bases only on separable extensions:

Corollary 1.1.30: Discriminant And Separability

Let F/k be a finite extension of degree n, and for $\alpha_1, \dots, \alpha_n \in F$ put

$$D(\alpha_1, \dots, \alpha_n) = \det(\mathrm{tr}_{F/k}(\alpha_i \alpha_j))$$

the discriminant of definition 50.1.24 whenever that is defined (proposition 50.1.26). Then D is not identically zero if and only if F/k is separable. More precisely, if F/k is separable then $D(\alpha_1, \dots, \alpha_n) \neq 0$ for every k-basis $\alpha_1, \dots, \alpha_n$ of F, and if F/k is inseparable then $\mathrm{tr}_{F/k}$ and D are identically zero.

If you recall that the resultant is zero if two polynomials share a root, then this shows how the discriminant is also related to the resultant.

***Proof*:**

Step 1: separable extensions. Suppose F/k is separable and let $\alpha_1, \dots, \alpha_n$ be a k-basis of F. The trace $\mathrm{tr}_{F/k}$ of definition 50.1.10 is the trace of the multiplication map, so the bilinear form $(x, y) \mapsto \mathrm{tr}_{F/k}(xy)$ is the trace form of [3, The Trace Form of a Separable Extension Is Nondegenerate], which is nondegenerate. Its matrix $(\mathrm{tr}_{F/k}(\alpha_i \alpha_j))$ is therefore invertible ([3, Nondegeneracy and the Matrix of the Form]), and an invertible matrix T has $\det(T)\det(T^{-1}) = 1$ ([3, Multiplicativity of the Determinant]). So $D(\alpha_1, \dots, \alpha_n) \neq 0$.

Step 2: inseparable extensions. Suppose F/k is inseparable and let $\beta \in F$, $d = [k(\beta) : k]$, $s = [F : k(\beta)]$ and $m_{\beta,k}(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_0$. Expanding $m_{\beta,k}(x) = \prod_j (x - r_j)$ over $\overline{k}$ gives $\sum_j r_j = -a_{d-1}$, so proposition 50.1.12 reads $\mathrm{tr}_{F/k}(\beta) = -s\, a_{d-1}$. We show that this is 0.

If β is inseparable over k, then $m'_{\beta,k} = 0$ by [3, Separability and irreducible polynomial], and $d \geq 2$. The coefficients of x^{d-1} and x^{d-2} in $m'_{\beta,k}$ give $d \cdot 1 = 0$ and $(d-1)a_{d-1} = 0$ in k. As $(d-1) \cdot 1 = -1 \neq 0$, we get $a_{d-1} = 0$.

If β is separable over k, then $k(\beta)/k$ is separable ([3, Finite Separable Extension Equivalence]). Since F/k is inseparable and F is generated over k by a k-basis, the same result shows that some element γ of that basis is inseparable over k. If γ were separable over $k(\beta)$, then $k(\beta, \gamma)/k(\beta)$ would be separable by the same result, hence $k(\beta, \gamma)/k$ would be separable by [3, Tower Law For Separable Extensions], and γ would be separable over k. So the minimal polynomial g of γ over $k(\beta)$ is irreducible and inseparable, $g' = 0$, and the leading coefficient of g' gives $e \cdot 1 = 0$ for $e = \deg g = [k(\beta, \gamma) : k(\beta)]$. By [3, Multiplicity of Finite Extensions (Tower Law)], $s = [F : k(\beta, \gamma)]\, e$, so $s \cdot 1 = 0$ and $\mathrm{tr}_{F/k}(\beta) = 0$.

Hence $\mathrm{tr}_{F/k}$ is identically zero, every matrix $(\mathrm{tr}_{F/k}(\alpha_i \alpha_j))$ is the zero matrix, and D is identically zero, completing the proof.

Example 1.1.31: Discriminant On Cyclotomic extension

Let p be a prime, ζ_p a primitive pth root of unity, and $K = \mathbb{Q}(\zeta_p)$. Then recall from [3, Cyclotomic Polynomials and Fields] that its minimal polynomial is:

$$f(x) = 1 + x + x^2 + \cdots + x^{p-1}$$

Then to compute $f'(\zeta_p)$, recall that

$$x^p - 1 = (x-1)f(x) \implies px^{p-1} = f(x) + (x-1)f'(x) \implies f'(\zeta_p) = \frac{p}{\zeta_p(\zeta_p - 1)}$$

Taking the norm, for p odd:

$$\mathrm{Nm}_{K/\mathbb{Q}}(f'(\zeta_p)) = \frac{\mathrm{Nm}_{K/\mathbb{Q}}(p)}{\mathrm{Nm}_{K/\mathbb{Q}}(\zeta_p)\mathrm{Nm}_{K/\mathbb{Q}}(\zeta_p - 1)} = \frac{p^{p-1}}{1 \cdot p}$$

where $\mathrm{Nm}_{K/\mathbb{Q}}(\zeta_p - 1) = \prod_{k=1}^{p-1}(\zeta_p^k - 1) = (-1)^{p-1}f(1) = p$ by corollary 50.1.13, as the embeddings of K send ζ_p to the roots $\zeta_p, \ldots, \zeta_p^{p-1}$ of f, and $p-1$ is even. Hence, by proposition 50.1.29 with $n = p-1$, and since $(p-1)(p-2)/2 \equiv (p-1)/2 \bmod 2$ because $p-2$ is odd:

$$\mathrm{disc}_{K/\mathbb{Q}}(\zeta_p) = (-1)^{(p-1)(p-2)/2}\mathrm{Nm}_{K/\mathbb{Q}}(f'(\zeta_p)) = (-1)^{(p-1)/2}p^{p-2}$$

For instance $\mathrm{disc}(\zeta_3) = -3$, the discriminant of $x^2 + x + 1$. Lemma 50.4.66 computes $\mathrm{disc}(\zeta_n)$ up to sign when n is a prime power. For arbitrary n, $\mathrm{disc}(\zeta_n) \big| n^{\varphi(n)}$, where φ is the Euler totient. If f is the [monic] irreducible polynomial for ζ_n over $\mathbb{Q}$ for arbitrary $n \in \mathbb{N}$, we get that

$$x^n - 1 = f(x)g(x) \qquad g(x) \in \mathbb{Z}[x]$$

Differentiating and evaluating at ζ_n, we get $n = \zeta_n f'(\zeta_n)g(\zeta_n)$. Taking the norm, we get:

$$n^{\varphi(n)} = \pm\mathrm{disc}(\zeta_n)\mathrm{Nm}(\zeta_n g(\zeta_n))$$

which since $\mathrm{Nm}(\zeta_n g(\zeta_n)) \in \mathbb{Z}$, we get that $\mathrm{disc}(\zeta_n) \big| n^{\varphi(n)}$.

Exercise 1.1.1

1. Give a second proof of [3, The Trace Form of a Separable Extension Is Nondegenerate], independent of the Vandermonde argument used there: show that if F/K is finite and separable, then $\mathrm{tr}_{F/K}(a)$ cannot be 0 for all $a \in F$ (Hint: use Dedekind's lemma, [3, linear independence of Automorphisms (Dedekind's Lemma)]). Deduce that the bilinear form $(x, y) \mapsto \mathrm{tr}_{F/K}(xy)$ is non-degenerate.
2. (series of question for Stickelberger's theorem, that $d_K \equiv 0, 1 \bmod 4$)

1.1.3 Abelian Group Structure

We now prove that $\mathcal{O}_K$ is a free $\mathbb{Z}$-module of finite rank. From the above discussion, we can start seeing how this is useful in finding elements of $\mathcal{O}_K$: given it is finitely generated and we know the rank, we can look for generating linearly independent sets, i.e. a basis, for $\mathcal{O}_K$ (since $\mathcal{O}_K$ is a module over a PID).

We in fact can prove something slightly stronger that shall be useful later in this chapter:

Proposition 1.1.32: Generation of Ring of Integers

Let R be a normal integral domain, ($\overline{R}^{K(R)} = R$), $K = K(R)$ the field of fraction of R, F/K a separable extension of degree n, and let $S = \overline{R}^{F}$ where F/K. Then there exists free R-submodules M' of F such that if $M = R^n$,

$$M \subseteq S \subseteq M'$$

Hence, S is a finitely generated R-module if R is Noetherian, and it is free of rank n if R is a Principal Ideal Domain.

Hence, the integral closure over a larger field that is a finite extension shall only extend the integral closure by a relatively controlled manner. The more condition on R the better the control.

We are usually considering $\mathcal{O}_F$ for some number field $F/\mathbb{Q}$, the generality is to work over the cases that are less nice (as shall be done in algebraic geometry[6], see [4]). To see this result more concretely, it is worth stating the following immediate corollary:

Corollary 1.1.33: $\mathcal{O}_K$ is a Free $\mathbb{Z}$-Module

Let $F/\mathbb{Q}$ be a number field of degree n. Then $\mathcal{O}_F$ is a free $\mathbb{Z}$-module of degree n.

***Proof*:**
Of proposition 50.1.32
Let $K = K(R)$ and let $B = \{\beta_1, \beta_2, ..., \beta_n\}$ be a basis for F over K. We shall use this basis to squish S between two finitely generated R-modules (note that S is an R-module since S/R is an integral extension)

To define the R-module M, proposition 50.1.7 gives, for each i, a nonzero $d_i \in R$ with $d_i\beta_i \in S$, and $d = d_1 \cdots d_n \neq 0$ satisfies $d\beta_i \in S$ for every i. As $d \neq 0$, $dB = \{d\beta_1, d\beta_2, ..., d\beta_n\}$ is still a K-basis for F. Now, consider

$$\langle d\beta_1, d\beta_2, ..., d\beta_n \rangle_R = M$$

Since S is naturally an R-module and $d\beta_i \in S$, we have that $M \subseteq S$. Without loss of generality, we may now assume that $\beta_i \in S$ for all $1 \leq i \leq n$, that is $B \subseteq S$ and is a K-basis for F.

For the second R-module, define: the *integral trace dual*

$$^*M = M^{\mathrm{tr}\vee} = \left\{ b \in F \;\middle|\; \forall \ell \in M, \mathrm{tr}_{F/K}(b\ell) \in R \right\}$$

The extension F/K is separable by hypothesis, so the trace pairing is nondegenerate ([3, The Trace Form of a Separable Extension Is Nondegenerate]) and there is a dual K-basis $B' = \{\beta_1', ..., \beta_n'\}$ of F given by $\mathrm{tr}(\beta_i\beta_j') = \delta_{ij}$. Both modules are free of rank n over R, so in particular $M \cong_R M^{\mathrm{tr}\vee}$. This $M^{\mathrm{tr}\vee}$ is the module called M' in the statement.

The key result we shall prove is:

$$M = R\beta_1 + R\beta_2 + \cdots + R\beta_n \subseteq S \subseteq R\beta_1' + R\beta_2' + \cdots + R\beta_n' = M^{\mathrm{tr}\vee} \tag{1.3}$$

The first inclusion has already been proved, hence we show the second inclusion. Let $b \in S$ be

[6] The integral points of curves shall often have more nuanced behaviour

arbitrary. Since $S \subseteq F$ and B' is a basis for F, we may write b as

$$b = \sum_i b_i \beta_i' \qquad b_i \in K$$

If we show that each $b_i \in R$, we are done. First, Since $\beta_i, b \in S$, their product is in S since S is a ring, $b\beta_i \in S$, hence $\mathrm{tr}_{F/K}(b\beta_i) \in R$ for each i since S is integral over R. Computing, we get:

$$\mathrm{tr}(b\beta_i) = \mathrm{tr}(\sum_j b_j \beta_j' \beta_i) = \sum_j b_j \mathrm{tr}(\beta_j' \beta_i) = \sum_j b_j \delta_{ij} = b_i$$

Hence, $b_i \in R$, showing equation (50.3). If R is Noetherian, then $M^{\mathrm{tr}\vee}$ is a Noetherian R-module, and hence S would be finitely generated. If furthermore R is a Principal Ideal Domain, then any submodule of a free module (over a PID) is free, and S is free of rank less than or equal to n. Since it contains a free R-module of rank n, it has to be free rank greater than or equal to n, and so it has rank n as an R-module, completing the proof.

Note that this implies that if $\mathcal{O}_K$ is a PID and L/K a separable extension where $K = K(\mathcal{O}_K)$, then *every* nonzero finitely generated $\mathcal{O}_L$-module is a free $\mathcal{O}_K$-module of rank $[L : K]$.

Corollary 1.1.34: Maximality of Ring Of Integers for Number Fields

The ring of integers of a number field L is the largest subring that is finitely generated as a $\mathbb{Z}$-module.

***Proof*:**
By the above proposition, $\mathcal{O}_L$ is finitely generated as a $\mathbb{Z}$-module If $B \subseteq L$ is a subring that is also a finitely generated $\mathbb{Z}$-submodule, then every element of B is integral over $\mathbb{Z}$ ([2, Integral Element Equivalences](3)). Hence, $B \subseteq \mathcal{O}_L$.

With 50.1.32, we now have an "upper bound" on how far away a ring of integers is form simply being $\mathbb{Z}[\alpha]$ where $K = \mathbb{Q}(\alpha)$.

Definition 1.1.35: Integral Trace Dual

Let R be a normal integral domain, $K = K(R)$, L/K a finite separable extension, $S = \overline{R}^L$. Then the *integral trace dual* of S is

$${}^*S = \left\{ b \in L \;\middle|\; \forall \ell \in S, \mathrm{tr}_{L/K}(b\ell) \in R \right\}$$

Example 1.1.36: Finding Integral Trace Dual

Let $K = \mathbb{Q}(\sqrt{5})$ considered as an extension $K/\mathbb{Q}$ so that $R = \mathbb{Z}$. Then we know that $L = \langle 1, \sqrt{5} \rangle \subseteq \mathcal{O}_K$. Let's try to bound by above the elements of $\mathcal{O}_K$. Let $b = r + s\sqrt{5} \in \mathbb{Q}(\sqrt{5})$ be an arbitrary element. Then:

$$\mathrm{tr}(b \cdot 1) = 2r \qquad r(b\sqrt{5}) = 10s$$

Hence, if $2r \in \mathbb{Z}$ and $10s \in \mathbb{Z}$, we get:

$$^*\mathcal{O}_K = \left\langle \frac{1}{2}, \frac{\sqrt{5}}{10} \right\rangle_{\mathbb{Z}}$$

Note that $^*\mathcal{O}_K$ is bigger than $\mathcal{O}_K$, for example $1/2 \notin \mathcal{O}_K$. By similar reasoning, we can check that $\frac{1}{\sqrt{5}} \in {}^*\mathcal{O}_K$, but it can be verified that it is not in $\mathcal{O}_K$, and so $\mathcal{O}_K^\vee \neq \mathcal{O}_K$. This does not contradict corollary 50.1.34 as $^*\mathcal{O}_K$ is *not* a subring (note that $1/2 \cdot 1/2 \notin {}^*\mathcal{O}_K$). Note however that they are both still rank 2.

Note that separability is a necessary hypothesis to insure the non-degeneracy of the trace-pairing, meaning this result works at least over over number fields, function fields, and perfect fields[7]. Naturally, if R is not a Principal Ideal Domain, then S need not be a free R-module. For now, we shall always let $R = \mathbb{Z}$ so that $K = \mathbb{Q}$ and $F/\mathbb{Q}$ is a number field. Note that we are only working with the addition structure of $\mathcal{O}_K$, while $\mathcal{O}_K$ is a ring. Thus, when talking about generators as a $\mathbb{Z}$-module, it must not be confounded with a set of generators for the ring. For example $(\alpha) \subseteq \mathcal{O}_K$ is a principal ideal and hence generated by a single element, but it may be generated by many elements as a $\mathbb{Z}$-module.

Definition 1.1.37: Integral Basis

Let $K/\mathbb{Q}$ be a number field. Then a $\mathbb{Z}$-basis for $\mathcal{O}_K$ is called an *integral basis* for K.

The previous proposition also gives more motivation for squaring the determinant in the definition of the discriminant. By corollary 50.1.27, if $(\alpha_1, \alpha_2, ..., \alpha_n)$ is a $\mathbb{Z}$-basis for $\mathcal{O}_F$, then $\text{disc}_{F/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n) \in \mathbb{Z}$ and any change of basis A must map to a unit, hence $\det(A) \in \mathbb{Z}^\times = \{-1, 1\}$. Squaring thus guarantees that the discriminant is basis independent for rings of integers. Now that we have explicitly found that $\mathcal{O}_K$ is a finitely generated $\mathbb{Z}$-module in the case of K being a number field, the term *field discriminant*, or just *discriminant* for short, shall imply that the input is an integral basis for $\mathcal{O}_K$, unless explicitly stated otherwise:

Definition 1.1.38: Discriminant For Ring of Integers

Let $K/\mathbb{Q}$ be a number field. Then if $A = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is an integral basis for $\mathcal{O}_K$, then:

$$d_K := \text{disc}(\mathcal{O}_K) := \text{disc}_{K/\mathbb{Q}}(A) = \text{disc}(\mathcal{O}_K)$$

is the *field discriminant* of K. It is sometimes also denoted $\text{disc}(K)$ if clear from context.

The discriminant can be thought of as capturing some information on how far $\mathcal{O}_K$ is from being
Thus, in example 50.1.25 where $K = \mathbb{Q}(\sqrt{m})$, we have:

$$d_K = \begin{cases} 4m & m \equiv 2, 3 \bmod 4 \\ m & m \equiv 1 \bmod 4 \end{cases}$$

A field K has many bases, not all of them bases for $\mathcal{O}_K$. The discriminants of a free $\mathbb{Z}$-module of rank n in K and of a subgroup of finite index in it differ by the square of the index:

[7] fields of positive characteristic may require some extra work

Proposition 1.1.39: Discriminant And Comparing Basis

Let K be a number field of degree n, let $B \subseteq K$ be a free $\mathbb{Z}$-module with $\mathbb{Z}$-basis $\beta_1, \dots, \beta_n$, and let $A \subseteq B$ be a $\mathbb{Z}$-submodule of finite index (i.e. $[B : A] < \infty$). Then A is free of rank n, and if $\alpha_1, \dots, \alpha_n$ is a $\mathbb{Z}$-basis of A and $T \in M_n(\mathbb{Z})$ is the matrix with $\alpha_i = \sum_j T_{ij}\beta_j$, then $[B : A] = |\det(T)|$ and

$$\operatorname{disc}(A) = [B : A]^2 \operatorname{disc}(B)$$

where $\operatorname{disc}(A) = \operatorname{disc}_{K/\mathbb{Q}}(\alpha_1, \dots, \alpha_n)$ and $\operatorname{disc}(B) = \operatorname{disc}_{K/\mathbb{Q}}(\beta_1, \dots, \beta_n)$.

***Proof*:**
The group A is free of rank at most n ([3, Submodule Of Free Module Over PID]), and its rank is n: if $m = [B : A]$, then $mB \subseteq A$ by [3, Lagrange Theorem], and mB is free of rank n. Let α, β and T be as in the statement. By [3, Smith Normal Form], applied over the Euclidean domain $\mathbb{Z}$, the matrix T is equivalent to $\operatorname{diag}(d_1, \dots, d_n)$ for integers d_i, some possibly 0. That book defines equivalence for matrices over a field, so we record what its proof gives over $\mathbb{Z}$. Read over $\mathbb{Z}$, the proof takes a nonzero entry of least valuation, here the absolute value, among the matrices reachable from T by integer row and column operations, that is, swaps of two rows or of two columns and additions of an integer multiple of one row or column to another, and it reaches the normal form by finitely many such operations. Each such operation is multiplication on the left or on the right by a matrix in $M_n(\mathbb{Z})$ whose inverse, the matrix of the reverse operation, also lies in $M_n(\mathbb{Z})$. So there are matrices $U, V \in M_n(\mathbb{Z})$, invertible over $\mathbb{Z}$, with $UTV = \operatorname{diag}(d_1, \dots, d_n)$. The determinants of U and V are ± 1, since $\det(U)\det(U^{-1}) = 1$ in $\mathbb{Z}$ ([3, Multiplicativity of the Determinant]). Put $\alpha'_i = \sum_j U_{ij}\alpha_j$ and $\beta'_i = \sum_j (V^{-1})_{ij}\beta_j$. As U and V^{-1} are invertible over $\mathbb{Z}$, these are $\mathbb{Z}$-bases of A and B, and $\alpha'_i = d_i\beta'_i$ for each i. Hence

$$B/A = \left(\bigoplus_i \mathbb{Z}\beta'_i\right) \Big/ \left(\bigoplus_i \mathbb{Z}d_i\beta'_i\right) \cong \bigoplus_i \mathbb{Z}/d_i\mathbb{Z}$$

This group is finite, so no d_i is 0, and it has $|d_1 \cdots d_n| = |\det(UTV)| = |\det(T)|$ elements ([3, Multiplicativity of the Determinant]). So $[B : A] = |\det(T)|$.

Let $\sigma_1, \dots, \sigma_n$ be the embeddings of K into $\mathbb{C}$. Applying σ_k to $\alpha_i = \sum_j T_{ij}\beta_j$ shows that $(\sigma_k(\alpha_i))_{k,i} = (\sigma_k(\beta_j))_{k,j}\, T^t$, so by definition 50.1.24 and [3, Multiplicativity of the Determinant],

$$\operatorname{disc}(A) = \det(T)^2 \operatorname{disc}(B) = [B : A]^2 \operatorname{disc}(B)$$

as we sought to show.

Corollary 1.1.40: Subset And Same Discriminant, Then Equal

Let $F/\mathbb{Q}$ be a number field of degree n, let $A \subseteq F$ be a free $\mathbb{Z}$-module of rank n with $\operatorname{disc}(A) \neq 0$, and let $B \subseteq A$ be a subgroup of finite index with $\operatorname{disc}(A) = \operatorname{disc}(B)$. Then $B = A$. Hence, one may use discriminants to identify free $\mathbb{Z}$-modules of finite rank.

***Proof*:**
By proposition 50.1.39, $\text{disc}(B) = [A : B]^2\text{disc}(A)$. As $\text{disc}(A) = \text{disc}(B) \neq 0$, $[A : B]^2 = 1$, so $[A : B] = 1$ and $B = A$, as claimed.

Corollary 1.1.41: Isomorphism Check Of Number Fields

Let K, K' be number fields. If $d_K \neq d_{K'}$, then $K \ncong K'$.

***Proof*:**
Let $\varphi\colon K \to K'$ be an isomorphism of number fields. An element of K and its image under φ are roots of the same polynomials with integer coefficients, so $\varphi(\mathcal{O}_K) \subseteq \mathcal{O}_{K'}$ and $\varphi^{-1}(\mathcal{O}_{K'}) \subseteq \mathcal{O}_K$ (definition 50.1.3). So if $\omega_1, \ldots, \omega_n$ is an integral basis of $\mathcal{O}_K$, then $\varphi(\omega_1), \ldots, \varphi(\omega_n)$ is an integral basis of $\mathcal{O}_{K'}$. The embeddings of K' into $\mathbb{C}$ are the maps $\sigma \circ \varphi^{-1}$ with σ an embedding of K, so by definition 50.1.24 and definition 50.1.38

$$d_{K'} = \det\left(\sigma_k\varphi^{-1}(\varphi(\omega_i))\right)^2 = \det\left(\sigma_k(\omega_i)\right)^2 = d_K$$

Hence isomorphic number fields have equal discriminants, which is the contrapositive of the statement.

Note that it is possible that there are multiple fields with the same discriminant:

Example 1.1.42: Same Discriminant, Different Fields
Take $\mathbb{Q}(\alpha), \mathbb{Q}(\beta), \mathbb{Q}(\gamma)$ where

$$a^3 - 18\alpha - 6 = 0 \qquad \beta^3 - 36\beta - 78 = 0 \qquad \gamma^3 - 58\gamma - 150 = 0$$

Then each has discriminant 22356, but these fields are not isomorphic (as shall be shown in example 50.4.20)

It should be noted that there are only finitely many fields for each discriminant value, however we shall not prove this fact[8]

The next couple of theorems give some computational tools on finding the ring of integers:

Proposition 1.1.43: Computation Of Dual For Ring Of Integers

Let K be a number field of degree n with a $\mathbb{Q}$-basis $\alpha_1, \alpha_2, ..., \alpha_n$ of algebraic integers, and let $d = \text{disc}_{K/\mathbb{Q}}(\alpha_1, \alpha_2, ..., \alpha_n)$. Then every $\alpha \in \mathcal{O}_K$ can be expressed as:

$$\frac{m_1\alpha_1}{d} + \cdots + \frac{m_n\alpha_n}{d}$$

where $m_i \in \mathbb{Z}$ and each m_i^2 is divisible by d $(d|m_i^2)$

[8] For those interested, it is the *Hermite-Minkowski's Theorem*

***Proof*:**
Step 1: the discriminant. Let $\sigma_1, \ldots, \sigma_n$ be the embeddings of K into $\mathbb{C}$ and $S = (\sigma_i(\alpha_j))_{ij}$, with $\delta = \det(S)$. By definition 50.1.24, $d = \delta^2$, which is a nonzero integer by corollary 50.1.28 and corollary 50.1.27.

Step 2: Cramer's rule. Write $\alpha = \sum_j x_j \alpha_j$ with $x_j \in \mathbb{Q}$. Applying σ_i gives $\sigma_i(\alpha) = \sum_j x_j \sigma_i(\alpha_j)$, so the column $s = (\sigma_i(\alpha))_i$ equals $\sum_k x_k S_k$, where S_k is the kth column of S. Let y_j be the determinant of S with its jth column replaced by s. By multilinearity and the alternating property ([3, Properties Of The Determinant]), $y_j = \sum_k x_k \det(S \text{ with column } j \text{ replaced by } S_k) = x_j \delta$, the terms with $k \neq j$ having two equal columns.

Step 3: integrality. Each σ_i maps algebraic integers to algebraic integers, as an element and its image are roots of the same polynomials over $\mathbb{Z}$, so the entries of S and s are algebraic integers. By the expansion of the determinant ([3, Determinant]), δ and y_j are sums of products of these entries, hence algebraic integers ([2, Integral Elements Form Rings]). Put $m_j = dx_j \in \mathbb{Q}$. Then $m_j = \delta^2 x_j = \delta y_j$ and $m_j^2/d = (\delta x_j)^2 = y_j^2$ are algebraic integers, and rational, so they lie in $\mathbb{Z}$ by lemma 50.1.5 and theorem 50.1.9(1). So $\alpha = \sum_j (m_j/d)\alpha_j$ with $m_j \in \mathbb{Z}$ and $d | m_j^2$, completing the proof.

Hence, for any $\mathcal{O}_K$, if $\alpha_1, \alpha_2, ..., \alpha_n$ is a basis for $K/\mathbb{Q}$

$$\bigoplus \alpha_i \mathbb{Z} \subseteq \mathcal{O}_K \subseteq \bigoplus \frac{\alpha_i}{d} \mathbb{Z}$$

And if $K = \mathbb{Q}(\alpha)$ so that $1, \alpha, ..., \alpha^{n-1}$ is a basis for K, then by proposition 50.1.29 $d = \prod_{i<j} (\alpha_i - \alpha_j)^2$ and:

$$\mathbb{Z}[\alpha] \subseteq \mathcal{O}_K \subseteq \frac{1}{d} \mathbb{Z}[\alpha]$$

Note that in general, $\mathbb{Z}[\alpha] \neq \mathcal{O}_K$. A counter-example shall be given in example 50.4.29. In the special case where $\mathbb{Z}[\alpha] = \mathcal{O}_K$, we give such a number ring a name:

Definition 1.1.44: Monogenic Number Ring

Let $K/\mathbb{Q}$ be a number field, $\mathcal{O}_K$ be the associated number ring, and α the primitive element so that $K = \mathbb{Q}(\alpha)$. Then if

$$\mathbb{Z}[\alpha] = \mathcal{O}_K$$

Then $\mathcal{O}_K$ is called *monogenic* and is said to be a *monogenic number ring.*

Since all number fields are monogenic, that is $K = \mathbb{Q}(\alpha)$ for some appropriate α, it is reasonable to ask if $\mathcal{O}_K$ can be expressed using α. The following theorem gives this structure:

Theorem 1.1.45: Basis For Ring Of Integers

Let $\alpha \in \mathcal{O}_k$ and suppose α has degree n over $\mathbb{Q}$. Then there exists $d_1, d_2, ..., d_{n-1} \in \mathbb{Z}$, and monic polynomial $f_i \in \mathbb{Z}[x]$ of degree i for $1 \leq i \leq n-1$ such that

$$\mathcal{O}_K = \mathbb{Z} \oplus \frac{f_1(\alpha)}{d_1}\mathbb{Z} \oplus \cdots \oplus \frac{f_{n-1}(\alpha)}{d_{n-1}}\mathbb{Z}$$

is an integral basis where

$$d_1 \big| d_2 \big| \cdots \big| d_{n-1}$$

and the d_i's are unique.

***Proof*:**

Since $\alpha \in \mathcal{O}_K$ has degree n over $\mathbb{Q}$, we have $K = \mathbb{Q}(\alpha)$ and $n = [K : \mathbb{Q}]$. Put $d = \operatorname{disc}_{K/\mathbb{Q}}(1, \alpha, \ldots, \alpha^{n-1})$, an integer by corollary 50.1.27 and nonzero by corollary 50.1.28. For $1 \leq k \leq n$ set

$$F_k = \mathbb{Z}\frac{1}{d} \oplus \mathbb{Z}\frac{\alpha}{d} \oplus \cdots \oplus \mathbb{Z}\frac{\alpha^{k-1}}{d} \qquad R_k = \mathcal{O}_K \cap F_k$$

The two extreme cases are $R_1 = \mathbb{Z}$, because $F_1 = \frac{1}{d}\mathbb{Z} \subseteq \mathbb{Q}$ and $\mathcal{O}_K \cap \mathbb{Q} = \mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$ by lemma 50.1.5, and $R_n = \mathcal{O}_K$, because proposition 50.1.43 applied to the basis $1, \alpha, \ldots, \alpha^{n-1}$ places every element of $\mathcal{O}_K$ inside F_n.

The construction is by induction on k: there are $d_1 \big| d_2 \big| \cdots \big| d_{k-1}$ in $\mathbb{Z}$ and monic $f_i \in \mathbb{Z}[x]$ with $\deg f_i = i$ such that $1, f_1(\alpha)/d_1, \ldots, f_{k-1}(\alpha)/d_{k-1}$ is a $\mathbb{Z}$-basis of R_k. For $k = 1$ the element 1 is a basis of $R_1 = \mathbb{Z}$, and we set $f_0 = 1$ and $d_0 = 1$. Assume the claim for some $k < n$ and let $\pi : F_{k+1} \to \mathbb{Z}\frac{\alpha^k}{d}$ be the projection onto the last summand. Restricted to R_{k+1} its kernel is $R_{k+1} \cap F_k = R_k$ and its image is a subgroup of an infinite cyclic group. The image is nonzero, since $\alpha^k \in \mathcal{O}_K$ and $\alpha^k = d \cdot \frac{\alpha^k}{d} \in F_{k+1}$ give $\alpha^k \in R_{k+1}$ with $\pi(\alpha^k) \neq 0$. So $\pi(R_{k+1})$ is infinite cyclic, generated by $\pi(\beta)$ for some $\beta \in R_{k+1}$, and the exact sequence

$$0 \to R_k \to R_{k+1} \xrightarrow{\pi} \pi(R_{k+1}) \to 0$$

has free right-hand term, hence splits. Therefore $R_{k+1} = R_k \oplus \mathbb{Z}\beta$, and adjoining β to the basis of R_k gives a $\mathbb{Z}$-basis of R_{k+1}.

It remains to put β in the stated shape. The element $\gamma := \alpha f_{k-1}(\alpha)/d_{k-1}$ is a product of two elements of $\mathcal{O}_K$, and it lies in the $\mathbb{Q}$-span of $1, \alpha, \ldots, \alpha^k$ because f_{k-1} is monic of degree $k-1$. Any element of $\mathcal{O}_K$ in that span lies in F_{k+1}: by $R_n = \mathcal{O}_K$ its coordinates against $1, \alpha, \ldots, \alpha^{n-1}$ lie in $\frac{1}{d}\mathbb{Z}$, and the coordinates in degrees above k vanish. So $\gamma \in R_{k+1}$ and $\pi(\gamma) = \alpha^k/d_{k-1}$. Hence $\alpha^k/d_{k-1} = m\pi(\beta)$ for some nonzero $m \in \mathbb{Z}$, and after replacing β by $-\beta$ if needed we may take $m > 0$. Put $d_k := md_{k-1}$, so that $d_{k-1} \big| d_k$ and $\pi(\beta) = \alpha^k/d_k$. Writing $\beta = (b_0 + b_1\alpha + \cdots + b_k\alpha^k)/d$ with $b_i \in \mathbb{Z}$ and reading off the top coordinate gives $b_k d_k = d$, so

$$f_k(x) := \frac{d_k}{d}\left(b_0 + b_1 x + \cdots + b_k x^k\right)$$

is monic of degree k with $f_k(\alpha) = d_k\beta$.

The coefficients of f_k are so far only rational. Since $f_k(\alpha)/d_{k-1} = m\beta \in \mathcal{O}_K$, the element

$$\gamma' := \frac{f_k(\alpha) - \alpha f_{k-1}(\alpha)}{d_{k-1}} = m\beta - \gamma$$

lies in R_{k+1} and satisfies $\pi(\gamma') = m\pi(\beta) - \pi(\gamma) = 0$, so $\gamma' \in R_k$. Expanding γ' in the inductive basis, $\gamma' = \sum_{i=0}^{k-1} a_i f_i(\alpha)/d_i$ with $a_i \in \mathbb{Z}$, and multiplying through by d_{k-1} using $d_i \mid d_{k-1}$ gives $\gamma' = g(\alpha)/d_{k-1}$ for

$$g(x) = \sum_{i=0}^{k-1} a_i \frac{d_{k-1}}{d_i} f_i(x) \in \mathbb{Z}[x] \qquad \deg g \le k-1$$

Now $f_k(x) - x f_{k-1}(x) - g(x)$ has degree at most $k-1 < n$ and vanishes at α, which has degree n over $\mathbb{Q}$, so it is the zero polynomial. Thus $f_k = x f_{k-1} + g \in \mathbb{Z}[x]$, monic of degree k, and $\beta = f_k(\alpha)/d_k$. Taking $k = n$ closes the induction and produces the stated decomposition of $\mathcal{O}_K$.

For uniqueness, let $V_k = \mathbb{Q} + \mathbb{Q}\alpha + \cdots + \mathbb{Q}\alpha^k$ and suppose $1, f_1(\alpha)/d_1, \ldots, f_{n-1}(\alpha)/d_{n-1}$ is any $\mathbb{Z}$-basis of $\mathcal{O}_K$ of the stated shape. Since $\deg f_i = i$ exactly, an integral combination $\sum_i a_i f_i(\alpha)/d_i$ lies in V_k if and only if $a_i = 0$ for $i > k$, so

$$\mathcal{O}_K \cap V_k = \mathbb{Z} \oplus \frac{f_1(\alpha)}{d_1}\mathbb{Z} \oplus \cdots \oplus \frac{f_k(\alpha)}{d_k}\mathbb{Z}$$

The left-hand side does not refer to the chosen basis. Applying to it the functional $\pi_k : V_k \to \mathbb{Q}$ that reads off the coefficient of α^k, and using that $\pi_k(f_i(\alpha)/d_i) = 0$ for $i < k$ while $\pi_k(f_k(\alpha)/d_k) = 1/d_k$, gives $\pi_k(\mathcal{O}_K \cap V_k) = \frac{1}{d_k}\mathbb{Z}$. So the ideal $d_k\mathbb{Z}$ depends only on $\mathcal{O}_K$ and α, which determines each d_k up to sign. Replacing d_k by $-d_k$ replaces the basis element by its negative and keeps f_k monic, so the d_k are unique once taken positive.

Example 1.1.46: Ring Of Integers Of Cubic Extension

Let $K = \mathbb{Q}(\sqrt[3]{2})$ and $\alpha = \sqrt[3]{2}$. We want to find $\mathcal{O}_K$. First $\operatorname{rank}\mathcal{O}_K = 3$ since $[K : \mathbb{Q}] = 3$, and $L = \mathbb{Z}[\sqrt[3]{2}] \subseteq \mathcal{O}_K$. If A is the change of basis from L to $\mathcal{O}_K$, then $\det(A) = [\mathcal{O}_K : \mathbb{Z}[\alpha]]$[a]. Thus, by doing a change of basis from $\mathbb{Z}[\alpha]$ to $\mathcal{O}_K$ by A, we get that:

$$\operatorname{disc}(\mathbb{Z}[\alpha]) = [\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 \operatorname{disc}(\mathcal{O}_K)$$

Thus, computing the discriminant of the left hand side we get:

$$\operatorname{disc}(\mathbb{Z}[\alpha]) = \det \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 6 \\ 0 & 6 & 0 \end{pmatrix} = -3 \cdot 36 = -(2^2 \cdot 3^2) \cdot 3 = -108$$

giving us that $[\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 \mid 108 \Rightarrow [\mathcal{O}_K : \mathbb{Z}[\alpha]] \mid 2 \cdot 3 = 6$. If we show the index is 1, the proof is done. We shall prove the contradiction for the case of $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 3$, the proof for $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 2$ is similar, and from that we can also conclude that $[\mathcal{O}_K : \mathbb{Z}[\alpha]] \nmid 6$.

Let's say $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 3$. Then $\mathcal{O}_K/\mathbb{Z}[\alpha] \cong \mathbb{Z}/3\mathbb{Z}$. Any element $\beta \in \mathcal{O}_K$ lands in one of the cosets of $\mathcal{O}_K/\mathbb{Z}[\alpha]$ which implies that $a, b, c \in \frac{1}{3}\mathbb{Z}$. We want to show that a, b, c must be integers (lands in the zero coset). Since the ring of integers is closed under subtraction, we may without loss of generality subtract the non-fractional part of any element β we choose so that $a, b, c \in \{-1, 0, 1\}$[b].

Hence we take the norm of $\beta = (a/3) + (b/3)\alpha + (c/3)\alpha^2$

$$N(\beta) = \det \begin{pmatrix} a/3 & 2c/3 & 2b/3 \\ b/3 & a/3 & 2c/3 \\ c/3 & b/3 & a/3 \end{pmatrix} = \frac{1}{27}(a^3 + 2b^2 + 4c^3 - 6abc)$$

Since β is algebraic, $N(\beta) \in \mathbb{Z}$ meaning $27 \big| (a^3 + 2b^2 + 4c^3 - 6abc)$. But, unless $a = b = c = 0$:

$$|a^3 + 2b^2 + 4c^3 - 6abc| \leq 15$$

showing that no such combination of a, b, c exist, meaning β is an algebraic integer if $a, b, c \in \mathbb{Z}$. A similar argument can be made in the case of $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 2$, hence we have that $\mathcal{O}_K = \mathbb{Z}[\alpha]$, completing the proof.

[a] since the index is finite, a basis for $\mathbb{Z}[\alpha]$ may be scaled to form a basis for $\mathcal{O}_K$, which translates to the determinant result

[b] Equivalently, any coset can be chosen to have such representatives

In some special cases, the ring of integers of more composite fields may be found given ring of integers of each respective field:

Theorem 1.1.47: Ring of Integers of Composites

Let $K_1/K, K_2/K$ be Galois with degree n and m and with coprime discriminants $d_{K_1} = d$ and $d_{K_2} = d'$ respectively. Let $L = K_1K_2$ and $K_1 \cap K_2 = K$. Let $\omega_1, \omega_2, ..., \omega_n$ and $\omega'_1, \omega'_2, ..., \omega'_m$ be an integral basis for K_1/K and K_2/K. Then $\omega_i\omega'_j$ is an integral basis for K_1K_2 with discriminant $(d_{K_1})^m(d_{K_2})^n = d^m(d')^n$, namely

$$\mathcal{O}_L = \mathcal{O}_{K_1}\mathcal{O}_{K_2}$$

***Proof*:**
As $K_1 \cap K_2 = K$, [3, Galois Composite Fields II] gives $\mathrm{Gal}_K(L) \cong \mathrm{Gal}_K(K_1) \times \mathrm{Gal}_K(K_2)$ through restriction, so $[L : K] = nm$. Under this isomorphism, $\mathrm{Gal}_{K_2}(L) = \{\sigma_1, \ldots, \sigma_n\}$ restricts bijectively onto $\mathrm{Gal}_K(K_1)$, $\mathrm{Gal}_{K_1}(L) = \{\tau_1, \ldots, \tau_m\}$ restricts bijectively onto $\mathrm{Gal}_K(K_2)$, and $\mathrm{Gal}_K(L)$ consists of the nm products $\sigma_k\tau_\ell$. The nm products $\omega_i\omega'_j$ span L over K ([3, Degree Of Composite Field]), so they form a basis. As d and d' are coprime, $yd + zd' = 1$ for some $y, z \in \mathcal{O}_K$.

Step 1: the integral basis. Let $\alpha \in \mathcal{O}_L$, write $\alpha = \sum_{i,j} a_{ij}\omega_i\omega'_j$ with $a_{ij} \in K$, and put $\beta_j = \sum_i a_{ij}\omega_i \in K_1$, so that $\alpha = \sum_j \beta_j\omega'_j$. Each τ_ℓ fixes K_1, so $\tau_\ell(\alpha) = \sum_j \tau_\ell(\omega'_j)\beta_j$ for $1 \leq \ell \leq m$. That is, the column $a = (\tau_\ell(\alpha))_\ell$ is Tb, for the matrix $T = (\tau_\ell(\omega'_j))_{\ell,j}$ and the column $b = (\beta_j)_j$. As the τ_ℓ restrict to the K-embeddings of K_2, $\det(T)^2 = d'$ by definition 50.1.24. Let $T^{(j)}$ be T with its jth column replaced by a, which is the sum of the columns of T weighted by the $\beta_{j'}$. Multilinearity in that column and the vanishing of determinants with two equal columns ([3, Properties Of The Determinant]) give $\det(T^{(j)}) = \beta_j \det(T)$. The entries of T and $T^{(j)}$ are algebraic integers, hence so are their determinants ([2, Integral Elements Form Rings]), and $d'\beta_j = \det(T)\det(T^{(j)})$ is an algebraic integer lying in K_1, that is, $d'\beta_j = \sum_i d'a_{ij}\omega_i \in \mathcal{O}_{K_1}$. As the ω_i form an integral basis of K_1/K, $d'a_{ij} \in \mathcal{O}_K$. Exchanging the roles of K_1 and K_2 gives

$da_{ij} \in \mathcal{O}_K$, so $a_{ij} = yda_{ij} + zd'a_{ij} \in \mathcal{O}_K$. As every $\omega_i\omega'_j$ lies in $\mathcal{O}_L$, the $\omega_i\omega'_j$ form an integral basis of L/K, and $\mathcal{O}_L = \mathcal{O}_{K_1}\mathcal{O}_{K_2}$.

Step 2: the discriminant. As τ_ℓ fixes $\omega_i \in K_1$ and σ_k fixes $\tau_\ell(\omega'_j) \in K_2$,

$$\sigma_k\tau_\ell(\omega_i\omega'_j) = \sigma_k(\omega_i)\tau_\ell(\omega'_j)$$

Index the rows of $M = (\sigma_k\tau_\ell(\omega_i\omega'_j))$ by the pairs (ℓ, k) and its columns by the pairs (j, i), so that M is an $m \times m$ array of $n \times n$ blocks, the (ℓ, j) block being $\tau_\ell(\omega'_j)Q$ with $Q = (\sigma_k(\omega_i))_{k,i}$. Then $M = \operatorname{diag}(Q, \dots, Q)\, P$, where the (ℓ, j) block of P is $\tau_\ell(\omega'_j)E$ and E is the $n \times n$ identity matrix. Reordering the rows and the columns of P by the pairs (k, ℓ) and (i, j) turns it into $\operatorname{diag}(T, \dots, T)$, with n blocks, and changes its determinant at most by a sign ([3, Properties Of The Determinant]). So by [3, Multiplicativity of the Determinant] and [3, Determinant of a Block Diagonal Matrix], and as $\det(Q)^2 = d$ by definition 50.1.24,

$$\operatorname{disc}(\omega_i\omega'_j) = \det(M)^2 = \det(Q)^{2m}\det(T)^{2n} = d^m(d')^n$$

as we sought to show.

For separable extensions that are not Galois, the hypothesis $K_1 \cap K_2 = K$ has to be replaced by the linear disjointness of K_1 and K_2 over K. If the discriminants are not relatively prime, then there is a partial generalization in the case where $K = \mathbb{Q}$:

Corollary 1.1.48: Ring of Integers of Composite, Partial Generalization

Let $K_1/\mathbb{Q}$ and $K_2/\mathbb{Q}$ be finite separable extensions with $L = K_1K_2$ of degree $[L : \mathbb{Q}] = [K_1 : \mathbb{Q}][K_2 : \mathbb{Q}]$, and let $d_1 = \operatorname{disc}(\mathcal{O}_{K_1})$ and $d_2 = \operatorname{disc}(\mathcal{O}_{K_2})$. Then

$$\mathcal{O}_L \subseteq \frac{1}{\gcd(d_1, d_2)}\mathcal{O}_{K_1}\mathcal{O}_{K_2}$$

***Proof*:**
Write $m = [K_1 : \mathbb{Q}]$ and $n = [K_2 : \mathbb{Q}]$, and fix integral bases $\alpha_1, \dots, \alpha_m$ of $\mathcal{O}_{K_1}$ and $\beta_1, \dots, \beta_n$ of $\mathcal{O}_{K_2}$ (corollary 50.1.33). By [3, Degree Of Composite Field] the mn products $\alpha_i\beta_j$ span L over $\mathbb{Q}$, so the degree hypothesis makes them a $\mathbb{Q}$-basis of L. They also span $\mathcal{O}_{K_1}\mathcal{O}_{K_2}$ as a $\mathbb{Z}$-module, since that ring consists of the finite sums of products ab with $a \in \mathcal{O}_{K_1}$ and $b \in \mathcal{O}_{K_2}$.

Let $\alpha \in \mathcal{O}_L$ and write

$$\alpha = \sum_{i,j} \frac{m_{ij}}{r}\alpha_i\beta_j$$

with $r \in \mathbb{Z}_{>0}$ the least common denominator of the rational coordinates, so that $m_{ij} \in \mathbb{Z}$ and $\gcd\left(r, \gcd_{i,j}(m_{ij})\right) = 1$. It suffices to prove $r \mid d_1$, since the same argument with the roles of K_1 and K_2 exchanged gives $r \mid d_2$, whence $r \mid \gcd(d_1, d_2)$ and $\alpha \in \frac{1}{\gcd(d_1,d_2)}\mathcal{O}_{K_1}\mathcal{O}_{K_2}$.

Restriction sends a $\mathbb{Q}$-embedding of L into $\mathbb{C}$ to the pair of its restrictions to K_1 and K_2, and this pair determines the embedding because L is generated by $K_1 \cup K_2$. Both sides are counted by [3, Finite Separable Extension Equivalence], giving $[L : \mathbb{Q}] = mn$ embeddings on the left and mn pairs on the right, so restriction is a bijection. In particular every embedding σ of K_1 extends to an embedding of L, still written σ, that fixes K_2 pointwise. Applying the m such

extensions to the display above gives the m equations

$$\sigma(\alpha) = \sum_{i=1}^{m} \sigma(\alpha_i) x_i \qquad x_i := \sum_{j=1}^{n} \frac{m_{ij}}{r} \beta_j \in K_2$$

Let $\delta = \det(\sigma_k(\alpha_i))$, so $\delta^2 = \operatorname{disc}_{K_1/\mathbb{Q}}(\alpha_1, \dots, \alpha_m) = d_1$ by definition 50.1.24 and definition 50.1.38, and $\delta \neq 0$ by corollary 50.1.28. Cramer's rule gives $x_i = \gamma_i/\delta$, where γ_i is the determinant obtained by replacing the ith column by the column of the $\sigma_k(\alpha)$. All entries of both determinants are algebraic integers, hence so are δ and every γ_i, and therefore

$$d_1 x_i = \delta^2 x_i = \delta \gamma_i$$

is an algebraic integer. It lies in K_2, so $d_1 x_i \in \mathcal{O}_{K_2}$ by lemma 50.1.5. Since the β_j are a $\mathbb{Z}$-basis of $\mathcal{O}_{K_2}$, every rational number $d_1 m_{ij}/r$ is an integer, that is r divides $d_1 m_{ij}$ for all i and j. Hence r divides $d_1 \gcd_{i,j}(m_{ij})$, and r is coprime to $\gcd_{i,j}(m_{ij})$, so $r \mid d_1$.

Example: Ring of Integers of Cyclotomic Field

Let's use the above results to show that if $K = \mathbb{Q}(\zeta_n)$ then $\mathcal{O}_K = \mathbb{Z}[\zeta_n]$, that is the ring of integers of K is monogenic. Besides quadratic extensions, this is so far the only other ring of integers we have classified (in particular since they are rather difficult to classify). We first require some lemmas:

Lemma 1.1.49: Cyclotomic Field Lemmas

Let $n = p^r$ where $n \geq 3$. Then:

1. $\mathbb{Z}[\zeta_n] = \mathbb{Z}[1 - \zeta_n]$ and $\operatorname{disc}(\zeta_n) = \operatorname{disc}(1 - \zeta_n)$
2. $\prod_{k \in (\mathbb{Z}/n\mathbb{Z})^\times} (1 - \zeta_n^k) = p$

***Proof*:**

1. Certainly $\mathbb{Z}[\zeta_n] = \mathbb{Z}[1 - \zeta_n]$ since $\zeta_n = 1 - (1 - \zeta_n)$, implying the discriminants are equal. Similarly, by proposition 50.1.29

$$\operatorname{disc}(\zeta_n) = \prod_{r<s} (\alpha_r - \alpha_s)^2 = \prod_{r<s} ((1 - \alpha_r) - (1 - \alpha_s))^2 = \operatorname{disc}(1 - \zeta_n)$$

2. By [3, Cyclotomic Polynomials], $\Phi_n(x) = \prod_{k \in (\mathbb{Z}/n\mathbb{Z})^\times} (x - \zeta_n^k)$. By [3, Cyclotomic Polynomial divisor Representation], $x^{p^r} - 1 = \prod_{j=0}^{r} \Phi_{p^j}(x)$ and $x^{p^{r-1}} - 1 = \prod_{j=0}^{r-1} \Phi_{p^j}(x)$, so

$$\Phi_n(x) = \frac{x^{p^r} - 1}{x^{p^{r-1}} - 1} = 1 + x^{p^{r-1}} + x^{2p^{r-1}} + \cdots + x^{(p-1)p^{r-1}}$$

Setting $x = 1$ gives $\prod_k (1 - \zeta_n^k) = \Phi_n(1) = p$.

Proposition 1.1.50: Ring Of Integers Of Prime-Power Cyclotomic Field

Let $n = p^r$ and $K = \mathbb{Q}(\zeta_n)$. Then:

$$\mathcal{O}_K = \mathbb{Z}[\zeta_n]$$

***Proof*:**
If $n \leq 2$, then $K = \mathbb{Q}$ and $\mathcal{O}_K = \mathbb{Z} = \mathbb{Z}[\zeta_n]$, so let $n \geq 3$. Write $\lambda = 1 - \zeta_n$ and $m = \varphi(p^r)$. By lemma 50.1.49, $\mathbb{Z}[\lambda] = \mathbb{Z}[\zeta_n]$ and $\text{disc}(\lambda) = \text{disc}(\zeta_n)$. The minimal polynomial of ζ_n is $\Phi_n(x) = \prod_{k\in(\mathbb{Z}/n\mathbb{Z})^\times}(x - \zeta_n^k)$ ([3, Cyclotomic Polynomials], [3, Irreducibility Of Cyclotomic Polynomials]), so $[K : \mathbb{Q}] = m$ and $1, \lambda, \ldots, \lambda^{m-1}$ is a $\mathbb{Q}$-basis of $K = \mathbb{Q}(\lambda)$ consisting of algebraic integers. By proposition 50.1.43, every $\alpha \in \mathcal{O}_K$ can be written as

$$\alpha = \frac{n_1 + n_2\lambda + \cdots + n_m\lambda^{m-1}}{d}$$

with $n_j \in \mathbb{Z}$ and $d = \text{disc}(\zeta_n)$. By example 50.1.31, d divides $n^{\varphi(n)} = p^{rm}$, so $d = \pm p^s$ for some $s \geq 0$.

Suppose that $\mathcal{O}_K \neq \mathbb{Z}[\lambda]$, and pick $\alpha \in \mathcal{O}_K \setminus \mathbb{Z}[\lambda]$, written as above. Not all n_j are divisible by p^s, so the largest power p^t dividing all of them has $t < s$. Then $\pm p^{s-1-t}\alpha \in \mathcal{O}_K$ equals $\frac{1}{p}\sum_j n'_j\lambda^{j-1}$ with $n'_j = n_j/p^t \in \mathbb{Z}$, not all divisible by p. Let $i \leq m$ be the least index with $p \nmid n'_i$. The terms $\frac{n'_j}{p}\lambda^{j-1}$ with $j < i$ lie in $\mathbb{Z}[\lambda]$, so subtracting them shows that $\mathcal{O}_K$ contains

$$\beta = \frac{n'_i\lambda^{i-1} + n'_{i+1}\lambda^i + \cdots + n'_m\lambda^{m-1}}{p}$$

whose leading coefficient n'_i is prime to p.

By lemma 50.1.49, $p = \prod_k(1 - \zeta_n^k)$, the product over the m classes $k \in (\mathbb{Z}/n\mathbb{Z})^\times$, represented by positive integers. Each factor is $1 - \zeta_n^k = \lambda(1 + \zeta_n + \cdots + \zeta_n^{k-1})$, so $p/\lambda^m \in \mathbb{Z}[\zeta_n]$, and $p/\lambda^i = (p/\lambda^m)\lambda^{m-i} \in \mathbb{Z}[\zeta_n]$ as $i \leq m$. Hence

$$\frac{p\beta}{\lambda^i} = \frac{n'_i}{\lambda} + n'_{i+1} + n'_{i+2}\lambda + \cdots + n'_m\lambda^{m-1-i}$$

lies in $\mathcal{O}_K$, and subtracting the terms in $\mathbb{Z}[\lambda] \subseteq \mathcal{O}_K$ gives $n'_i/\lambda \in \mathcal{O}_K$, whose norm is an integer by corollary 50.1.14. The embeddings of K send ζ_n to the roots ζ_n^k of Φ_n, so $\text{Nm}_{K/\mathbb{Q}}(\lambda) = \prod_k(1 - \zeta_n^k) = p$ by corollary 50.1.13, and $\text{Nm}_{K/\mathbb{Q}}(n'_i/\lambda) = (n'_i)^m/p$. So $p\big|(n'_i)^m$, hence $p\big|n'_i$, contrary to the choice of i. Therefore $\mathcal{O}_K = \mathbb{Z}[\lambda] = \mathbb{Z}[\zeta_n]$, as we sought to show.

We now look to generalize to any cyclotomic field. The key idea is that the ring of integers of cyclotomic fields will be the composite of simpler ring of integers

Proposition 1.1.51: Ring of Integers of Cyclotomic Field

Let $K = \mathbb{Q}(\zeta_n)$ for any $n \in \mathbb{N}$. Then:

$$\mathcal{O}_K \cong \mathbb{Z}[\zeta_n]$$

***Proof*:**
We shall do a form of induction. This was proven if n is a prime power, so let's assume it's not so that we may write $n = n_1 n_2$ for relatively prime $n_1, n_2 > 1$. We shall work by induction on n_1, n_2. The base case was proposition 50.1.50, hence set:

$$\begin{aligned} \omega_1 &= \zeta_{n_1} & \omega_2 &= \zeta_{n_2} \\ K_1 &= \mathbb{Q}(\omega_1) & K_2 &= \mathbb{Q}(\omega_2) \\ \mathcal{O}_{K_1} &= \mathbb{Z}[\omega_1] & \mathcal{O}_{K_2} &= \mathbb{Z}[\omega_2] \end{aligned}$$

As $\zeta_n = e^{2\pi i/n}$ and $n = n_1 n_2$, we have $\zeta_n^{n_1} = \omega_2$ and $\zeta_n^{n_2} = \omega_1$. As $\gcd(n_1, n_2) = 1$, there are $r, s \in \mathbb{Z}$ with $rn_2 + sn_1 = 1$, and then

$$\zeta_n = \zeta_n^{rn_2 + sn_1} = \omega_1^r \omega_2^s$$

Thus $K = K_1 K_2$. Next, as we saw in example 50.1.31,

$$\operatorname{disc}(\omega_1) \big| n_1^{\varphi(n_1)} \qquad \operatorname{disc}(\omega_2) \big| n_2^{\varphi(n_2)}$$

and as $\gcd(n_1, n_2) = 1$, the discriminants of the integral bases $1, \omega_l, \ldots, \omega_l^{\varphi(n_l)-1}$ of $\mathcal{O}_{K_l} = \mathbb{Z}[\omega_l]$ are coprime. The fields K_1 and K_2 are Galois over $\mathbb{Q}$, being the splitting fields of $x^{n_1} - 1$ and $x^{n_2} - 1$ ([3, Finite Normal Extension Equivalences], [3, Separability Over Characteristic 0]). By [3, Irreducibility Of Cyclotomic Polynomials], $[K : \mathbb{Q}] = \varphi(n) = \varphi(n_1)\varphi(n_2) = [K_1 : \mathbb{Q}][K_2 : \mathbb{Q}]$. With $E = K_1 \cap K_2$, an E-basis of K_2 spans $K = K_1 K_2$ over K_1 ([3, Degree Of Composite Field]), so $[K : K_1] \leq [K_2 : E]$, and the tower law ([3, Multiplicity of Finite Extensions (Tower Law)]) gives $[K : \mathbb{Q}] \leq [K_1 : \mathbb{Q}][K_2 : \mathbb{Q}]/[E : \mathbb{Q}]$. Hence $K_1 \cap K_2 = \mathbb{Q}$, and theorem 50.1.47 gives

$$\mathcal{O}_K = \mathcal{O}_{K_1} \mathcal{O}_{K_2} = \mathbb{Z}[\omega_1]\mathbb{Z}[\omega_2] = \mathbb{Z}[\zeta_n]$$

as we sought to show.

Exercise 1.1.2

1. Generalize proposition 50.1.39 for modules over $\mathcal{O}_K$.
2. Give a counter-example where $\mathcal{O}_{KL} = \mathcal{O}_K \mathcal{O}_L$?

1.1.4 Lattice Structure: Geometry of Number Rings

An important word of warning must be given about free $\mathbb{Z}$-modules due to the existence of non-invertible elements. It is possible that A, B are two free $\mathbb{Z}$-module of rank 2, one contains the basis of the other, however $A \subsetneq B$! This is in contract to the case of modules over fields where if A, B are two

k-vector-spaces of the same dimension and $A \subseteq B$, then $A = B$[9]. Even more behavior can happen: consider $\mathbb{Q} \otimes_{\mathbb{Z}} A$. Then this is a vector-space with a finite dimension. However, the dimension need not match the rank of A:

Example 1.1.52: Behavior of $\mathbb{Z}$-modules

1. Let $A = \langle (1,0), (0,1) \rangle_{\mathbb{Z}}$ and $B = \langle (1/2,0), (0,1/2) \rangle_{\mathbb{Z}}$. Then by construction they are both isomorphic as abelian groups to $\mathbb{Z}^2$, however $A \subsetneq B$.
2. Let A be a free $\mathbb{Z}$-module of rank 2. Then we may extend scalar to make A into a free $\mathbb{R}$-module. Either take $\mathbb{R} \otimes_{\mathbb{Z}} A$ or consider $\langle a_1, a_1 \rangle_{\mathbb{R}}$ for some free generating set $\{a_1, a_2\}$. Then $\mathbb{R} \otimes_{\mathbb{Z}} A$ *need not be dimension 2.* For example, take
$$A = \langle (1,1), (\sqrt{2}, \sqrt{2}) \rangle_{\mathbb{Z}}$$
Then $\mathbb{R} \otimes_{\mathbb{Z}} A$ is one dimensional
3. By construction of $\mathcal{O}_k$, if $\langle \alpha_1, \alpha_2, ..., \alpha_n \rangle_{\mathbb{Z}} = \mathcal{O}_k$, then $\langle \alpha_1, \alpha_2, ..., \alpha_n \rangle_{\mathbb{Q}} = k$, Thus $\mathcal{O}_k$ is a little more than just a $\mathbb{Z}$-module. We shall explore this in this section.

The third example shows that the rank of the ring of integers more coincides with the dimension of the natural vector-space when extending the actions to a field. As was explored in [3, Vector Space], [finite dimensional] vector spaces have easy visualizations using orthonormal bases. In [3, Exploring Integral Domains: Factorization and Valuation], this was used implicitly when it was intuited that $\mathbb{Z}[i]$ forms a nice square lattice in $\mathbb{Q}^2$. More generally, ring of integer $\mathcal{O}_K$ will form a lattice when embedded into an appropriate vector-space. To better visualize this lattice, it is easer to see it embedded into a direct sum of $\mathbb{R}$ and $\mathbb{C}$[10]. This has the extra advantage that these spaces have a well-defined notion of *measure* that gives us an area. We shall take advantage of tools from analysis to further enlighten the structure of rings of integers, however as a consequence an elementary level of topology and analysis will be assumed.

Let $n = [K : \mathbb{Q}]$ so that $\mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{C}) = n$. Then every embedding of K into $\mathbb{C}$ is either embeds into $\mathbb{R}$ or does not (that is, either $K \cap (\mathbb{C} \setminus \mathbb{R}) = \emptyset$ or $\neq 0$). Let's call the former *real embeddings* and the latter *complex embeddings.* For a complex embedding $\tau \in \mathrm{Hom}_{\mathbb{Q}}(K, \mathbb{C})$, we may always find another embedding $\overline{\tau}$ since for any α where $p(\alpha) = 0$ for appropriate $p(x) \in \mathbb{Q}[x]$
$$0 = \overline{\tau}(0) = \overline{\tau}(p(\alpha)) = \tau(\overline{p(\alpha)}) = \tau(p(\overline{\alpha}))$$
showing that $\overline{\tau}$ maps roots to roots. Hence the degree n may be split into $n = r_1 + 2r_2$ where r_1 represents the amount of embeddings where the image σ_i is real:
$$\alpha_1, \alpha_2, ..., \alpha_{r_1}$$
while r_2 represents the embeddings that embedded into $\mathbb{C}$ is not strictly in $\mathbb{R}$, giving us a total list of embeddings:
$$\sigma_1, \sigma_2, ..., \sigma_{r_1}, \tau_1, \tau_2, ..., \tau_{r_2}, \overline{\tau_1}, ..., \overline{\tau_{r_2}}$$

[9] $b \in B$ can be appropriately shrunk to a nonzero size so that its in A (since the dimensions are the same), and then we use closure of scalar multiplication

[10] For those who may prefer $\overline{\mathbb{Q}}$ over $\mathbb{R}$ or $\mathbb{C}$, I would stipulate $\mathbb{R}$ and $\mathbb{C}$ are is a very natural object in this context: $\mathbb{R}$ is perhaps the only "natural" object for which a finite extension is an algebraic closure, this will have algebraic and geometric consequence when counting the number of embeddings of a field into the algebraically closed field $\mathbb{C}$

Definition 1.1.53: Archimedean Embedding

Let $K/\mathbb{Q}$ be a number field. Then its *Archimedean embedding* is defined as

$$\begin{aligned}\sigma :&K \hookrightarrow \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}\\ &\alpha \mapsto (\sigma_1(\alpha), ..., \tau_{r_2}(\alpha))\end{aligned}$$

The space $\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ is called the *Minkowski space*, and will be labeled $K_{\mathbb{R}}$.

Example 1.1.54: Archimedean Embedding

1. Trivially, $\mathbb{Q}_{\mathbb{R}} = \mathbb{R}$, and $\mathbb{C}_{\mathbb{R}}$ is not defined since $\mathbb{C}/\mathbb{Q}$ is not a finite extension.
2. Take $K = \mathbb{Q}(i)$ so that $\mathcal{O}_K = \mathbb{Z}[i]$. Then $\text{Gal}_{\mathbb{Q}}(\mathbb{Q}(i)) = \{\text{id}, c\}$ where c is the conjugate. Then K is simply the embedding of K into $\mathbb{C}$

   $$\sigma : K \hookrightarrow \mathbb{C}^1 \qquad k \mapsto k$$

   Notice the gaussian integers form the usual grid in $\mathbb{C}$.
3. Take $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ Then $\text{Gal}_{\mathbb{Q}}(K) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ corresponds to the conjugating each root separately. Then

   $$\begin{gathered}\sigma : K \mapsto \mathbb{R}^4\\ (a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6})\\ \mapsto\\ (a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}, a - b\sqrt{2} + c\sqrt{3} + d\sqrt{6},\\ a + b\sqrt{2} - c\sqrt{3} + d\sqrt{6}, a - b\sqrt{2} - c\sqrt{3} + d\sqrt{6})\end{gathered}$$

Our goal is to show that $\sigma(\mathcal{O}_K)$ behaves like the just like $\mathbb{Z}[i]$ when embedded in $\mathbb{C}$. To characterize a lattice in higher dimensions, we define it in the following way:

Definition 1.1.55: [Integral] Lattice

Let V be a real vector space of dimension n with an inner product, for instance $\mathbb{R}^n$ with its standard inner product. Then a *lattice* in V is a subgroup $\Lambda \subseteq V$ satisfying any two of the following 3 properties:

1. Λ spans V as an $\mathbb{R}$-vector space ($\langle \Lambda \rangle_{\mathbb{R}} = V$)
2. Λ is discrete in the topological space V: there is an open ball around the origin that meets Λ only in the origin
3. Λ is a free $\mathbb{Z}$-module of rank n[a]

[a] some authors call such lattices *complete lattices*, and call the subgroups generated by fewer than n linearly independent vectors *lattices*

Lemma 1.1.56: Equivalence Of Conditions

Any two of these conditions implies the third. Moreover, every bounded subset of V contains only finitely many points of a lattice $\Lambda \subseteq V$.

***Proof*:**

Let $|\cdot|$ denote the norm of the inner product. Expanding $0 \leq |v - tw|^2$ at $t = \langle v, w \rangle / |w|^2$ gives the Cauchy-Schwarz inequality $|\langle v, w \rangle| \leq |v||w|$ for $w \neq 0$, and expanding $|v + w|^2$ then gives the triangle inequality $|v + w| \leq |v| + |w|$.

Step 1: (1) and (3) imply (2). Let $\ell_1, \ldots, \ell_n$ be a $\mathbb{Z}$-basis of Λ. These n vectors span the n-dimensional space V, so they form a basis. An inner product is nondegenerate, since $\langle y, y \rangle > 0$ for $y \neq 0$, so the matrix $G = (\langle \ell_i, \ell_j \rangle)$ is invertible ([3, Nondegeneracy and the Matrix of the Form]). The vectors $\ell_j^* = \sum_k (G^{-1})_{kj} \ell_k$ satisfy $\langle \ell_i, \ell_j^* \rangle = (GG^{-1})_{ij}$, which is 1 for $i = j$ and 0 otherwise. So the coordinates of $v = \sum_i x_i \ell_i$ are $x_i = \langle v, \ell_i^* \rangle$, and $|x_i| \leq |v||\ell_i^*|$ by the Cauchy-Schwarz inequality. A nonzero $\gamma \in \Lambda$ has some coordinate x_i equal to a nonzero integer, so $1 \leq |\gamma||\ell_i^*|$. Hence the open ball of radius $1/\max_i |\ell_i^*|$ around 0 meets Λ only in 0.

Step 2: (2) and (3) imply (1). Let $\ell_1, \ldots, \ell_n$ be a $\mathbb{Z}$-basis of Λ and suppose they do not span V. Then they are linearly dependent over $\mathbb{R}$: $\sum_i a_i \ell_i = 0$ with $a_i \in \mathbb{R}$ not all zero. Let $M \geq 1$ be an integer with $1/M < \max_i |a_i|$. Consider the $M^n + 1$ points $(\{ka_1\}, \ldots, \{ka_n\}) \in [0, 1)^n$ for $k = 0, 1, \ldots, M^n$, where $\{y\} = y - \lfloor y \rfloor$. Dividing $[0, 1)^n$ into the M^n boxes $\prod_i [j_i/M, (j_i + 1)/M)$, two of these points, for some $k < k'$, lie in the same box. With $N = k' - k$ and $b_i = \lfloor k' a_i \rfloor - \lfloor k a_i \rfloor \in \mathbb{Z}$, we get $|Na_i - b_i| = |\{k' a_i\} - \{k a_i\}| < 1/M$ for every i. Not every b_i is 0, since otherwise $|a_i| < 1/(NM) \leq 1/M$ for every i. So $\gamma = \sum_i b_i \ell_i$ is a nonzero element of Λ, the ℓ_i being independent over $\mathbb{Z}$, and

$$|\gamma| = \left| \sum_i (b_i - Na_i) \ell_i \right| \leq \frac{1}{M} \sum_i |\ell_i|$$

As M can be taken arbitrarily large, every ball around 0 contains a nonzero element of Λ, so Λ is not discrete.

Step 3: (1) and (2) imply (3). Choose $r > 0$ with $|\gamma| \geq r$ for every $0 \neq \gamma \in \Lambda$. Then distinct elements of Λ are at distance at least r. As Λ spans V, it contains a basis $u_1, \ldots, u_n$ of V. Let $\Lambda_0 = \bigoplus_i \mathbb{Z}u_i \subseteq \Lambda$ and $P = \{\sum_i x_i u_i \mid 0 \leq x_i < 1\}$. Choose an integer $N \geq 1$ with $\frac{1}{N}\sum_i |u_i| < r$, and divide $[0,1)^n$ into the N^n boxes $\prod_i [j_i/N, (j_i+1)/N)$. If two elements $\sum_i x_i u_i$ and $\sum_i x_i' u_i$ of $\Lambda \cap P$ have coordinate vectors in the same box, the triangle inequality bounds the norm of their difference by $\sum_i |x_i - x_i'||u_i| < \frac{1}{N}\sum_i |u_i| < r$, so they are equal. Hence $\Lambda \cap P$ has at most N^n elements. Every $\gamma = \sum_i x_i u_i \in \Lambda$ equals $\sum_i \lfloor x_i \rfloor u_i + \sum_i \{x_i\} u_i$, where the second summand lies in $\Lambda \cap P$, a finite set. So Λ/Λ_0 is finite, of order q say, and $q\Lambda \subseteq \Lambda_0$ by [3, Lagrange Theorem]. Thus $\Lambda \subseteq \frac{1}{q}\Lambda_0$, a free $\mathbb{Z}$-module of rank n, so Λ is free of rank at most n ([3, Submodule Of Free Module Over PID]), and of rank n as it contains Λ_0.

Step 4: bounded sets. A lattice Λ satisfies (1) and (3), so it has a $\mathbb{Z}$-basis $\ell_1, \ldots, \ell_n$ that is a basis of V, with vectors ℓ_i^* as in Step 1. A bounded subset of V lies in a ball $\{v \mid |v| \leq \rho\}$, and the coordinates of an element of Λ in that ball are integers x_i with $|x_i| \leq \rho|\ell_i^*|$, by Step 1. There are finitely many such coordinate vectors, completing the proof.

The discrete condition need only be checked at the origin due to linearity[11]. An example of a free $\mathbb{Z}$-module in $\mathbb{R}^2$ that is not discrete is the one elaborated in example 50.1.52

$$\langle (1,1), (\sqrt{2}, \sqrt{2}) \rangle \subseteq \mathbb{R}^2$$

In particular, we may get as close as we want to $(0,0)$ by adding integer-multiples these two elements.

Proposition 1.1.57: Archimedean Embedding of $\mathcal{O}_K$ is Lattice

Let $K/\mathbb{Q}$ be a number field of degree n and σ its Archimedean embedding. Identify $K_\mathbb{R} = \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ with $\mathbb{R}^n$ by writing each complex coordinate z as $(\mathrm{Re}z, \mathrm{Im}z)$, with the standard inner product of $\mathbb{R}^n$. Then $\sigma(\mathcal{O}_K)$ is a lattice in $K_\mathbb{R}$, and $\sigma(\alpha_1), \ldots, \sigma(\alpha_n)$ is a $\mathbb{Z}$-basis of it for every integral basis $\alpha_1, \ldots, \alpha_n$ of $\mathcal{O}_K$. In particular

$$\langle \sigma(\mathcal{O}_K) \rangle_\mathbb{R} = \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$$

***Proof*:**
Let $\alpha_1, \ldots, \alpha_n$ be an integral basis of $\mathcal{O}_K$, which exists by corollary 50.1.33.

Step 1: a nonzero determinant. List the embeddings of K into $\mathbb{C}$ as $\sigma_1, \ldots, \sigma_{r_1}, \tau_1, \overline{\tau_1}, \ldots, \tau_{r_2}, \overline{\tau_{r_2}}$, and let A be the $n \times n$ matrix whose ith row lists their values at α_i in this order. Up to transposing and reordering columns, which changes the determinant at most by a sign ([3, Properties Of The Determinant]), A is the matrix of definition 50.1.24, so $\det(A)^2 = \mathrm{disc}_{K/\mathbb{Q}}(\alpha_1, \ldots, \alpha_n) = d_K$ by definition 50.1.38. The α_i are linearly independent over $\mathbb{Z}$, hence over $\mathbb{Q}$ after clearing denominators, so $d_K \neq 0$ by corollary 50.1.28. Let B be the real matrix obtained from A by replacing each pair of columns $(\tau_j(\alpha_i), \overline{\tau_j}(\alpha_i))$ by $(\mathrm{Re}\tau_j(\alpha_i), \mathrm{Im}\tau_j(\alpha_i))$. In the coordinates of $\mathbb{R}^n$, the ith row of B is $\sigma(\alpha_i)$. Column by column $\tau_j = \mathrm{Re}\tau_j + i\mathrm{Im}\tau_j$ and $\overline{\tau_j} = \mathrm{Re}\tau_j - i\mathrm{Im}\tau_j$, so expanding both columns by multilinearity, and using that the determinant vanishes when two columns are equal and changes sign when two columns are swapped ([3, Properties Of The

[11] This is reminiscent to how many conditions for topological groups need only be checked at the origin, including many proofs in real and complex analysis

Determinant]), gives

$$\det(A) = (-2i)^{r_2} \det(B)$$

Hence $\det(B) \neq 0$.

Step 2: a basis of $K_{\mathbb{R}}$. Suppose $\sum_i c_i \sigma(\alpha_i) = 0$ with $c_i \in \mathbb{R}$ and some $c_k \neq 0$. Replacing the kth row of B by $\sum_i c_i \sigma(\alpha_i)$ produces a matrix with a zero row, whose determinant is 0 by multilinearity in rows. By multilinearity and the alternating property, the same determinant equals $c_k \det(B) \neq 0$, a contradiction. So $\sigma(\alpha_1), \ldots, \sigma(\alpha_n)$ are n linearly independent vectors of the n-dimensional space $K_{\mathbb{R}}$, hence a basis of $K_{\mathbb{R}}$.

Step 3: the lattice. Every embedding of K is a ring homomorphism, so σ is additive and

$$\sigma(\mathcal{O}_K) = \mathbb{Z}\sigma(\alpha_1) + \cdots + \mathbb{Z}\sigma(\alpha_n)$$

Vectors linearly independent over $\mathbb{R}$ are linearly independent over $\mathbb{Z}$, so $\sigma(\mathcal{O}_K)$ is a free $\mathbb{Z}$-module of rank n with basis $\sigma(\alpha_1), \ldots, \sigma(\alpha_n)$, which is condition (3) of definition 50.1.55. It spans $K_{\mathbb{R}}$ by Step 2, which is condition (1). By lemma 50.1.56 it is also discrete, so $\sigma(\mathcal{O}_K)$ is a lattice in $K_{\mathbb{R}}$, completing the proof.

Corollary 1.1.58: Sign Of Discriminant

Let $K/\mathbb{Q}$ be a number field, $\mathcal{O}_K$ its ring of integers, its discriminant. Then:

$$\operatorname{sign}(d_K) = (-1)^{r_2}$$

***Proof*:**
Let $n = [K : \mathbb{Q}]$, let $\alpha_1, \ldots, \alpha_n$ be an integral basis of $\mathcal{O}_K$ (corollary 50.1.33), list the embeddings of K into $\mathbb{C}$ as $\sigma_1, \ldots, \sigma_{r_1}, \tau_1, \overline{\tau_1}, \ldots, \tau_{r_2}, \overline{\tau_{r_2}}$, and let A be the $n \times n$ matrix whose ith row lists their values at α_i in this order. By definition 50.1.24 and definition 50.1.38, $d_K = \det(A)^2$, the square making the order of the embeddings irrelevant. Let B be the real matrix obtained from A by replacing each pair of columns $(\tau_j(\alpha_i), \overline{\tau_j}(\alpha_i))$ by $(\mathrm{Re}\tau_j(\alpha_i), \mathrm{Im}\tau_j(\alpha_i))$. Column by column $\tau_j = \mathrm{Re}\tau_j + i\mathrm{Im}\tau_j$ and $\overline{\tau_j} = \mathrm{Re}\tau_j - i\mathrm{Im}\tau_j$, so expanding both columns by multilinearity, and using that the determinant vanishes when two columns are equal and changes sign when two columns are swapped ([3, Properties Of The Determinant]), gives

$$\det(A) = (-2i)^{r_2} \det(B)$$

Hence $d_K = (-2i)^{2r_2} \det(B)^2 = (-1)^{r_2} 4^{r_2} \det(B)^2$. Here $\det(B)$ is a real number, nonzero because $d_K \neq 0$ (corollary 50.1.28), so $\det(B)^2 > 0$ and $\operatorname{sign}(d_K) = (-1)^{r_2}$, completing the proof.

By proposition 50.1.57, the ring of integers $\mathcal{O}_K$ forms a lattice in $K_{\mathbb{R}}$, and the same will hold for every nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}_K$. Since $K_{\mathbb{R}}$ is an inner product space and a lattice is discrete, it is meaningful to define the volume of a lattice:

Definition 1.1.59: Volume of Lattice

Let V be a real vector space of dimension m with an inner product $\langle\,,\rangle$, and let $\Gamma \subseteq V$ be a complete lattice with $\mathbb{Z}$-basis $v_1, \ldots, v_m$. Then the *volume* of Γ is

$$\mathrm{vol}(\Gamma) := \mathrm{vol}(V/\Gamma) = \mathrm{coVol}(\Gamma) = |\det(\langle v_i, v_j\rangle)|^{1/2}$$

For $V = K_{\mathbb{R}} = \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$, identify $K_{\mathbb{R}}$ with $\mathbb{R}^n$ by writing each complex coordinate z as $(\mathrm{Re}z, \mathrm{Im}z)$ and use the standard inner product of $\mathbb{R}^n$. Then $\mathrm{vol}(\Gamma) = |\det(B)|$, where B is the matrix whose rows are the coordinates of $v_1, \ldots, v_n$, because $(\langle v_i, v_j\rangle) = BB^t$. This is the volume of $v_1, \ldots, v_n$ in the sense of [3, Volume], the Lebesgue volume of the fundamental mesh $\{\sum_i a_i v_i \mid 0 \le a_i < 1\}$. The notation $\mathrm{vol}(\mathcal{O}_K)$ means $\mathrm{vol}(\sigma(\mathcal{O}_K))$.

If you know some differential geometry, we may equivalently take the canonical differential form on $\mathbb{R}^n \cong \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$:

$$\omega = dx_1 \wedge \cdots \wedge dx_n$$

and write:

$$\mathrm{vol}(\mathcal{O}_K) = \mathrm{coVol}(K_{\mathbb{R}}/\mathcal{O}_K) = \left|\int_{K_{\mathbb{R}}/\mathcal{O}_K} \omega\right| = \left|\int_P \omega\right|$$

where $x_1, \ldots, x_n$ are the real coordinates of $K_{\mathbb{R}} \cong \mathbb{R}^n$ fixed in definition 50.1.59 and:

$$P = \left\{\sum a_i v_i \;\middle|\; 0 \le a_i \le 1\right\}$$

This also motivates the vocabulary of covolume. Note that any different choice of a basis will produce the same result since the change of basis matrix B will have determinant ± 1 which in the absolute value becomes 1, and hence it is well-defined. Note that every ideal $\alpha \subseteq \mathcal{O}_K$ will also produce a lattice, since it's a sublattice of $\mathcal{O}_K$, and hence we may find the volume of any sublattice. The way we shall find the volume of the lattice is by taking advantage of the discriminant and the index of the ideal:

Proposition 1.1.60: Volume of Ideal Lattice

Let $0 \neq \mathfrak{a} \subseteq \mathcal{O}_K$ be an ideal. Then $\Gamma_{\mathfrak{a}} = \sigma(\mathfrak{a})$ is a [complete] lattice in $K_{\mathbb{R}}$ and its fundamental mesh has volume

$$\mathrm{vol}(\Gamma_{\mathfrak{a}}) = \frac{1}{2^{r_2}}\sqrt{|\mathrm{disc}(K)|}[\mathcal{O}_K : \mathfrak{a}]$$

As a consequence:

$$\mathrm{vol}(\mathcal{O}_K) = \frac{1}{2^{r_2}}\sqrt{|\mathrm{disc}(\mathcal{O}_K)|}$$

Thus, the discriminant provides volume information for the lattice given by the ring of integers.

In terms of the form ω above, the volume in the proposition is:

$$\mathrm{vol}(\Gamma_{\mathfrak{a}}) = \left|\int_{(\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2})/\Gamma_{\mathfrak{a}}} \omega\right|$$

***Proof*:**
First, $\mathfrak{a}$ has finite index in $\mathcal{O}_K$. Let $0 \neq \beta \in \mathfrak{a}$ and $m = \mathrm{Nm}_{K/\mathbb{Q}}(\beta)$, a nonzero integer by corollary 50.1.14 and corollary 50.1.13. In a Galois closure of K, m/β is the product of the images of β under the embeddings of K other than the inclusion. Each of them is a root of a monic polynomial in $\mathbb{Z}[x]$ that β satisfies, so m/β is an algebraic integer ([2, Integral Elements Form Rings]) lying in K, hence in $\mathcal{O}_K$ (lemma 50.1.5). So $m \in \beta\mathcal{O}_K \subseteq \mathfrak{a}$, and $\mathcal{O}_K/\mathfrak{a}$ is a quotient of $\mathcal{O}_K/m\mathcal{O}_K \cong (\mathbb{Z}/m\mathbb{Z})^n$, using that $\mathcal{O}_K$ is free of rank n (corollary 50.1.33). A subgroup of finite index of $\mathcal{O}_K$ is free of rank n ([3, Submodule Of Free Module Over PID], the rank being n because the quotient is finite). Let $\alpha_1, \dots, \alpha_n$ be a $\mathbb{Z}$-basis of $\mathfrak{a}$, so that $\Gamma_\mathfrak{a} = \mathbb{Z}\sigma(\alpha_1) + \cdots + \mathbb{Z}\sigma(\alpha_n)$. List all n embeddings $K \to \mathbb{C}$ as $\sigma_1, \dots, \sigma_{r_1}, \tau_1, \overline{\tau_1}, \dots, \tau_{r_2}, \overline{\tau_{r_2}}$, and let A be the $n \times n$ matrix whose ith row lists their values at α_i in this order. By definition 50.1.24 and proposition 50.1.39,

$$\det(A)^2 = \mathrm{disc}(\alpha_1, \dots, \alpha_n) = [\mathcal{O}_K : \mathfrak{a}]^2 \mathrm{disc}(\mathcal{O}_K) = [\mathcal{O}_K : \mathfrak{a}]^2 d_K$$

Identify $K_\mathbb{R} = \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ with $\mathbb{R}^n$ by writing each complex coordinate z as $(\mathrm{Re}z, \mathrm{Im}z)$, and measure volumes with Lebesgue measure on $\mathbb{R}^n$. Then $\sigma(\alpha_i)$ is the ith row of the real matrix B obtained from A by replacing each pair of columns $(\tau_j(\alpha_i), \overline{\tau_j}(\alpha_i))$ by $(\mathrm{Re}\tau_j(\alpha_i), \mathrm{Im}\tau_j(\alpha_i))$. Column by column $\tau_j = \mathrm{Re}\tau_j + i\mathrm{Im}\tau_j$ and $\overline{\tau_j} = \mathrm{Re}\tau_j - i\mathrm{Im}\tau_j$, so expanding both columns by multilinearity of the determinant leaves $-2i$ times the determinant with columns $\mathrm{Re}\tau_j, \mathrm{Im}\tau_j$, and

$$\det(A) = (-2i)^{r_2} \det(B)$$

Hence $|\det(B)| = 2^{-r_2}\sqrt{|d_K|}\,[\mathcal{O}_K : \mathfrak{a}]$, which is nonzero since $d_K \neq 0$ (corollary 50.1.28). So $\sigma(\alpha_1), \dots, \sigma(\alpha_n)$ are linearly independent over $\mathbb{R}$: if $\sum_i c_i\sigma(\alpha_i) = 0$ with $c_k \neq 0$, replacing the kth row of B by this combination gives a matrix with a zero row, whose determinant is both 0 and $c_k \det(B)$ by multilinearity and the alternating property ([3, Properties Of The Determinant]). Being n independent vectors of the n-dimensional space $K_\mathbb{R}$, they form a basis, $\Gamma_\mathfrak{a}$ is free of rank n and spans $K_\mathbb{R}$, and it is a complete lattice by lemma 50.1.56. The volume of its fundamental mesh $\{\sum_i a_i\sigma(\alpha_i) \mid 0 \leq a_i < 1\}$ is $|\det(B)|$, which is definition 50.1.59 in these coordinates, since $|\det(\langle\sigma(\alpha_i), \sigma(\alpha_j)\rangle)|^{1/2} = |\det(BB^t)|^{1/2} = |\det(B)|$. This is the first formula, and $\mathfrak{a} = \mathcal{O}_K$ gives the second, as we sought to show.

Thus, if $K = \mathbb{Q}(\sqrt{m})$ where m is square-free, then $\mathrm{vol}(\mathcal{O}_K) = 2^{-r_2}\sqrt{|d_K|}$ with $d_K = m$ or $4m$ (example 50.1.25), where $r_2 = 0$ if $m > 0$ and $r_2 = 1$ if $m < 0$:

$$\mathrm{vol}(\mathcal{O}_K) = \begin{cases} \sqrt{m} & m > 0,\ m \equiv 1 \bmod 4 \\ 2\sqrt{m} & m > 0,\ m \equiv 2, 3 \bmod 4 \\ \frac{1}{2}\sqrt{|m|} & m < 0,\ m \equiv 1 \bmod 4 \\ \sqrt{|m|} & m < 0,\ m \equiv 2, 3 \bmod 4 \end{cases}$$

Absolute Norm and Geometric Inequalities

Besides introducing a geometric intuition for number rings and their ideals, the key reason to introduce volumes is that they will allow us make many bounding arguments on ideals using the lattice structure of $\mathcal{O}_K$. To do this, we introduce a new "norm" on the ideals of $\mathcal{O}_K$ that shall

represent the size of the ideal with respect to the lattice. This will be helpful in many algebraic arguments, but also in upgrading many clasical results about primes (such as their distribution[12], see [8]).

Definition 1.1.61: Absolute Norm

Let $0 \neq \mathfrak{a} \subseteq \mathcal{O}_K$ be a nonzero ideal of $\mathcal{O}_K$. Then the *absolute norm* of $\mathfrak{a}$ is given by

$$\mathfrak{N}_K(\mathfrak{a}) := [\mathcal{O}_K : \mathfrak{a}]$$

If unambiguous, the notation $\mathfrak{N}(\mathfrak{a})$ is also used.

The absolute norm is inspired by the picture of an ideal of $\mathcal{O}_K$ as a sublattice of a certain sparsity inside the lattice $\mathcal{O}_K$: the index $[\mathcal{O}_K : \mathfrak{a}]$ is the measure of that sparsity. Note that the notation $\mathfrak{N}_{K/\mathbb{Q}}$ should not be used, for it shall have a different meaning than $\mathfrak{N}$[13]

For the definition to be of any use, the index $[\mathcal{O}_K : \mathfrak{a}]$ had better be finite. Note that proposition 50.1.39 does not give this on its own, for it *assumes* the submodule has finite index. The finiteness deserves its own proof:

Proposition 1.1.62: Finiteness of Residue Rings

Let $K/\mathbb{Q}$ be a number field of degree n and let $0 \neq \mathfrak{a} \subseteq \mathcal{O}_K$ be a nonzero ideal. Then the quotient ring $\mathcal{O}_K/\mathfrak{a}$ is finite. In particular, $\mathfrak{N}(\mathfrak{a}) = [\mathcal{O}_K : \mathfrak{a}]$ is finite for every nonzero ideal $\mathfrak{a}$.

***Proof*:**
First, we show that $\mathfrak{a}$ contains a nonzero rational integer. Let $0 \neq \alpha \in \mathfrak{a}$ and let $m = \mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$. By corollary 50.1.14, $m \in \mathbb{Z}$, and by corollary 50.1.13, $m = \prod_i \sigma_i(\alpha)$ is a product of nonzero elements of a field, hence $m \neq 0$. Furthermore, since $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = \alpha \prod_{\sigma_i \neq \mathrm{id}} \sigma_i(\alpha)$, we have that m/α is a product of Galois conjugates of α. Each conjugate is integral over $\mathbb{Z}$ (being a root of the same monic integer minimal polynomial as α), hence m/α is an algebraic integer by [2, Integral Elements Form Rings]. Since $m/\alpha \in K$, we get $m/\alpha \in \mathcal{O}_K$, and so:

$$m = \alpha \cdot \frac{m}{\alpha} \in \alpha\mathcal{O}_K \subseteq \mathfrak{a}$$

Next, since $m\mathcal{O}_K \subseteq \mathfrak{a} \subseteq \mathcal{O}_K$, the quotient map $\mathcal{O}_K/m\mathcal{O}_K \to \mathcal{O}_K/\mathfrak{a}$ is a surjection of abelian groups. By corollary 50.1.33, $\mathcal{O}_K$ is a free $\mathbb{Z}$-module of rank n, so choosing an integral basis:

$$\mathcal{O}_K/m\mathcal{O}_K \cong (\mathbb{Z}/m\mathbb{Z})^n$$

which has $|m|^n$ elements. Being a quotient of a finite group, $\mathcal{O}_K/\mathfrak{a}$ is finite, with $[\mathcal{O}_K : \mathfrak{a}]$ dividing $|m|^n$, as we sought to show.

[12] the number of primes p where $|p| \leq X$ tends asymptotically to $\frac{X}{\log(X)}$
[13] see definition 50.3.13

Lemma 1.1.63: Norm Of Principal Ideal

Let $0 \neq \alpha \in \mathcal{O}_K$. Then:

$$\mathfrak{N}((\alpha)) = \left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right|$$

***Proof*:**
Let $\beta_1, \ldots, \beta_n$ be an integral basis of $\mathcal{O}_K$ (corollary 50.1.33). Multiplication by α is an injective additive map from $\mathcal{O}_K$ onto $\alpha\mathcal{O}_K$, so $\alpha\beta_1, \ldots, \alpha\beta_n$ is a $\mathbb{Z}$-basis of $(\alpha) = \alpha\mathcal{O}_K$. Let $A = (a_{ij}) \in M_n(\mathbb{Z})$ be the matrix with $\alpha\beta_i = \sum_j a_{ij}\beta_j$. The ideal (α) is nonzero, so it has finite index in $\mathcal{O}_K$ (proposition 50.1.62), and proposition 50.1.39 gives $\mathfrak{N}((\alpha)) = [\mathcal{O}_K : \alpha\mathcal{O}_K] = |\det(A)|$. The matrix of multiplication by α in the basis $\beta_1, \ldots, \beta_n$ has as jth column the coordinates of $\alpha\beta_j$, so it is A^t, and by definition 50.1.10 and [3, Properties Of The Determinant], $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = \det(A^t) = \det(A)$. Hence $\mathfrak{N}((\alpha)) = |\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)|$, completing the proof.

Lemma 1.1.64: Ideal Divides Absolute Norm

Let $0 \neq I \subseteq \mathcal{O}_K$ be an ideal. Then $\mathfrak{N}(I) \in I$, that is:

$$\mathfrak{N}(I)\mathcal{O}_K \subseteq I$$

***Proof*:**
The index $\mathfrak{N}(I) = [\mathcal{O}_K : I]$ is finite (proposition 50.1.62). In the additive group $\mathcal{O}_K/I$ of order $\mathfrak{N}(I)$, the element $1 + I$ satisfies $\mathfrak{N}(I) \cdot (1 + I) = 0$ by [3, Lagrange Theorem], so $\mathfrak{N}(I) \in I$ and $\mathfrak{N}(I)\mathcal{O}_K \subseteq I$, completing the proof.

Lemma 1.1.65: Volume Form Reformulation

Let K be a number field and $0 \neq I \subseteq \mathcal{O}_K$ an ideal, and write $\mathrm{vol}(I)$ for $\mathrm{vol}(\sigma(I))$. Then:

$$\mathrm{vol}(I) = \mathfrak{N}(I)\mathrm{vol}(\mathcal{O}_K)$$

***Proof*:**
By proposition 50.1.60, $\mathrm{vol}(I) = 2^{-r_2}\sqrt{|d_K|}\,[\mathcal{O}_K : I]$ and $\mathrm{vol}(\mathcal{O}_K) = 2^{-r_2}\sqrt{|d_K|}$. Since $\mathfrak{N}(I) = [\mathcal{O}_K : I]$ by definition 50.1.61, the two formulas combine to $\mathrm{vol}(I) = \mathfrak{N}(I)\mathrm{vol}(\mathcal{O}_K)$, completing the proof.

Theorem 1.1.66: Bounding Ideals

Let $K/\mathbb{Q}$ be a number field and $\mathcal{O}_K$ its number ring. Then there exists $\lambda \in \mathbb{R}_{>0}$ such that for every nonzero ideal $I \subseteq \mathcal{O}_K$, there exists a nonzero $\alpha \in I$ such that

$$|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| \leq \lambda\mathfrak{N}(I)$$

***Proof*:**
Let $\alpha_1, \dots, \alpha_n$ be an integral basis of $\mathcal{O}_K$ (corollary 50.1.33), let $\sigma_1, \dots, \sigma_n$ be the embeddings of K into $\mathbb{C}$, and put

$$\lambda = \prod_{i=1}^{n} \left(\sum_{j=1}^{n} |\sigma_i(\alpha_j)| \right)$$

Each factor is positive, since σ_i is injective and $\alpha_1 \neq 0$. Let I be a nonzero ideal. Then $\mathfrak{N}(I)$ is a positive integer (proposition 50.1.62), so there is a unique integer $m \geq 1$ with

$$m^n \leq \mathfrak{N}(I) < (m+1)^n$$

The $(m+1)^n$ elements $\sum_j m_j \alpha_j$ with $m_j \in \mathbb{Z}$ and $0 \leq m_j \leq m$ are distinct, as the α_j form a $\mathbb{Z}$-basis, and there are more of them than the $\mathfrak{N}(I) = |\mathcal{O}_K/I|$ cosets of I. So two of them are congruent modulo I, and their difference is a nonzero element of I of the form

$$\alpha = \sum_{j=1}^{n} m_j \alpha_j \qquad m_j \in \mathbb{Z} \qquad |m_j| \leq m$$

By corollary 50.1.13 and the triangle inequality in $\mathbb{C}$,

$$\left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| = \prod_{i=1}^{n} |\sigma_i(\alpha)| \leq \prod_{i=1}^{n} \left(\sum_{j=1}^{n} |m_j||\sigma_i(\alpha_j)| \right) \leq m^n \lambda \leq \mathfrak{N}(I)\lambda$$

completing the proof.

Proposition 1.1.67: Image Of Norm Is Finite Up To Units

Let K be a number field of degree n, $\mathcal{O}_K$ its ring of integers, and $a \in \mathbb{Z}_{>0}$. Then up to multiplication by units, there are only finitely many elements $\alpha \in \mathcal{O}_K$ such that

$$\left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| = a$$

More precisely, there are at most a^n of them.

***Proof*:**
Step 1: one associate class per coset. Let $\alpha, \beta \in \mathcal{O}_K$ with $|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)| = |\mathrm{Nm}_{K/\mathbb{Q}}(\beta)| = a$ and $\beta - \alpha \in a\mathcal{O}_K$, say $\beta = \alpha + a\gamma$ with $\gamma \in \mathcal{O}_K$. Both are nonzero, as their norms are. By lemma 50.1.63, the additive group $\mathcal{O}_K/\beta\mathcal{O}_K$ has a elements, so $a \cdot 1 \in \beta\mathcal{O}_K$ by [3, Lagrange Theorem], that is, $a/\beta \in \mathcal{O}_K$. In the same way $a/\alpha \in \mathcal{O}_K$. Hence

$$\frac{\alpha}{\beta} = 1 - \frac{a}{\beta}\gamma \in \mathcal{O}_K \qquad \frac{\beta}{\alpha} = 1 + \frac{a}{\alpha}\gamma \in \mathcal{O}_K$$

so α/β is a unit of $\mathcal{O}_K$, and α and β are associates.

Step 2: counting. By corollary 50.1.33, $\mathcal{O}_K/a\mathcal{O}_K \cong (\mathbb{Z}/a\mathbb{Z})^n$ has a^n cosets. By Step 1, the elements of absolute norm a in one coset form at most one class up to units, so there are at most

a^n such classes, completing the proof.

For computational purposes, we add a tighter bound on the norms of ideals. We start with the following lemma:

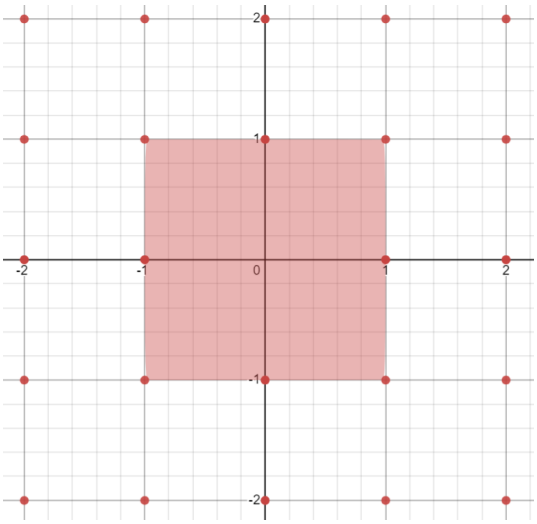
Lemma 1.1.68: Minkowski's Lattice Point Theorem

Let Γ be a discrete complete lattice in $\mathbb{R}^n$, with its standard inner product, and let X be a centrally symmetric convex subset of $\mathbb{R}^n$. Then if

$$\text{vol}(X) > 2^n \text{vol}(\Gamma)$$

then X contains at least one nonzero lattice point $\gamma \in \Gamma$

Before proving this result, consider $\sigma(\mathbb{Z}[i]) = \mathbb{Z}^2$ in $\mathbb{C} = \mathbb{R}^2$, with $\text{vol}(\mathbb{Z}^2) = 1$. The open square $(-1, 1)^2$ is convex and centrally symmetric, has area $4 = 2^2\text{vol}(\mathbb{Z}^2)$, and meets $\mathbb{Z}^2$ only in 0, so the constant 2^n cannot be lowered:



***Proof*:**
Volumes are Lebesgue measure on $\mathbb{R}^n$, and we use these facts of the analysis assumed in this chapter: convex sets are Lebesgue measurable, Lebesgue measure is countably additive and translation invariant, $\text{vol}(\frac{1}{2}X) = 2^{-n}\text{vol}(X)$, and the fundamental mesh $\Phi = \{\sum_i a_i v_i \mid 0 \leq a_i < 1\}$ of a $\mathbb{Z}$-basis $v_1, \ldots, v_n$ of Γ has volume $\text{vol}(\Gamma)$ (definition 50.1.59).

Step 1: a tiling. The v_i form a basis of $\mathbb{R}^n$, as Γ is complete. Every $v = \sum_i x_i v_i \in \mathbb{R}^n$ equals $\sum_i (x_i - \lfloor x_i \rfloor)v_i + \sum_i \lfloor x_i \rfloor v_i$, and this is the only way to write $v = \varphi + \gamma$ with $\varphi \in \Phi$ and $\gamma \in \Gamma$. So $\mathbb{R}^n$ is the disjoint union of the translates $\Phi + \gamma$, $\gamma \in \Gamma$.

Step 2: two translates of $\frac{1}{2}X$ meet. Suppose the sets $\frac{1}{2}X + \gamma$, $\gamma \in \Gamma$, are pairwise disjoint. Then

so are the sets $\Phi \cap (\frac{1}{2}X + \gamma) \subseteq \Phi$, and as $\Gamma \cong \mathbb{Z}^n$ is countable,

$$\sum_{\gamma \in \Gamma} \mathrm{vol}\left(\Phi \cap \left(\tfrac{1}{2}X + \gamma\right)\right) \leq \mathrm{vol}(\Phi) = \mathrm{vol}(\Gamma)$$

Translating by $-\gamma$ gives $\mathrm{vol}(\Phi \cap (\frac{1}{2}X + \gamma)) = \mathrm{vol}((\Phi - \gamma) \cap \frac{1}{2}X)$, and by Step 1 the sets $(\Phi - \gamma) \cap \frac{1}{2}X$, $\gamma \in \Gamma$, partition $\frac{1}{2}X$. So the left side is $\mathrm{vol}(\frac{1}{2}X) = 2^{-n}\mathrm{vol}(X) > \mathrm{vol}(\Gamma)$, a contradiction.

Step 3: a lattice point. By Step 2 there are $\gamma_1 \neq \gamma_2$ in Γ and $x_1, x_2 \in X$ with $\frac{1}{2}x_1 + \gamma_1 = \frac{1}{2}x_2 + \gamma_2$. Then $\gamma = \gamma_1 - \gamma_2$ is a nonzero element of Γ, and $\gamma = \frac{1}{2}x_2 + \frac{1}{2}(-x_1)$. As X is centrally symmetric, $-x_1 \in X$, and as X is convex, the midpoint γ of x_2 and $-x_1$ lies in X, completing the proof.

Theorem 1.1.69: Minkowski's Bound

Let K be a number field of degree $n = r_1 + 2r_2$, with r_1 real embeddings and r_2 pairs of complex conjugate embeddings, and let $0 \neq \mathfrak{a} \subseteq \mathcal{O}_K$ be an ideal. Then $\mathfrak{a}$ contains an element $\alpha \neq 0$ with

$$\left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| \leq \frac{n!}{n^n}\left(\frac{4}{\pi}\right)^{r_2}\sqrt{|d_K|}\,\mathfrak{N}(\mathfrak{a})$$

The constant $M_K = \frac{n!}{n^n}\left(\frac{4}{\pi}\right)^{r_2}\sqrt{|d_K|}$ is called the *Minkowski bound* of K.

***Proof*:**
Let $\sigma_1, \ldots, \sigma_{r_1}$ be the real embeddings of K and $\tau_1, \ldots, \tau_{r_2}$ one complex embedding from each conjugate pair, as in definition 50.1.53. As in definition 50.1.59, identify $K_{\mathbb{R}} = \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ with $\mathbb{R}^n$ by writing each complex coordinate z as $(\mathrm{Re}z, \mathrm{Im}z)$, and measure volumes with Lebesgue measure on $\mathbb{R}^n$. By proposition 50.1.60 and definition 50.1.61, $\Gamma_{\mathfrak{a}} = \sigma(\mathfrak{a})$ is a complete lattice in $K_{\mathbb{R}}$ with

$$\mathrm{vol}(\Gamma_{\mathfrak{a}}) = \frac{1}{2^{r_2}}\sqrt{|d_K|}\,\mathfrak{N}(\mathfrak{a})$$

For $(x, z) \in \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$ put $\|(x, z)\| = \sum_{i=1}^{r_1} |x_i| + 2\sum_{j=1}^{r_2} |z_j|$, and for $t \geq 0$ let

$$X_t = \{(x, z) \in \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2} \mid \|(x, z)\| \leq t\}$$

Step 1: X_t is convex and centrally symmetric. The triangle inequality for absolute values on $\mathbb{R}$ and on $\mathbb{C}$ gives $\|u + v\| \leq \|u\| + \|v\|$, and $\|\lambda u\| = |\lambda|\,\|u\|$ for $\lambda \in \mathbb{R}$. So $\|-u\| = \|u\|$, and for $u, v \in X_t$ and $0 \leq \lambda \leq 1$ we get $\|\lambda u + (1-\lambda)v\| \leq \lambda\|u\| + (1-\lambda)\|v\| \leq t$.

Step 2: $\mathrm{vol}(X_t) = 2^{r_1}\left(\frac{\pi}{2}\right)^{r_2}\frac{t^n}{n!}$. Write $V_{r_1,r_2}(t)$ for this volume. By Fubini's theorem the volume of a closed region is the integral, over one coordinate in $\mathbb{R}$ or in $\mathbb{C} \cong \mathbb{R}^2$, of the volumes of its slices, and in polar coordinates $w = \rho e^{i\theta}$ on $\mathbb{C}$ the area element is $\rho\, d\rho\, d\theta$. We induct on $n \geq 1$. For $(r_1, r_2) = (1, 0)$, $X_t = [-t, t]$ has length $2t$. For $(r_1, r_2) = (0, 1)$, X_t is the disc $|z| \leq t/2$, of area $\pi t^2/4 = \frac{\pi}{2} \cdot \frac{t^2}{2!}$. Let $n \geq 2$ with $(r_1, r_2) \neq (0, 1)$, and assume the formula in all smaller dimensions and for all $t \geq 0$.

If $r_1 \geq 1$, slice along the first real coordinate $x_1 = y$. For $|y| \leq t$ the slice is the region $X_{t-|y|}$ for $(r_1 - 1, r_2)$, of dimension $n - 1 \geq 1$, and for $|y| > t$ it is empty. Since $\int_{-t}^{t} (t - |y|)^{n-1}\, dy = 2t^n/n$,

$$V_{r_1,r_2}(t) = \int_{-t}^{t} 2^{r_1-1} \left(\frac{\pi}{2}\right)^{r_2} \frac{(t-|y|)^{n-1}}{(n-1)!}\, dy = 2^{r_1} \left(\frac{\pi}{2}\right)^{r_2} \frac{t^n}{n!}$$

If $r_1 = 0$, then $r_2 \geq 2$. Slice along the last complex coordinate $z_{r_2} = w$. For $2|w| \leq t$ the slice is the region $X_{t-2|w|}$ for $(0, r_2 - 1)$, of dimension $n - 2 \geq 2$, and otherwise it is empty. In polar coordinates, followed by the substitution $u = 2\rho$,

$$V_{0,r_2}(t) = \int_0^{2\pi} \int_0^{t/2} \left(\frac{\pi}{2}\right)^{r_2-1} \frac{(t-2\rho)^{n-2}}{(n-2)!}\, \rho\, d\rho\, d\theta = \left(\frac{\pi}{2}\right)^{r_2-1} \frac{2\pi}{4\,(n-2)!} \int_0^t (t-u)^{n-2} u\, du$$

Writing $u = t - (t - u)$ gives $\int_0^t (t-u)^{n-2} u\, du = \frac{t^n}{n-1} - \frac{t^n}{n} = \frac{t^n}{n(n-1)}$, so $V_{0,r_2}(t) = \left(\frac{\pi}{2}\right)^{r_2} \frac{t^n}{n!}$, which completes the induction.

Step 3: the inequality between the means. For real numbers $c_1, \ldots, c_n \geq 0$ with mean $\mu = (c_1 + \cdots + c_n)/n$ we have $c_1 \cdots c_n \leq \mu^n$. For $n = 1$ this is an equality. For $n \geq 2$, reorder so that $c_1 \leq \mu \leq c_n$ and put $c' = c_1 + c_n - \mu$, so that $c' \geq c_1 \geq 0$. Then $c_1 c_n \leq \mu c'$, because $\mu c' - c_1 c_n = (\mu - c_1)(c_n - \mu) \geq 0$, and the $n - 1$ numbers $c', c_2, \ldots, c_{n-1}$ have mean μ. By induction on n, $c' c_2 \cdots c_{n-1} \leq \mu^{n-1}$, hence $c_1 \cdots c_n \leq \mu\, c' c_2 \cdots c_{n-1} \leq \mu^n$.

Step 4: a nonzero element of small norm. Let $\varepsilon > 0$ and choose $t > 0$ with

$$t^n = n! \left(\frac{4}{\pi}\right)^{r_2} \sqrt{|d_K|}\, \mathfrak{N}(\mathfrak{a}) + n^n \varepsilon$$

By Step 2, and since $n - r_2 = r_1 + r_2$,

$$\mathrm{vol}(X_t) > 2^{r_1} \left(\frac{\pi}{2}\right)^{r_2} \left(\frac{4}{\pi}\right)^{r_2} \sqrt{|d_K|}\, \mathfrak{N}(\mathfrak{a}) = 2^{r_1+r_2} \sqrt{|d_K|}\, \mathfrak{N}(\mathfrak{a}) = 2^n \mathrm{vol}(\Gamma_{\mathfrak{a}})$$

By Step 1 and lemma 50.1.68, there is $\alpha \in \mathfrak{a}$ with $0 \neq \sigma(\alpha) \in X_t$, so $\alpha \neq 0$. By corollary 50.1.13, in which τ_j and $\overline{\tau_j}$ together contribute $\tau_j(\alpha)\overline{\tau_j(\alpha)} = |\tau_j(\alpha)|^2$,

$$\left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| = \prod_{i=1}^{r_1} |\sigma_i(\alpha)| \prod_{j=1}^{r_2} |\tau_j(\alpha)|^2$$

This is the product of the n numbers $|\sigma_i(\alpha)|$ and $|\tau_j(\alpha)|$, the latter each taken twice, whose sum is $\|\sigma(\alpha)\| \leq t$. So Step 3 gives

$$\left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| \leq \frac{t^n}{n^n} = M_K \mathfrak{N}(\mathfrak{a}) + \varepsilon$$

Step 5: removing ε. Put $C = M_K \mathfrak{N}(\mathfrak{a})$ and apply Step 4 with $\varepsilon = \frac{1}{2}(\lfloor C \rfloor + 1 - C) > 0$. The resulting α satisfies $\left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| \leq C + \varepsilon < \lfloor C \rfloor + 1$, and $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$ is an integer (corollary 50.1.14), so $\left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| \leq \lfloor C \rfloor \leq C$, as we sought to show.

Applied to $\mathfrak{a} = \mathcal{O}_K$, theorem 50.1.69 gives some $\alpha \neq 0$ in $\mathcal{O}_K$ with $\left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| \leq M_K$. Here $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$ is an integer (corollary 50.1.14) and is nonzero, being a product of the nonzero numbers $\sigma_i(\alpha)$ (corollary 50.1.13), so $1 \leq M_K$, that is

$$\frac{n^n}{n!}\left(\frac{\pi}{4}\right)^{r_2} \leq \sqrt{|d_K|}$$

Thus, we get a result that may seem obvious, but is in general rather difficult to prove:

Corollary 1.1.70: Discriminant Nonzero

Let $\mathcal{O}_K \neq \mathbb{Z}$. Then:

$$|d_K| > 1$$

***Proof*:**
If $K = \mathbb{Q}$ then $\mathcal{O}_K = \mathcal{O}_K \cap \mathbb{Q} = \mathbb{Z}$ by theorem 50.1.9(1), so $\mathcal{O}_K \neq \mathbb{Z}$ forces $n \geq 2$. As $0 < \pi/4 < 1$ and $r_2 \leq n/2$, the inequality above gives $\sqrt{|d_K|} \geq a_n$, where $a_m = \frac{m^m}{m!}\left(\frac{\pi}{4}\right)^{m/2}$. For every $m \geq 1$,

$$\frac{a_{m+1}}{a_m} = \left(1 + \frac{1}{m}\right)^m \frac{\sqrt{\pi}}{2} \geq 2 \cdot \frac{\sqrt{\pi}}{2} > 1$$

because the binomial expansion of $(1 + 1/m)^m$ has nonnegative terms, the first two of which sum to 2. Hence $a_n \geq a_2 = \pi/2 > 1$, so $|d_K| > 1$, as claimed.

1.2 Ring Structure and Dedekind Domains

So far, we have shown that rings of integers are lattices whose rank corresponds to the dimension of their corresponding field extension. Since they are rings, we may naturally ask what type of ring are they. Using example 50.1.17, we can look at the particular example of $\mathcal{O}_{\mathbb{Q}(\sqrt{-5})} = \mathbb{Z}[\sqrt{-5}]$. Is this ring a Euclidean Domain, Principal Ideal Domain, or Unique Factorization Domain? It is certainly Noetherian, being a finitely generated $\mathbb{Z}$-module (where $\mathbb{Z}$ is Noetherian). However, not all irreducible elements are prime, for example:

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

showing that 2 is not prime. This may seem troubling since we may have to keep track of multiple factorizations showing there is more structure or tools that would need to be added/used to keep track of this information. Fortunately, there is a way to recover factorization. The goal of this section is to define a new type of ring for which the notion of Unique factorization is recovered, and show that all rings of integers are this type of ring. First, we may use the norm to see that a factorization into irreducible is always certainly possible in $\mathcal{O}_k$. For, if $\alpha = \beta\gamma$, then if β, γ are non-units:

$$1 < |N_{K/\mathbb{Q}}(\beta)| < |N_{K/\mathbb{Q}}(\alpha)| \qquad 1 < |N_{K/\mathbb{Q}}(\gamma)| < |N_{K/\mathbb{Q}}(\alpha)|$$

showing that it is a factorization domain. However this factorization need not be unique as we have demonstrated with $\mathcal{O}_{\mathbb{Q}(\sqrt{-5})} = \mathbb{Z}[\sqrt{-5}]$, another example being $21 = 3 \cdot 7 = (1 + 2\sqrt{-5})(1 - 2\sqrt{-5})$. We may see these are irreducible elements by using the norm. Doing the case of 3, if $3 = \alpha\beta$ for

non-units α, β, then $9 = N_{K/\mathbb{Q}}(\alpha)N_{K/\mathbb{Q}}(\beta)$ would give us that $N_{K/\mathbb{Q}}(\alpha) = \pm 3$. Given $\alpha = a + b\sqrt{-5}$ we get

$$N_{K/\mathbb{Q}}(a + b\sqrt{-5}) = a^2 + 5b^2 = \pm 3$$

But this certainly has no solutions in $\mathbb{Z}$, and hence 3 is irreducible in $\mathcal{O}_K$. A similar argument can be made for $7, 1 \pm 2\sqrt{-5}$. Furthermore, the numbers 3 and 5 are not associated to $1 \pm 2\sqrt{-5}$, for we would require that

$$\frac{1 \pm 2\sqrt{-5}}{3}, \frac{1 \pm 2\sqrt{-5}}{7} \in \mathcal{O}_K$$

both of which we can check are not using the norm. Hence, we certainly have different factorizations.

The failure of unique factorization has lead Eduard Kummer to try to embed the ring of integers of K into a bigger domain of "ideal numbers" where unique factorization into "ideal prime numbers" would hold (which was inspired by the discovery of complex numbers somehow being "ideal" for roots polynomial roots). As an example, our original decomposition in $\mathbb{Z}[\sqrt{-5}]$

$$21 = 3 \cdot 7 = (1 + 2\sqrt{-5})(1 - 2\sqrt{-5})$$

would factor into ideal prime numbers $\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3, \mathfrak{p}_4$ with the relations

$$3 = \mathfrak{p}_1\mathfrak{p}_2 \qquad 7 = \mathfrak{p}_3\mathfrak{p}_4 \qquad 1 + 2\sqrt{-5} = \mathfrak{p}_1\mathfrak{p}_3 \qquad 1 - 2\sqrt{-5} = \mathfrak{p}_2\mathfrak{p}_4$$

Then we would have fixed non-uniqueness since we would get the factorization:

$$21 = (\mathfrak{p}_1\mathfrak{p}_2)(\mathfrak{p}_3\mathfrak{p}_4) = (\mathfrak{p}_1\mathfrak{p}_3)(\mathfrak{p}_2\mathfrak{p}_4)$$

Some key properties that these ideals numbers should have is that for $a, b, \lambda \in \mathcal{O}_K$, an ideal number $\mathfrak{a}$ should satisfy the following natural divisibility restrictions:

$$\mathfrak{a} \big| a \text{ and } \mathfrak{a} \big| b \Rightarrow \mathfrak{a} \big| a \pm b \qquad a\mathfrak{a} \big| a \Rightarrow \mathfrak{a} \big| \lambda a$$

An furthermore an ideal number $\mathfrak{a}$ ought to be determinable by the set of its divisors in $\mathcal{O}_K$, that is

$$\underline{\mathfrak{a}} = \{a \in \mathcal{O}_K \mid \mathfrak{a} \big| a\}$$

But given these rules, this set is simply an ideal of $\mathcal{O}_K$ as we know the notion of ideal. It was Richard Dedekind that took the notion of ideal numbers that Kummer invented and invented the notion of *ideals* of $\mathcal{O}_K$ giving the modern definition of ideals. Then division $\mathfrak{a} \big| a$ is replaced by belonging $a \in \mathfrak{a}$, and divisibility $\mathfrak{a} \big| \mathfrak{b}$ is replaced by containment $\mathfrak{b} \subseteq \mathfrak{a}$.

Hence, we shall seek a unique factorization of *ideals* into *prime ideals* in $\mathcal{O}_K$. The 3 key properties of $\mathcal{O}_K$ that require this factorization was singled out by Dedekind and form a class of rings larger than rings of integers and are worth taking a look at on their own:

Definition 1.2.1: Dedekind Domain

Let R be an integral domain. Then if:

1. R is a Noetherian ring
2. every nonzero prime ideal of R is maximal
3. R is normal ($\overline{R}^{K(R)} = R$)

then R is called a *Dedekind Domain*

The Noetherian condition can be thought of as a natural eliminator of any infinites lurking around, while the prime implies maximal condition can be thought of as making sure that there can be a prime ideal that further contains prime ideals. In particular, we shall see that $(a) \subseteq (b)$ if and only if $b|a$, which does not make sense if we allow for prime ideals being contained in larger prime ideals.

1.2.2 Foreshadowing in Algebraic Geometry Notice that we have already seen a ring with these conditions before, namely if V is a smooth 1-dimensional variety, then the associated k-algebra is Noetherian, 1 [Krull] dimensional and is normal (normal implies smoothness[a]). In Algebraic Geometry, Dedekind domains can be thought of as representing 1-dimensional smooth schemes (in particular, 1-dimensional smooth $\mathbb{Z}$-curves)

[a]But only necessarily in 1 dimension; in higher dimension normalness need not imply smoothness, see [1]

Lemma 1.2.3: PID's, UFD's, and Dedekind Domain

Let R be a ring. Then:

1. If R is a PID, R is a Dedekind Domain
2. If R is a UFD, R need not be a Dedekind Domain, and if R is a Dedekind domain, R need not be a UFD
3. If R is a Dedekind domain and a UFD, it is a PID

The following image demonstrates the realation:

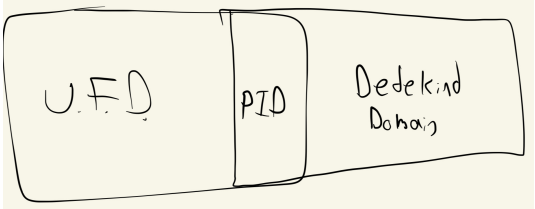


Figure 1.1: Venn diagram

***Proof*:**

1. Let R be a PID. Every ideal of R is principal, hence finitely generated, so R is Noetherian ([2, Equivalence of Noetherian Ring]). Every nonzero prime ideal of R is maximal by [3, In PID, Prime Ideal iff Maximal Ideal]. Finally, R is a UFD by [3, Prime If And Only If Irreducible In PID], and a UFD is integrally closed in its fraction field ([2, A Unique Factorization Domain Is Integrally Closed]). So R satisfies the three conditions of definition 50.2.1.
2. The ring $\mathbb{Z}[x]$ is a UFD by [3, R UFD If And Only If $R[x]$ UFD], since $\mathbb{Z}$ is one ([3, $\mathbb{Z}$ is a UFD (Fundamental Theorem of Arithmetics)]). Evaluation at 0 is a surjective ring homomorphism $\mathbb{Z}[x] \to \mathbb{Z}$ with kernel (x), so $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, an integral domain that is not a field. Hence (x) is a nonzero prime ideal that is not maximal, and $\mathbb{Z}[x]$ is not a Dedekind

domain.

For the second claim let $R = \mathbb{Z}[\sqrt{-13}]$. Since -13 is square-free and $-13 \equiv 3 \bmod 4$, $R = \mathcal{O}_K$ for $K = \mathbb{Q}(\sqrt{-13})$ (example 50.1.17). It is a Dedekind domain. It is Noetherian: R is a free $\mathbb{Z}$-module of rank 2 (corollary 50.1.33), so every ideal, being a $\mathbb{Z}$-submodule, is a finitely generated $\mathbb{Z}$-module ([3, Submodule Of Free Module Over PID]), hence a finitely generated ideal. Every nonzero prime $\mathfrak{p}$ of R is maximal, because $R/\mathfrak{p}$ is a finite integral domain (proposition 50.1.62), hence a field ([3, Finite Integral Domain is a Field]). And R, being the integral closure of $\mathbb{Z}$ in K, is integrally closed in K ([2, Integral Elements Form Rings]).

For $\gamma = a + b\sqrt{-13}$ with $a, b \in \mathbb{Z}$, corollary 50.1.13 gives $\mathrm{Nm}(\gamma) = \mathrm{Nm}_{K/\mathbb{Q}}(\gamma) = (a + b\sqrt{-13})(a - b\sqrt{-13}) = a^2 + 13b^2$, a nonnegative integer. By proposition 50.1.18, γ is a unit exactly when $\mathrm{Nm}(\gamma) = 1$, that is when $b = 0$ and $a = \pm 1$. In R,

$$14 = 2 \cdot 7 = (1 + \sqrt{-13})(1 - \sqrt{-13})$$

with $\mathrm{Nm}(2) = 4$, $\mathrm{Nm}(7) = 49$ and $\mathrm{Nm}(1 \pm \sqrt{-13}) = 14$. Each of these four factors is irreducible. Let γ be one of them and suppose $\gamma = \beta\delta$ with β, δ nonunits. Both are nonzero, so $\mathrm{Nm}(\beta)$ and $\mathrm{Nm}(\delta)$ are positive integers, both different from 1, with product $\mathrm{Nm}(\gamma) \in \{4, 49, 14\}$. The product formula holds because the norm is the determinant of multiplication (definition 50.1.10) and the determinant is multiplicative ([3, Multiplicativity of the Determinant]). Hence $\mathrm{Nm}(\beta) \in \{2, 7\}$. But $a^2 + 13b^2 \in \{2, 7\}$ has no solution in integers: $b \neq 0$ gives $a^2 + 13b^2 \geq 13$, and $b = 0$ would make 2 or 7 a square. The two factorizations are not related by reordering and units: associates differ by a unit ± 1 and so have equal norms, while $4 \neq 14$ and $49 \neq 14$, so neither 2 nor 7 is associate to $1 + \sqrt{-13}$ or to $1 - \sqrt{-13}$. By [3, Unique Factorization Domain (UFD)], R is not a UFD.

3. Let R be a Dedekind domain and a UFD. First, every prime ideal $\mathfrak{p}$ of R is principal. The zero ideal is (0). If $\mathfrak{p} \neq 0$, pick $0 \neq a \in \mathfrak{p}$, which is not a unit, and write $a = q_1 \cdots q_k$ with irreducible q_i ([3, Unique Factorization Domain (UFD)]). As $\mathfrak{p}$ is prime, some $q_i \in \mathfrak{p}$, and q_i is a prime element ([3, Prime If And Only If Irreducible In UFD]), so (q_i) is a nonzero prime ideal. It is maximal by definition 50.2.1, and $(q_i) \subseteq \mathfrak{p} \neq R$, so $\mathfrak{p} = (q_i)$.

   Next, suppose some ideal of R is not principal. Since R is Noetherian, the nonempty set of non-principal ideals has a maximal element I ([2, Equivalence of Noetherian Ring]). Then $I \neq R$, and I is not prime by the previous paragraph, so there are $a, b \in R \setminus I$ with $ab \in I$. The ideal $I + (a)$ strictly contains I, so $I + (a) = (c)$ for some c, by maximality. Let $J = \{x \in R \mid xc \in I\}$, an ideal containing I. Writing $c = i + ra$ with $i \in I$ and $r \in R$ gives $bc = bi + r(ab) \in I$, so $b \in J \setminus I$ and $J = (d)$ for some d, again by maximality. Now $cd \in I$ because $d \in J$, and every $x \in I \subseteq (c)$ has the form $x = yc$ with $y \in R$, where $yc \in I$ gives $y \in J = (d)$, so $x \in (cd)$. Hence $I = (cd)$ is principal, a contradiction. So every ideal of R is principal, and R is a PID, completing the proof.

For algebro-geometric purposes, the following characterizes Dedekind domains locally:

Lemma 1.2.4: Dedekind Domain As DVR

Let R be a Noetherian integral domain. Then R is a Dedekind domain if and only if for each nonzero prime P, R_P is a DVR. Thus, Dedekind domains are locally DVR's.

***Proof*:**
If R is a field, then both sides hold: R satisfies the three conditions of definition 50.2.1 (the second vacuously), and R has no nonzero prime. So assume R is not a field.

($\Rightarrow$) Let $P \neq 0$ be a prime of R and localize at $D = R \setminus P$. By [2, Localization Property Preservation], R_P is a Noetherian integrally closed domain with fraction field $K(R)$. By [2, Ideals of Localization], the primes of R_P correspond bijectively to the primes of R contained in P. These are 0 and P alone, for a nonzero prime contained in P is maximal by definition 50.2.1 and hence equals P. Thus R_P is local of Krull dimension 1 with unique nonzero prime PR_P (nonzero because the correspondence is a bijection and $P \neq 0$). By [2, Characterizations of a DVR], an integrally closed Noetherian local domain of dimension 1 is a DVR.

($\Leftarrow$) We verify the three conditions of definition 50.2.1. R is Noetherian by hypothesis. For the second condition, let $P \neq 0$ be prime and choose a maximal ideal $\mathfrak{m} \supseteq P$ ([3, Ring with $1 \neq 0$ have Maximal ideals]). Since R is not a field, $\mathfrak{m} \neq 0$, so $R_{\mathfrak{m}}$ is a DVR, and by [2, Characterizing DVR] it has a unique nonzero prime ideal. Under the correspondence of [2, Ideals of Localization], P and $\mathfrak{m}$ give two nonzero primes of $R_{\mathfrak{m}}$, so $PR_{\mathfrak{m}} = \mathfrak{m}R_{\mathfrak{m}}$ and hence $P = \mathfrak{m}$ is maximal. For normalness, let $x \in K(R)$ be integral over R. A monic equation for x over R is a monic equation over each $R_{\mathfrak{m}}$, and each $R_{\mathfrak{m}}$, being a DVR and in particular a valuation ring, is integrally closed ([2, Valuation Ring Properties](5)), so $x \in R_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$. By [2, Local Properties On Modules], $R = \bigcap_{\mathfrak{m}} R_{\mathfrak{m}}$ inside $K(R)$, hence $x \in R$, as we sought to show.

Lemma 1.2.5: Dedekind Domains And Localization

Let R be a Dedekind domain and $S \subseteq R \setminus \{0\}$ is a multiplicative subset. Then $S^{-1}R$ is also a Dedekind Domain.

***Proof*:**
The localization of Noetherian rings is Noetherian, and by the correspondence of ideals in localizations, we have that prime implies maximal in $S^{-1}R$. Finally, $S^{-1}R$ is integrally closed since if $x \in K = K(R)$ satisfies:

$$x^n + \frac{a_1}{s_1}x^{n-1} + \cdots + \frac{a_n}{s_n} = 0 \qquad \frac{a_i}{s_i} \in S^{-1}R$$

Then multiplying with the nth power of $s_1 \cdots s_n$ gives that sx is integral over R, hence $sx \in R$, and so $x \in S^{-1}R$, as we sought to show.

Importantly for number theory, all rings of integers are Dedekind domains:

Theorem 1.2.6: Ring Of Integers Are Dedekind Domains

Let $\mathcal{O}_K$ be a ring of integers. Then it is a Dedekind domain.

***Proof*:**

By proposition 50.1.32, $\mathcal{O}_K$ is Noetherian, and by definition $\mathcal{O}_K$ is normal, namely $\mathcal{O}_K = \overline{\mathbb{Z}}^K$ meaning by [2, Integral Elements Form Rings]:

$$\overline{\mathcal{O}_K}^K = \overline{\overline{\mathbb{Z}}^K}^K = \overline{\mathbb{Z}}^K$$

What's left to show is that every prime ideal is maximal, which is equivalent to showing $\mathcal{O}_K/\mathfrak{p}$ is a field. In particular, since $\mathcal{O}_K/\mathfrak{p}$ is always an integral domain, if we show that it is finite then by [3, Finite Integral Domain is a Field] we shall have that it's a field. The key intuition comes from the fact that $\mathcal{O}_K$ and its ideals form a lattice, meaning when we quotient we necessarily get a finite ring. Furthermore, the volume of the ideal is determined by the discriminant and the index, which will determine the cardinality of the quotient ring[a]

Let $0 \neq \alpha \in \mathfrak{p}$. Then $(p) = \mathfrak{p} \cap \mathbb{Z}$ is also a prime ideal. Since $\mathfrak{p} \subseteq \mathcal{O}_K$, we know α is associated to a minimal polynomial $m_{\alpha,\mathbb{Z}}(x)$ where

$$m_{\alpha,\mathbb{Z}}(\alpha) = \alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0 \qquad a_i \in \mathbb{Z}$$

Then $a_0 = -(\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha) \in \mathfrak{p} \subseteq \mathcal{O}_K$, and since $m_{\alpha,\mathbb{Q}}(x)$ is irreducible with integer coefficients, $a_0 \neq 0$ and $a_0 \in (p) = p\mathbb{Z}$. From here, we shall give two proofs: first, we may directly appeal to the fact that

$$(\mathcal{O}_K/\mathfrak{p})/(\mathbb{Z}/p\mathbb{Z})$$

is an integral extension (see [2, Integral Extensions]) where the base ring is a field, hence $\mathcal{O}_K/\mathfrak{p}$ is a field by [2, Integral Extension over Fields]. Another is the following: we have a non-trivial ideal with chains of inclusions $(a_0) \subseteq \mathfrak{p} \subseteq \mathcal{O}_K$ implying

$$|\mathcal{O}_K/\mathfrak{p}| \leq |\mathcal{O}_K/(a_0)| = |\mathcal{O}_K/a_0\mathcal{O}_K|$$

Now, since we are quotienting ideals, this is a $\mathbb{Z}$-module quotient, and since we are quotienting n-rank free $\mathbb{Z}$-modules, we get that

$$|\mathcal{O}_K/(a_0)| = |a_0|^n$$

But then $\mathcal{O}_K/\mathfrak{p}$ is finite (its order divides $|a_0|^n$). Thus, $\mathfrak{p}$ must be maximal, and since $\mathfrak{p}$ was arbitrary every prime ideal is maximal, completing the proof.

[a] We shall soon see that the index can be thought of as the "norm" of a general ideal; this will mean the constant term of a minimal polynomial represent the "size", sheading some light on the proof

Since in general $\mathcal{O}_K$ is not a Unique Factorization Domain, rings of integers represent a class of rings where prime implies maximal, but factorization is not unique. In this way, we see that the integral closure of $\mathbb{Z}$ with respect to K may be far from having the nice element-wise properties we are used to in PIDs, but this shall be salvaged through having an understandable ideal structure and a way to measure how far away our ring is from being a PID.

Another important example of a Dedekind domain that is not a number ring is the ring is $k[x]$ (since it's a PID). We shall later see that integral extensions of Dedekind domains are Dedekind domains, an so these types of rings will form a basis for another large class of Dedekind domains that are not

number rings.

1.2.1 Fractional Ideals in Rings

Following from the idea that ideals are the generalisation of numbers, we may define the following divisibility notions usually used for the [algebraic] lattice of integers induced by divisibility. These concepts shall be defined for general commutative rings:

Definition 1.2.7: GCD, LCM, Divisibility Of Ideals

let R be a commutative ring, $\mathfrak{a}, \mathfrak{b} \subseteq R$ ideals of R. Then:

$$\gcd(\mathfrak{a}, \mathfrak{b}) = (\mathfrak{a}, \mathfrak{b}) = \mathfrak{a} + \mathfrak{b} \qquad \operatorname{lcm}(\mathfrak{a}, \mathfrak{b}) = \mathfrak{a} \cap \mathfrak{b}$$

And we can say that $\mathfrak{a}, \mathfrak{b}$ are relatively prime when $\gcd(\mathfrak{a}, \mathfrak{b}) = (1)$. Divisibility $\mathfrak{a} \big| \mathfrak{b}$ is defined as $\mathfrak{b} \subseteq \mathfrak{a}$, and hence we have an order and the ideals similar to that given by divisibility.

As we have seen in [3, Exploring Integral Domains: Factorization and Valuation], gcd and lcm are usually well-behaved in special classes of rings, usually UFDs. We shall show they are well-behaved in Dedekind domains since the ideals shall have unique factorizations. Let's now prove some lemmas for this result.

One finiteness fact is needed throughout, and it is the reason the Noetherian hypothesis appears in the definition of a Dedekind domain at all: in a Noetherian domain every nonzero proper ideal *contains* a product of nonzero prime ideals, each of which contains it ([2, Nonzero Ideals Contain Products of Nonzero Primes]). An ideal need not factor into primes (that is the theorem this section works toward, and it needs the full strength of a Dedekind domain), but it always contains such a product, and that much is available from the Noetherian condition alone.

For the following lemmas, we bring back the notion of fractional ideal that was defined in [2, Noetherian Rings, Hilbert's Basis Theorem, and Localization]. These ideals were at the time a way of extending ideals into the ring of fractions, and were shown to work well with localization. Due to all the properties of Dedekind domains, the product of ideals can be made into a *group*, even an *abelian group*. Since the fractional ideal structure will have such a strong restriction, this will produce incredible results on the relations between ideals. To begin, the following lemma shows that no prime ideal acts like the identity:

Lemma 1.2.8: Fractional Ideals

Let $P \subseteq R$ be a nonzero prime ideal of the Dedekind domain R, $K = K(R)$. Define

$$(R :_K P) = P^{-1} = \{x \in K \mid xP \subseteq R\}$$

Then

$$AP^{-1} = \left\{ \sum_i a_i x_i \;\middle|\; a_i \in A,\ x_i \in P^{-1} \right\} \neq A$$

for every ideal $0 \neq A \subseteq R$

Note that by definition $R \subseteq P^{-1}$ and so P^{-1} is not in general an ideal. The simplest "fractional

ideal" would be those of $\mathbb{Z}$, namely $(n)^{-1} = \frac{1}{n}\mathbb{Z}$.

***Proof*:**
Step 1: $P^{-1} \neq R$. Let $0 \neq a \in P$. The ideal aR is nonzero and proper, so it contains a product $P_1 \cdots P_r$ of nonzero primes ([2, Nonzero Ideals Contain Products of Nonzero Primes]). Choose such a product with r minimal. As $P_1 \cdots P_r \subseteq aR \subseteq P$ and P is prime, some P_i lies in P: otherwise elements $a_i \in P_i \setminus P$ would give $a_1 \cdots a_r \in P$. Renumber so that $P_1 \subseteq P$. Nonzero primes of R are maximal (definition 50.2.1), so $P_1 = P$. By minimality of r, the product $P_2 \cdots P_r$ (which is R when $r = 1$) is not contained in aR, so there is $b \in P_2 \cdots P_r$ with $b \notin aR$. Then $bP = bP_1 \subseteq P_1 \cdots P_r \subseteq aR$, so $(b/a)P \subseteq R$ and $b/a \in P^{-1}$, while $b/a \notin R$. Hence $P^{-1} \neq R$.

Step 2: $AP^{-1} \neq A$. Suppose $AP^{-1} = A$ for an ideal $A \neq 0$, and let $x \in P^{-1}$. Then $xA \subseteq A$, so $x^k A \subseteq A$ for every $k \geq 0$. Fix $0 \neq c \in A$. Then $x^k c \in A \subseteq R$ for every k, so $cR[x]$ is an R-submodule of R, that is, an ideal, and it is finitely generated because R is Noetherian ([2, Equivalence of Noetherian Ring]). Then $R[x] = c^{-1}(cR[x])$ is a finitely generated R-module, so x is integral over R ([2, Integral Element Equivalences]), and $x \in R$ because R is integrally closed (definition 50.2.1). So $P^{-1} \subseteq R$, contradicting Step 1, completing the proof.

Definition 1.2.9: Fractional Ideal and Quotient ideal

Let R be an integral domain and $K = K(R)$. Then a *fractional ideal* J of K is a finitely generated non-trivial R-submodule of K where there exists a nonzero $r \in R$ such that $rJ \subseteq R$ is an ideal.
If $I, J \subseteq R$ are ideals, then the *quotient ideal* of I with respect to J is:

$$(I : J) = \{x \in K \mid xJ \subseteq I\}$$

If $I = R$,then $(R : J)$ is denoted J^{-1}

Fractional ideals are multiplied in the similar way ideals are multiplied: IJ is the set of finite sums of products xy with $x \in I$ and $y \in J$. Every nonzero finitely generated ideal of R is a fractional ideal, with $r = 1$. For those who are motivated by algebraic geometry, fractional ideals play the role of line bundles over the smooth $\mathbb{Z}$-curves $\mathcal{O}_K$[14].

Quotient ideals have a lot of interesting theory, for example to motivate their name if J is principle and $I \subseteq J$, then $I = J \cdot (I : J)$. There is no room in this chapter to explore these in detail, hence they are left as exercises 50.2.1(50.2.1 (6)).

Fractional ideals that are ideals are given a name:

Definition 1.2.10: Integral Ideals

Let R be an integral domain and $K = K(R)$. A fractional ideal $I \subseteq K$ with $I \subseteq R$ is called an *integral ideal*. Equivalently, an integral ideal is a nonzero finitely generated ideal of R, and when R is Noetherian every nonzero ideal of R is integral.

[14]See section 50.7 for rings of integers as smooth $\mathbb{Z}$-curves, and for invertible fractional ideals and the Picard group of an order

Example 1.2.11: Fractional Ideal

1. Since $\mathbb{Z}$ is a Principal Ideal Domain, every ideal is of the form $a\mathbb{Z}$ for $a \in \mathbb{Z}$. Then $(a)^{-1} = a^{-1}\mathbb{Z}$. Hence, the fractional ideals are in bijective correspondence with $\mathbb{Q}^{\times}_{>0}$. Naturally, all ideals of $\mathbb{Z}$ are fractional ideals, in particular they are integral ideals.
2. More generally, if $a \in K^{\times}$, then $(a) = aR$ is a fractional principal ideal, and $aR = bR$ exactly when $a/b \in R^{\times}$, giving a bijective correspondence between the principal fractional ideals and $K^{\times}/R^{\times}$.

In a Noetherian domain the two conditions of definition 50.2.9 imply each other:

Corollary 1.2.12: Fractional Ideal Representative

Let R be a Noetherian domain, $K = K(R)$, and $\mathfrak{a} \subseteq K$ a nonzero R-submodule. Then $\mathfrak{a}$ is finitely generated if and only if $r\mathfrak{a} \subseteq R$ for some $r \in R \setminus \{0\}$. In that case $\mathfrak{a}$ is a fractional ideal, $I = r\mathfrak{a}$ is a nonzero ideal of R with

$$\mathfrak{a} = r^{-1}I$$

and multiplication by r is an isomorphism of R-modules $\mathfrak{a} \cong_R I$. If $R = \mathcal{O}_K$ for a number field K, then r may be taken in $\mathbb{Z}_{>0}$.

***Proof*:**
Step 1: generators give a denominator. Let $\mathfrak{a} = R\alpha_1 + \cdots + R\alpha_n$ and write $\alpha_i = a_i/b_i$ with $a_i, b_i \in R$ and $b_i \neq 0$. For $r = b_1 \cdots b_n \neq 0$, every $r\alpha_i$ lies in R, so $r\mathfrak{a} \subseteq R$.

Step 2: a denominator gives generators. If $r\mathfrak{a} \subseteq R$ with $r \neq 0$, then $I = r\mathfrak{a}$ is an R-submodule of R, that is, an ideal, nonzero because $\mathfrak{a} \neq 0$ and R is a domain. It is finitely generated as R is Noetherian ([2, Equivalence of Noetherian Ring]), say by $y_1, \ldots, y_m$, and then $\mathfrak{a} = r^{-1}I$ is generated by $r^{-1}y_1, \ldots, r^{-1}y_m$. Multiplication by r is R-linear, and bijective from $\mathfrak{a}$ onto I since r is invertible in K.

Step 3: integer denominators. Let $R = \mathcal{O}_K$ and $r\mathfrak{a} \subseteq \mathcal{O}_K$ with $r \neq 0$. The index $d = [\mathcal{O}_K : r\mathcal{O}_K]$ is a positive integer (proposition 50.1.62), and $d \cdot 1 \in r\mathcal{O}_K$ by [3, Lagrange Theorem], say $d = ru$ with $u \in \mathcal{O}_K$. Then $d\mathfrak{a} = u(r\mathfrak{a}) \subseteq \mathcal{O}_K$, completing the proof.

Notice that this generalizes the definition of ideals, if R is a Dedekind ring over itself, then the ideals of R are the finitely generated sub-modules. Then if $K = \mathrm{Frac}(R)$, the fractional ideals are the finitely generated K-modules.

Corollary 1.2.13: Inverse of Prime Ideals

Let R be a Dedekind domain and $\mathfrak{p} \subseteq R$ be a nonzero prime ideal. Then:

$$\mathfrak{p}\mathfrak{p}^{-1} = (1)$$

***Proof*:**
If $x \in \mathfrak{p}\mathfrak{p}^{-1}$, then $x = \sum p_i k_i$ for $p_i \in \mathfrak{p}$ and $k_i \in \mathfrak{p}^{-1}$. But by definition of $\mathfrak{p}^{-1}$, we have that $p_i k_i \in R$, hence $x \in R$, and so $\mathfrak{p}\mathfrak{p}^{-1} \subseteq R$. Next, $1 \in \mathfrak{p}^{-1}$ gives $\mathfrak{p} \subseteq \mathfrak{p}\mathfrak{p}^{-1} \subseteq R$. Since $\mathfrak{p}$ is maximal (definition 50.2.1), either $\mathfrak{p}\mathfrak{p}^{-1} = \mathfrak{p}$ or $\mathfrak{p}\mathfrak{p}^{-1} = R$. The former is a contradiction by lemma 50.2.8, and hence $\mathfrak{p}\mathfrak{p}^{-1} = (1)$, completing the proof.

Lemma 1.2.14: Invariance Number Ring Multiplication

Let R be a Dedekind domain with fraction field K, $\mathfrak{a} \subseteq K$ a fractional ideal and $t \in K$. If $t\mathfrak{a} \subseteq \mathfrak{a}$, then $t \in R$.

***Proof*:**
From $t\mathfrak{a} \subseteq \mathfrak{a}$ we get $t^k\mathfrak{a} \subseteq \mathfrak{a}$ for every $k \geq 0$. Fix $0 \neq c \in \mathfrak{a}$ and a nonzero $r \in R$ with $r\mathfrak{a} \subseteq R$ (definition 50.2.9). Then $rct^k \in r\mathfrak{a} \subseteq R$ for every k, so $rcR[t]$ is an R-submodule of R, that is, an ideal, finitely generated as R is Noetherian ([2, Equivalence of Noetherian Ring]). Hence $R[t] = (rc)^{-1}(rcR[t])$ is a finitely generated R-module, t is integral over R ([2, Integral Element Equivalences]), and $t \in R$ because R is integrally closed (definition 50.2.1), completing the proof.

1.2.2 Prime Decomposition in Dedekind Domain

With these properties, we may show the factorization of prime ideals in Dedekind domains.

Theorem 1.2.15: Unique Factorization Of Ideals In Dedekind Domains

Let R be a Dedekind Domain and $\mathfrak{a} \subseteq R$ a non-trivial proper ideal. Then there exists unique $\mathfrak{p}_1, \mathfrak{p}_2, ..., \mathfrak{p}_n$ such that

$$\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_n$$

***Proof*:**
We start with the existence of such a factorization. Let $\mathfrak{M}$ be the set of all non-trivial proper ideals which do not admit a prime ideal decomposition. If $\mathfrak{M}$ is non-empty, then since R is Noetherian there exists a maximal element, say $\mathfrak{m}$. Then this ideal is contained in some maximal ideal $\mathfrak{p}$. Since $R \subseteq \mathfrak{p}^{-1}$, we get the following chain of inclusions:

$$\mathfrak{m} \subseteq \mathfrak{m}\mathfrak{p}^{-1} \subseteq \mathfrak{p}\mathfrak{p}^{-1} \subseteq R$$

by lemma 50.2.8, we have $\mathfrak{m} \subsetneq \mathfrak{m}\mathfrak{p}^{-1}$ and $\mathfrak{p} \subsetneq \mathfrak{p}\mathfrak{p}^{-1} \subseteq R$. By maximality of $\mathfrak{p}$, $\mathfrak{p}\mathfrak{p}^{-1} = R$. If $\mathfrak{m}\mathfrak{p}^{-1} = R$, then $\mathfrak{m} = \mathfrak{m}\mathfrak{p}^{-1}\mathfrak{p} = \mathfrak{p}$ would be its own prime decomposition. So $\mathfrak{m}\mathfrak{p}^{-1}$ is a nonzero proper ideal strictly containing $\mathfrak{m}$, and by maximality of $\mathfrak{m}$ in $\mathfrak{M}$ it admits a prime ideal decomposition, say

$$\mathfrak{m}\mathfrak{p}^{-1} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$$

But then, since $\mathfrak{p}\mathfrak{p}^{-1} = R$, $\mathfrak{m}R = \mathfrak{m}$ and so

$$\mathfrak{m} = \mathfrak{m}\mathfrak{p}^{-1}\mathfrak{p} = \mathfrak{p}_1 \cdots \mathfrak{p}_r\mathfrak{p}$$

contradicting the definition of $\mathfrak{m}$. Hence $\mathfrak{M}$ is an empty-set

For uniqueness, let

$$\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2\cdots\mathfrak{p}_r = \mathfrak{q}_1\mathfrak{q}_2\cdots\mathfrak{q}_s$$

be two prime ideal factorizations of $\mathfrak{a}$. Then by the repeated application of property of prime ideals ($\mathfrak{ab} \subseteq P \Rightarrow \mathfrak{a} \subseteq P$ or $\mathfrak{b} \subseteq P$), we have that $\mathfrak{p}_1$ divides a factor of the right hand side, say $\mathfrak{q}_1$, and by maximality is thus equal to $\mathfrak{q}_1$. Then we can multiply both sides by $\mathfrak{p}_1^{-1}$ and since $\mathfrak{p}_1 \neq \mathfrak{p}_1\mathfrak{p}_1^{-1} = R$ we get

$$\mathfrak{p}_2\cdots\mathfrak{p}_r = \mathfrak{q}_2\cdots\mathfrak{q}_s$$

Then proceeding inductively, we will get that $r = s$ and $\mathfrak{p}_i = \mathfrak{q}_i$, up to renumbering, completing the proof.

1.2.16 Note The converse holds as well: an integral domain in which every nonzero proper ideal factors uniquely into prime ideals is a Dedekind domain, so this theorem characterizes Dedekind domains. The converse is not used in this book.

From now on, we shall group the prime ideals to write

$$\mathfrak{a} = \mathfrak{p}_1^{n_1}\cdots\mathfrak{p}_r^{n_r} \qquad n_i > 0$$

If $\mathfrak{a} = (a)$ is a principal ideal, we shall use the slight abuse of notation and write

$$a = \mathfrak{p}_1^{n_1}\cdots\mathfrak{p}_r^{n_r} \qquad n_i > 0$$

Similarly, we shall often write $\mathfrak{a}\big|a$ instead of $\mathfrak{a}\big|(a)$. An important result worth emphasizing is the structure of R/I for an ideal $I \subseteq R$. It rests on the following lemma:

Lemma 1.2.17: Powers Of Comaximal Ideals

Let R be a commutative ring and $\mathfrak{a}, \mathfrak{b} \subseteq R$ ideals with $\mathfrak{a} + \mathfrak{b} = R$. Then $\mathfrak{a}^m + \mathfrak{b}^l = R$ for all $m, l \geq 1$. In particular, in a Dedekind domain, powers of distinct nonzero primes are comaximal.

***Proof*:**
Write $1 = x + y$ with $x \in \mathfrak{a}$ and $y \in \mathfrak{b}$. In the binomial expansion of $1 = (x+y)^{m+l-1}$, the term containing $x^j y^{m+l-1-j}$ lies in $\mathfrak{a}^m$ if $j \geq m$, and otherwise $m + l - 1 - j \geq l$ and it lies in $\mathfrak{b}^l$. So $1 \in \mathfrak{a}^m + \mathfrak{b}^l$. Distinct nonzero primes $\mathfrak{p} \neq \mathfrak{q}$ of a Dedekind domain are maximal (definition 50.2.1), so $\mathfrak{q} \not\subseteq \mathfrak{p}$, the ideal $\mathfrak{p} + \mathfrak{q}$ strictly contains $\mathfrak{p}$, and $\mathfrak{p} + \mathfrak{q} = R$. The first part then applies, completing the proof.

Then by lemma 50.2.17 and the Chinese Remainder Theorem:

$$R/I = \bigoplus_i R/P_i^{n_i}$$

By the 4th isomorphism theorem, $\overline{I} \subseteq R/P_i^{n_i}$ if and only if $P_i^{n_i} \subseteq I$, that is $I\big|P_i^{n_i}$. But by unique factorization, that means that $I = P_i^k$ for $1 \leq k \leq n_i$. Hence, every quotient has only finitely many ideals, each ideal I corresponding to a set of prime power of the decomposition of I via the isomorphism given by the CRT. This matches with the result that $R/(a)$ is a finite ring, showing there can only be finitely many ideals, and the prime factorization theorem for ideals gives us the

structure of the ideals of $R/(a)$ through the correspondence given by the 4th isomorphism theorem. This also immediately shows that R/I is *Artinian*[15].

Using the fact that ideals factor and "behave" like the positive rational numbers ($-\mathfrak{p} = \mathfrak{p}$ since -1 is a unit), we can conclude some powerful results. We have shown that every ideal has a unique inverse. This immediately shows that:

$$\mathfrak{ab} = \mathfrak{ac} \quad \Longrightarrow \quad \mathfrak{b} = \mathfrak{c}$$

by the Cancellation law. Next, it is easy to see is that we may generalize the notion of finding GCD and LCM of UFD's to Dedekind domains:

Corollary 1.2.18: GCD, LCM, and Divisibility In Dedekind Domains

Let R be a Dedekind domain and $\mathfrak{a}, \mathfrak{b} \subseteq R$ nonzero ideals, written $\mathfrak{a} = P_1^{a_1} \cdots P_k^{a_k}$ and $\mathfrak{b} = P_1^{b_1} \cdots P_k^{b_k}$ with distinct nonzero primes P_i and exponents $a_i, b_i \geq 0$. Then

1. $\operatorname{lcm}(\mathfrak{a}, \mathfrak{b}) = P_1^{\max(a_1,b_1)} \cdots P_k^{\max(a_k,b_k)}$
2. $\gcd(\mathfrak{a}, \mathfrak{b}) = P_1^{\min(a_1,b_1)} \cdots P_k^{\min(a_k,b_k)}$
3. $\mathfrak{a} \big| \mathfrak{b}$ if and only if there exists an ideal $\mathfrak{c} \subseteq R$ such that $\mathfrak{ac} = \mathfrak{b}$, if and only if $a_i \leq b_i$ for every i

***Proof*:**

Step 1: divisibility and products. If $\mathfrak{ac} = \mathfrak{b}$ for an ideal $\mathfrak{c}$, then $\mathfrak{b} = \mathfrak{ac} \subseteq \mathfrak{a}R = \mathfrak{a}$, that is, $\mathfrak{a} \big| \mathfrak{b}$ (definition 50.2.7). Conversely let $\mathfrak{b} \subseteq \mathfrak{a}$. Write $\mathfrak{a} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$ with nonzero primes $\mathfrak{p}_j$ (theorem 50.2.15, the empty product if $\mathfrak{a} = R$) and put $\mathfrak{a}' = \mathfrak{p}_1^{-1} \cdots \mathfrak{p}_r^{-1}$. Multiplication of R-submodules of $K(R)$ is associative and commutative, so $\mathfrak{aa}' = \prod_j \mathfrak{p}_j \mathfrak{p}_j^{-1} = R$ by corollary 50.2.13. Then $\mathfrak{c} = \mathfrak{a}'\mathfrak{b} \subseteq \mathfrak{a}'\mathfrak{a} = R$ is an ideal with $\mathfrak{ac} = \mathfrak{aa}'\mathfrak{b} = \mathfrak{b}$.

Step 2: the exponent criterion. Let $\mathfrak{x} = \prod_i P_i^{x_i}$ and $\mathfrak{y} = \prod_i P_i^{y_i}$ be nonzero ideals with exponents $x_i, y_i \geq 0$. If $x_i \leq y_i$ for every i, then $\mathfrak{y} = \mathfrak{x} \prod_i P_i^{y_i - x_i}$, so $\mathfrak{x} \big| \mathfrak{y}$ by Step 1. Conversely, let $\mathfrak{z} \big| \mathfrak{y}$ for a nonzero ideal $\mathfrak{z}$. Step 1 gives $\mathfrak{y} = \mathfrak{zc}$ with $\mathfrak{c} \neq 0$, and concatenating factorizations of $\mathfrak{z}$ and $\mathfrak{c}$ gives one of $\mathfrak{y}$. By the uniqueness in theorem 50.2.15, every prime occurring in $\mathfrak{z}$ or $\mathfrak{c}$ is one of the P_i, so $\mathfrak{z} = \prod_i P_i^{z_i}$ and $\mathfrak{c} = \prod_i P_i^{c_i}$, and $y_i = z_i + c_i \geq z_i$. So every nonzero divisor of $\mathfrak{y}$ is a product of the P_i, and $\mathfrak{x} \big| \mathfrak{y}$ exactly when $x_i \leq y_i$ for every i. With $\mathfrak{x} = \mathfrak{a}$ and $\mathfrak{y} = \mathfrak{b}$ this proves (3).

Step 3: the gcd. The ideal $\mathfrak{g} = \mathfrak{a} + \mathfrak{b}$ contains $\mathfrak{a}$, so $\mathfrak{g} \big| \mathfrak{a}$, and by Step 2 it is a product $\mathfrak{g} = \prod_i P_i^{g_i}$. As $\mathfrak{g} \big| \mathfrak{a}$ and $\mathfrak{g} \big| \mathfrak{b}$, Step 2 gives $g_i \leq \min(a_i, b_i)$. The ideal $\mathfrak{d} = \prod_i P_i^{\min(a_i,b_i)}$ divides $\mathfrak{a}$ and $\mathfrak{b}$ by Step 2, that is, $\mathfrak{a} \subseteq \mathfrak{d}$ and $\mathfrak{b} \subseteq \mathfrak{d}$, so $\mathfrak{g} = \mathfrak{a} + \mathfrak{b} \subseteq \mathfrak{d}$ and $\mathfrak{d} \big| \mathfrak{g}$, giving $\min(a_i, b_i) \leq g_i$. So $\mathfrak{g} = \mathfrak{d}$, which is (2).

Step 4: the lcm. The ideal $\mathfrak{l} = \mathfrak{a} \cap \mathfrak{b}$ contains $\mathfrak{ab} = \prod_i P_i^{a_i + b_i} \neq 0$, so $\mathfrak{l} \big| \mathfrak{ab}$, and by Step 2 it is a product $\mathfrak{l} = \prod_i P_i^{l_i}$. As $\mathfrak{l} \subseteq \mathfrak{a}$ and $\mathfrak{l} \subseteq \mathfrak{b}$, Step 2 gives $l_i \geq \max(a_i, b_i)$. The ideal $\mathfrak{m} = \prod_i P_i^{\max(a_i,b_i)}$ is divisible by $\mathfrak{a}$ and by $\mathfrak{b}$, that is, $\mathfrak{m} \subseteq \mathfrak{a} \cap \mathfrak{b} = \mathfrak{l}$, so $\mathfrak{l} \big| \mathfrak{m}$ and $l_i \leq \max(a_i, b_i)$. So $\mathfrak{l} = \mathfrak{m}$, which is (1), completing the proof.

[15] Equivalently, $\operatorname{kdim}(R/I) = 0$, and hence it is Artinian

We know that every ideal has an inverse. However, that inverse is a fractional ideal. Fortunately, there always exists an integral ideal that is "almost" the inverse, in that the product is a principal ideal[16]:

Corollary 1.2.19: Exists J such that IJ is Principal

Let R be a Dedekind Domain and $I \subseteq R$ an ideal. Then there exists an ideal $J \subseteq R$ such that $IJ = (a)$ for some element $a \in I$. In other words, there exists an ideal J such that the product IJ is principal. If $I \neq 0$, then J and a may be taken nonzero.

***Proof*:**
If $I = 0$, take $J = R$ and $a = 0$, and if $I = R$, take $J = R$ and $a = 1$. Otherwise let $I = P_1^{n_1} \cdots P_k^{n_k}$ with distinct nonzero primes P_i and $n_i \geq 1$. By unique factorization $P_i^{n_i} \neq P_i^{n_i+1}$, so we may choose $r_i \in P_i^{n_i} \setminus P_i^{n_i+1}$. The ideals $P_i^{n_i+1}$ are pairwise comaximal (lemma 50.2.17), so by [3, Chinese Remainder Theorem] there exists an $a \in R$ such that

$$a \equiv r_i \bmod P_i^{n_i+1} \qquad 1 \leq i \leq k$$

Thus, $a \in P_i^{n_i} \setminus P_i^{n_i+1}$ for each i, so $a \neq 0$, and by corollary 50.2.18(3) the power of P_i in the factorization of (a) is exactly n_i. Now, taking the prime factorization of (a), we get:

$$(a) = P_1^{n_1} \cdots P_k^{n_k} P_{k+1}^{n_{k+1}} \cdots P_m^{n_m}$$

with further distinct primes $P_{k+1}, \ldots, P_m$. Letting $J = P_{k+1}^{n_{k+1}} \cdots P_m^{n_m}$ gives us

$$(a) = IJ$$

and $a \in I$ because $(a) = IJ \subseteq I$, completing the proof.

In section 50.3.1, we shall find a more systematic way of finding the prime ideals that multiply to get a principal ideal using Galois groups. As seen with rings of integers, not all Dedekind domains are PIDs since they are not all UFDs. With the factorization theorem, we get that they are the next best thing:

Corollary 1.2.20: PIR, and Generation Of Ideals In Dedekind Domains

Let R be a Dedekind domain.

1. If $0 \neq I \subseteq J$ are ideals of R, then $J = I + aR$ for some $a \in J$.
2. For every nonzero ideal $I \subseteq R$, the ring R/I is a principal ideal ring.
3. Every ideal of R is generated by *at most* 2 elements.

[16] this almost inverse can be made into a concrete inverse when we show the principal ideals form a normal subgroup formed by the group of fractional ideals

***Proof*:**
Step 1: proof of (1). If $I = R$, take $a = 0$. Otherwise write $I = P_1^{n_1} \cdots P_s^{n_s}$ with distinct nonzero primes P_i and $n_i \geq 1$ (theorem 50.2.15). As $J \big| I$, corollary 50.2.18(3) gives $J = P_1^{k_1} \cdots P_s^{k_s}$ with $0 \leq k_i \leq n_i$. By unique factorization $P_i^{k_i} \neq P_i^{k_i+1}$, so we may choose $r_i \in P_i^{k_i} \setminus P_i^{k_i+1}$. The ideals $P_i^{k_i+1}$ are pairwise comaximal (lemma 50.2.17), and [3, Chinese Remainder Theorem] gives $a \in R$ with $a - r_i \in P_i^{k_i+1}$ for every i. Then $a \in P_i^{k_i} \setminus P_i^{k_i+1}$, so by corollary 50.2.18(3) the exponent of P_i in aR is exactly k_i. By corollary 50.2.18(2), $I + aR = \gcd(I, aR)$ has exponent $\min(n_i, k_i) = k_i$ at each P_i and exponent 0 at every other prime, so $I + aR = J$, and $a \in J$.

Step 2: proof of (2). By [3, 2nd, 3rd, 4th Isomorphism Theorem for Rings], every ideal of R/I has the form J/I for an ideal $J \supseteq I$ of R. By (1), $J = I + aR$, so J/I is generated by $a + I$.

Step 3: proof of (3). The zero ideal is generated by 0. If $J \neq 0$, choose $0 \neq \alpha \in J$ and apply (1) to $I = \alpha R \subseteq J$: then $J = \alpha R + aR = (\alpha, a)$, completing the proof.

In particular, if $\mathfrak{p} \big| (p)$ for a prime number p, then corollary 50.2.20(1) with $I = (p)$ and $J = \mathfrak{p}$ gives an element $\alpha \in \mathfrak{p}$ such that $\mathfrak{p} = (p, \alpha)$. Hence the primes above p are recovered by adding an appropriate element to (p). Our final result shows all pairs of ideals are, up to a multiple of $K(R)$, relatively prime. This mirrors the property that if $\gcd(a, b) = c$, we can divide a by c to get $\gcd(a/c, b) = 1$:

Corollary 1.2.21: Always Almost Relatively Prime

Let R be a Dedekind domain with fraction field K, A a fractional ideal and B a nonzero ideal of R. Then there is $c \in K^\times$ such that cA is an ideal of R with $cA + B = R$. In particular, for fractional ideals A, B there are $x, y \in K^\times$ such that xA and yB are relatively prime ideals of R.

***Proof*:**
Step 1: integral A. Suppose first that $A \subseteq R$, a nonzero ideal. By corollary 50.2.19 there are a nonzero ideal I and $0 \neq a \in A$ with $AI = aR$. The ideal BI is nonzero and contained in I, so corollary 50.2.20(1) gives $b \in I$ with $I = BI + bR$. Multiplying by A, and using that multiplication of ideals distributes over sums,

$$aR = AI = ABI + bA = aB + bA$$

Multiplying by a^{-1} gives $R = B + (b/a)A$, where $(b/a)A \subseteq a^{-1}(aR) = R$. If $b \neq 0$, then $c = b/a$ works. If $b = 0$, then $B = R$, and $c = 1$ works.

Step 2: the general case. For a fractional ideal A, choose $0 \neq r \in R$ with $rA \subseteq R$ (corollary 50.2.12). Step 1 applied to rA gives c_0 with $c_0 rA \subseteq R$ and $c_0 rA + B = R$, so $c = c_0 r$ works. For the last claim, choose $y \neq 0$ with $yB \subseteq R$ (corollary 50.2.12) and apply the first claim to A and the nonzero ideal yB, completing the proof.

With the division of ideals defined and mirroring the structure of $K^\times / R^\times$ ($\mathbb{Q}^\times_{>0}$ when working with $\mathbb{Z}$), it is tempting to say that they form a group structure, and this is indeed the case and is the focus in the remainder of this section.

One of the main metrics to measure is how many more "ideal numbers" need to be added to recover unique factorization into primes. If $\mathfrak{p}$ and $\mathfrak{q}$ are added, then if for some $t \in K$, $t\mathfrak{p} = \mathfrak{q}$, then $\mathfrak{p}$ and $\mathfrak{q}$ do not actually differ much in structure: it can be shown that $I = aJ$ $(a \in K(R))$ if and only if $I \cong_{\mathcal{O}_K} J$). Hence, we may classify "different" classes of ideals up to multiplication by a principal ideal. The more classes there are, the more a ring R fails to be a unique factorization domain. When R is a number ring, we shall show that there are only finitely many such classes.

The first result to prove is that the collection of fractional ideals of a Dedekind domain does indeed form a group:

Theorem 1.2.22: Ideal Group

Let R be a Dedekind domain, $K = K(R)$, and let J_K be the collection of fractional ideals. Then $(J_K, \cdot)$ has an abelian group structure via fractional ideal multiplication, and is called the *ideal group* of K. The inverse of $\mathfrak{a} \in J_K$ is $\mathfrak{a}^{-1} = (R : \mathfrak{a}) = \{x \in K \mid x\mathfrak{a} \subseteq R\}$.

***Proof*:**
Step 1: a commutative monoid. Let $\mathfrak{a}, \mathfrak{b} \in J_K$, generated by $\alpha_1, \dots, \alpha_m$ and $\beta_1, \dots, \beta_l$, with $r\mathfrak{a} \subseteq R$ and $t\mathfrak{b} \subseteq R$ for nonzero $r, t \in R$. Then $\mathfrak{ab}$ is the R-submodule generated by the products $\alpha_i\beta_j$, it is nonzero because K is a field, and $rt\mathfrak{ab} \subseteq R$. So $\mathfrak{ab} \in J_K$. Multiplication of submodules is associative and commutative because multiplication in K is, and $R\mathfrak{a} = \mathfrak{a}$, so R is an identity.

Step 2: integral ideals are invertible. For a nonzero prime $\mathfrak{p}$, the set $\mathfrak{p}^{-1}$ is a nonzero R-submodule of K containing 1. For $0 \neq a \in \mathfrak{p}$, $a\mathfrak{p}^{-1} \subseteq R$ is an ideal, finitely generated as R is Noetherian, so $\mathfrak{p}^{-1} = a^{-1}(a\mathfrak{p}^{-1})$ is finitely generated and $\mathfrak{p}^{-1} \in J_K$. A nonzero ideal $\mathfrak{a} \subseteq R$ is $\mathfrak{p}_1 \cdots \mathfrak{p}_r$ with nonzero primes $\mathfrak{p}_j$ (theorem 50.2.15, the empty product for $\mathfrak{a} = R$), and $\mathfrak{b} = \mathfrak{p}_1^{-1} \cdots \mathfrak{p}_r^{-1} \in J_K$ satisfies $\mathfrak{ab} = \prod_j \mathfrak{p}_j\mathfrak{p}_j^{-1} = R$ by corollary 50.2.13.

Step 3: all fractional ideals are invertible. Let $\mathfrak{a} \in J_K$ and $0 \neq r \in R$ with $r\mathfrak{a} \subseteq R$. The ideal $r\mathfrak{a}$ is nonzero, so it has an inverse $\mathfrak{c}$ by Step 2. Then $(r\mathfrak{c})\mathfrak{a} = \mathfrak{c}(r\mathfrak{a}) = R$, and $r\mathfrak{c} = (rR)\mathfrak{c} \in J_K$ is an inverse of $\mathfrak{a}$.

Step 4: the formula for the inverse. Let $\mathfrak{b}$ be an inverse of $\mathfrak{a}$. As $\mathfrak{ba} = R$, every $x \in \mathfrak{b}$ satisfies $x\mathfrak{a} \subseteq R$, so $\mathfrak{b} \subseteq (R : \mathfrak{a})$. Conversely, if $x\mathfrak{a} \subseteq R$, then $x \in xR = x\mathfrak{ab} \subseteq R\mathfrak{b} = \mathfrak{b}$. So $\mathfrak{b} = (R : \mathfrak{a})$, which also shows that the inverse is unique, completing the proof.

Corollary 1.2.23: Fractional Ideal Prime Decomposition

Let R be a Dedekind domain. Every fractional ideal $\mathfrak{a}$ admits a unique representation as the product

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_\mathfrak{p}}$$

over the nonzero primes $\mathfrak{p}$ of R, where $n_\mathfrak{p} \in \mathbb{Z}$ and $n_\mathfrak{p} = 0$ for almost all $\mathfrak{p}$. Thus, J_K is the free abelian group generated by the set of nonzero prime ideals $\mathfrak{p}$ of R.

***Proof*:**
Step 1: existence. Choose $0 \neq r \in R$ with $I = r\mathfrak{a} \subseteq R$, a nonzero ideal. In the group J_K (theorem 50.2.22), $\mathfrak{a} = (rR)^{-1}I$. Both rR and I are products of nonzero primes (theorem 50.2.15), say $rR = \prod_{\mathfrak{p}} \mathfrak{p}^{c_\mathfrak{p}}$ and $I = \prod_{\mathfrak{p}} \mathfrak{p}^{d_\mathfrak{p}}$, so $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{d_\mathfrak{p} - c_\mathfrak{p}}$.

Step 2: uniqueness. Suppose $\prod_{\mathfrak{p}} \mathfrak{p}^{n_\mathfrak{p}} = \prod_{\mathfrak{p}} \mathfrak{p}^{m_\mathfrak{p}}$, with finitely many nonzero exponents. For each $\mathfrak{p}$ put $e_\mathfrak{p} = \max(0, -n_\mathfrak{p}, -m_\mathfrak{p})$ and multiply both sides by $\prod_{\mathfrak{p}} \mathfrak{p}^{e_\mathfrak{p}}$. Both sides become products of nonzero primes with the exponents $n_\mathfrak{p} + e_\mathfrak{p} \geq 0$ and $m_\mathfrak{p} + e_\mathfrak{p} \geq 0$, that is, the same nonzero ideal of R factored in two ways. By the uniqueness in theorem 50.2.15, $n_\mathfrak{p} + e_\mathfrak{p} = m_\mathfrak{p} + e_\mathfrak{p}$ for every $\mathfrak{p}$.

Step 3: freeness. The map from the free abelian group on the nonzero primes to J_K sending $(n_\mathfrak{p})_\mathfrak{p}$ to $\prod_{\mathfrak{p}} \mathfrak{p}^{n_\mathfrak{p}}$ is a homomorphism, as J_K is abelian. It is surjective by Step 1 and injective by Step 2, completing the proof.

The quotients of consecutive prime powers are all isomorphic to the residue field:

Lemma 1.2.24: Quotients Of Prime Powers

Let R be a Dedekind domain, $\mathfrak{p} \subseteq R$ a nonzero prime and $i \geq 0$. Then $\mathfrak{p}^i/\mathfrak{p}^{i+1} \cong R/\mathfrak{p}$ as R-modules.

***Proof*:**
If $\mathfrak{p}^i = \mathfrak{p}^{i+1}$, multiplying by the inverse of $\mathfrak{p}^i$ in the group J_K (theorem 50.2.22) gives $R = \mathfrak{p}$, which is false. So there is $a \in \mathfrak{p}^i \setminus \mathfrak{p}^{i+1}$. The ideal $\mathfrak{b} = aR + \mathfrak{p}^{i+1}$ satisfies $\mathfrak{p}^{i+1} \subseteq \mathfrak{b} \subseteq \mathfrak{p}^i$, so by definition 50.2.7 and corollary 50.2.18(3) there are ideals $\mathfrak{c}, \mathfrak{d}$ with $\mathfrak{bc} = \mathfrak{p}^{i+1}$ and $\mathfrak{p}^i\mathfrak{d} = \mathfrak{b}$. By theorem 50.2.15, the first equation forces $\mathfrak{b} = \mathfrak{p}^j$ with $j \leq i+1$, and the second forces $j \geq i$. As $a \in \mathfrak{b} \setminus \mathfrak{p}^{i+1}$, we get $j = i$, that is, $\mathfrak{p}^i = aR + \mathfrak{p}^{i+1}$. Hence the R-module homomorphism

$$\varphi \colon R \to \mathfrak{p}^i/\mathfrak{p}^{i+1} \qquad x \mapsto xa + \mathfrak{p}^{i+1}$$

is surjective. Its kernel is an ideal of R. It contains $\mathfrak{p}$, because $\mathfrak{p}a \subseteq \mathfrak{p}\mathfrak{p}^i = \mathfrak{p}^{i+1}$, and it does not contain 1, because $a \notin \mathfrak{p}^{i+1}$. The prime $\mathfrak{p}$ is maximal (definition 50.2.1), so the kernel is $\mathfrak{p}$, and [3, Isomorphism Theorems For Modules] gives $\mathfrak{p}^i/\mathfrak{p}^{i+1} \cong R/\mathfrak{p}$, completing the proof.

Definition 1.2.25: Ideal Group

Let R be a Dedekind Domain. Then J_R denotes the *ideal group* given by ideal multiplication.

We shall often write J_R interchangeably with J_K

Example 1.2.26: Ideal Group

1. The canonical example would be $J_\mathbb{Z} \cong \mathbb{Q}_+^\times$ with the bijection
$$q\mathbb{Z} = (q) \mapsto q$$
Note that it is important to think of the multiplicative structure of $\mathbb{Q}$, $(\mathbb{Q}, +)$ is not a free

abelian group, having no two distinct elements be linearly dependent and not being cyclic.

2. More generally, let R be a Principal Ideal Domain. Then:

$$J_R \cong K^\times / R^\times$$

This isomorphism does not work in general, namely since not all ideals of R are principal.

A natural subgroup of J_K is formed by the principal ideal (a) where $a \in K^\times$. This subgroup can be thought of as representing our usual notion of factorization into elements instead of ideals As some foreshadowing, notice that the principal ideal The quotient of J_K by the subgroup of principle ideals can be thought of as measuring how far away the ideal group is from being a PID:

Definition 1.2.27: Ideal Class Group

Let J_K be the class group of K. Then

$$Cl(K) = J_K / P_K$$

is called the *ideal class group*, *class group*, or *Picard group* of K.

1.2.28 Note on Nomenclature The term Picard group shall not be used in this book, as the nomenclature is more so associated within algebraic geometry. It is pointed out here that some call the class group the Picard group as the class group is a special case of the Picard group.

These naturally fit in the following exact sequence:

$$1 \to R^\times \to K^\times \to J_K \to Cl(K) \to 1$$

where $K^\times \to J_K$ is the map $a \mapsto (a)$. Hence, we may think of the class group $Cl(K)$ as a measure of the expansion that happens when we pass from numbers to ideals, where the unit group $R^\times$ measure the contraction of the same process. Thus, we may want to understand the groups $R^\times$ and $Cl(K)$ more thoroughly. In general, these can be arbitrary abelian groups[17], but for rings of integers $\mathcal{O}_K$ of number fields K, we get that the size of the class group is always finite and that $\mathcal{O}_K^\times$ is a free $\mathbb{Z}$-module along with roots of unity contained in K, as will soon be shown.

1.2.3 Class Group of Ring of Integer

Recall that ideals $\mathfrak{a} \subseteq \mathcal{O}_K$ form a sub-lattice of $\mathcal{O}_K$ in the Archimedean embedding. The following theorem is key bound on the covolume of lattice. Using theorem 50.1.66, we may rather easily show that the class group is finite. The way we shall find the size of the class group is by showing that every element in a class group will have absolute norm less than a bound (say M), and that only finitely many ideals can have norm equal to $m < M$. Then, by considering the principal ideals that have norm less than M, we shall be able to find all prime ideals that compose these principal ideals, which shall give us an estimate on the size of the group.

[17] There are "enough" 1-dimensional smooth schemes to make arbitrary class groups on Dedekind domains

Lemma 1.2.29: Absolute Norm of Prime

Let $0 \neq \mathfrak{p} \subseteq \mathcal{O}_K$ be a prime ideal of $\mathcal{O}_K$ and let $\mathcal{O}_K$ be rank n. Then there is a prime number p such that $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$, the quotient $\mathcal{O}_K/\mathfrak{p}$ is a finite field of characteristic p, and:

$$\mathfrak{N}(\mathfrak{p}) = p^f \qquad \text{where} \qquad f = [\mathcal{O}_K/\mathfrak{p} : \mathbb{F}_p]$$

Furthermore, $f \leq n$, with equality if and only if $\mathfrak{p} = p\mathcal{O}_K$, that is, if and only if p stays prime in $\mathcal{O}_K$.

The number f shall be given a name in definition 50.4.1: the *inertia degree* of $\mathfrak{p}$ over p. Note that f may be strictly smaller than n: in $\mathbb{Z}[i] = \mathcal{O}_{\mathbb{Q}(i)}$, the ideal $(2+i)$ is prime with $\mathbb{Z}[i]/(2+i) \cong \mathbb{F}_5$, hence $\mathfrak{N}((2+i)) = 5$ and not $5^2 = 25$.

***Proof*:**

By proposition 50.1.62, $\mathcal{O}_K/\mathfrak{p}$ is finite, and since $\mathfrak{p}$ is prime, $\mathcal{O}_K/\mathfrak{p}$ is an integral domain. Hence, $\mathcal{O}_K/\mathfrak{p}$ is a finite field by [3, Finite Integral Domain is a Field].

Next, consider the ring homomorphism $\mathbb{Z} \to \mathcal{O}_K/\mathfrak{p}$ given by composing the inclusion with the quotient map. Its kernel $\mathfrak{p} \cap \mathbb{Z}$ is a prime ideal of $\mathbb{Z}$, and it is nonzero, for otherwise $\mathbb{Z}$ would embed into the finite ring $\mathcal{O}_K/\mathfrak{p}$. Hence, $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$ for some prime number p, and $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ embeds into $\mathcal{O}_K/\mathfrak{p}$, which is therefore of characteristic p. Thus, $\mathcal{O}_K/\mathfrak{p}$ is a finite field extension of $\mathbb{F}_p$, and setting $f = [\mathcal{O}_K/\mathfrak{p} : \mathbb{F}_p]$, counting the elements of an f-dimensional $\mathbb{F}_p$-vector space gives:

$$\mathfrak{N}(\mathfrak{p}) = |\mathcal{O}_K/\mathfrak{p}| = p^f$$

For the bound on f, let $\alpha_1, \alpha_2, ..., \alpha_n$ be an integral basis for $\mathcal{O}_K$ (corollary 50.1.33). Since $\mathcal{O}_K = \mathbb{Z}\alpha_1 + \cdots + \mathbb{Z}\alpha_n$, the images $\overline{\alpha_1}, ..., \overline{\alpha_n}$ span $\mathcal{O}_K/\mathfrak{p}$ as an $\mathbb{F}_p$-vector space, hence $f \leq n$. If $\mathfrak{p} = p\mathcal{O}_K$, then $\mathcal{O}_K/\mathfrak{p} = \mathcal{O}_K/p\mathcal{O}_K \cong (\mathbb{Z}/p\mathbb{Z})^n$ by freeness, which has p^n elements, so $f = n$. Conversely, if $f = n$, then since $p \in \mathfrak{p}$ we have $p\mathcal{O}_K \subseteq \mathfrak{p}$, so $\mathcal{O}_K/\mathfrak{p}$ is a quotient of $\mathcal{O}_K/p\mathcal{O}_K$. Both have exactly p^n elements, forcing $\mathfrak{p} = p\mathcal{O}_K$, completing the proof.

Proposition 1.2.30: Absolute Norm for Ring of Integers Is Multiplicative

Let $\mathfrak{a} = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_s^{n_s}$ be the prime factorization of the nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}_K$, with the $\mathfrak{p}_j$ distinct. Then:

$$\mathfrak{N}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{p}_1)^{n_1} \cdots \mathfrak{N}(\mathfrak{p}_s)^{n_s}$$

Consequently, $\mathfrak{N}(\mathfrak{ab}) = \mathfrak{N}(\mathfrak{a})\mathfrak{N}(\mathfrak{b})$ for all nonzero ideals $\mathfrak{a}, \mathfrak{b} \subseteq \mathcal{O}_K$.

***Proof*:**

The ring $\mathcal{O}_K$ is a Dedekind domain (theorem 50.2.6), so its nonzero primes are maximal (definition 50.2.1).

Step 1: $\mathfrak{p}^i/\mathfrak{p}^{i+1} \cong \mathcal{O}_K/\mathfrak{p}$. For a nonzero prime $\mathfrak{p}$ and $i \geq 0$, this is lemma 50.2.24.

Step 2: $\mathfrak{N}(\mathfrak{p}^n) = \mathfrak{N}(\mathfrak{p})^n$. We induct on n, the case $n = 0$ being $\mathfrak{N}(\mathcal{O}_K) = 1$. The group $\mathcal{O}_K/\mathfrak{p}^{n+1}$ is finite (proposition 50.1.62) and contains the subgroup $\mathfrak{p}^n/\mathfrak{p}^{n+1}$, with quotient isomorphic to

$\mathcal{O}_K/\mathfrak{p}^n$ by [3, Isomorphism Theorems For Modules]. By [3, Lagrange Theorem] and Step 1,

$$\mathfrak{N}(\mathfrak{p}^{n+1}) = |\mathcal{O}_K/\mathfrak{p}^n| \cdot |\mathfrak{p}^n/\mathfrak{p}^{n+1}| = \mathfrak{N}(\mathfrak{p})^n\mathfrak{N}(\mathfrak{p}) = \mathfrak{N}(\mathfrak{p})^{n+1}$$

Step 3: the product formula. The ideals $\mathfrak{p}_j^{n_j}$ are pairwise comaximal (lemma 50.2.17). By [3, Chinese Remainder Theorem],

$$\mathcal{O}_K/\mathfrak{a} \cong \mathcal{O}_K/\mathfrak{p}_1^{n_1} \times \cdots \times \mathcal{O}_K/\mathfrak{p}_s^{n_s}$$

so Step 2 gives $\mathfrak{N}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{p}_1^{n_1}) \cdots \mathfrak{N}(\mathfrak{p}_s^{n_s}) = \mathfrak{N}(\mathfrak{p}_1)^{n_1} \cdots \mathfrak{N}(\mathfrak{p}_s)^{n_s}$.

Step 4: multiplicativity. If $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{a_\mathfrak{p}}$ and $\mathfrak{b} = \prod_{\mathfrak{p}} \mathfrak{p}^{b_\mathfrak{p}}$, then $\mathfrak{ab} = \prod_{\mathfrak{p}} \mathfrak{p}^{a_\mathfrak{p}+b_\mathfrak{p}}$ is the prime factorization of $\mathfrak{ab}$, and Step 3 gives

$$\mathfrak{N}(\mathfrak{ab}) = \prod_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{a_\mathfrak{p}+b_\mathfrak{p}} = \mathfrak{N}(\mathfrak{a})\mathfrak{N}(\mathfrak{b})$$

completing the proof.

Recall from lemma 50.2.29 that the absolute norm of a prime is $\mathfrak{N}(\mathfrak{p}) = p^f$ where $\mathfrak{p}$ lies over p and $f = [\mathcal{O}_K/\mathfrak{p} : \mathbb{F}_p]$. In corollary 50.3.16, we shall prove the relative counterpart: if $\mathfrak{B}$ is a prime of $\mathcal{O}_L$ lying over $\mathfrak{p}$ in an extension of number fields L/K, then $\mathfrak{N}_{L/K}(\mathfrak{B}) = \mathfrak{p}^f$ where $f = [\mathcal{O}_L/\mathfrak{B} : \mathcal{O}_K/\mathfrak{p}]$.

Every fractional ideal factors uniquely as $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_\mathfrak{p}}$ with $n_\mathfrak{p} \in \mathbb{Z}$, almost all zero (corollary 50.2.23), so J_K is free abelian on the nonzero primes, and a homomorphism out of J_K may be prescribed freely on the primes:

Definition 1.2.31: Absolute Norm Of A Fractional Ideal

Let K be a number field. The *absolute norm* on fractional ideals is the group homomorphism

$$\mathfrak{N}_K : J_K \to \mathbb{R}_{>0}^\times \qquad \mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_\mathfrak{p}} \mapsto \prod_{\mathfrak{p}} \mathfrak{N}(\mathfrak{p})^{n_\mathfrak{p}}$$

On a nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ every $n_\mathfrak{p}$ is non-negative, and by proposition 50.2.30 the value is the index $[\mathcal{O}_K : \mathfrak{a}]$ of definition 50.1.61, so the two definitions agree.

Corollary 1.2.32: Volume Form Reformulation

Let $I \subseteq K$ be a fractional ideal of a number field K. Then $\sigma(I)$ is a complete lattice in $K_\mathbb{R}$, and, writing $\mathrm{vol}(I)$ for $\mathrm{vol}(\sigma(I))$:

$$\mathrm{vol}(I) = \mathfrak{N}(I)\mathrm{vol}(\mathcal{O}_K)$$

***Proof*:**
Step 1: an integral multiple. By definition 50.2.9 there is $0 \neq r \in \mathcal{O}_K$ with $rI \subseteq \mathcal{O}_K$. The ideal $r\mathcal{O}_K$ is nonzero, so $d = [\mathcal{O}_K : r\mathcal{O}_K]$ is finite (proposition 50.1.62), and $d \cdot 1 \in r\mathcal{O}_K$ by [3, Lagrange Theorem] applied to the additive group $\mathcal{O}_K/r\mathcal{O}_K$. Writing $d = ru$ with $u \in \mathcal{O}_K$, we

get $dI = u(rI) \subseteq \mathcal{O}_K$, so dI is a nonzero ideal of $\mathcal{O}_K$.

Step 2: scaling the lattice. Let $\beta_1, \dots, \beta_n$ be a $\mathbb{Z}$-basis of dI, so that $\sigma(\beta_1), \dots, \sigma(\beta_n)$ is a basis of $K_{\mathbb{R}}$ and a $\mathbb{Z}$-basis of the complete lattice $\sigma(dI)$ (proposition 50.1.60). Multiplication by d is a group isomorphism $I \to dI$, so $\beta_1/d, \dots, \beta_n/d$ is a $\mathbb{Z}$-basis of I, and $\sigma(\beta_i/d) = d^{-1}\sigma(\beta_i)$ because every embedding fixes $\mathbb{Q}$. So $\sigma(I) = d^{-1}\sigma(dI)$ is a complete lattice, and its basis matrix is d^{-1} times that of $\sigma(dI)$. Scaling each of the n rows by d^{-1} scales the determinant by d^{-n} ([3, Properties Of The Determinant]), so definition 50.1.59 gives

$$\operatorname{vol}(dI) = d^n \operatorname{vol}(I)$$

Step 3: comparing norms. As $dI = (d\mathcal{O}_K)I$ in J_K and $\mathfrak{N}$ is a homomorphism on J_K (definition 50.2.31), $\mathfrak{N}(dI) = \mathfrak{N}(d\mathcal{O}_K)\mathfrak{N}(I)$. Here $\mathfrak{N}(d\mathcal{O}_K) = [\mathcal{O}_K : d\mathcal{O}_K] = d^n$, since $\mathcal{O}_K/d\mathcal{O}_K \cong (\mathbb{Z}/d\mathbb{Z})^n$ for an integral basis (corollary 50.1.33). Applying lemma 50.1.65 to the nonzero ideal dI, on which the two definitions of $\mathfrak{N}$ agree, and using Step 2,

$$d^n \operatorname{vol}(I) = \operatorname{vol}(dI) = \mathfrak{N}(dI)\operatorname{vol}(\mathcal{O}_K) = d^n \mathfrak{N}(I)\operatorname{vol}(\mathcal{O}_K)$$

Dividing by d^n gives $\operatorname{vol}(I) = \mathfrak{N}(I)\operatorname{vol}(\mathcal{O}_K)$, completing the proof.

Theorem 1.2.33: Class Number

Let K be a number field and let $Cl(K) = J_K/P_K$ be the class group of K. Then $Cl(K)$ is finite.

***Proof*:**
Step 1: finitely many ideals of bounded norm. Fix $M > 0$. A nonzero prime $\mathfrak{p}$ satisfies $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$ and $\mathfrak{N}(\mathfrak{p}) = p^f \geq p$ for a prime number p (lemma 50.2.29). As $p\mathcal{O}_K \subseteq \mathfrak{p}$, the prime $\mathfrak{p}$ divides $p\mathcal{O}_K$ (definition 50.2.7), so $p\mathcal{O}_K = \mathfrak{p}\mathfrak{c}$ for an ideal $\mathfrak{c}$ (corollary 50.2.18(3)), and by theorem 50.2.15 the prime $\mathfrak{p}$ is one of the finitely many primes in the factorization of $p\mathcal{O}_K$. Every prime with $\mathfrak{N}(\mathfrak{p}) \leq M$ lies over one of the finitely many primes $p \leq M$, so there are finitely many such $\mathfrak{p}$. A nonzero ideal $\mathfrak{a} = \mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_s^{n_s}$ has $\mathfrak{N}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{p}_1)^{n_1} \cdots \mathfrak{N}(\mathfrak{p}_s)^{n_s}$ (proposition 50.2.30) with every $\mathfrak{N}(\mathfrak{p}_i) \geq 2$. So if $\mathfrak{N}(\mathfrak{a}) \leq M$, then each $\mathfrak{N}(\mathfrak{p}_i) \leq M$ and each $n_i \leq \log_2 M$, and only finitely many nonzero ideals have $\mathfrak{N}(\mathfrak{a}) \leq M$.

Step 2: a small ideal in each class. Let λ be as in theorem 50.1.66 and let $[\mathfrak{b}] \in Cl(K)$. By theorem 50.2.22, $\mathfrak{b}$ has an inverse $\mathfrak{b}^{-1}$, and by corollary 50.2.12 there are $0 \neq r \in \mathcal{O}_K$ and an ideal $I \subseteq \mathcal{O}_K$ with $\mathfrak{b}^{-1} = r^{-1}I$. So $I = (r)\mathfrak{b}^{-1}$ is a nonzero integral ideal with $[I] = [\mathfrak{b}]^{-1}$. By theorem 50.1.66 there is $0 \neq \alpha \in I$ with $\left|\operatorname{Nm}_{K/\mathbb{Q}}(\alpha)\right| \leq \lambda\mathfrak{N}(I)$. Since $(\alpha) \subseteq I$, corollary 50.2.18(3) gives an integral ideal $\mathfrak{a}$ with $I\mathfrak{a} = (\alpha)$, so $[\mathfrak{a}] = [I]^{-1} = [\mathfrak{b}]$. By proposition 50.2.30 and lemma 50.1.63,

$$\mathfrak{N}(I)\mathfrak{N}(\mathfrak{a}) = \mathfrak{N}((\alpha)) = \left|\operatorname{Nm}_{K/\mathbb{Q}}(\alpha)\right| \leq \lambda\mathfrak{N}(I)$$

and dividing by $\mathfrak{N}(I) > 0$ gives $\mathfrak{N}(\mathfrak{a}) \leq \lambda$.

Step 3: finiteness. Choosing such an ideal $\mathfrak{a}$ in each class defines an injective map from $Cl(K)$

to the set of nonzero ideals of absolute norm at most λ, which is finite by Step 1. So $Cl(K)$ is finite, completing the proof.

Corollary 1.2.34: Minkowski's Bound For Ideal Classes

Let K be a number field with Minkowski bound M_K (theorem 50.1.69). Then every ideal class $[\mathfrak{b}] \in Cl(K)$ contains an integral ideal $\mathfrak{a}$ with $\mathfrak{N}(\mathfrak{a}) \leq M_K$.

***Proof*:**
By theorem 50.2.22, the fractional ideal $\mathfrak{b}$ has an inverse $\mathfrak{b}^{-1}$, and by corollary 50.2.12 there are $0 \neq r \in \mathcal{O}_K$ and an ideal $I \subseteq \mathcal{O}_K$ with $\mathfrak{b}^{-1} = r^{-1}I$. So $I = (r)\mathfrak{b}^{-1}$ is a nonzero integral ideal with $[I] = [\mathfrak{b}]^{-1}$ in $Cl(K)$. By theorem 50.1.69 there is $0 \neq \alpha \in I$ with $\left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| \leq M_K\mathfrak{N}(I)$. Since $(\alpha) \subseteq I$, that is $I \big| (\alpha)$, corollary 50.2.18(3) gives an integral ideal $\mathfrak{a}$ with $I\mathfrak{a} = (\alpha)$. In $Cl(K)$ this reads $[I][\mathfrak{a}] = 1$, so $[\mathfrak{a}] = [I]^{-1} = [\mathfrak{b}]$. By proposition 50.2.30 and lemma 50.1.63,

$$\mathfrak{N}(I)\mathfrak{N}(\mathfrak{a}) = \mathfrak{N}((\alpha)) = \left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right| \leq M_K\mathfrak{N}(I)$$

and dividing by $\mathfrak{N}(I) > 0$ gives $\mathfrak{N}(\mathfrak{a}) \leq M_K$, as claimed.

Definition 1.2.35: The Class Number h_K

Let $Cl(K)/ = J_K/P_k$ be the class group of K. Then $h_K = [J_K : P_K]$ is called the *class number* of $Cl(K)$ or K.

A consequence of there being a finite class number is that for every ideal $I \subseteq \mathcal{O}_K$, there exists a minimal number n such that $I^n = (\alpha)$ for some $\alpha \in \mathcal{O}_K$. There is also a minimal number n' such that for *any* ideal of $\mathcal{O}_K$, the ideal to the power of n' is principal. This may encourage thoughts of converting the problem on ideals into one on principal ideals, given there is a way of going back to the original ideals. We shall see that this shall sometimes be the case.

Example 1.2.36: Class Groups

1. Naturally, $|Cl(\mathbb{Q})| = 1$ since $\mathbb{Z}$ is a PID
2. Similarly, $|Cl(\mathbb{Q}(i)| = |Cl(\mathbb{Q}(\sqrt{-3})| = 1$ since $\mathbb{Z}[i]$ and $\mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]$ are PID.
3. Every DVR has ideals of the form (π^n) for appropriate uniformizer π. Thus, the ideal group is isomorphic to $\mathbb{Z}$, and the class group is trivial.
4. Consider $Cl(\mathbb{Q}(\sqrt{-5})$. Since $\mathbb{Z}[\sqrt{-5}]$ is not a Unique Factorization Domain (hence Principal Ideal Domain), we will get a non-trivial class group. Let $K = \mathbb{Q}(\sqrt{-5})$. Choose an integral basis
$$\mathcal{O}_K = \langle 1, \sqrt{-5}\rangle_{\mathbb{Z}}$$
so that $D_k = -20$. Hence, the Minkowski bound gives us:
$$M_k = \sqrt{20}\frac{2}{2^2}\frac{4}{\pi} < 4$$

In fact, we have < 3, but we shall purposefully miss-guess the bound to show how to determine when two ideals belong in the same class group. Consider the ideal $(2), (3)$. Assume that $\mathfrak{p}_2 \big| (2)$. Then we know $\mathfrak{N}(\mathfrak{p}_2) \big| \mathfrak{N}((2)) = 4$. To find $\mathfrak{p}_2 \subseteq \mathcal{O}_K$ consider the map $\varphi : \mathcal{O}_K \to \mathcal{O}_K/(2)$. Since $\mathcal{O}_K/(2)$ is finite the pre-image of a prime ideal of a surjective map is prime, we may find prime ideals of $\mathcal{O}_K/(2)$. By the 4th isomorphism theorem, if $\overline{I} \subseteq \mathcal{O}_K/(2)$ if and only if $(2) \subseteq I$, that is $I \big| (2)$, and hence any prime ideal corresponding to prime ideal of $\mathcal{O}_K/(2)$ divides (2). Thus we look for $\varphi^{-1}(P) = \mathfrak{p}_2$. To find such a $\mathfrak{p}_2$, think of $\mathcal{O}_K$ as $\mathcal{O}_K \cong \mathbb{Z}[x]/(x^2+5)$[a] so that we can take:

$$\mathcal{O}_K/(2) \cong \mathbb{Z}[x]/(x^2+5, 2) \cong \mathbb{F}_2[x]/(x^2-1) = \mathbb{F}_2[x]/((x+1)^2)$$

Recall that the prime ideals of a ring are in corresponding with the prime ideals of the ring quotiented by the nilradical (see [7, Radical Ideals]). But then we immediately see that the only prime ideal in $\mathcal{O}_K/(2)$ is $(1+x)$. Hence

$$\mathfrak{p}_2 = (1+\sqrt{-5}, 2)$$

Every prime ideal dividing (2) contains 2, so by the computation of $\mathcal{O}_K/(2)$ above it is $\mathfrak{p}_2$, and $\mathfrak{p}_2 \neq (2)$ because $1+\sqrt{-5} \notin (2)$. Hence $(2) = \mathfrak{p}_2^e$ for some $e \geq 2$, and indeed:

$$\mathfrak{p}_2^2 = (4, 2+2\sqrt{-5}, -4+2\sqrt{-5}) = (2)$$

Is $\mathfrak{p}_2$ principal? If it was, then $\mathfrak{p}_2 = (a+b\sqrt{-5})$. Then:

$$2 = \mathfrak{N}(\mathfrak{p}_2) = |\mathrm{Nm}_{K/\mathbb{Q}}(a+b\sqrt{-5})| = a^2+5b^2$$

but it's evident this is not possible, hence it is not principal! Similarly with (3), we get:

$$\mathcal{O}_K/(3) \cong \mathbb{Z}[x]/(x^2+5, 3) = \mathbb{F}_3[x]/(x^2-1) \cong \mathbb{F}_3 \times \mathbb{F}_3$$

From this, we can see that $3 = \mathfrak{p}_3\mathfrak{p}_3'$ where

$$\mathfrak{p}_3 = (3, 1+\sqrt{-5}) \qquad \mathfrak{p}_3' = (3, 1-\sqrt{-5})$$

Finally, notice that $[\mathfrak{p}_2] = [\mathfrak{p}_3]$. To see this, notice their product is in fact principal:

$$\mathfrak{p}_2\mathfrak{p}_3 = (2, 1+\sqrt{-5})(3, 1+\sqrt{-5}) = (6, 1+\sqrt{-5}, (1+\sqrt{-5})^2) = (1+\sqrt{-5})$$

Hence, $\mathfrak{p}_3 = (1+\sqrt{-5})^{-1}\mathfrak{p}_2$. Evidently, $\mathfrak{p}_3, \mathfrak{p}_3'$ are in the same class. Since there are only two elements, it must be that $Cl(K) \cong \mathbb{Z}/2\mathbb{Z}$.

5. Let $K = \mathbb{Q}(\sqrt{-23})$. Choose $\mathcal{O}_K = \langle 1, \frac{1+\sqrt{-23}}{2} \rangle_{\mathbb{Z}}$ so that it's discriminant is $d_k = -23$ and the Minkowski bound is $M_K = \sqrt{23}\frac{4}{\pi}\frac{1}{2} < 4$. Hence, we again check the prime ideals (2) and (3), noting that the minimal polynomial of the generator is $x^2 - x + 6$:

$$\mathcal{O}_K/(2) \cong \mathbb{F}_2[x]/(t^2-t)$$

Thus, if $t = \frac{1+\sqrt{-23}}{2}$:

$$(2) = (2, t-1)(2, t)$$

and similarly

$$(3) = (3, t-1)(3, t)$$

Certainly, each prime multiple of (2) and (3) are in the same equivalence class. Is $[(2,t)] = [(3,t-1)]$? Choosing good representatives and multiplying, we get

$$(3,t)(2,t) = (6,t) = (t)$$

hence, we again get that $Cl(K) = \mathbb{Z}/2\mathbb{Z}$.

6. Let $K = \mathbb{Q}(\sqrt{15})$ and $\mathcal{O}_K = \langle 1, \sqrt{15}\rangle_{\mathbb{Z}}$. Then doing the same procedure, we get:

$$(3, \sqrt{15})^2 = (3)$$

How do we know that $(3, \sqrt{15})$ is principal or not? We can see that $a^2 - 15b^2 = 3$ has no solutions using modular arithmetic's, but it will not always be feasible to use this method, so we introduce another. Let's say $(3, \sqrt{15}) = (\alpha)$. Then

$$(\alpha^2) = (3)$$

meaning they are associate, hence $\alpha^2 = 3u$ for $u \in \mathcal{O}_K^\times$. Does there exist such a unit? In section 50.2.4, we shall find this structure. For now, it can be taken for granted that $\mathcal{O}_K^\times = \pm\langle 4 + \sqrt{15}\rangle$ (notice the norm of all such numbers is 1). Then from this we see that no such unit exist, hence $(3, \sqrt{15})$ is not principal.

Notice that all of these examples are for quadratic extensions. It is much more difficult to find prime-factorization for non-quadratic extension since the ring of integers is not easy to find. In the next section, we shall see how we may get some information on the factorization of primes when $\mathcal{O}_K$ is not available

[a] Note how this relied on knowing the structure of $\mathcal{O}_K$! See comment at the end of this example

In general, we have $h_K > 1$. If $h_K = 1$ then we can think of the prime factorization of ideals being the same as the prime factorization of elements as is already well-understood. If $K = \mathbb{Q}(\sqrt{d})$ where d is square-free and negative, then the quadradic extensions in which the class number is 1 is

$$-1, -2, -3, -7, -11, -19, -43, -67, -163$$

For real quadratic extensions, there are a lot more, for example:

$$2, 3, 5, 6, 7, 11, 13, 14, 17, 19, 22, 23, 29, 31, 33, \cdots$$

it is unknown where there are infinitely many real quadratic fields of class number 1, or even if there are infinitely many algebraic number fields (of any degree) with class number 1. In general, it is in fact very difficult to determine some structure in the class group given some class of number fields, with an important notable exception cyclotomic extensions which due to the work of Kenkichi Iwasawa we know behave in a regular way.

The importance of studying cyclotomic extensions links back to Fermat's last theorem which the reader surely recalls states that the equation $x^p + y^p = z^p$ has no nonzero integer solutions for $p \geq 3$. For $p = 2$, we take the cyclotomic extension $\mathbb{Q}(\zeta_4) = \mathbb{Q}(i)$ to get

$$x^2 + y^2 = (x + iy)(x - iy)$$

and then we get the Fermat's twin prime theorem that fully solved the problem in this case. If p is

prime, then considering the equation $y^p = z^p - x^p$ over $\mathbb{Q}(\zeta_p)$, we may factor this in $\mathbb{Z}[\zeta_p]$ as

$$\begin{aligned} y^p &= x^p - z^p \\ &= z^p \left(\left(\frac{x}{z} \right)^p - 1 \right) \\ &= z^p \prod_{k=0}^{p-1} \left(\frac{x}{z} - \omega_k \right) \\ &= \prod_{k=0}^{p-1} z \left(\frac{x}{z} - \omega_k \right) \\ &= \prod_{k=0}^{p-1} (x - \omega_k z) \end{aligned}$$

where $\omega_k = \zeta_p^k$. Thus, if we assume an existence of a solution, we obtain two multiplicative decompositions of the same number in $\mathbb{Z}[\zeta_p]$. Using this, we can show it contradicts unique factorization *provided* that $\mathbb{Z}[\zeta_p]$ is a unique factorization domain. It was due to some mathematician erroneously assuming unique factorization (equivalently for us, that the class number is always 1) that lead to many fake proofs of Fermat's last theorem[18]. Kummer fixed this by introducing the notion of ideal primes, and translated the statement into one where we must show that $p \nmid h_p$ instead of assuming $h_p = 1$. If this is the case, such a prime is called *regular*, and *irregular* otherwise. Unfortunately, not all primes are regular. Of the first 25 primes less than 100, the primes 37, 59, and 67 are irregular. As of writing this book, it is unknown whether there are infinitely many regular primes.

The regular and irregular prime are going to play important roles in our coming theory, but they do not play much of a role in the eventual proof of Fermat's Last theorem. Due to the work of Gerhard Frey, there was a reformulation of Fermat's last theorem into a statement about elliptic curves. This lead to a new development of mathematics which is beyond the scope of this book, but the interested reader may feel free to search up the *Taniyama-Shimura-Weil Conjecture* to see how Fermat's Last Theorem got translated into a problem of elliptic curves.

We end this section with showing a certain Diophantine equation has no integer solution as it combines nicely all the machinery we have done, and some generalizations that is motivation for class field theory. Starting with the Diophantine equation:

Theorem 1.2.37: No Solution To Diophantine Equation

The equation

$$y^2 = x^3 - 5$$

has no integer solutions.

***Proof*:**

Suppose the pair (x, y) is an integer solution. If x were even, then $4 \mid x^3$ and $y^2 = x^3 - 5 \equiv 3 \bmod 4$, which is impossible because every square is congruent to 0 or 1 modulo 4. So x is odd. Next, x and y are coprime: a prime p dividing both would give $p^2 \mid y^2$ and $p^3 \mid x^3$, hence $p^2 \mid x^3 - y^2 = 5$, which is impossible.

[18] It is speculated that this is the proof Fermat was referring to as being too long for the margins of his book

Let $K = \mathbb{Q}(\sqrt{-5})$. Since $-5 \not\equiv 1 \bmod 4$, example 50.1.17 gives $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$, which is a Dedekind domain by theorem 50.2.6. In $\mathcal{O}_K$ the equation factors as $(y+\sqrt{-5})(y-\sqrt{-5}) = y^2+5 = x^3$, so as ideals

$$(y+\sqrt{-5})(y-\sqrt{-5}) = (x)^3$$

and the two factors on the left are coprime. Suppose a prime ideal $\mathfrak{p}$ contains both. Then $\mathfrak{p}$ contains x^3, hence x, and it contains their difference $2\sqrt{-5}$, hence $2\sqrt{-5}\cdot\sqrt{-5} = -10$. Let p be the prime with $\mathfrak{p}\cap\mathbb{Z} = p\mathbb{Z}$. Then p divides both x and 10, and x is odd, so $p = 5$. Now $(\sqrt{-5})^2 = -5 \in \mathfrak{p}$, and $\mathfrak{p}$ is prime, so $\sqrt{-5} \in \mathfrak{p}$ and $y = (y+\sqrt{-5}) - \sqrt{-5} \in \mathfrak{p}\cap\mathbb{Z} = 5\mathbb{Z}$. Then 5 divides both x and y, contradicting their coprimality.

By theorem 50.2.15, every prime ideal occurs in $(x)^3$ with exponent divisible by 3. The two factors are coprime, so each prime ideal of $(x)^3$ occurs in exactly one of them, with its full exponent. Hence $(y+\sqrt{-5}) = \mathfrak{a}^3$ for an ideal $\mathfrak{a}$ of $\mathcal{O}_K$. The class group of K has order 2 (example 50.2.36, the case $\mathbb{Q}(\sqrt{-5})$), so $[\mathfrak{a}]^2 = 1$, while $[\mathfrak{a}]^3 = [(y+\sqrt{-5})] = 1$. Hence $[\mathfrak{a}] = [\mathfrak{a}]^3[\mathfrak{a}]^{-2} = 1$, and $\mathfrak{a} = (a+b\sqrt{-5})$ for some $a, b \in \mathbb{Z}$. Then $y+\sqrt{-5} = u(a+b\sqrt{-5})^3$ for a unit u of $\mathcal{O}_K$. A unit $c+d\sqrt{-5}$ has norm $c^2+5d^2 = \pm 1$, which forces $d = 0$ and $c = \pm 1$. So $u = \pm 1 = (\pm 1)^3$, and replacing a, b by $-a, -b$ if $u = -1$ gives

$$y+\sqrt{-5} = (a+b\sqrt{-5})^3 = (a^3 - 15ab^2) + (3a^2b - 5b^3)\sqrt{-5}$$

and comparing the coefficients of $\sqrt{-5}$ gives $1 = b(3a^2 - 5b^2)$, so $b = \pm 1$. If $b = 1$, then $3a^2 - 5 = 1$ and $a^2 = 2$. If $b = -1$, then $3a^2 - 5 = -1$ and $3a^2 = 4$. Neither has an integer solution, so no integer solution (x, y) exists.

Note that a few things lined up well here:

- There were only two units, if there are more then the considerations are more complicated.
- The class group happened to let us deduce the ideal is principal. If the ideal was not principal there would be more to consider.
- The equation was easily factorizable in the extension (ex. $y^2 = x^5 + x + 1$ is a good deal more difficult)
- The factors are coprime.

Exercise 1.2.1

1. Let $A, B \subseteq R$ with $B \subseteq A$ and $A \neq R$. Show there exists a $\gamma \in K$ such that $\gamma B \subseteq R$, but $\gamma B \not\subseteq A$
2. Show that $IJ = (I+J)(I\cap J) = \gcd(I,J)\mathrm{lcm}(I,J)$
3. Let $\alpha \in \mathcal{O}_K \setminus \{0\}$. Show that there exists a $\beta \in \mathcal{O}_K$ st $\alpha\beta \in \mathbb{Z}\setminus\{0\}$
4. Let $K = \mathbb{Q}(\sqrt{-3})$. Show that $\mathcal{O}_K \neq \mathbb{Z}[\sqrt{-3}]$. Show that
5. Show that $3 = x^2 + 5y^2$ has no integer solutions. Note that 3 splits in $\mathbb{Z}[\sqrt{-3}]$ showing that if the ringer of integers is not UFD then it is more complicated to use factorizations.
6. exercise for quotient ideals

a) here

1.2.4 Dirichlet's Unit Theorem

(Current intuition when it comes to solving equations: we are working with ideals, but equations use elements. We must be concious of the units at play to reduce back to elements)

As we saw in example 50.2.36, it is important to know the unit group of $\mathcal{O}_K$ to find elements in the class group. We start this section by showing that we may embed $\mathcal{O}_K^\times$ to form a lattice, and then use this to find the structure of $\mathcal{O}_K$.

The group $\mathcal{O}_K^\times$ may have some torsion elements, namely all roots of unity that lie in K. This collection is labeled as $\mu(K)$. This group will in fact be determined by the number of embeddings into Minkowski space. On a high-level, the structure of this group is within this diagram:

$$\begin{array}{ccccc} & & \overset{\sigma^\times}{\frown} & & \\ K^\times & \xrightarrow{\sigma} & K_{\mathbb{R}}^\times & \xrightarrow{\log} & [\prod_\tau \mathbb{R}]^+ \\ \downarrow{\scriptstyle N_{K/\mathbb{Q}}} & & \downarrow{\scriptstyle \mathrm{nm}} & & \downarrow{\scriptstyle \mathrm{Tr}} \\ \mathbb{Q}^* & \hookrightarrow & \mathbb{R}^* & \xrightarrow{\log|\ |} & \mathbb{R} \end{array}$$

where we shall write $K_{\mathbb{R}}^\times = (\mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2})^\times$. In particular, the top row of the above commutative diagram can be replaced by:

$$\begin{aligned} \mathcal{O}_K^\times &= \left\{ r \in \mathcal{O}_K \;\middle|\; N_{K/\mathbb{Q}(n)=\pm 1} \right\} \\ S &= \left\{ y \in K_{\mathbb{R}}^\times \;\middle|\; N(y) = \pm 1 \right\} && \text{norm-one surface} \\ H &= \left\{ x \in \left[\prod_\tau \mathbb{R} \right]^+ \;\middle|\; \mathrm{tr}(x) = 0 \right\} && \text{trace-zero hyperplane} \end{aligned}$$

Then we get the homomorphism:

$$\mathcal{O}_K^\times \xrightarrow{\sigma} S \xrightarrow{\log} H$$

where $\sigma^\times = \log \circ \sigma$.

Proposition 1.2.38: Roots Of Unity In Group Of Units

Let $\sigma^\times(x) = (\log|\sigma_1(x)|, \dots, \log|\sigma_{r_1}(x)|, \log(|\tau_1(x)|^2), \dots, \log(|\tau_{r_2}(x)|^2)) \in \mathbb{R}^{r_1+r_2}$ for $x \in K^\times$, let $H = \{x \in \mathbb{R}^{r_1+r_2} \mid x_1 + \cdots + x_{r_1+r_2} = 0\}$ be the trace-zero hyperplane, and let $\Gamma_K^\times = \sigma^\times(\mathcal{O}_K^\times)$. Then $\Gamma_K^\times \subseteq H$, the group $\mu(K)$ of roots of unity in K is finite, and the sequence

$$1 \to \mu(K) \to \mathcal{O}_K^\times \xrightarrow{\sigma^\times} \Gamma_K^\times \to 0$$

is exact.

***Proof*:**
Step 1: the hyperplane. The map $\sigma^\times$ is a homomorphism from $K^\times$ to the additive group $\mathbb{R}^{r_1+r_2}$, as each embedding is multiplicative and log turns products into sums. For $u \in \mathcal{O}_K^\times$, $\mathrm{Nm}_{K/\mathbb{Q}}(u) = \pm 1$ (proposition 50.1.18). By corollary 50.1.13, and as $|\overline{\tau_j}(u)| = |\tau_j(u)|$, $|\mathrm{Nm}_{K/\mathbb{Q}}(u)| = \prod_i |\sigma_i(u)| \prod_j |\tau_j(u)|^2$, so the coordinates of $\sigma^\times(u)$ sum to $\log|\mathrm{Nm}_{K/\mathbb{Q}}(u)| = 0$, and $\Gamma_K^\times \subseteq H$.

Step 2: the kernel. A root of unity ζ with $\zeta^m = 1$ is a root of $x^m - 1$, so $\zeta \in \mathcal{O}_K$, and $\zeta^{-1} = \zeta^{m-1} \in \mathcal{O}_K$, so $\mu(K) \subseteq \mathcal{O}_K^\times$. Every embedding φ of K into $\mathbb{C}$ has $|\varphi(\zeta)|^m = 1$, so $|\varphi(\zeta)| = 1$ and $\sigma^\times(\zeta) = 0$. Conversely, if $r \in \mathcal{O}_K^\times$ has $\sigma^\times(r) = 0$, then $|\varphi(r)| = 1$ for every embedding φ, so $\sigma(r)$ lies in the bounded subset of $K_\mathbb{R}$ where every coordinate has absolute value at most 1. That set contains only finitely many points of the lattice $\sigma(\mathcal{O}_K)$ (proposition 50.1.57 and lemma 50.1.56), and σ is injective, so the kernel of $\sigma^\times$ is a finite group, of order m say. By [3, Lagrange Theorem], $r^m = 1$ for every r in the kernel, so the kernel is $\mu(K)$, which is therefore finite. The map onto $\Gamma_K^\times$ is surjective by definition, completing the proof.

We now build towards the structure of $\Gamma_K^\times$. Under the right adjustments $\mathcal{O}_K^\times$ can be embedded in a similar way to $\mathcal{O}_K$. Take $\sigma(\mathcal{O}_K^\times) \subseteq \mathbb{R}^{r_1} \oplus \mathbb{C}^{r_2}$. the way a mesh is formed is by taking the logarithm, for $\log(ab) = \log(a) + \log(b)$

$$\sigma^\times : K^\times \to \mathbb{R}^{r_1} \oplus \mathbb{R}^{r_2} \qquad \sigma^\times = \log \circ \sigma$$

Proposition 1.2.39: Multiplicative Group Forms Lattice

Let $\sigma^\times$ and the trace-zero hyperplane H be as in proposition 50.2.38. Then $\sigma^\times(\mathcal{O}_K^\times) \subseteq H$, and $\sigma^\times(\mathcal{O}_K^\times)$ is discrete: some open ball around 0 meets it only in 0.

***Proof*:**
Step 1: the hyperplane. This is proposition 50.2.38.

Step 2: discreteness. Let B be the open ball of radius 1 around 0 in $\mathbb{R}^{r_1+r_2}$. If $\sigma^\times(u) \in B$, every coordinate has absolute value less than 1, so $|\sigma_i(u)| < e$ and $|\tau_j(u)| < e$ for all i, j. Then $\sigma(u)$ lies in the bounded subset of $K_\mathbb{R}$ where every coordinate has absolute value less than e, which contains only finitely many points of the lattice $\sigma(\mathcal{O}_K)$ (proposition 50.1.57 and lemma 50.1.56). As σ is injective, only finitely many $u \in \mathcal{O}_K^\times$ have $\sigma^\times(u) \in B$. If some of these images are nonzero, let r be the least norm among them, and otherwise put $r = 1$. The open ball of radius r around 0 meets $\sigma^\times(\mathcal{O}_K^\times)$ only in 0, completing the proof.

Theorem 1.2.40: Dirichlet Unit Theorem

The group of units $\mathcal{O}_K^\times$ is the direct product of the finite cyclic group $\mu(K)$ and a free abelian group of rank $r_1 + r_2 - 1$. The free-generators of $\mathcal{O}_K^\times$ are called the *fundamental units* of $\mathcal{O}_K^\times$:

$$\mathcal{O}_K^\times \cong_\mathbb{Z} \mathbb{Z}^{r_1+r_2-1} \oplus \mu(K)$$

We may also characterize $\mathcal{O}_K^\times$ by the following short exact sequence:

$$1 \to \mu(K) \to \mathcal{O}_K^\times \to \Gamma_K^\times \to 0$$

where since the right term is a free $\mathbb{Z}$-module, the sequence splits giving us $\mathcal{O}_K^\times \cong \mu(K) \oplus \Gamma_K^\times$.

***Proof*:**
Write $r = r_1 + r_2$ and record the coordinates of $\sigma^\times(x)$, for $x \in K^\times$, as

$$\ell_i(x) = \log|\sigma_i(x)| \quad (1 \le i \le r_1) \qquad \ell_{r_1+j}(x) = \log\left(|\tau_j(x)|^2\right) \quad (1 \le j \le r_2)$$

In these coordinates the trace on $[\prod_\tau \mathbb{R}]^+$ is $\sum_{i=1}^r \ell_i$, and corollary 50.1.13 gives

$$\sum_{i=1}^{r} \ell_i(x) = \log\left|\mathrm{Nm}_{K/\mathbb{Q}}(x)\right|$$

so H is the sum-zero hyperplane in $\mathbb{R}^r$ and $\dim_{\mathbb{R}} H = r - 1$.

By proposition 50.2.38 the sequence $1 \to \mu(K) \to \mathcal{O}_K^\times \to \Gamma_K^\times \to 0$ is exact and $\mu(K)$ is finite. A finite subgroup of the multiplicative group of a field is cyclic ([3, Subgroup of Multiplicative Group is Cyclic]), so $\mu(K)$ is finite cyclic. By proposition 50.2.39 the group $\Gamma_K^\times$ is discrete in H. Let $W \subseteq H$ be its $\mathbb{R}$-span. A discrete subgroup spanning W is a lattice in W by definition 50.1.55 and lemma 50.1.56, hence free of rank $\dim_{\mathbb{R}} W$. The right-hand term of the sequence is then free, so the sequence splits and

$$\mathcal{O}_K^\times \cong \mu(K) \oplus \mathbb{Z}^{\dim_{\mathbb{R}} W}$$

What remains is $\dim_{\mathbb{R}} W = r - 1$, and since $W \subseteq H$ it is enough to produce $r - 1$ elements of $\Gamma_K^\times$ that are linearly independent over $\mathbb{R}$.

A box of bounded norm. For positive reals $c_1, \dots, c_r$ put

$$X(c) = \left\{ x \in K_{\mathbb{R}} \;\middle|\; |x_i| \le c_i \ (1 \le i \le r_1),\ |z_j|^2 \le c_{r_1+j} \ (1 \le j \le r_2) \right\}$$

where $x_1, \dots, x_{r_1}$ and $z_1, \dots, z_{r_2}$ are the real and complex coordinates of $K_{\mathbb{R}}$. This set is convex and centrally symmetric, and its volume equals $\kappa\, c_1 \cdots c_r$ for a constant $\kappa > 0$ depending only on r_1 and r_2, because the real coordinates range over intervals of length $2c_i$ and the complex coordinates over discs of radius $\sqrt{c_{r_1+j}}$. By proposition 50.1.60 applied to the ideal $\mathcal{O}_K$, the set $\Gamma = \sigma(\mathcal{O}_K)$ is a complete lattice in $K_{\mathbb{R}}$. Fix once and for all a real number

$$M > \frac{2^n \mathrm{vol}(\Gamma)}{\kappa}$$

which depends only on K. If $c_1 \cdots c_r = M$ then $\mathrm{vol}(X(c)) > 2^n \mathrm{vol}(\Gamma)$, so lemma 50.1.68 produces a nonzero $\beta \in \mathcal{O}_K$ with $\sigma(\beta) \in X(c)$. For such a β,

$$\left|\mathrm{Nm}_{K/\mathbb{Q}}(\beta)\right| = \prod_{i=1}^{r_1} |\sigma_i(\beta)| \prod_{j=1}^{r_2} |\tau_j(\beta)|^2 \le c_1 \cdots c_r = M \qquad \ell_i(\beta) \le \log c_i \text{ for every } i$$

A unit negative away from one coordinate. Fix k with $1 \le k \le r$. Set $\beta_1 = 1$, and given a nonzero $\beta_p \in \mathcal{O}_K$ apply the previous paragraph to

$$c_i = \tfrac{1}{2} e^{\ell_i(\beta_p)} \ (i \ne k) \qquad c_k = M \Big/ \prod_{i \ne k} c_i$$

to obtain a nonzero $\beta_{p+1} \in \mathcal{O}_K$ with $\ell_i(\beta_{p+1}) \le \ell_i(\beta_p) - \log 2$ for all $i \neq k$ and $\left|\mathrm{Nm}_{K/\mathbb{Q}}(\beta_{p+1})\right| \le M$. Each $\mathrm{Nm}_{K/\mathbb{Q}}(\beta_p)$ is a nonzero integer by corollary 50.1.14, so for $p \ge 2$ these norms take only finitely many values, and for each value proposition 50.1.67 leaves only finitely many elements of $\mathcal{O}_K$ up to multiplication by units. Hence $\beta_q = \beta_p u$ for some $p < q$ and some $u \in \mathcal{O}_K^\times$, and then

$$\ell_i(u) = \ell_i(\beta_q) - \ell_i(\beta_p) \le -(q-p)\log 2 < 0 \qquad (i \neq k)$$

Rank. Choose units $u_1, \dots, u_r$ so that $\ell_i(u_k) < 0$ for all $i \neq k$. By proposition 50.1.18 the norm $\mathrm{Nm}_{K/\mathbb{Q}}(u_k)$ is ± 1, so $\sum_i \ell_i(u_k) = 0$ and therefore $\ell_k(u_k) = -\sum_{i \neq k} \ell_i(u_k) > 0$. Let $A = (a_{ki})$ be the $r \times r$ real matrix with $a_{ki} = \ell_i(u_k)$, so that $a_{kk} > 0$, $a_{ki} < 0$ for $i \neq k$, and every row sums to 0. The row sums give $A(1, \dots, 1)^t = 0$, so $\operatorname{rank} A \le r - 1$. For the reverse bound, suppose the first $r-1$ columns $v_1, \dots, v_{r-1}$ satisfy $t_1 v_1 + \cdots + t_{r-1} v_{r-1} = 0$ with real t_j not all zero. Pick $k \le r-1$ with $|t_k|$ maximal, then divide by t_k and change the overall sign if needed, so that $t_k = 1$ and $|t_j| \le 1$ for every j. Since $a_{kj} < 0$ and $t_j \le 1$ for $j \neq k$, each term satisfies $t_j a_{kj} \ge a_{kj}$, and row k gives

$$0 = \sum_{j=1}^{r-1} t_j a_{kj} \ge \sum_{j=1}^{r-1} a_{kj} = -a_{kr} > 0$$

using $a_{kr} < 0$, which holds because $k \neq r$. This contradiction makes $v_1, \dots, v_{r-1}$ linearly independent, so $\operatorname{rank} A = r - 1$.

The rows of A are the vectors $\sigma^\times(u_1), \dots, \sigma^\times(u_r)$ of $\Gamma_K^\times$, so $\Gamma_K^\times$ contains $r-1$ linearly independent vectors and $\dim_{\mathbb{R}} W = r - 1$. Therefore $\Gamma_K^\times \cong \mathbb{Z}^{r_1+r_2-1}$ and $\mathcal{O}_K^\times \cong \mu(K) \oplus \mathbb{Z}^{r_1+r_2-1}$.

Thus, there exist units $u_1, u_2, ..., u_t$, $t = r_1 + r_2 - 1$ such that for any other unit $u \in \mathcal{O}_K^\times$, it can be written as

$$u = \zeta u_1^{n_1} \cdots u_t^{n_t}$$

where $\zeta \in \mu(K)$ is a root of unity.

Example 1.2.41: Unit Group And Class Group

1. Let $K = \mathbb{Q}(\sqrt{2})$. Then $1 + \sqrt{2}$ generates the free part of the group of units. Similarly, for $K = \mathbb{Q}(\sqrt{3})$, $2 + \sqrt{3}$ generates the free part of the group of units.

2. Let us compute the class group and unit group of $\mathbb{Q}(\sqrt[3]{7})$ Let $K = \mathbb{Q}(\sqrt[3]{7})$ and $s = \sqrt[3]{7}$. We again look for the Minkowski bound. Doing a similar computation as before, we may workout

   $$\mathcal{O}_K = \langle 1, \sqrt[3]{7}, \sqrt[3]{7}^2 \rangle$$

   Hence, the discriminant is:

   $$\det \begin{pmatrix} 1 & \sqrt[3]{7} & \sqrt[3]{7}^2 \\ 1 & \sqrt[3]{7}\zeta_3 & \sqrt[3]{7}^2\zeta_3^2 \\ 1 & \sqrt[3]{7}\zeta_3^2 & \sqrt[3]{7}^2\zeta_3 \end{pmatrix} = -1323$$

   Thus:

   $$M_K = \sqrt{1323}\frac{6}{27}\frac{4}{\pi} < 11$$

Hence, we check prime ideal $(2), (3), (5), (7)$ of $\mathbb{Z}$. Since the minimal polynomial of $\sqrt[3]{7}$ is $x^3 - 7$, we see that in $\mathbb{F}_p[x]/(\overline{x^3 - 7_p})$ we get:

(2) $(x+1)(x^2+x+1)$, hence

$$\mathfrak{p}_2 = (2, s+1) \qquad \mathfrak{p}_2' = (2, s^2+s+1)$$

(3) $(x-1)^3$, hence

$$\mathfrak{p}_3 = (3, s-1)$$

(5) $(x+2)(x^2+3x+4)$, hence

$$\mathfrak{p}_5 = (5, s+2) \qquad \mathfrak{p}_5' = (5, s^2+3s+4)$$

(7) x^3, hence

$$\mathfrak{p}_7 = (7, s) = (s)$$

Thus, just like before we have that $[\mathfrak{p}_2]$, $[\mathfrak{p}_3]$, and $[\mathfrak{p}_5]$ generate $Cl(K)$ (since $[\mathfrak{p}_7] = [1]$). Next, we see that

$$\mathfrak{p}_2\mathfrak{p}_3 = (6) \qquad \mathfrak{p}_3^2\mathfrak{p}_5 = (15)$$

Hence, $[\mathfrak{p}_3] = [\mathfrak{p}_2^{-1}]$ and $[\mathfrak{p}_3^2] = [\mathfrak{p}_5^{-1}]$. Then, with some computation we see that:

$$[\mathfrak{p}_3^3] = [\mathfrak{p}_3]^3 = [\mathrm{id}]$$

Thus, we see that $Cl(K)$ is generated by $[\mathfrak{p}_3]$ and has order 3, hence $CL(K) \cong \mathbb{Z}/3\mathbb{Z}$.

However, what if $\mathfrak{p}_3$ is in fact principal? For the sake of contradiction, let's say

$$\mathfrak{p}_3 = (\alpha) \implies \mathfrak{p}_3^3 = (3) = (\alpha^3)$$

Then $\alpha^3 = 3u$ for some unit u. To find u, we find the units of the ring of integers. To find its units, first notice that if:

$$1 = x^3 - 7 = (x-s)(x^2+sx+s^2)$$

Then $x - s$ is a unit. But then, $x = 2$ works since $8 - 7 = 1$. Then by Dirichlet's Unit Theorem, we have that the rank of the unit group is:

$$r_1 + r_2 - 1 = 1 + 1 - 1 = 1$$

Hence, the group of units is

$$\langle 2 - \sqrt[7]{2} \rangle$$

which is the second object asked of us to find. Using this, we can finally conclude that there is no unit u that satisfies the equation $\alpha^3 = 3u$ (if $\alpha = a + bs + cs^2$, in the process of finding the coefficients a, b, c we shall be forced to have $b = c = 0$). Hence, we may properly conclude that

$$Cl(K) \cong \mathbb{Z}/3\mathbb{Z}$$

completing the proof.

In general, there is an algorithm for computing the ring of units that involved continued fractions. It is not covered in this book, but for those interested see *Borevich and Shafarevich, Number Theory* section 7.3 of chapter 2.

Proposition 1.2.42: Volume Of Unit Lattice

Let $\mathcal{O}_K$ be a ring of integers with unit group $\mathcal{O}_K^*$, let $t = r_1 + r_2 - 1$, and write the coordinates of $\sigma^\times(x)$ as $\sigma_i^\times(x) = \log|\sigma_i(x)|$ for $1 \le i \le r_1$ and $\sigma_{r_1+j}^\times(x) = \log\left(|\tau_j(x)|^2\right)$ for $1 \le j \le r_2$. Let $u_1, \dots, u_t \in \mathcal{O}_K^\times$ be units whose images $\sigma^\times(u_1), \dots, \sigma^\times(u_t)$ form a $\mathbb{Z}$-basis of the unit lattice $\sigma^\times(\mathcal{O}_K^\times)$. Then the volume of the unit lattice $\sigma^\times(\mathcal{O}_K^\times)$ in the hyper-plane H is:

$$\operatorname{vol}\left(\sigma^\times(\mathcal{O}_K^\times)\right) = \sqrt{r_1 + r_2}R$$

where R is the absolute value of the determinant of the $t \times t$ matrix obtained by deleting any one row of the following $(t+1) \times t$ matrix:

$$\begin{pmatrix} \sigma_1^\times(u_1) & \cdots & \sigma_1^\times(u_t) \\ \vdots & \ddots & \vdots \\ \sigma_{t+1}^\times(u_1) & \cdots & \sigma_{t+1}^\times(u_t) \end{pmatrix}$$

In particular, R does not depend on which row is deleted.

***Proof*:**

By proposition 50.2.38 and theorem 50.2.40, $\Gamma = \sigma^\times(\mathcal{O}_K^\times)$ is a free abelian group of rank t, so units $u_1, \dots, u_t$ as in the statement exist. Put $v_j = \sigma^\times(u_j) \in \mathbb{R}^{t+1}$. By proposition 50.1.18, $\operatorname{Nm}_{K/\mathbb{Q}}(u_j) = \pm 1$, and by corollary 50.1.13 the coordinates of v_j sum to $\log\left|\operatorname{Nm}_{K/\mathbb{Q}}(u_j)\right| = 0$, each pair $\tau_l, \overline{\tau_l}$ of complex embeddings contributing $\log\left(|\tau_l(u_j)|^2\right)$. So every v_j lies in the hyperplane

$$H = \left\{x \in \mathbb{R}^{t+1} \;\middle|\; x_1 + \cdots + x_{t+1} = 0\right\}$$

of $\mathbb{R}^{t+1}$ with its standard inner product. The group $\Gamma \subseteq H$ is discrete (proposition 50.2.39) and free of rank t, and $\dim H = t$, as H is the kernel of the nonzero linear form $x \mapsto x_1 + \cdots + x_{t+1}$. So Γ spans H by lemma 50.1.56, and it is a complete lattice in H. By definition 50.1.59, applied to Γ in the space H with the inner product of $\mathbb{R}^{t+1}$, $\operatorname{vol}(\Gamma) = |\det(G)|^{1/2}$ with $G = (\langle v_i, v_j\rangle)$. Another $\mathbb{Z}$-basis of Γ replaces G by $U^t G U$ with $\det(U) = \pm 1$, so this number depends only on Γ.

Let $w = (t+1)^{-1/2}(1, \dots, 1)$, a unit vector orthogonal to H, and let P be the $(t+1) \times (t+1)$ matrix with columns $v_1, \dots, v_t, w$. Since $\langle v_j, w\rangle = 0$ and $\langle w, w\rangle = 1$, the matrix $P^t P$ is block diagonal with blocks G and (1). By [3, Properties Of The Determinant](2), [3, Multiplicativity of the Determinant] and [3, Determinant of a Block Diagonal Matrix],

$$\det(P)^2 = \det(P^t P) = \det(G)$$

so $\operatorname{vol}(\Gamma) = |\det(P)|$.

Fix $k \in \{1, \dots, t+1\}$ and add every other row of P to row k, which does not change the determinant ([3, Elementary Matrices And Determinant](2)). The new row k has entry $\sum_i (v_j)_i = 0$ in each column $j \le t$, because $v_j \in H$, and entry $(t+1)(t+1)^{-1/2} = \sqrt{t+1}$ in the last column. Subtracting multiples of this row from the other rows clears the rest of the last column without changing the first t columns or the determinant, and moving row k to the bottom multiplies the determinant by ± 1 ([3, Properties Of The Determinant](5)). The result is block diagonal with

blocks P_k and $(\sqrt{t+1})$, where P_k is the matrix $(\sigma_i^\times(u_j))$ with row k deleted. By [3, Determinant of a Block Diagonal Matrix],

$$\operatorname{vol}(\Gamma) = |\det(P)| = \sqrt{t+1}\,|\det(P_k)|$$

The left-hand side does not depend on k, so neither does $R = |\det(P_k)|$, and $\operatorname{vol}(\Gamma) = \sqrt{r_1+r_2}\,R$, as we sought to show.

Definition 1.2.43: Regulator

The value of R in the above proof is called the *regulator* of $\mathcal{O}_K$.

Exercise 1.2.2

1. What number rings $\mathcal{O}_K$ have finitely many units?

1.3 Extension of Dedekind Domains and Number Fields

In [2, Integral Extensions and Prime Ideals], we covered how prime ideals of a base ring correspond to prime ideals in an integral extension. Since all extensions that we are working with are integral extensions, the same theory applies. Working over Dedekind domains, we may further ask if the integral extension of a Dedekind domain is a Dedekind domain. This will be the case, and so we may further factor prime ideals in the extended ring, namely if a prime ideal $\mathfrak{p} \subseteq R$ factors in S where S/R is an integral extension. When focusing on rings of integers $\mathcal{O}_L$, the exploration of how primes factor into larger primes has a lot of consequences on the structure of $\mathcal{O}_L$ itself, and perhaps a bit surprisingly at first sight on the field extension L/K itself. We shall see in the next section that we can use the theory of factorization of prime ideals to classify all abelian extension over $\mathbb{Q}$, this is the *Kronecker-Weber Theorem* promised in [3, Composite Fields Determining Abelian Extensions].

First, if R is a Dedekind domain with field of fraction K, $I \subseteq R$ an ideal, through them map $K \hookrightarrow L$ we get $I^e \subseteq \overline{R}^L$. We further generalize this for fractional ideals:

Definition 1.3.1: Extension Of Fractional Ideals

Let R be a Dedekind domain, K it's field of fractions, and L/K be a finite field extension. Then if $I \subseteq K$ is a fractional ideal and $\mathcal{O} = \overline{R}^L$,

$$I_L := I\mathcal{O} = \left\{ \sum a_i b_i \;\middle|\; a_i \in I,\ b_i \in \mathcal{O} \right\}$$

When necessary, the notation $I_{L/K}$ will be used to specify the base field over which I is taken.

It can be verified that $I_L \subseteq L$ is the minimal fractional ideal containing I, that if $M/L/K$, then $(I_L)_M = I_M$, and that $I_L = I \otimes_{\mathcal{O}_K} \mathcal{O}$. The results of [2, Extension And Contraction With Common Operations] follow quickly giving all the same results. Furthermore since $(I+J)_L = I_L + J_L$:

$$(\underset{K}{\gcd}(I,J))_L = \underset{L}{\gcd}(I_L, J_L)$$

However, by [2, Extension And Contraction With Common Operations], we so far only have

$$(\text{lcm}_K(I,J))_L \subseteq \text{lcm}_M(I_L, J_L)$$

Note that for $\alpha \in K$, $(\alpha)_L = (\alpha\mathcal{O}_K)_L = \alpha\mathcal{O}_L$, hence principal ideals extend to principal ideals. For reasons that shall become clearer later[19], the map

$$J_K \to J_L \qquad I \mapsto I_L$$

is called the *base change map*. To study the fractional ideals of L, the first immediate result we may want is that $\overline{R}^L$ remains a Dedekind domain:

Lemma 1.3.2: Extending Dedekind Domains

Let R be a Dedekind domain with field of fraction K and let L/K be a finite field extension of K. Then $\mathcal{O} = \overline{R}^L$ is a Dedekind domain.

***Proof*:**
By definition, $\mathcal{O}$ is integrally closed, and if $\mathfrak{p} \subseteq \mathcal{O}$ is a nonzero prime, then $P = \mathfrak{p} \cap R$ is also prime (the contraction of a prime ideal is prime). It is nonzero: for $0 \neq x \in \mathfrak{p}$, an equation $x^m + a_{m-1}x^{m-1} + \cdots + a_0 = 0$ with $a_i \in R$ and m least has $a_0 \neq 0$, since otherwise dividing by x would give one of smaller degree, and $a_0 = -x(x^{m-1} + \cdots + a_1) \in \mathfrak{p} \cap R$. So P is maximal in R. Hence $(\mathcal{O}/\mathfrak{p})/(R/P)$ is an integral extension over a field meaning $\mathcal{O}/\mathfrak{p}$ is a field, thus $\mathfrak{p}$ is maximal. It remains to show that $\mathcal{O}$ is Noetherian.

The separable case has the self-contained proof below, and it is all that number fields need. The general case is [2, Integral Closure of a One-Dimensional Noetherian Domain]: a Dedekind domain is a Noetherian domain of dimension one, and that corollary makes $\mathcal{O}$ a Noetherian normal domain of dimension one with fraction field L, which together with the maximality of primes shown above is the Dedekind property. Let $\alpha_1, \alpha_2, ..., \alpha_n$ be a K -basis for L. By multiplying by an appropriate constant, we may let $\alpha_1, \alpha_2, ..., \alpha_n \in \mathcal{O}$ (see proposition 50.1.7). By proposition 50.1.32,

$$\mathcal{O} \subseteq \frac{R\alpha_1}{d} + \cdots + \frac{R\alpha_n}{d}$$

Now, for any ideal $I \subseteq \mathcal{O}$, by the above it is contained in a finitely generated R-module, hence since R is Noetherian I (as a R-module) is finitely generated. But then *a fortiori*, it is a finitely generated $\mathcal{O}$-module, hence a finitely generated ideal, making $\mathcal{O}$ Noetherian.

Example 1.3.3: Dedekind Domains Via Extension
The result above shows that all rings of integers of a finite extension $L/K/\mathbb{Q}$ is a ring of integers. For an example that is not a number ring, take $R = k[x]$ for some field k. As R is a PID, it is a Dedekind domain. Then if $L/k(x)$ is an integral extension, the integral closure of R over L, $S = \overline{R}^L$, is a Dedekind domain. As $L \cong k(x)[y]/(p(y))$ for some irreducible polynomial over $k[x]$ L represents 1-dimensional curves. Section 50.7 returns to the correspondence between Dedekind domains and smooth curves.

[19] we shall possibly increase the number of primes, hence change the "basis"

Extending Number Fields

An immediate corollary of lemma 50.3.2 is the following:

Corollary 1.3.4: Extending Number Fields

Let L/K be a finite extension of numbers fields so that $\mathcal{O}_K = \overline{\mathbb{Z}}^K$ and $\mathcal{O} = \overline{\mathcal{O}_K}^L$. Then

$$\mathcal{O} = \mathcal{O}_L = \overline{\mathbb{Z}}^L$$

***Proof*:**
Every element of $\mathcal{O}_L$ is integral over $\mathbb{Z} \subseteq \mathcal{O}_K$, so $\mathcal{O}_L \subseteq \mathcal{O}$. Conversely, let $\alpha \in \mathcal{O}$. Every element of $\mathcal{O}_K[\alpha]$ is integral over $\mathcal{O}_K$ ([2, Integral Elements Form Rings]), and $\mathcal{O}_K$ is integral over $\mathbb{Z}$, so α is integral over $\mathbb{Z}$ ([2, Transitivity of Integrality]) and $\alpha \in \overline{\mathbb{Z}}^L = \mathcal{O}_L$, completing the proof.

Thus, when working with finite extensions of number ring L/K, there is a corresponding integral extensions are those of the ring of integers $\mathcal{O}_L/\mathcal{O}_K$ creating the diagram:

$$\begin{array}{ccc} K & \xrightarrow{\iota} & L \\ \downarrow & & \downarrow \\ \mathcal{O}_K & \xrightarrow{\iota} & \mathcal{O}_L \end{array}$$

As we saw, principal ideals extend to principal ideals. As it turns out, all non-principal ideals extend to be principal given a large enough extension. This result shall underpin a lot of future simplification and is good to see early on:

Lemma 1.3.5: Number Fields, Extension To Principal

Let $I \subseteq K$ be a fractional ideal where $K/\mathbb{Q}$ is a number field. Then in some finite extension L/K, I_L becomes a principal ideal.

***Proof*:**
As $K/\mathbb{Q}$ is a number fields, by theorem 50.2.33, $Cl(K)$ is finite. Thus, for every $I \in J_K$, there exists an $n \in \mathbb{N}$ such that $I^n = \alpha\mathcal{O}_K$. Let $L = K(\sqrt[n]{\alpha})$. letting $\beta = \sqrt[n]{\alpha}$, we get

$$(I_L)^n = (I_L^n) = (I^n\mathcal{O}_L) = (\alpha\mathcal{O}_L) = (\beta\mathcal{O}_L)^n$$

Thus, by unique factorization:

$$I_L = \beta\mathcal{O}_L$$

as we sought to show.

This result is quite useful. In the right cases we can pass to the field in which all our ideals become principal ideals, proof what we want for principal ideals, then try to come back down (coming back down is often called the descent argument). The following lemma gives a very common descent argument, and is important for computing the absolute norm of extended ideals: since the norm of

an ideal gives a measure how space a given ideal is within its Dedekind, extending the domain shall make the given ideal sparser[20]:

Lemma 1.3.6: Number Fields, Extension And Absolute Norm

Let L/K be a finite extension of number fields, $I \subseteq K$ a fractional ideal, and $I_L = I\mathcal{O}_L$ the $\mathcal{O}_L$-submodule of L generated by I. Then I_L is a fractional ideal of L and:

$$\mathfrak{N}_L(I_L) = \mathfrak{N}_K(I)^{[L:K]}$$

***Proof*:**
Step 1: fractional ideals extend. If I is generated by $\alpha_1, \dots, \alpha_m$ and $d \in \mathbb{Z}_{>0}$ satisfies $dI \subseteq \mathcal{O}_K$ (corollary 50.2.12), then I_L is generated over $\mathcal{O}_L$ by the same elements, is nonzero, and $dI_L \subseteq \mathcal{O}_L$. For fractional ideals I, J of K, both $(IJ)_L$ and $I_L J_L$ are the $\mathcal{O}_L$-submodule generated by the products xy with $x \in I$ and $y \in J$, so $(IJ)_L = I_L J_L$.

Step 2: principal fractional ideals. For a number field F and $\alpha \in F^\times$, we show $\mathfrak{N}_F(\alpha\mathcal{O}_F) = |\mathrm{Nm}_{F/\mathbb{Q}}(\alpha)|$. The norm is multiplicative, $\mathrm{Nm}_{F/\mathbb{Q}}(xy) = \det(L_x L_y) = \mathrm{Nm}_{F/\mathbb{Q}}(x)\mathrm{Nm}_{F/\mathbb{Q}}(y)$ by [3, Multiplicativity of the Determinant], and $\mathrm{Nm}_{F/\mathbb{Q}}(d) = d^{[F:\mathbb{Q}]}$ for $d \in \mathbb{Z}$. Choose $d \in \mathbb{Z}_{>0}$ with $d\alpha \in \mathcal{O}_F$ (corollary 50.2.12). In J_F, $\alpha\mathcal{O}_F = (d\alpha\mathcal{O}_F)(d\mathcal{O}_F)^{-1}$, so definition 50.2.31 and lemma 50.1.63 give

$$\mathfrak{N}_F(\alpha\mathcal{O}_F) = \frac{|\mathrm{Nm}_{F/\mathbb{Q}}(d\alpha)|}{|\mathrm{Nm}_{F/\mathbb{Q}}(d)|} = |\mathrm{Nm}_{F/\mathbb{Q}}(\alpha)|$$

Step 3: the norm of an element of K. Let $\alpha \in K^\times$ and let r run over the roots of $m_{\alpha,\mathbb{Q}}$ with multiplicity. By proposition 50.1.12, $\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) = (\prod r)^{[L:\mathbb{Q}(\alpha)]}$ and $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha) = (\prod r)^{[K:\mathbb{Q}(\alpha)]}$. Since $[L:\mathbb{Q}(\alpha)] = [L:K][K:\mathbb{Q}(\alpha)]$ ([3, Multiplicity of Finite Extensions (Tower Law)]), $\mathrm{Nm}_{L/\mathbb{Q}}(\alpha) = \mathrm{Nm}_{K/\mathbb{Q}}(\alpha)^{[L:K]}$. With Step 2 for $F = L$ and for $F = K$,

$$\mathfrak{N}_L(\alpha\mathcal{O}_L) = \left|\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)\right|^{[L:K]} = \mathfrak{N}_K(\alpha\mathcal{O}_K)^{[L:K]}$$

Step 4: the general case. The class group $Cl(K)$ is finite (theorem 50.2.33), of order h say, so $[I]^h = 1$ by [3, Lagrange Theorem], that is, $I^h = \alpha\mathcal{O}_K$ for some $\alpha \in K^\times$. By Step 1, $(I_L)^h = (I^h)_L = \alpha\mathcal{O}_L$. As $\mathfrak{N}_L$ and $\mathfrak{N}_K$ are homomorphisms (definition 50.2.31), Step 3 gives

$$\mathfrak{N}_L(I_L)^h = \mathfrak{N}_L(\alpha\mathcal{O}_L) = \mathfrak{N}_K(\alpha\mathcal{O}_K)^{[L:K]} = \mathfrak{N}_K(I)^{h[L:K]}$$

and taking positive hth roots completes the proof.

1.3.1 Ideal Correspondence in Extension

Let L/K be a finite extension, J_K and J_L be the corresponding ideal group. Then we have a map $J_K \to J_L$ given by $I \mapsto I_L$. Do we have a converse map, and if so is it an inverse of the natural

[20] This means that the ideal does not "extend enough" to cancel the added density given by extending the ring

extension map? If not, to what degree is it an inverse map? Recall in [2, Properties of Extensions and Contractions] that $I \subseteq SI \cap R$, in other words $I \subseteq I_S \cap R$. This can be made an equality for when extending number fields, and further generalized to fractional ideals. Recall that this is not true in general (a counter-example for the general case is $(2) \subsetneq \mathbb{Z} = (2)_{\mathbb{Q}} \cap \mathbb{Z}$), hence this is special for extension of Dedekind domains:

Proposition 1.3.7: Ideal Correspondence In Extensions Of Dedekind Domains

let L/K be a finite field extension with Dedekind domain $\mathcal{O}$ and $\mathfrak{o}$ respectively. Let $I \subseteq K$ a fractional ideal. Then:

$$I_L \cap K = (I^e)^c = I$$

Hence, unlike for general integral extensions, the contraction is an inverse map to the extension.

***Proof*:**
Let $\beta \in J = I\mathcal{O} \cap K$ so that $\beta = \sum_i a_i k_i$ where $a_i \in I_L$ and $k_i \in \mathfrak{o}$ (since $\mathcal{O} \cap K = \mathfrak{o}$). Then the following equality holds:

$$\beta^n = \mathrm{Nm}_{L/K}(\beta) = \prod_\tau \tau(\beta) = \prod_\tau \left(\sum_i k_i \tau(a_i) \right)$$

where the first equality comes from $\beta \in K$, the second comes from the equivalence definition of norm (given an appropriate normal closure N/L), and the third is expanding the definition of β. On the right hand side, we have that the sum is invariant under the Galois group $\mathrm{Gal}_K(N)$ for appropriate normal closure N/L from which we took each τ. Thus, the sum is an element in $\mathfrak{o}$, so the product is in $\mathfrak{o}$. Then we have $\beta^n \in I^n$, so $\beta \in I$, completing the proof.

Due to the importance of this new paradigm of thinking, two other proofs shall be presented. Notice that $IJ^{-1} \subseteq \mathfrak{o}$ and:

$$\mathfrak{N}(IJ^{-1}) = 1$$

giving that $IJ^{-1} = \mathfrak{o}$, i.e. $I = J$.

Another proof goes as follows: Since $I \subseteq J$, $I_L \subseteq J_L$. Since $I_L \subseteq J_L \subseteq I_L$ by properties of extensions, $I_L = J_L$. Thus:

$$\mathfrak{N}(J) = \mathfrak{N}(I_L)^{1/[L:K]} = \mathfrak{N}(I)$$

and since $I \subseteq J$, the norms being equal means the index between the two are the same, hence $I = J$.

Note that in the case where $\mathfrak{p}$ is prime, the result can be essentially immediately proven since by [2, Properties of Extensions and Contractions] $\mathfrak{p} \subseteq \mathfrak{p}\mathcal{O}_L \cap K$, but since $\mathfrak{p}$ is maximal we get equality.

Corollary 1.3.8: Homomorphism between Extensions of Ideal Group

Let J_K be the ideal group of a Dedekind domain R and let L/K be a finite field extension. Then the map $J_K \to J_L$ mapping $I \mapsto I_L$ is an injective group homomorphism that preserves gcd and lcm.

On the other hand, the map $Cl(K) \to Cl(L)$ is a homomorphism that need not be injective.

For class groups of number fields:

$$\varinjlim_{\substack{K \\ [K:\mathbb{Q}]<\infty}} (Cl(K)) = 0$$

***Proof*:**
By the previous proposition, $I_L = J_K$ if $I = J$, hence the map $\varphi : \mathscr{F}(K) \to \mathscr{F}(L)$ mapping $I \mapsto I_L$ is injective and is certainly a homomorphism by a simple generalization of [2, Extension And Contraction With Common Operations] (recall that $(\mathfrak{ab})^e = \mathfrak{a}^e\mathfrak{b}^e$ for ideals).

For the map between class groups of number fields, since the extension for every fractional ideal $I \subseteq \mathbb{Q}$, there exists an extension in which it is principle, we see that the colimit can only be satisfied by the trivial group, hence $\varinjlim_{\substack{K \\ [K:\mathbb{Q}]<\infty}} (Cl(K)) = 0$, as we sought to show.

A side consequence of the colimit definition is that we may define

$$Cl(\overline{\mathbb{Q}}) := \varinjlim(Cl(K)) = 0$$

This shows that to some degree, the lack of principal ideals is tied to what algebraic extension we choose. Though the extension map is injective, it is not in general surjective:

Example 1.3.9: Lack of Surjectivity
Let $L = \mathbb{Q}(\sqrt{-5})$ and $K = \mathbb{Q}$ so $\mathcal{O}_L = \mathbb{Z}[\sqrt{-5}]$. Then the (non-principal) ideal $I = (1+\sqrt{-5}, 1-\sqrt{-5})$ are all numbers of the form $m+n\sqrt{-5}$ where m, n are even, so $I \cap \mathbb{Q} = (2)$, hence I lies over (2). However, $I \neq (2)_L$ since I is not principal (recall $(-)_L$ maps principals to principals), and no other ideal can map to I or else I would lie over some different prime, hence I is not an extension. The ideal (2) is still related to I, namely $I^2 = (2)$ (this can be used to show I is not principal).

With these results, we have better control over the extension of primes and primes lying over (resp. lying under) relate:

Proposition 1.3.10: Dedekind Domain, Lying Over And Lying Under

Let $\mathcal{O}/o$ be a finite extension of Dedekind domains given by L/K, $0 \neq \mathfrak{p} \subseteq o$ a prime, and $\mathcal{O} = \overline{o}^L$, $\mathfrak{p}$, where $\mathfrak{p}$ may factor further. Then $\mathfrak{p}$ will never be trivialized in an extension, namely

$$\mathfrak{p}\mathcal{O} \neq \mathcal{O} = (1)$$

***Proof*:**
Let $p \in \mathfrak{p} \setminus \mathfrak{p}^2$. Then

$$(p) = \mathfrak{p}\mathfrak{a} \qquad \mathfrak{p} \nmid \mathfrak{a}$$

Hence, $\mathfrak{p} + \mathfrak{a} = (1)$ ($\gcd(\mathfrak{p}, \mathfrak{a}) = (1)$), and so $a + b = 1$ for $a \in \mathfrak{p}$ and $b \in \mathfrak{a}$. Certainly $b \notin \mathfrak{p}$ so $b\mathfrak{p} \subseteq \mathfrak{p}\mathfrak{a} = (p)$. Now, if $\mathfrak{p}\mathcal{O} = \mathcal{O}$, then $b\mathcal{O} = b\mathfrak{p}\mathcal{O} \subseteq p\mathcal{O}$. hence $b\mathcal{O}_K \subseteq \mathfrak{p}$ so $b = px$ for some $x \in \mathcal{O} \cap K = \mathcal{O}_K$, implying $b \in \mathfrak{p}$, a contradiction.

Proposition 1.3.11: Equivalence Of Prime Division In Dedekind Domain

Let $\mathfrak{p} \subseteq R$, $\mathfrak{B} \subseteq \mathcal{O}$, $\mathcal{O}/R$ an integral extension where R is a Dedekind Domain. Then the following are equivalent:

1. $\mathfrak{B} \big| \mathfrak{p}\mathcal{O}$
2. $\mathfrak{B} \supseteq \mathfrak{p}\mathcal{O}$
3. $\mathfrak{B} \supseteq \mathfrak{p}$
4. $\mathfrak{B} \cap \mathcal{O} = \mathfrak{p}$
5. $\mathfrak{B} \cap K = \mathfrak{p}$

***Proof*:**
Since we are in a Dedekind domain, $A \big| B$ if and only if $A \supseteq b$, giving us $1 \Leftrightarrow 2$. Since $\mathfrak{B}$ is an ideal of $\mathcal{O}_L$, certainly $2 \Leftrightarrow 3$. We got $3 \Leftarrow 4 \Leftrightarrow 5$ by the above injectivity argument. For $3 \implies 4$, since $\mathfrak{p}$ is maximal and $\mathfrak{B} \cap \mathcal{O}_K \supseteq \mathfrak{p}$, we have either equality or $\mathcal{O}_K$, if $\mathfrak{p} \cap \mathcal{O}_K = \mathcal{O}_K$, then $1 \in Q$, implying $Q = \mathcal{O}_L$, contradicting it's maximality.

Corollary 1.3.12: Lying Over Unique Prime

Let $\mathfrak{p}, \mathfrak{q} \subseteq \mathcal{O}$, $\mathfrak{p} \neq \mathfrak{q}$. Let:

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_n^{e_n} \qquad \mathfrak{q}\mathcal{O}_L = \mathfrak{C}_1^{f_1} \cdots \mathfrak{C}_m^{f_m}$$

Then:

$$\{\mathfrak{B}_1, ..., \mathfrak{B}_n\} \cap \{\mathfrak{C}_1, ..., \mathfrak{C}_n\} = \emptyset$$

that is, extended prime ideals lie over a single prime

***Proof*:**
Let's say the intersection is non-empty, so that $\mathfrak{a}$ is a common factor. Then write:

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{a}^{e_1} \cdots \mathfrak{B}_n^{e_n} \qquad \mathfrak{q}\mathcal{O}_L = \mathfrak{a}^{f_1} \cdots \mathfrak{C}_m^{f_m}$$

Then:

$$(\mathfrak{p})_L^{-1}\mathfrak{p}^{e_1-1}\mathfrak{B}_2^{-e_2} \cdots \mathfrak{B}_n^{-e_n} = \mathfrak{a} = (\mathfrak{q})_L^{-1}\mathfrak{p}^{f_1-1}\mathfrak{C}_2^{-f_2} \cdots\cdots \mathfrak{C}_m^{-f_m}$$

Thus we get:

$$\mathfrak{q}\mathcal{O}_L = (\mathfrak{p})_L\mathfrak{p}^{e_1-1}\mathfrak{B}_2^{-e_2} \cdots \mathfrak{B}_n^{-e_n}\mathfrak{p}^{f_1-1} \cdots \mathfrak{C}_m^{f_m}$$

And in particular:

$$\mathfrak{C}_1^{f_1} \cdots \mathfrak{C}_m^{f_m} = q\mathcal{O}_L \subseteq p\mathcal{O}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_n^{e_n}$$

But then, since $\mathfrak{p}\mathcal{O}_L \cap K = \mathfrak{p}$ and $\mathfrak{q}\mathcal{O}_L \cap K = \mathfrak{q}$, we have by corollary 50.3.8 that:

$$\mathfrak{q} \subseteq \mathfrak{p}$$

which since $\mathfrak{q}$ is maximal and $\mathfrak{p} \neq \mathcal{O}_K$ we have $\mathfrak{q} = \mathfrak{p}$, contradicting the fact that $\mathfrak{p} \neq \mathfrak{q}$, hence, it must be that they share no common factors.

Another proof goes as follows: if $\mathfrak{p} \subseteq \mathcal{O}_K$ is a prime and $\mathfrak{p}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_k^{e_k}$ is the factorization of $\mathfrak{p}_L$ in $\mathcal{O}_L$, then if there was a prime ideal $\mathfrak{q}$ such that $\mathfrak{q} \mapsto \mathfrak{B}_i$ so that by the above for any $1 \leq i \leq k$, then since $\mathfrak{p} \subseteq \mathfrak{B}_i \cap \mathcal{O}_k$, by maximality $\mathfrak{p} = \mathfrak{B}_i \cap \mathcal{O}_K$. Hence if $\mathfrak{q} = \mathfrak{B}_I \cap \mathcal{O}_k$, by injectivity $\mathfrak{p} = \mathfrak{q}$.

Another consequence of the above theorem is that if two prime factorizations of a prime share a factor in an extended ring of integers, they are equal.

Relative Norm

Since the ideals of Dedekind domains share factorization structure to numbers, we may hope there is an equivalent of the field norm $\mathrm{Nm}_{L/K}$ for ideals. by lemma 50.3.5, we may hope that our new relative norm map gives us some information on the power to which we may rise our ideal to get a principal ideal. We now endeavour to find a map $J_L \to J_K$ that behaves more like the field norm and generalizes the absolute norm, which can be thought of as a map:

$$\mathfrak{N}_{L/\mathbb{Q}} : J_L \to \mathbb{Q}_L \cong \mathbb{Q}_{>0}^{\times}$$

In particular, if we work of *Galois* extensions L/K, then we may hope to translate some of the fact that

$$\mathrm{Nm}_{L/K}(\alpha) = \prod_{\sigma} \sigma(\alpha)$$

to prime ideals! In particular, this allows us to bring in the Galois theory in the study of prime ideals!

We start by showing the case for Galois extensions, the general case can be inferred from the Galois case. Given a Galois extension L/K with Galois group $G = \mathrm{Gal}_K(N)$, then we can recover L from K using G by taking $L^G = K$, hence $\mathcal{O}_L^G = \mathcal{O}_K$. Certainly, if $\tau \in G$ and $\mathfrak{a} \subseteq L$, then $\tau(\mathfrak{a}) \subseteq L$ Since L/K is normal, and hence $\tau(\mathfrak{a})$ is well-defined. It may be tempting to then say that $J_L^G = J_K$ and hence defining $J_L \to J_K$ by taking

$$I \mapsto \prod_{\tau} \tau(I)$$

However, it is not the case that $J_L^G = J_K$. The ideal in example 50.3.9 is invariant under the Galois group, but certainly an ideal of $\mathbb{Z}$. We fix this by the following:

Definition 1.3.13: Relative Norm

Let L/K be a finite field extension and $I \subseteq K$ a fractional ideal in L. Then the *relative norm* or *ideal norm* of L with respect to K is:

$$\mathfrak{N}_{L/K}(I) = \left(\left\{\mathrm{Nm}_{L/K}(\alpha) \;\middle|\; \alpha \in I\right\}\right) \in J_K$$

namely, $\mathfrak{N}_{L/K}(I)$ is the ideal generated by all norms of elements in I.

Note that this is well-defined since $\mathrm{Nm}_{L/K}(\alpha) \in K$ as every minimal polynomial $m_{\alpha,K}(x) \in K[x]$. The norm as defined above gives us a map $\mathfrak{N}_{L/K} : J_L \to J_K$. The field norm can be thought of as the ideal norm, for example take

$$\mathfrak{N}_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}((\sqrt{2})) = \left(\left\{\mathrm{Nm}_{L/K}\left(\sqrt{2}b\right) \;\middle|\; \sqrt{2}b \in \sqrt{2}\mathbb{Z}\right\}\right) = (2)$$

suggesting that $(\mathrm{Nm}_{L/K}(\alpha)) = \mathfrak{N}_{L/K}((\alpha))$, which will indeed be the case and shall be proved after the next theorem. An important reason to choose this definition is that it works well with extension: Let $J_L \xrightarrow{\mathrm{Nm}_{L/K}} J_K \xrightarrow{I_L} J_L$. What will be the image of an ideal in this chain? In the simplest case, we may take principal ideal $\alpha\mathcal{O}_L \subseteq \mathcal{O}_L$ where $\alpha \in \mathcal{O}_K$. Then it's image is $\alpha\mathcal{O}_K$, and it's extension is $\alpha\mathcal{O}_L$. Thus for the simplest type of integral principal ideal, the map is a right-inverse. For more general cases, it is not necessarily a right inverse (it cannot be, since I_L is not surjective). However, the behavior of the image is well-behaved, and in fact tied with the "almost" definition of the ideal norm presented earlier:

Theorem 1.3.14: Ideal Correspondence in Extension

Let $N/L/K$ be finite field extensions where N/K is normal, let $I \subseteq L$ be a fractional ideal, and let $\sigma_1, \ldots, \sigma_n$ be the distinct K-embeddings $\sigma_i : L \to N$, each extended to an automorphism of N (possible since N/K is normal: [3, Embedding Into Algebraic Extension] extends each embedding into $\overline{N}$, and normality forces the image to be N). Then:

$$(\mathfrak{N}_{L/K}(I))_N = \prod_i (\sigma_i I)_{N/\sigma_i L} =: \mathrm{Nm}'_{L/K}(I)$$

In particular

$$\left\{\begin{matrix}\text{Fractional ideals of} \\ \mathcal{O}_K\end{matrix}\right\} \underset{\mathfrak{N}_{L/K}}{\overset{I_L}{\rightleftarrows}} \left\{\begin{matrix}\text{Fractional Ideals of} \\ \mathcal{O}_L\end{matrix}\right\}$$

***Proof*:**
Without loss of generality, we may make I integral. Then, certainly an ideal is a subset of its extension (given the natural embedding) $\sigma_i I \subseteq (\sigma_i I)_{N/\sigma_i L}$, and so by definition of $\mathrm{Nm}'_{L/K}(I)$, for all $\alpha \in I$:

$$\mathrm{Nm}_{L/K}(\alpha) = \prod_i \sigma_i(\alpha) \in \mathrm{Nm}'_{L/K}(I)$$

Hence $\mathfrak{N}_{L/K}(I) \subseteq \mathrm{Nm}'_{L/K}(I)$. Extending both sides, we get $(\mathfrak{N}_{L/K}(I))_N \subseteq (\mathrm{Nm}'_{L/K}(I))_N = \mathrm{Nm}'_{L/K}(I)$. Hence, since we are in a Dedekind domain, there exists an [integral] ideal $J \subseteq \mathcal{O}_N$

such that

$$(\mathfrak{N}_{L/K}(I))_N = J\mathrm{Nm}'_{L/K}(I)$$

We want to show that $J = (1)$. By corollary 50.2.21, applied in $\mathcal{O}_L$ to the fractional ideal I^{-1} (theorem 50.2.22) and the nonzero ideal $J \cap L$, there is $b \in L^\times$ such that $B = bI^{-1}$ is an ideal of $\mathcal{O}_L$ relatively prime to $J \cap L$. Then $IB = (b)$, and $b \in IB \subseteq I$. Then since J is fixed under $\mathrm{Gal}_K(N)$, each $(\sigma_i B)_{N/\sigma_i L}$ is relatively prime to J for each i. Then since $b \in I$:

$$\begin{aligned}\left(\mathfrak{N}_{L/K}(I)\right)_N &\supseteq \left(\prod_i \sigma_i(b)\right) \\ &= \left(\prod_i (\sigma_i B)_{N/\sigma_i L}\right)\left(\prod_i (\sigma_i I)_{N/\sigma_i L}\right)\end{aligned}$$

Thus, $\prod_i (\sigma_i B)_{N/\sigma_i L}$ is a multiple of J. However, each $(\sigma_i B)_{N/\sigma_i L}$ is relatively prime to J, hence it must be that $J = (1)$, as we sought to show.

Lemma 1.3.15: Ideal Norm Properties

Let $M/L/K$ be algebraic number fields, $\mathfrak{a}, \mathfrak{b} \subseteq M$ be ideals, $I \subseteq K$ an ideal, and $a \in M$ an element. Then:

1. $\mathfrak{N}_{M/K}(a) = (\mathrm{Nm}_{M/K}(a)) = (\prod_\sigma \sigma(a))$, where σ ranges over the distinct K-embeddings of M into a normal extension of K containing M
2. $\mathfrak{N}_{L/K}(\mathfrak{N}_{M/L}(\mathfrak{a})) = \mathfrak{N}_{M/K}(\mathfrak{a})$
3. $\mathfrak{N}_{M/K}(\mathfrak{ab}) = \mathfrak{N}_{M/K}(\mathfrak{a})\mathfrak{N}_{M/K}(\mathfrak{b})$
4. $I^{[M:K]} = \mathfrak{N}_{M/K}((I)_M)$
5. $(\mathfrak{N}_K(I)) = \mathfrak{N}_{K/\mathbb{Q}}(I)$

***Proof*:**
Note that for all of these equalities, we immediately have $\supseteq$, hence we want to show $\subseteq$ for each of these.

1. By definition 50.3.13, the ideal $\mathfrak{N}_{M/K}(a\mathcal{O}_M)$ is generated by the elements $\mathrm{Nm}_{M/K}(ab)$ with $b \in \mathcal{O}_M$. The norm is multiplicative because the determinant is, by [3, Multiplicativity of the Determinant], so $\mathrm{Nm}_{M/K}(ab) = \mathrm{Nm}_{M/K}(a)\mathrm{Nm}_{M/K}(b)$. The norms $\mathrm{Nm}_{M/K}(b)$ with $b \in \mathcal{O}_M$ lie in $\mathcal{O}_K$ (corollary 50.1.14) and include $\mathrm{Nm}_{M/K}(1) = 1$, so they generate $\mathcal{O}_K$, and $\mathfrak{N}_{M/K}(a\mathcal{O}_M) = \mathrm{Nm}_{M/K}(a)\mathcal{O}_K$. The second expression is corollary 50.1.13.
2. Let N/K be a finite normal extension of K containing M, so that N/L is normal as well. Let $\tau_1, \ldots, \tau_n$ be the distinct K-embeddings of L into N and $\sigma_1, \ldots, \sigma_m$ the distinct L-embeddings of M into N. Each of these extends to an automorphism of N: it extends to a homomorphism $N \to \overline{N}$ by [3, Embedding Into Algebraic Extension], and the image is N by normality of N/K, respectively N/L, via [3, Finite Normal Extension Equivalences]. Any automorphism τ of N fixes $\mathbb{Q}$ pointwise, and an element of N lies in $\mathcal{O}_N$ exactly

when its minimal polynomial over $\mathbb{Q}$ has integer coefficients. An element and its image under τ share this minimal polynomial, so $\tau(\mathcal{O}_N) \subseteq \mathcal{O}_N$, and the same applied to τ^{-1} gives $\tau(\mathcal{O}_N) = \mathcal{O}_N$. Since τ is a ring isomorphism it carries products of ideals to products of their images, and since $\tau(\mathcal{O}_N) = \mathcal{O}_N$ the image of a fractional ideal of $\mathcal{O}_N$ is again a fractional ideal of $\mathcal{O}_N$. For every fractional ideal J of a subfield $F \subseteq N$, we write $(\tau J)_N = (\tau J)_{N/\tau F}$ for the extension of τJ to $\mathcal{O}_N$, so that:

$$(\tau J)_N = \tau(J)\mathcal{O}_N = \tau(J)\tau(\mathcal{O}_N) = \tau(J_N)$$

Applying theorem 50.3.14 in the tower $N/L/K$ to the fractional ideal $\mathfrak{N}_{M/L}(\mathfrak{a})$ of L, and then in the tower $N/M/L$ to $\mathfrak{a}$:

$$\begin{aligned}\left(\mathfrak{N}_{L/K}(\mathfrak{N}_{M/L}(\mathfrak{a}))\right)_N &= \prod_i \left(\tau_i\, \mathfrak{N}_{M/L}(\mathfrak{a})\right)_N = \prod_i \tau_i \left(\left(\mathfrak{N}_{M/L}(\mathfrak{a})\right)_N\right) \\ &= \prod_i \tau_i \left(\prod_j (\sigma_j \mathfrak{a})_N\right) = \prod_{i,j} (\tau_i \sigma_j \mathfrak{a})_N\end{aligned}$$

The restrictions of the composites $\tau_i\sigma_j$ to M are K-embeddings of M into N, and they are distinct: if $\tau_i\sigma_j = \tau_{i'}\sigma_{j'}$ on M, then restricting to L, where every σ_j is the identity, gives $\tau_i = \tau_{i'}$ on L, so $i = i'$, and composing with τ_i^{-1} gives $\sigma_j = \sigma_{j'}$ on M, so $j = j'$. Number fields are separable over $\mathbb{Q}$, so the number of K-embeddings of L into $\overline{K}$ equals $[L : K]$ by [3, Finite Separable Extension Equivalence], and each of them lands inside N by the extension argument above. Hence $n = [L : K]$, and likewise $m = [M : L]$ and there are $[M : K]$ distinct K-embeddings of M into N. As the composites give $nm = [M : L][L : K] = [M : K]$ distinct ones, they are exactly the distinct K-embeddings of M into N. Applying theorem 50.3.14 once more, now in the tower $N/M/K$:

$$\prod_{i,j} (\tau_i \sigma_j \mathfrak{a})_N = \left(\mathfrak{N}_{M/K}(\mathfrak{a})\right)_N$$

Since the extension map $J_K \to J_N$ is injective by corollary 50.3.8, it follows that $\mathfrak{N}_{L/K}(\mathfrak{N}_{M/L}(\mathfrak{a})) = \mathfrak{N}_{M/K}(\mathfrak{a})$. In particular the remaining containment $\subseteq$ holds.

3. Let N/K be a finite normal extension of K containing M and let $\tau_1, \ldots, \tau_n$ be the distinct K-embeddings of M into N. Each τ_i is a ring isomorphism onto $\tau_i M$, so $\tau_i(\mathfrak{ab}) = \tau_i(\mathfrak{a})\tau_i(\mathfrak{b})$. The minimal polynomial argument of the previous item, applied to the elements of M, gives $\tau_i(\mathcal{O}_M) = \mathcal{O}_{\tau_i M}$, so $\tau_i\mathfrak{a}$ is an ideal of $\mathcal{O}_{\tau_i M}$ and its extension $(\tau_i\mathfrak{a})_N$ is defined. Extension of ideals is multiplicative by corollary 50.3.8. Applying theorem 50.3.14 in the tower $N/M/K$ to $\mathfrak{ab}$, then to $\mathfrak{a}$ and $\mathfrak{b}$ separately:

$$\left(\mathfrak{N}_{M/K}(\mathfrak{ab})\right)_N = \prod_i (\tau_i(\mathfrak{a})\tau_i(\mathfrak{b}))_N = \prod_i (\tau_i\mathfrak{a})_N \prod_i (\tau_i\mathfrak{b})_N = \left(\mathfrak{N}_{M/K}(\mathfrak{a})\mathfrak{N}_{M/K}(\mathfrak{b})\right)_N$$

Since the extension map $J_K \to J_N$ is injective by corollary 50.3.8, the stated equality follows. In particular the containment $\subseteq$ holds.

4. Let N and $\tau_1, \ldots, \tau_{[M:K]}$ be as in item (3) above, so that the τ_i are the $[M : K]$ distinct K-embeddings of M into N, as counted in item (2) above, each extended to an automorphism

of N. Each τ_i fixes $I \subseteq K$ pointwise, so $\tau_i(I_M) = I\tau_i(\mathcal{O}_M)$ and $(\tau_i I_M)_N = I\mathcal{O}_N = I_N$. By theorem 50.3.14 in the tower $N/M/K$,

$$\left(\mathfrak{N}_{M/K}(I_M)\right)_N = \prod_i (\tau_i I_M)_N = (I_N)^{[M:K]} = \left(I^{[M:K]}\right)_N$$

and injectivity of the extension map $J_K \to J_N$ (corollary 50.3.8) gives $\mathfrak{N}_{M/K}(I_M) = I^{[M:K]}$.

5. Here I is a nonzero ideal of $\mathcal{O}_K$. If $I = a\mathcal{O}_K$ is principal, then (1) gives $\mathfrak{N}_{K/\mathbb{Q}}(I) = \mathrm{Nm}_{K/\mathbb{Q}}(a)\mathbb{Z}$, while $\mathfrak{N}_K(I) = |\mathrm{Nm}_{K/\mathbb{Q}}(a)|$ by lemma 50.1.63. In general, the class group is finite (theorem 50.2.33), of order h say, so $I^h = a\mathcal{O}_K$ for some a by [3, Lagrange Theorem]. Write $\mathfrak{N}_{K/\mathbb{Q}}(I) = c\mathbb{Z}$ with $c > 0$. By (3), the principal case and proposition 50.2.30,

$$c^h\mathbb{Z} = \mathfrak{N}_{K/\mathbb{Q}}(I^h) = \mathfrak{N}_K(I^h)\mathbb{Z} = \mathfrak{N}_K(I)^h\mathbb{Z}$$

so $c^h = \mathfrak{N}_K(I)^h$, and $c = \mathfrak{N}_K(I)$ as both are positive, completing the proof.

Corollary 1.3.16: Relative Norm Of Prime Ideal

Let L/K be a finite extension of number fields and let $\mathfrak{B}$ be a nonzero prime of $\mathcal{O}_L$ lying above the prime $\mathfrak{p}$ of $\mathcal{O}_K$. Then if $f = [\mathcal{O}_L/\mathfrak{B} : \mathcal{O}_K/\mathfrak{p}]$, then:

$$\mathfrak{N}_{L/K}(\mathfrak{B}) = \mathfrak{p}^f$$

***Proof*:**
Step 1: a power of $\mathfrak{p}$. As $\mathfrak{p}\mathcal{O}_L \subseteq \mathfrak{B}$, corollary 50.2.18(3) gives an ideal $\mathfrak{c}$ of $\mathcal{O}_L$ with $\mathfrak{p}\mathcal{O}_L = \mathfrak{B}\mathfrak{c}$. By lemma 50.3.15(3) and (4), $\mathfrak{N}_{L/K}(\mathfrak{B})\mathfrak{N}_{L/K}(\mathfrak{c}) = \mathfrak{N}_{L/K}(\mathfrak{p}\mathcal{O}_L) = \mathfrak{p}^{[L:K]}$. Both factors are ideals of $\mathcal{O}_K$, generated by norms of elements of $\mathcal{O}_L$ (corollary 50.1.14), so by theorem 50.2.15, $\mathfrak{N}_{L/K}(\mathfrak{B}) = \mathfrak{p}^j$ for some $j \geq 0$.

Step 2: the exponent. The field $k = \mathcal{O}_K/\mathfrak{p}$ embeds into $\mathcal{O}_L/\mathfrak{B}$, which is a k-vector space of dimension f, so $\mathfrak{N}_L(\mathfrak{B}) = |k|^f = \mathfrak{N}_K(\mathfrak{p})^f$, with $\mathfrak{N}_K(\mathfrak{p}) \geq 2$ (lemma 50.2.29). By lemma 50.3.15(2) and (5), and proposition 50.2.30,

$$\mathfrak{N}_K(\mathfrak{p})^f\mathbb{Z} = \mathfrak{N}_L(\mathfrak{B})\mathbb{Z} = \mathfrak{N}_{L/\mathbb{Q}}(\mathfrak{B}) = \mathfrak{N}_{K/\mathbb{Q}}(\mathfrak{N}_{L/K}(\mathfrak{B})) = \mathfrak{N}_{K/\mathbb{Q}}(\mathfrak{p}^j) = \mathfrak{N}_K(\mathfrak{p})^j\mathbb{Z}$$

so $j = f$, completing the proof.

The relative norm can in fact be defined to be the unique group homomorphism where

$$\mathfrak{N}(\mathfrak{B}) = \mathfrak{p}^{[\mathcal{O}_L/\mathfrak{B}:\mathcal{O}_K/\mathfrak{p}]}$$

which is left as an exercise. By multiplicity, we get that in general:

$$\mathfrak{N}_{L/K}(\mathfrak{p}\mathcal{O}_L) = \prod_i \mathfrak{N}_{L/K}(\mathfrak{B}_i)^{e_i} = \prod_i \mathfrak{p}^{f_i e_i}$$

Corollary 1.3.17: Ideal Norm Correspondence Not Galois

Let $M/L/K$ be algebraic number fields with M/K normal but not necessarily L/K, and let $I \subseteq L$ be a fractional ideal. Then

$$(\mathfrak{N}_{L/K}(I))_M = \prod_{\tau \in \mathrm{Gal}_K(M)/\mathrm{Gal}_L(M)} \tau(I_M)$$

***Proof*:**
Let $\sigma_1, \ldots, \sigma_n$ be the distinct K-embeddings $\sigma_i : L \to M$, each extended to an automorphism of M as in theorem 50.3.14, which applied with $N = M$ gives:

$$(\mathfrak{N}_{L/K}(I))_M = \prod_i (\sigma_i I)_{M/\sigma_i L}$$

Every $\tau \in \mathrm{Gal}_K(M)$ restricts to a K-embedding $L \to M$, and two automorphisms τ, τ' have the same restriction exactly when $\tau^{-1}\tau'$ fixes L pointwise, that is, when $\tau\mathrm{Gal}_L(M) = \tau'\mathrm{Gal}_L(M)$. Since every K-embedding $L \to M$ extends to an automorphism of M (as recorded in theorem 50.3.14), restriction is a bijection from $\mathrm{Gal}_K(M)/\mathrm{Gal}_L(M)$ onto $\{\sigma_1, \ldots, \sigma_n\}$.

Now let $\tau \in \mathrm{Gal}_K(M)$ extend σ_i. Any automorphism τ of M fixes $\mathbb{Q}$ pointwise, an element of M lies in $\mathcal{O}_M$ exactly when its minimal polynomial over $\mathbb{Q}$ has integer coefficients, and an element and its image under τ share this minimal polynomial, so $\tau(\mathcal{O}_M) = \mathcal{O}_M$. Therefore

$$(\sigma_i I)_{M/\sigma_i L} = \tau(I)\mathcal{O}_M = \tau(I)\tau(\mathcal{O}_M) = \tau(I_M)$$

This value depends only on the coset $\tau\mathrm{Gal}_L(M)$: for $\rho \in \mathrm{Gal}_L(M)$, ρ fixes $I \subseteq L$ pointwise and $\rho(\mathcal{O}_M) = \mathcal{O}_M$, so $\rho(I_M) = I_M$ and $\tau\rho(I_M) = \tau(I_M)$. Substituting $\tau(I_M)$ for each factor of the displayed product gives the claimed equality, as we sought to show.

With the relative norm defined, it is perhaps natural to think there is a relative discriminant:

Definition 1.3.18: Relative Discriminant

Let L/K be a finite extension of number fields. Then:

$$\mathrm{disc}'_{L/K}(\mathcal{O}_L) = \left(\left\{\mathrm{disc}_{L/K}(\alpha_1, \alpha_2, ..., \alpha_n) \;\middle|\; \alpha_1, \alpha_2, ..., \alpha_n \text{ basis for } \mathcal{O}_L \right\}\right)$$

is the *relative discriminant* of L with respect to K.

This will not come up much, but is useful in a couple of tower arguments later on.

Though we have characterized a correspondence of the ideals, it is more difficult to characterize I_L. In the following section, we characterize prime extensions, and take advantage of the fact that the factorization to give us a characterization of I_L.

1.4 Prime Factorization in Extension

Since $\mathcal{O}$ is a Dedekind domain by lemma 50.3.2, we have that $\mathfrak{p}\mathcal{O}$ decomposes uniquely into a product of prime ideals:

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_r^{e_r}$$

If it is clear from context, we shall often write $\mathfrak{p}$ instead of $\mathfrak{p}\mathcal{O}$. Sometimes, we shall also write $\mathfrak{p}_L$ for $\mathfrak{p}\mathcal{O}_L$. We shall *almost always* find exactly how $\mathfrak{p}\mathcal{O}$ decomposes, sometimes without requiring knowledge of the ring of integers. When the extension L/K is Galois, we shall have even more control, and if L/K is not Galois, we can take $M/L/K$ to be a Galois extension, work within the Galois, and see if the results can be translated back to L/K.

Let's say $\mathfrak{B}$ lies over $\mathfrak{p}$ and that $\mathfrak{B}$ is in the prime decomposition of $\mathfrak{B}$ [21]. Then we have two important quantities associated to $\mathfrak{B}$: it's power in the prime decomposition, and the index $[\mathcal{O}/\mathfrak{B} : \mathcal{O}_K/\mathfrak{p}]$, that is the degree of the extension of the residue field. These quantities are given a name:

Definition 1.4.1: Ramification Index and Inertia Degree

Let R be a Dedekind domain, $K = \text{Frac}(R)$, L/K be a finite field extension, $\mathcal{O} = \overline{R}^L$ and $\mathfrak{p} \subseteq R$ a nonzero prime ideal. If $\mathfrak{B}_i$ is a prime in $\mathcal{O}$ lying above $\mathfrak{p}$, then the exponent of $\mathfrak{B}_i$ in $\mathfrak{p}\mathcal{O}$ is called the *ramification index* and is often denoted

$$e(\mathfrak{B}_i/\mathfrak{p}) \qquad \text{where} \qquad \mathfrak{B}^{e(\mathfrak{B}_i/\mathfrak{p})} \| \mathfrak{p}\mathcal{O}$$

or e_i for short, where $\|$ means *maximally divides*. The degree of the field extension

$$f(\mathfrak{B}_i/\mathfrak{p}) = [\mathcal{O}/\mathfrak{B}_i : R/\mathfrak{p}]$$

or f_i for short, is called the *inertia degree* or *residue field degree* of $\mathfrak{B}_i$ over $\mathfrak{p}$.

For future reference, it will be important to keep track to "how much" a prime ideal splits. This will share strong similarities to the splitting of polynomials[22] (recall how "rare" an irreducible polynomial is inseparable)? This prompts the following vocabulary:

[21] we shall soon show all primes lying above $\mathfrak{p}$ will be part of the prime decomposition

[22] Though there shall be important failures when this comparison will *not* hold! This shall be elaborated soon

Definition 1.4.2: Split, Unramified, and Totally Ramified Primes

Let $\mathfrak{p} \subseteq R$ be a prime ideal of a Dedekind domain R, $K = K(R)$, L/K a finite extension, and $\mathcal{O} = \overline{R}^L$. Then given

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_g^{e_g}$$

If $g = [L : K]$, then $\mathfrak{p}$ is said to *split completely* ($e_i = f_i = 1$), giving:

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1 \cdots \mathfrak{B}_n$$

If $e_i = 1$ for all i (but some f_i are not necessarily 1), then we sy that $\mathfrak{p}$ is *unramified* and (if the residue field is separable):

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1 \cdots \mathfrak{B}_g$$

for some $g \leq n$, where $n = [L : K]$. If $g = 1$, so that $\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1}$ and $\mathfrak{B}_1$ is the only prime ideal lying over $\mathfrak{p}$, then $\mathfrak{p}$ is said to be *indecomposable* or *nonsplit*. If moreover

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^n$$

then $\mathfrak{p}$ is said to be *totally ramified*.

Most of the time, we shall work in the case of character zero extensions, hence L/K is separable. In this case, the values e_i and f_i are connected. If $\mathfrak{p}$ factors in an extension, then there is a simple equation the ramification and inertia degree must satisfy which equals the degree of the extension:

Proposition 1.4.3: Fundamental Identity

Let $\mathfrak{o}$ be a Dedekind domain with fraction field K, L/K a finite separable extension of degree n, $\mathcal{O} = \overline{\mathfrak{o}}^L$ the integral closure of $\mathfrak{o}$ in L, and $0 \neq \mathfrak{p} \subseteq \mathfrak{o}$ a prime ideal. If

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_g^{e_g}$$

is the prime decomposition of $\mathfrak{p}\mathcal{O}$, with the $\mathfrak{B}_i$ distinct, and f_i is the inertia degree of $\mathfrak{B}_i$ over $\mathfrak{p}$, then

$$\sum_{i=1}^{g} e_i f_i = n$$

For an extension L/K of number fields this applies with $\mathfrak{o} = \mathcal{O}_K$ and $\mathcal{O} = \mathcal{O}_L$ (corollary 50.3.4).

***Proof*:**

The ring $\mathcal{O}$ is a Dedekind domain (lemma 50.3.2), so $\mathfrak{p}\mathcal{O}$ has the stated factorization. Put $\kappa = \mathfrak{o}/\mathfrak{p}$, a field as $\mathfrak{p}$ is maximal (definition 50.2.1), and $S = \mathfrak{o} \setminus \mathfrak{p}$. Each $\mathfrak{B}_i$ contains $\mathfrak{p}$, so $\mathfrak{B}_i \cap \mathfrak{o}$ is a prime of $\mathfrak{o}$ containing the maximal ideal $\mathfrak{p}$, hence equal to it, and every quotient below is a κ-vector space through $\mathfrak{o} \to \kappa$.

Step 1: a free localization. By lemma 50.2.4, $\mathfrak{o}_\mathfrak{p} = S^{-1}\mathfrak{o}$ is a DVR, hence a principal ideal domain ([2, Characterizing DVR]). The ring $S^{-1}\mathcal{O} \subseteq L$ is the integral closure of $\mathfrak{o}_\mathfrak{p}$ in L. An element x/s with $x \in \mathcal{O}$ and $s \in S$ is integral over $\mathfrak{o}_\mathfrak{p}$: dividing an equation $x^m + a_{m-1}x^{m-1} + \cdots + a_0 = 0$

over o by s^m gives a monic equation for x/s with coefficients $a_i/s^{m-i} \in o_\mathfrak{p}$. Conversely, if $y \in L$ satisfies $y^m + (b_{m-1}/t_{m-1})y^{m-1} + \cdots + b_0/t_0 = 0$ with $b_i \in o$ and $t_i \in S$, then for $t = t_0 \cdots t_{m-1}$, multiplying by t^m shows that ty is integral over o, so $ty \in \mathcal{O}$ and $y = ty/t \in S^{-1}\mathcal{O}$. The ring $o_\mathfrak{p}$ has fraction field K and is integrally closed in it, being the localization of the normal domain o ([2, Normalization Commutes with Localization]). As L/K is separable and $o_\mathfrak{p}$ is a principal ideal domain, proposition 50.1.32 shows that $S^{-1}\mathcal{O}$ is a free $o_\mathfrak{p}$-module of rank n.

Step 2: $\dim_\kappa \mathcal{O}/\mathfrak{p}\mathcal{O} = n$. Let $M \subseteq L$ be an o-submodule. For $s \in S$ we have $o = \mathfrak{p} + so$, by maximality of $\mathfrak{p}$, so there is $t \in o$ with $st - 1 \in \mathfrak{p}$. We claim that the map $M/\mathfrak{p}M \to S^{-1}M/\mathfrak{p}S^{-1}M$ induced by inclusion is an isomorphism. It is surjective, since for $x \in M$ we have $x/s - tx = x(1-st)/s \in \mathfrak{p}S^{-1}M$. It is injective: if $x \in M$ lies in $\mathfrak{p}S^{-1}M$, then $x = y/s$ with $y \in \mathfrak{p}M$, so $sx = y \in \mathfrak{p}M$, and $x = t(sx) - (st-1)x \in \mathfrak{p}M$. With $M = o$ this gives $\kappa \cong o_\mathfrak{p}/\mathfrak{p}o_\mathfrak{p}$. With $M = \mathcal{O}$, and an isomorphism $S^{-1}\mathcal{O} \cong o_\mathfrak{p}^n$ of $o_\mathfrak{p}$-modules from Step 1, which carries $\mathfrak{p}S^{-1}\mathcal{O}$ onto $(\mathfrak{p}o_\mathfrak{p})^n$,

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong S^{-1}\mathcal{O}/\mathfrak{p}S^{-1}\mathcal{O} \cong (o_\mathfrak{p}/\mathfrak{p}o_\mathfrak{p})^n \cong \kappa^n$$

All these maps are o-linear, hence κ-linear, so $\dim_\kappa \mathcal{O}/\mathfrak{p}\mathcal{O} = n$.

Step 3: the Chinese remainder theorem. The ideals $\mathfrak{B}_i^{e_i}$ are pairwise comaximal (lemma 50.2.17), so [3, Chinese Remainder Theorem] gives a ring isomorphism, which is o-linear and hence κ-linear,

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \mathcal{O}/\mathfrak{B}_1^{e_1} \times \cdots \times \mathcal{O}/\mathfrak{B}_g^{e_g}$$

Dimensions add over finite products ([3, Dimension Of Quotient Space]), so by Step 2, $n = \sum_i \dim_\kappa \mathcal{O}/\mathfrak{B}_i^{e_i}$.

Step 4: $\dim_\kappa \mathcal{O}/\mathfrak{B}^e = ef$. Let $\mathfrak{B} = \mathfrak{B}_i$, $e = e_i$ and $f = f_i = [\mathcal{O}/\mathfrak{B} : \kappa]$ (definition 50.4.1), finite because $\mathcal{O}/\mathfrak{B}$ is a quotient of $\mathcal{O}/\mathfrak{p}\mathcal{O}$. For $0 \le j < e$, the κ-space $\mathcal{O}/\mathfrak{B}^{j+1}$ contains $\mathfrak{B}^j/\mathfrak{B}^{j+1}$, which is isomorphic to $\mathcal{O}/\mathfrak{B}$ by lemma 50.2.24 and so has dimension f, with quotient isomorphic to $\mathcal{O}/\mathfrak{B}^j$ ([3, Isomorphism Theorems For Modules]). So $\dim_\kappa \mathcal{O}/\mathfrak{B}^{j+1} = \dim_\kappa \mathcal{O}/\mathfrak{B}^j + f$ ([3, Dimension Of Quotient Space]), and induction on j gives $\dim_\kappa \mathcal{O}/\mathfrak{B}^e = ef$. With Step 3, $n = \sum_i e_i f_i$, completing the proof.

For number fields, absolute norms give a second proof. Let $k = \mathcal{O}_K/\mathfrak{p}$, a finite field with $|k| = \mathfrak{N}_K(\mathfrak{p}) \ge 2$ (lemma 50.2.29). Each $\mathfrak{B}_i \cap \mathcal{O}_K$ is the maximal ideal $\mathfrak{p}$, so k embeds into $\mathcal{O}_L/\mathfrak{B}_i$, which has $|k|^{f_i}$ elements, and $\mathfrak{N}_L(\mathfrak{B}_i) = |k|^{f_i}$. Then proposition 50.2.30, applied in L, gives $\mathfrak{N}_L(\mathfrak{p}\mathcal{O}_L) = \prod_i \mathfrak{N}_L(\mathfrak{B}_i)^{e_i} = |k|^{\sum_i e_i f_i}$, while lemma 50.3.6 gives $\mathfrak{N}_L(\mathfrak{p}\mathcal{O}_L) = \mathfrak{N}_K(\mathfrak{p})^n = |k|^n$.

Corollary 1.4.4: Multiplicity of Ramification And Inertia

Let $M/L/K$ be finite extensions of number fields, let $\mathfrak{B}$ be a nonzero prime of $\mathcal{O}_M$, and put $\mathfrak{q} = \mathfrak{B} \cap \mathcal{O}_L$ and $\mathfrak{p} = \mathfrak{B} \cap \mathcal{O}_K$. Then:

$$e(\mathfrak{B}/\mathfrak{p}) = e(\mathfrak{B}/\mathfrak{q})\, e(\mathfrak{q}/\mathfrak{p}) \qquad f(\mathfrak{B}/\mathfrak{p}) = f(\mathfrak{B}/\mathfrak{q})\, f(\mathfrak{q}/\mathfrak{p})$$

that is, in short, $e_{M/K} = e_{M/L}e_{L/K}$ and $f_{M/K} = f_{M/L}f_{L/K}$.

***Proof*:**
The ideal $\mathfrak{q}$ is a prime of $\mathcal{O}_L$ and $\mathfrak{p} = \mathfrak{q} \cap \mathcal{O}_K$ is a prime of $\mathcal{O}_K$, both nonzero since $\mathfrak{B} \cap \mathbb{Z} \neq 0$ (lemma 50.2.29). The rings $\mathcal{O}_L$ and $\mathcal{O}_M$ are the integral closures of $\mathcal{O}_K$ and $\mathcal{O}_L$ in L and M (corollary 50.3.4) and are Dedekind domains (theorem 50.2.6). By proposition 50.3.11, $\mathfrak{B}$ divides $\mathfrak{q}\mathcal{O}_M$ and $\mathfrak{p}\mathcal{O}_M$, and $\mathfrak{q}$ divides $\mathfrak{p}\mathcal{O}_L$, so the three ramification indices and inertia degrees are defined (definition 50.4.1).

The inclusions $\mathcal{O}_K \subseteq \mathcal{O}_L \subseteq \mathcal{O}_M$ induce homomorphisms of fields $\mathcal{O}_K/\mathfrak{p} \to \mathcal{O}_L/\mathfrak{q} \to \mathcal{O}_M/\mathfrak{B}$, injective because $\mathfrak{q} \cap \mathcal{O}_K = \mathfrak{p}$ and $\mathfrak{B} \cap \mathcal{O}_L = \mathfrak{q}$. So [3, Multiplicity of Finite Extensions (Tower Law)] gives $f(\mathfrak{B}/\mathfrak{p}) = f(\mathfrak{B}/\mathfrak{q})f(\mathfrak{q}/\mathfrak{p})$.

For the ramification indices, factor $\mathfrak{p}\mathcal{O}_L = \mathfrak{q}^a\mathfrak{a}$ with $a = e(\mathfrak{q}/\mathfrak{p})$ and $\mathfrak{q} \nmid \mathfrak{a}$, and $\mathfrak{q}\mathcal{O}_M = \mathfrak{B}^b\mathfrak{b}$ with $b = e(\mathfrak{B}/\mathfrak{q})$ and $\mathfrak{B} \nmid \mathfrak{b}$ (theorem 50.2.15). Extension of ideals is multiplicative, the two sides being generated by the same products, so

$$\mathfrak{p}\mathcal{O}_M = (\mathfrak{p}\mathcal{O}_L)\mathcal{O}_M = (\mathfrak{q}\mathcal{O}_M)^a(\mathfrak{a}\mathcal{O}_M) = \mathfrak{B}^{ab}\mathfrak{b}^a(\mathfrak{a}\mathcal{O}_M)$$

The prime $\mathfrak{B}$ does not divide $\mathfrak{a}\mathcal{O}_M$: otherwise $\mathfrak{B} \supseteq \mathfrak{a}$, so the prime $\mathfrak{B} \cap \mathcal{O}_L = \mathfrak{q}$ contains $\mathfrak{a}$ and divides it (corollary 50.2.18), contrary to $\mathfrak{q} \nmid \mathfrak{a}$. So the exponent of $\mathfrak{B}$ in $\mathfrak{p}\mathcal{O}_M$ is ab, that is $e(\mathfrak{B}/\mathfrak{p}) = e(\mathfrak{B}/\mathfrak{q})e(\mathfrak{q}/\mathfrak{p})$, as we sought to show.

Hence, we have a measure of how much a prime ideal splits in an extension.

Example 1.4.5: Decomposing Prime

1. It is possible that $\mathfrak{p}$ is totally ramified where $e_1 > 1$ and $f_1 = 1$: take $L = \mathbb{Q}(\sqrt{3})$ and $K = \mathbb{Q}$ with $I = (3)$. Then $(3)\mathcal{O} = (\sqrt{3})^2$
2. It is possible that $\mathfrak{p}$ is indecomposable with $e_1 = 1$ and $f_1 > 1$, so that $\mathfrak{p}\mathcal{O}$ is prime: take $K = \mathbb{Q}$, $L = \mathbb{Q}(\sqrt{3})$, but $I = (5)$. Here $x^2 - 3$ is irreducible modulo 5, as 3 is not a square modulo 5.
3. Let us consider the inertia degree to get some structure of the Dedekind domain. Let $L = \mathbb{Q}(\sqrt[3]{2})$ and $L/\mathbb{Q}$ be a finite extension. Since it is always the case that $\mathbb{Z}[\sqrt[3]{2}] \subseteq \mathcal{O}_L$[a], Take $I = (\sqrt[3]{2}) \subseteq \mathcal{O}_L$, $I \cap \mathbb{Q} = (2)$, and $\mathfrak{N}((\sqrt[3]{2})) = |\mathrm{Nm}_{L/\mathbb{Q}}(\sqrt[3]{2})| = 2$, and $2 = (\sqrt[3]{2})^3$. Hence, $f_1 = 1$, meaning
$$[\mathcal{O}_L/\sqrt[3]{2}\mathcal{O}_L : \mathbb{Z}/2\mathbb{Z}] = 1$$
implying both fields only have two elements, hence
$$\mathcal{O}_L/\sqrt[3]{2}\mathcal{O}_L \cong \mathbb{Z}/2\mathbb{Z}$$
Notice how this also gave information on the ring of integers,
4. Let $\mathcal{O}_K = \mathbb{Z}[i]$ so that $2 = (1+i)(1-i)$. Then $\mathcal{O}_K/2\mathcal{O}_K$ has order 4 (look at it'snorm), and certainly $2\mathcal{O}_K \subseteq (1-i)$, hence $|\mathcal{O}_K/(1-i)|\big|4$, which by direct computation we may see that $|\mathcal{O}_K/(1-i)| = 2$, hence $f = 1$. On the other hand, we can see that 3 is prime in $\mathcal{O}_K$, and $|\mathcal{O}_K/3\mathcal{O}_K| = 9$, giving $f_{3\mathcal{O}_K/3} = 2$.

[a] In fact, they are equal in this case, which is not needed at the moment

Since the factorization of $\mathfrak{p} \subseteq K$ in L/K is tied to the degree of L/K, we may venture into hoping that the structure of L/K may reflect something about the factorization of $\mathfrak{p}$. A natural question we may ask here is how do we find the prime factors in the extension? Finding them is split into two cases: the "good" primes and the "bad' primes. We shall see that the power of the exponentials has a lot to do with it. This is because we shall see that most primes shall decompose with ramification index 1, since it will correspond to the splitting of a polynomial in separable extension (note that all number field extensions are separable). However, as we saw it is possible that prime may have higher ramification index, however it is impossible that a polynomial splits irreducibly with multiplicity higher than 1 over a separable extension. Hence, we may think of such prime as "bad" primes. In the following we shall show how to find primes whose factorization mirrors that of a minimal polynomial, and how to detect when it is not the case.

let L/K be a finite separable extension. Then by the primitive extension theorem, there exists a $\theta \in K$

$$L = K(\theta)$$

Usually, we shall let L be Galois so that we may computationally find this element, namely it must not be fixed by any element of $\mathrm{Gal}_K(L)$ (hence, not in any intermediate extension). In fact, we may find θ so that $\theta \in \mathcal{O}_K$: if $L = K(\alpha)$, since $L = \overline{\mathcal{O}_K}^L$ we may multiply out the denominator by d to get $\alpha = \theta$. Let us simply re-define θ to be in $\mathcal{O}_L$. Now we have a single element, θ with minimal polynomial $m_{K,\theta}(x)$ which characterizes L. It would be nice to say something about the nature of the decomposition of $\mathfrak{p}$ in $\mathcal{O}_L$ given this information. In the good case, we shall be able to do so, and in the bad case we shall require a bit more work. In particular, a finite number of primes ideals will be exclude. Only those that are relatively prime to an ideal known as the *conductor* of the ring $\mathcal{O}_K[\theta]$ will be considered:

Definition 1.4.6: Conductor

Let $\mathcal{O}_K[\theta]$ be the ring of integers extended by θ. Then the conductor of $\mathcal{O}_K[\theta]$, is the largest ideal $\mathscr{F}_\theta \subseteq \mathcal{O}_L$ which is contained in $\mathcal{O}_K[\theta]$:

$$\mathscr{F}_\theta = \{\alpha \in \mathcal{O}_L \mid \alpha\mathcal{O}_L \subseteq \mathcal{O}_K[\theta]\}$$

In the case where $K = \mathbb{Q}$ and $\mathcal{O}_\mathbb{Q} = \mathbb{Z}$, then $\mathscr{F}_\theta = \{\alpha \in \mathcal{O}_L \mid (\alpha)_L \subseteq \mathbb{Z}[\theta]\}$. Since $\mathcal{O}_L$ is a finitely generated $\mathcal{O}_K$ module, $\mathscr{F}_\theta \neq 0$. The conductor is a measure of the "bad primes" (hence the ramification). In our case, the conductor is never 0, but it may be that $\mathscr{F}_\theta = (1)$, in which case the bad primes will be "tame" (or the field is "tamely ramified")

Theorem 1.4.7: Dedekind-Kummer Theorem

Let $\mathcal{O}_K$ be a Dedekind domain with fraction field K, let L/K be a finite separable extension, and let $\mathcal{O}$ be the integral closure of $\mathcal{O}_K$ in L. Let $L = K(\theta)$ where θ is a primitive element whose minimal polynomial $m(x) = m_{\theta,K}(x)$ lies in $\mathcal{O}_K[x]$. Let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$ which is relatively prime to the conductor $\mathscr{F}$ of $\mathcal{O}_K[\theta]$, that is, $\mathfrak{p}\mathcal{O} + \mathscr{F} = \mathcal{O}$, and let

$$\overline{m}(x) = \overline{m}_1(x)^{e_1} \cdots \overline{m}_r(x)^{e_r}$$

be the factorization of the polynomial $\overline{m}(x) = m(x) \bmod \mathfrak{p}$ into distinct monic irreducible polynomials over the residue field $\mathcal{O}_K/\mathfrak{p}$. Let $m_i(x) \in \mathcal{O}_K[x]$ be monic polynomials reducing to $\overline{m}_i(x)$. Then

$$\mathfrak{B}_i = \mathfrak{p}\mathcal{O} + m_i(\theta)\mathcal{O} = (\mathfrak{p}\mathcal{O}, m_i(\theta))_L$$

are different prime ideals of $\mathcal{O}$ over $\mathfrak{p}$, and they are all the primes of $\mathcal{O}$ over $\mathfrak{p}$. Moreover $\mathcal{O}/\mathfrak{B}_i \cong (\mathcal{O}_K/\mathfrak{p})[x]/(\overline{m}_i)$ as $\mathcal{O}_K/\mathfrak{p}$-algebras, so the inertia degree f_i of $\mathfrak{B}_i$ is the degree of $\overline{m}_i(x)$, and

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_r^{e_r}$$

If $\mathfrak{p}$ is not relatively prime to $\mathscr{F}$, the theorem gives no information: the factorization of $\overline{m}$ need not reflect that of $\mathfrak{p}\mathcal{O}$.

***Proof*:**
Write $\mathcal{O}' = \mathcal{O}_K[\theta]$, $\kappa = \mathcal{O}_K/\mathfrak{p}$ and $n = [L : K] = \deg m$. The ring $\mathcal{O}$ is a Dedekind domain (lemma 50.3.2), and $\theta \in \mathcal{O}$ because m is monic with coefficients in $\mathcal{O}_K$, so $\mathcal{O}' \subseteq \mathcal{O}$. The conductor $\mathscr{F}$ is an ideal of $\mathcal{O}$ contained in $\mathcal{O}'$: if $\alpha \in \mathscr{F}$ and $\beta \in \mathcal{O}$ then $\alpha\beta\mathcal{O} \subseteq \alpha\mathcal{O} \subseteq \mathcal{O}'$, and $\alpha = \alpha \cdot 1 \in \mathcal{O}'$. A nonzero x in a nonzero ideal I of $\mathcal{O}$ satisfies a monic equation $x^d + a_{d-1}x^{d-1} + \cdots + a_0 = 0$ over $\mathcal{O}_K$ of least degree, and then $a_0 \neq 0$, since otherwise dividing by x gives one of smaller degree. Hence $a_0 = -x(x^{d-1} + \cdots + a_1)$ is a nonzero element of $I \cap \mathcal{O}_K$. We use this observation twice.

Step 1: $\mathfrak{p} + (\mathscr{F} \cap \mathcal{O}_K) = \mathcal{O}_K$. We have $\mathscr{F} \neq 0$, since otherwise $\mathfrak{p}\mathcal{O} = \mathcal{O}$, contrary to proposition 50.3.10. Factor $\mathscr{F} = \prod_{\mathfrak{Q}} \mathfrak{Q}^{c_{\mathfrak{Q}}}$ (theorem 50.2.15). For each prime $\mathfrak{Q}$ dividing $\mathscr{F}$, the prime $\mathfrak{q} = \mathfrak{Q} \cap \mathcal{O}_K$ of $\mathcal{O}_K$ is nonzero by the observation above, hence maximal, and $\mathfrak{q} \neq \mathfrak{p}$: otherwise $\mathfrak{Q}$ contains $\mathfrak{p}\mathcal{O}$ and $\mathscr{F}$, hence $\mathfrak{p}\mathcal{O} + \mathscr{F} = \mathcal{O}$. Let $\mathfrak{q}_1, \ldots, \mathfrak{q}_s$ be the distinct primes so obtained, and let c be the largest exponent $c_{\mathfrak{Q}}$. The ideals $\mathfrak{q}_1^c, \ldots, \mathfrak{q}_s^c, \mathfrak{p}$ are pairwise comaximal. A maximal ideal containing $\mathfrak{q}_j^c$ contains $\mathfrak{q}_j$, being prime, and so equals $\mathfrak{q}_j$, and a maximal ideal containing $\mathfrak{p}$ equals $\mathfrak{p}$. So no maximal ideal contains two of these ideals, and an ideal containing two of them is $\mathcal{O}_K$ ([3, Ring with $1 \neq 0$ have Maximal ideals]). So [3, Chinese Remainder Theorem] gives $a \in \mathcal{O}_K$ with $a \in \mathfrak{q}_j^c$ for every j and $a - 1 \in \mathfrak{p}$. Then $a \notin \mathfrak{p}$, and for each $\mathfrak{Q} \big| \mathscr{F}$ we have $a\mathcal{O} \subseteq \mathfrak{q}^c\mathcal{O} \subseteq \mathfrak{Q}^c \subseteq \mathfrak{Q}^{c_{\mathfrak{Q}}}$, as $\mathfrak{q}\mathcal{O} \subseteq \mathfrak{Q}$. By unique factorization the exponent of every $\mathfrak{Q}$ in $a\mathcal{O}$ is at least $c_{\mathfrak{Q}}$, so $\mathscr{F}$ divides $a\mathcal{O}$, that is $a \in \mathscr{F}$ (corollary 50.2.18). So $a \in (\mathscr{F} \cap \mathcal{O}_K) \setminus \mathfrak{p}$, and as $\mathfrak{p}$ is maximal, $\mathfrak{p} + (\mathscr{F} \cap \mathcal{O}_K) = \mathcal{O}_K$.

Step 2: $\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \mathcal{O}'/\mathfrak{p}\mathcal{O}' \cong \kappa[x]/(\overline{m})$. Since $\mathscr{F} \subseteq \mathcal{O}'$, the hypothesis gives $\mathcal{O} = \mathfrak{p}\mathcal{O} + \mathcal{O}'$, so the ring homomorphism $\mathcal{O}' \to \mathcal{O}/\mathfrak{p}\mathcal{O}$ is surjective, with kernel $\mathfrak{p}\mathcal{O} \cap \mathcal{O}'$. This kernel is $\mathfrak{p}\mathcal{O}'$. The inclusion $\mathfrak{p}\mathcal{O}' \subseteq \mathfrak{p}\mathcal{O} \cap \mathcal{O}'$ holds as $\mathcal{O}' \subseteq \mathcal{O}$. Conversely, let $x \in \mathfrak{p}\mathcal{O} \cap \mathcal{O}'$ and use Step 1 to write $1 = a + c$ with $a \in \mathfrak{p}$ and $c \in \mathscr{F} \cap \mathcal{O}_K$. Then $ax \in \mathfrak{p}\mathcal{O}'$, as $x \in \mathcal{O}'$. Writing $x = \sum_j a_j y_j$

with $a_j \in \mathfrak{p}$ and $y_j \in \mathcal{O}$, we get $cx = \sum_j a_j(cy_j)$ with $cy_j \in \mathscr{F} \subseteq \mathcal{O}'$, so $cx \in \mathfrak{p}\mathcal{O}'$, and $x = ax + cx \in \mathfrak{p}\mathcal{O}'$. By [3, The first isomorphism theorem for Rings], $\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \mathcal{O}'/\mathfrak{p}\mathcal{O}'$.

Next, evaluation at θ is a surjective ring homomorphism $\mathcal{O}_K[x] \to \mathcal{O}'$, whose kernel is (m): if $g(\theta) = 0$, division by the monic polynomial m gives $g = qm + h$ with $q, h \in \mathcal{O}_K[x]$ and $\deg h < n$, and $h(\theta) = 0$ forces $h = 0$ since $1, \theta, \dots, \theta^{n-1}$ are linearly independent over K. Reducing coefficients modulo $\mathfrak{p}$ and then passing to $\kappa[x]/(\overline{m})$ is a surjective ring homomorphism $\mathcal{O}_K[x] \to \kappa[x]/(\overline{m})$ with kernel $\mathfrak{p}\mathcal{O}_K[x] + (m)$: if $\overline{g} = \overline{m}\,\overline{u}$ with $u \in \mathcal{O}_K[x]$, then $g - mu$ has coefficients in $\mathfrak{p}$. Since $\mathfrak{p}\mathcal{O}_K[x] + (m)$ is the preimage of $\mathfrak{p}\mathcal{O}'$ under evaluation, we get $\mathcal{O}'/\mathfrak{p}\mathcal{O}' \cong \kappa[x]/(\overline{m})$ ([3, The first isomorphism theorem for Rings]). Composing gives an isomorphism of rings

$$\psi\colon \mathcal{O}/\mathfrak{p}\mathcal{O} \to R := \kappa[x]/(\overline{m}) \qquad h(\theta) + \mathfrak{p}\mathcal{O} \mapsto \overline{h}(x) + (\overline{m})$$

for $h \in \mathcal{O}_K[x]$. It is $\mathcal{O}_K$-linear, hence κ-linear.

Step 3: the primes above $\mathfrak{p}$. By [3, 2nd, 3rd, 4th Isomorphism Theorem for Rings], ideals of $\mathcal{O}$ containing $\mathfrak{p}\mathcal{O}$ correspond, through ψ, to ideals of $\kappa[x]$ containing $(\overline{m})$, by an inclusion-preserving bijection matching quotient rings, hence prime ideals with prime ideals. Under it, the ideal generated by $\mathfrak{p}\mathcal{O}$ and some elements $h(\theta)$ corresponds to the ideal generated by $\overline{m}$ and the $\overline{h}$. The ring $\kappa[x]$ is a principal ideal domain ([3, Field Then $\mathbb{F}[x]$ PID And UFD]), so a prime ideal containing $\overline{m} \neq 0$ is (g) for an irreducible monic g dividing $\overline{m}$, that is one of the $(\overline{m}_i)$, and each $(\overline{m}_i)$ is prime ([3, Prime If And Only If Irreducible In PID]). Hence the primes of $\mathcal{O}$ containing $\mathfrak{p}\mathcal{O}$, which are the primes over $\mathfrak{p}$ (proposition 50.3.11), are exactly the ideals $\mathfrak{B}_i = \mathfrak{p}\mathcal{O} + m_i(\theta)\mathcal{O}$, which are distinct. Moreover $\mathcal{O}/\mathfrak{B}_i \cong \kappa[x]/(\overline{m}_i)$ as κ-algebras, and $1, x, \dots, x^{\deg \overline{m}_i - 1}$ is a κ-basis of the latter, so $f_i = \deg \overline{m}_i$.

Step 4: the exponents. By Step 3 and theorem 50.2.15, $\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{a_1} \cdots \mathfrak{B}_r^{a_r}$ with $a_i = e(\mathfrak{B}_i/\mathfrak{p}) \geq 1$. Fix i and $k \geq 1$. The ideal $\mathfrak{B}_i^k$ is generated by products of k elements of $\mathfrak{p}\mathcal{O}$ or of $m_i(\theta)\mathcal{O}$, and such a product lies in $\mathfrak{p}\mathcal{O}$ unless all its factors lie in $m_i(\theta)\mathcal{O}$, so $\mathfrak{B}_i^k + \mathfrak{p}\mathcal{O} = m_i(\theta)^k\mathcal{O} + \mathfrak{p}\mathcal{O}$. By corollary 50.2.18(2) this ideal is $\mathfrak{B}_i^{\min(k,a_i)}$. Under the correspondence of Step 3 it matches $(\overline{m}_i^k, \overline{m}) = (\overline{m}_i^{\min(k,e_i)})$, the generator being a greatest common divisor in the principal ideal domain $\kappa[x]$. As the correspondence is a bijection,

$$\mathfrak{B}_i^{\min(k,a_i)} = \mathfrak{B}_i^{\min(k+1,a_i)} \iff (\overline{m}_i^{\min(k,e_i)}) = (\overline{m}_i^{\min(k+1,e_i)})$$

By unique factorization in $\mathcal{O}$ and in $\kappa[x]$, the left side holds exactly when $k \geq a_i$ and the right side exactly when $k \geq e_i$. Hence $a_i = e_i$, and $\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_r^{e_r}$, completing the proof.

Corollary 1.4.8: Factorization For Number Rings (Dedekind-Kummer Theorem)

Let L/K be a finite extension of number fields, $L = K(\theta)$ for a primitive element θ where $m(x) = m_{\theta,K}(x) \in \mathcal{O}_K[x]$. Let $\mathfrak{p} \subseteq \mathcal{O}_K$ be a nonzero prime ideal lying over $p \in \mathbb{Z}$, let $\overline{m}(x) = \overline{m}_1(x)^{e_1} \cdots \overline{m}_r(x)^{e_r}$ be the factorization of m modulo $\mathfrak{p}$ into distinct monic irreducible polynomials over $\mathcal{O}_K/\mathfrak{p}$, and let $m_i(x) \in \mathcal{O}_K[x]$ be monic polynomials reducing to $\overline{m}_i(x)$. If $p \nmid [\mathcal{O}_L : \mathcal{O}_K[\theta]]$, then

$$\mathfrak{B}_i = \mathfrak{p}\mathcal{O}_L + m_i(\theta)\mathcal{O}_L = (\mathfrak{p}\mathcal{O}_L, m_i(\theta))_L$$

are different prime ideals of $\mathcal{O}_L$ over $\mathfrak{p}$. The inertia degree f_i of $\mathfrak{B}_i$ is the degree of $\overline{m}_i(x)$, and

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_r^{e_r}$$

***Proof*:**
The ring $\mathcal{O}_L$ is the integral closure of the Dedekind domain $\mathcal{O}_K$ in L (corollary 50.3.4, theorem 50.2.6), and L/K is separable as both fields have characteristic 0. We check that $\mathfrak{p}$ is relatively prime to the conductor $\mathscr{F}$ of $\mathcal{O}' = \mathcal{O}_K[\theta]$, and then theorem 50.4.7 gives the conclusion. The index $N = [\mathcal{O}_L : \mathcal{O}']$ is finite: if $\omega_1, \ldots, \omega_d$ is an integral basis of $\mathcal{O}_K$, the products $\omega_j\theta^i$ with $0 \leq i < [L : K]$ are algebraic integers forming a $\mathbb{Q}$-basis of L and spanning $\mathcal{O}'$ over $\mathbb{Z}$, so $\mathcal{O}' \subseteq \mathcal{O}_L \subseteq \frac{1}{D}\mathcal{O}'$ for their discriminant D, which is nonzero (corollary 50.1.28, proposition 50.1.43), and $\mathcal{O}_L/\mathcal{O}'$ is a quotient of the finite group $\mathcal{O}_L/D\mathcal{O}_L$. By [3, Lagrange Theorem], N annihilates $\mathcal{O}_L/\mathcal{O}'$, so $N\mathcal{O}_L \subseteq \mathcal{O}'$, that is $N \in \mathscr{F}$. Since $p \nmid N$ there are $u, v \in \mathbb{Z}$ with $1 = uN + vp$, and $p \in \mathfrak{p}$, so $1 \in \mathscr{F} + \mathfrak{p}\mathcal{O}_L$ and $\mathfrak{p}\mathcal{O}_L + \mathscr{F} = \mathcal{O}_L$, as we sought to show.

Corollary 1.4.9: Monogenic Number Rings

Let L/K be a finite extension of number rings such that $\mathcal{O}_L = \mathcal{O}_K[\alpha]$ for some $\alpha \in \mathcal{O}_L$. Then all primes split as given in the above

***Proof*:**
Choose $\theta = \alpha$. Then $[\mathcal{O}_L : \mathcal{O}_K[\alpha]] = [\mathcal{O}_K[\alpha] : \mathcal{O}_K[\alpha]] = 1$. Since certainly no prime divides 1, we have that all primes factor in given their minimal polynomials and modular factorization, as we sought to show.

Note that not all number rings are of the form $\mathcal{O}_L = \mathcal{O}_K[\alpha]$, that is not all number rings are monogenic. We shall show a counter-example in example 50.4.29. This contrasts to finite extension of field where every separable finite extension of is a simple extension by [3, Primitive Element Theorem]. Note how this is dependent on finding a θ and $\mathfrak{p}$ being relatively prime to $\mathscr{F}_\theta$ (or $[\mathcal{O}_L : \mathcal{O}_K[\theta]]$ in the case of number fields)[23].

As it turns out, the power of an extended prime has important geometric and algebraic consequences. It can be thought of as the measure of how much the Galois group moves a field around. We hence give it an name:

[23]However, if K is a local field, as defined in [5], every extension *is* monogenic. This property, among many other similar niceties, makes developing number theory over local fields much more streamlined. The interested reader should see [5] as a follow up to this chapter

Definition 1.4.10: Ramified and Unramified Extensions

Let $\mathfrak{p} \subseteq R$ be a prime ideal, $K = K(R)$, L/K a finite extension, and $\mathcal{O} = \overline{K}^L$. Then given

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1^{e_1} \cdots \mathcal{B}_r^{e_r}$$

Then the prime ideal $\mathfrak{B}_i$ is called *unramified* over $\mathcal{O}_k$ (or over K) if $e_i = 1$ and if the residue fields $(\mathcal{O}/\mathfrak{B}_i)/(\mathcal{O}_K/\mathfrak{p})$ is separable[a]:

$$\mathfrak{p}\mathcal{O} = \mathfrak{B}_1 \cdots \mathfrak{B}_r$$

Otherwise, $\mathfrak{B}_i$ is called *ramified*. The prime $\mathfrak{B}_i$ is called *totally ramified* if $\mathfrak{p}\mathcal{O} = \mathfrak{B}_i^{[L:K]}$, that is, if $\mathfrak{p}$ is totally ramified in the sense of definition 50.4.2 with $\mathfrak{B}_i$ the prime above it. The prime ideal $\mathfrak{p}$ is called *unramified* if all $\mathfrak{B}_i$ are unramified, and $\mathfrak{p}$ is ramified otherwise. The extension L/K is called *unramified* or an *unramified extension* if all prime ideals $\mathfrak{p}$ of K are unramified.

[a] which for number fields this is always the case

It is in fact rare that a prime ideal $\mathfrak{p}$ of K is ramified in L[24], however every [nontrivial] extension of $\mathbb{Q}$ is ramified (i.e. some primes ramify). Later, we shall see that there do exist [nontrivial] unramified extension L/K where $K \neq \mathbb{Q}$. To understand ramification in extension, we first show how we can factor most $\mathfrak{B}_i$. The term "ramification" comes from complex analysis, where if a holomorphic function f has a root, $f(a) = 0$, the order of the root is the ramification index[25].

Theorem 1.4.11: Counting Ramified Primes

Let R be a Dedekind domain with fraction field K, L/K a finite separable extension of degree n, $\mathcal{O}$ the integral closure of R in L, and $\theta \in \mathcal{O}$ a primitive element of L/K with minimal polynomial $m = m_{\theta,K} \in R[x]$. Let $d = \det(\mathrm{tr}_{L/K}(\theta^{i+j}))_{0 \leq i,j < n}$ and let $\mathscr{F}_\theta$ be the conductor of $R[\theta]$. Then every nonzero prime $\mathfrak{p}$ of R with $d \notin \mathfrak{p}$ and $\mathfrak{p}\mathcal{O} + \mathscr{F}_\theta = \mathcal{O}$ is unramified in L (definition 50.4.10): every ramification index in $\mathfrak{p}\mathcal{O}$ equals 1, and every residue field extension over $R/\mathfrak{p}$ is separable. In particular, only finitely many primes of R ramify in L, and they lie among the primes dividing dR or not relatively prime to $\mathscr{F}_\theta$.

Note that it is still possible that primes factor as given in theorem 50.4.7 since a prime may be relatively prime to $\mathscr{F}_\theta$ but not the discriminant of $m_{K,\theta}(x)$.

***Proof*:**

Step 1: finitely many exceptions. The traces $\mathrm{tr}_{L/K}(\theta^k)$ lie in R (corollary 50.1.14), so $d \in R$, and $d \neq 0$ by corollary 50.1.30, as $1, \theta, \ldots, \theta^{n-1}$ is a K-basis of L. So only the finitely many primes in the factorization of dR contain d (theorem 50.2.15). The conductor $\mathscr{F}_\theta$ (definition 50.4.6) is a nonzero ideal of $\mathcal{O}$: by proposition 50.1.32, $\mathcal{O}$ is generated as an R-module by finitely many $y_1, \ldots, y_s$, each y_i is a K-combination of $1, \theta, \ldots, \theta^{n-1}$, and clearing the denominators of all these coefficients gives $0 \neq c \in R$ with $c\mathcal{O} \subseteq R[\theta]$, that is, $c \in \mathscr{F}_\theta$. A prime $\mathfrak{p}$ with $\mathfrak{p}\mathcal{O} + \mathscr{F}_\theta \neq \mathcal{O}$ lies below a maximal ideal of $\mathcal{O}$ containing $\mathfrak{p}\mathcal{O} + \mathscr{F}_\theta$, which is one of the finitely many primes

[24] This correspond to the algebro-geometric fact that all but finitely many points of curve are *nonsingular*

[25] if $f(a) = 0$ has order n, there exists a g such that $f = g^n$ in some neighbourhood, which shows how much f "winds" around the root

dividing $\mathscr{F}_\theta$, and a prime of $\mathcal{O}$ lies over only one prime of R. This gives the last sentence once the first is shown.

Step 2: the trace form modulo $\mathfrak{p}$. Let $\mathfrak{p}$ be as in the statement, $\kappa = R/\mathfrak{p}$, and $A = \kappa[x]/(\overline{m})$, with κ-basis $1, \overline{x}, \dots, \overline{x}^{n-1}$. The matrix of multiplication by θ on L in the basis $1, \theta, \dots, \theta^{n-1}$ is the companion matrix C of m, with entries in R, and its reduction $\overline{C}$ is the matrix of multiplication by $\overline{x}$ on A. So $\mathrm{tr}_{A/\kappa}(\overline{x}^k) = \mathrm{tr}(\overline{C}^k)$ is the reduction of $\mathrm{tr}_{L/K}(\theta^k) = \mathrm{tr}(C^k)$, and the matrix $T = (\mathrm{tr}_{A/\kappa}(\overline{x}^{i+j}))_{ij}$ has determinant $d + \mathfrak{p} \neq 0$.

Step 3: $\overline{m}$ *has no repeated factor.* Suppose $\overline{m} = g^2 h$ with g nonconstant, and let $\nu \neq 0$ be the class of gh in A. Then $\nu^2 = \overline{m}h = 0$ in A, so for each $y \in A$ the multiplication by νy is a κ-linear map N with $N^2 = 0$. In $M_n(\kappa[x])$, $(xI - N)(xI + N) = x^2 I$, so by [3, Multiplicativity of the Determinant] the characteristic polynomial $\det(xI - N)$, monic of degree n, divides x^{2n} in the unique factorization domain $\kappa[x]$ ([3, Field Then $\mathbb{F}[x]$ PID And UFD]), and so equals x^n. By [3, Properties Of The Characteristic Polynomial], $\mathrm{tr}_{A/\kappa}(\nu y) = 0$. Writing $\nu = \sum_k c_k \overline{x}^k$ with $(c_k) \neq 0$ gives $\sum_k c_k \mathrm{tr}_{A/\kappa}(\overline{x}^{k+j}) = \mathrm{tr}_{A/\kappa}(\nu \overline{x}^j) = 0$ for every j, so (c_k) is a nonzero vector killed by T, and $\det(T) = 0$, a contradiction.

Step 4: the residue fields are separable. By Step 3, $\overline{m} = \overline{m}_1 \cdots \overline{m}_r$ with distinct monic irreducible $\overline{m}_i$, which are pairwise comaximal in the principal ideal domain $\kappa[x]$, so [3, Chinese Remainder Theorem] gives a κ-algebra isomorphism $A \cong E_1 \times \cdots \times E_r$ with $E_i = \kappa[x]/(\overline{m}_i)$, a field. In a basis adapted to this product, multiplication by an element $(y_1, \dots, y_r)$ is block diagonal, so $\mathrm{tr}_{A/\kappa}$ of it is $\sum_i \mathrm{tr}_{E_i/\kappa}(y_i)$. Suppose some E_i, say E_1, is inseparable over κ. Then $\mathrm{tr}_{E_1/\kappa}$ is identically zero (corollary 50.1.30), so the nonzero element $\nu \in A$ corresponding to $(1, 0, \dots, 0)$ has $\mathrm{tr}_{A/\kappa}(\nu y) = \mathrm{tr}_{E_1/\kappa}(y_1) = 0$ for every $y \in A$. As in Step 3, this gives a nonzero vector killed by T, contradicting $\det(T) \neq 0$. So every E_i is separable over κ.

Step 5: conclusion. As $\mathfrak{p}\mathcal{O} + \mathscr{F}_\theta = \mathcal{O}$, theorem 50.4.7 shows that the exponents of the primes $\mathfrak{B}_i$ of $\mathcal{O}$ in $\mathfrak{p}\mathcal{O}$ are the exponents in the factorization of $\overline{m}$ into irreducibles, which are all 1 by Step 3, and that $\mathcal{O}/\mathfrak{B}_i \cong E_i$, which is separable over κ by Step 4. So $\mathfrak{p}$ is unramified, completing the proof.

We can in fact be more precise than the above result: the ramified primes ideals will be given by the *relative discriminant* of $\mathcal{O}_L/\mathcal{O}_K$, which will be given by the ideal $\mathfrak{o} \subseteq \mathcal{O}_K$ which is generated by the discriminants $d(\omega_1, \omega_2, ..., \omega_n)$ for all bases $\omega_1, \omega_2, ..., \omega_n$ of L/K contained in $\mathcal{O}_L$. A special case is that of the base-field being $K = \mathbb{Q}$.

Proposition 1.4.12: Detecting Ramification In Number Fields

Let p be a prime in $\mathbb{Z}$. Suppose p is ramified. Then:

$$p \big| \mathrm{disc}(\mathcal{O}_K)$$

In proposition 50.4.16, we will show the converse, showing this is an iff.

***Proof*:**
Let $\mathfrak{B}$ lie over $\mathfrak{p}$ and $e_{\mathfrak{B}/\mathfrak{p}} > 1$. Thus

$$p\mathcal{O}_K = \mathfrak{B}I$$

where I is divisible by *all* primes of $\mathcal{O}_K$ lying over p. Let $\sigma_1, \sigma_2, ..., \sigma_n$ denote the embeddings of K in $\mathbb{C}$, and extend each σ_i to automorphisms of some extension of L/K which is normal over $\mathbb{Q}$. Let $\alpha_1, \alpha_2, ..., \alpha_n$ be some integral basis for R. What we shall do is replace one of the α_i be a suitable element that will show that $p \big| \text{disc}(\mathcal{O}_K)$.

Take $\alpha \in I - p\mathcal{O}_K$. Then certainly α is in every prime ideal lying over p but *not* in $p\mathcal{O}_K$. writing

$$\alpha = m_1\alpha_1 + \cdots m_n\alpha_n$$

for appropriate $m_i \in \mathbb{Z}$, then since $\alpha \notin p\mathcal{O}_K$, we have that p does not divide each m_i. Re-arrange the terms so that $p \nmid m_1$. Then if d is the discriminant of $\mathcal{O}_K$, through direct computation, we can see that

$$\text{disc}(\alpha, \alpha_2, ..., \alpha_n) = m_1^2 d$$

Since $p \nmid m_1$, we shall show that $p \big| \text{disc}(\alpha, \alpha_2, ..., \alpha_n)$

Given $L/K/\mathbb{Q}$, since α is in every prime of $\mathcal{O}_K$ lying over p, it certainly is in every prime in $\mathcal{O}_L$ lying over p. Fixing any $\mathfrak{B}$ of S lying over p, we see that for any automorphism σ_i, $\sigma(\alpha) \in \mathfrak{B}$. To see this ,$\sigma^{-1}(\alpha)$ is a prime of $\sigma^{-1}(S) = S$ lying over p, hence must contain α. In particular, this gives $\sigma_i(\alpha) \in \mathfrak{B}$ for all i. But then $\mathfrak{B}$ contains $\text{disc}(\alpha, \alpha_2, ..., \alpha_n)$. Since the discriminant is in $\mathbb{Z}$, we get that:

$$\mathfrak{B} \cap \mathbb{Z} = p\mathbb{Z}$$

but then $p \big| \text{disc}(\mathcal{O}_K)$, as we sought to show.

Corollary 1.4.13: Ramification Over Number Fields

Let $\alpha \in R$ so that $K = \mathbb{Q}(\alpha)$ and let f be a monic polynomial over $\mathbb{Z}$ such that $f(\alpha) = 0$. Then if p is a prime such that

$$p \not| \text{Nm} f'(\alpha)$$

Then p is unramified in K

***Proof*:**
This follows from the above an the fact that if α is an algebraic integer and $f \in \mathbb{Z}[x]$ monic such that $f(\alpha) = 0$, then by proposition 50.1.29:

$$\text{disc}(\alpha) \big| \text{Nm}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(f'(\alpha))$$

Using this result, we have an easier verification for ramification of primes of $\mathbb{Z}$ to number fields, since they must divide the discriminant, but the discriminant may only have finitely many factors. In fact, it is easy to generalize this result for extension of number fields

Corollary 1.4.14: Ramification In Number Fields

Let L/K be an extension of number fields Then only finitely many primes of $\mathcal{O}_K$ are ramified in $\mathcal{O}_L$.

***Proof*:**
Let $\mathfrak{p}$ be a prime of $\mathcal{O}_K$ which is ramified in $\mathcal{O}_L$. Then $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$ is ramified in S. Hence there are only finitely many possible primes p that ramify, meaning there are only finitely many possible $\mathfrak{p}$.

The converse is also true, namely

$$p \big| \text{disc}(\mathcal{O}_K) \Leftrightarrow \ p \text{ ramifies in } K$$

The full proof shall be given in greater generality in 50.4.3 where it is generalized for primes $\mathfrak{p} \subseteq \mathcal{O}_K$ being ramified in $\mathcal{O}_L$ for algebraic extension L/K. However, it is instructive for the reader to prove it in the following special case that is particularly important for the study of local fields (see [5]):

Lemma 1.4.15: Partial Converse For Ramification

Let $K = \mathbb{Q}(\alpha)$ where $\alpha \in \mathcal{O}_K$ and $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Then if $p \big| \text{disc}(\mathcal{O}_K)$, p ramifies in K

***Proof*:**
Let $n = [K : \mathbb{Q}]$ and let $m \in \mathbb{Q}[x]$ be the minimal polynomial of α. Since α is a root of a monic $g \in \mathbb{Z}[x]$ and m divides g, every root of m in $\mathbb{C}$ is an algebraic integer. The coefficients of m are, up to sign, elementary symmetric polynomials in these roots, so they are rational algebraic integers ([2, Integral Elements Form Rings]), that is integers by lemma 50.1.5 and theorem 50.1.9(1). So $m \in \mathbb{Z}[x]$. Write its reduction modulo p as

$$\overline{m} = \overline{m}_1^{e_1} \cdots \overline{m}_r^{e_r}$$

with $\overline{m}_1, \ldots, \overline{m}_r$ distinct monic irreducible polynomials in $\mathbb{F}_p[x]$, and let $m_i \in \mathbb{Z}[x]$ be monic lifts of the $\overline{m}_i$. Since $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$, corollary 50.4.8, applied over the base field $\mathbb{Q}$ with $\theta = \alpha$ and this factorization (as in theorem 50.4.7), gives

$$p\mathcal{O}_K = \mathfrak{B}_1^{e_1} \cdots \mathfrak{B}_r^{e_r} \qquad \mathfrak{B}_i = p\mathcal{O}_K + m_i(\alpha)\mathcal{O}_K$$

with distinct prime ideals $\mathfrak{B}_1, \ldots, \mathfrak{B}_r$. We prove the contrapositive: if p is unramified in K, then $p \nmid d_K$. So assume p is unramified, which makes $e_1 = \cdots = e_r = 1$ by definition 50.4.10.

Step 1: membership in $\mathfrak{B}_i$. Consider the ring homomorphism $\varphi_i \colon \mathbb{Z}[x] \to \mathcal{O}_K/\mathfrak{B}_i$, $h \mapsto h(\alpha) + \mathfrak{B}_i$. Its kernel contains p and $m_i(x)$, because $p, m_i(\alpha) \in \mathfrak{B}_i$, and it is proper because $\mathfrak{B}_i \neq \mathcal{O}_K$. The composite $\mathbb{Z}[x] \to \mathbb{F}_p[x] \to \mathbb{F}_p[x]/(\overline{m}_i)$ of reduction modulo p and the quotient map is surjective with kernel $(p, m_i(x))$: if $\overline{h} = \overline{m}_i\overline{u}$ with $u \in \mathbb{Z}[x]$, then $h - m_i u \in p\mathbb{Z}[x]$. The ideal $(\overline{m}_i)$ is maximal in $\mathbb{F}_p[x]$, a principal ideal domain ([3, Field Then $\mathbb{F}[x]$ PID And UFD]) in which the irreducible $\overline{m}_i$ is prime ([3, Prime If And Only If Irreducible In PID]) and nonzero prime ideals are maximal ([3, In PID, Prime Ideal iff Maximal Ideal]). So $\mathbb{Z}[x]/(p, m_i(x)) \cong \mathbb{F}_p[x]/(\overline{m}_i)$ is a field, $(p, m_i(x))$ is a maximal ideal of $\mathbb{Z}[x]$ contained in the proper ideal $\ker \varphi_i$, and $\ker \varphi_i = (p, m_i(x))$.

Hence, for $h \in \mathbb{Z}[x]$,

$$h(\alpha) \in \mathfrak{B}_i \iff h \in (p, m_i(x)) \iff \overline{m}_i \text{ divides } \overline{h} \text{ in } \mathbb{F}_p[x]$$

Step 2: $m'(\alpha)$ lies in no $\mathfrak{B}_i$. Suppose $m'(\alpha) \in \mathfrak{B}_i$. Reduction modulo p commutes with the formal derivative, so Step 1 shows that $\overline{m}_i$ divides $(\overline{m})'$. Since every $e_j = 1$, $\overline{m} = \overline{m}_i g$ with $g = \prod_{j \neq i} \overline{m}_j$, and $(\overline{m})' = \overline{m}_i' g + \overline{m}_i g'$, so $\overline{m}_i$ divides $\overline{m}_i' g$. The polynomial $\overline{m}_i$ is prime in $\mathbb{F}_p[x]$, and it does not divide g: otherwise it would divide some $\overline{m}_j$ with $j \neq i$, and two monic irreducibles dividing one another are equal. So $\overline{m}_i$ divides $\overline{m}_i'$, which has smaller degree, and $\overline{m}_i' = 0$. But $\mathbb{F}_p$ is perfect ([3, Finite Fields Are Perfect]), so the irreducible $\overline{m}_i$ is separable ([3, Perfect Field]) and relatively prime to its derivative ([3, Separable Polynomial Test]), which fails when $\overline{m}_i' = 0$, as $\deg \overline{m}_i \geq 1$. Hence $m'(\alpha) \notin \mathfrak{B}_i$ for every i.

Step 3: p does not divide $\mathrm{Nm}_{K/\mathbb{Q}}(m'(\alpha))$. The element $m'(\alpha)$ is nonzero, since $m' \neq 0$ has degree less than $n = \deg m$. Factor $m'(\alpha)\mathcal{O}_K = \prod_{\mathfrak{q}} \mathfrak{q}^{a_\mathfrak{q}}$ (theorem 50.2.15). By lemma 50.1.63, proposition 50.2.30 and lemma 50.2.29,

$$\left|\mathrm{Nm}_{K/\mathbb{Q}}(m'(\alpha))\right| = \mathfrak{N}(m'(\alpha)\mathcal{O}_K) = \prod_{\mathfrak{q}} \mathfrak{N}(\mathfrak{q})^{a_\mathfrak{q}}$$

where $\mathfrak{N}(\mathfrak{q})$ is a power of the prime q with $\mathfrak{q} \cap \mathbb{Z} = q\mathbb{Z}$. If p divided this product, some $\mathfrak{q}$ with $\mathfrak{q} \cap \mathbb{Z} = p\mathbb{Z}$ would have $a_\mathfrak{q} \geq 1$, that is $m'(\alpha) \in \mathfrak{q}$. Such a $\mathfrak{q}$ divides $p\mathcal{O}_K$ (proposition 50.3.11), so it is one of the $\mathfrak{B}_i$, contradicting Step 2.

Step 4: conclusion. Let $D = \mathrm{disc}_{K/\mathbb{Q}}(1, \alpha, \ldots, \alpha^{n-1})$. By proposition 50.1.29, applied to $K = \mathbb{Q}(\alpha)$, $D = (-1)^{n(n-1)/2}\mathrm{Nm}_{K/\mathbb{Q}}(m'(\alpha))$, so $p \nmid D$ by Step 3. The elements $1, \alpha, \ldots, \alpha^{n-1}$ are algebraic integers forming a $\mathbb{Q}$-basis of K and a $\mathbb{Z}$-basis of $\mathbb{Z}[\alpha]$, so $D\mathcal{O}_K \subseteq \mathbb{Z}[\alpha]$ by proposition 50.1.43, and $\mathbb{Z}[\alpha]$ has finite index in $\mathcal{O}_K$. By proposition 50.1.39, $D = [\mathcal{O}_K : \mathbb{Z}[\alpha]]^2 d_K$, so d_K divides D and $p \nmid d_K$. Hence if $p \mid d_K$, then p ramifies in K, as we sought to show.

For the result when *only* considering primes of $\mathbb{Z}$, the result is easier:

Proposition 1.4.16: Characterizing Ramification In Number Fields

Let p be a prime in $\mathbb{Z}$. Suppose $p \mid \mathrm{disc}(\mathcal{O}_K)$. Then p is ramified.

***Proof*:**
We prove the contrapositive: if p is unramified in K, then $p \nmid d_K$. Let $n = [K : \mathbb{Q}]$, let $\omega_1, \ldots, \omega_n$ be an integral basis of $\mathcal{O}_K$ (corollary 50.1.33), and let $A = \mathcal{O}_K / p\mathcal{O}_K$, with $\overline{u}$ the class of $u \in \mathcal{O}_K$.

Step 1: the trace form of A. Since $\mathcal{O}_K = \bigoplus_i \mathbb{Z}\omega_i$, the classes $\overline{\omega}_1, \ldots, \overline{\omega}_n$ form an $\mathbb{F}_p$-basis of A. For $x \in \mathcal{O}_K$, the matrix M_x of multiplication by x in the basis $\omega_1, \ldots, \omega_n$ has integer entries, because $x\omega_j \in \mathcal{O}_K$, and $\mathrm{tr}_{K/\mathbb{Q}}(x)$ is its trace (definition 50.1.10). The reduction of M_x modulo p is the matrix of the $\mathbb{F}_p$-linear map $A \to A$, $\overline{u} \mapsto \overline{xu}$, in the basis $\overline{\omega}_1, \ldots, \overline{\omega}_n$. Writing $\mathrm{tr}_A(\overline{x})$ for the trace of this map ([3, Trace Of Linear Map]), we get $\mathrm{tr}_A(\overline{x}) = \mathrm{tr}_{K/\mathbb{Q}}(x) \bmod p$. Let b

be the symmetric $\mathbb{F}_p$-bilinear form $b(\overline{u}, \overline{v}) = \mathrm{tr}_A(\overline{uv})$ on A. Its matrix in the basis $\overline{\omega}_1, \ldots, \overline{\omega}_n$ is the reduction modulo p of $(\mathrm{tr}_{K/\mathbb{Q}}(\omega_i\omega_j))$, whose determinant is d_K by proposition 50.1.26 and definition 50.1.38. So $\mathrm{disc}(b) = d_K \bmod p$, and by proposition 50.1.23 the form b is nondegenerate if and only if $p \nmid d_K$.

Step 2: A is a product of finite fields. As p is unramified, $p\mathcal{O}_K = \mathfrak{B}_1 \cdots \mathfrak{B}_g$ with distinct prime ideals $\mathfrak{B}_i$ (definition 50.4.10). These are maximal (theorem 50.2.6), hence pairwise comaximal, and [3, Chinese Remainder Theorem] gives a ring isomorphism

$$\psi\colon A = \mathcal{O}_K/\mathfrak{B}_1 \cdots \mathfrak{B}_g \to F_1 \times \cdots \times F_g \qquad F_i = \mathcal{O}_K/\mathfrak{B}_i$$

which is $\mathbb{F}_p$-linear. Each F_i is a finite field containing $\mathbb{F}_p$ (lemma 50.2.29).

Step 3: b is nondegenerate. Let $\overline{u} \in A$ with $\psi(\overline{u}) = (u_1, \ldots, u_g)$. Multiplication by $(u_1, \ldots, u_g)$ on $F_1 \times \cdots \times F_g$ maps each factor F_i, embedded as the tuples vanishing outside the ith slot, into itself, and acts on it as multiplication by u_i. In an $\mathbb{F}_p$-basis made of $\mathbb{F}_p$-bases of the factors its matrix is therefore block diagonal, with blocks the matrices of multiplication by u_i on F_i. Since ψ carries multiplication by $\overline{u}$ to multiplication by $\psi(\overline{u})$, taking traces gives

$$\mathrm{tr}_A(\overline{u}) = \sum_{i=1}^{g} \mathrm{tr}_{F_i/\mathbb{F}_p}(u_i)$$

with $\mathrm{tr}_{F_i/\mathbb{F}_p}$ the trace of [3, Trace of a Finite Field Extension]. Now suppose $b(\overline{u}, \overline{v}) = 0$ for every $\overline{v} \in A$. Fix i and $y \in F_i$, and let $\overline{v} \in A$ be the element with $\psi(\overline{v}) = (0, \ldots, y, \ldots, 0)$, the entry y in the ith slot. Then $\psi(\overline{uv}) = (0, \ldots, u_i y, \ldots, 0)$, and the display gives $\mathrm{tr}_{F_i/\mathbb{F}_p}(u_i y) = b(\overline{u}, \overline{v}) = 0$. The extension $F_i/\mathbb{F}_p$ is separable because $\mathbb{F}_p$ is perfect ([3, Finite Fields Are Perfect], [3, Perfect Field]), so its trace form is nondegenerate ([3, The Trace Form of a Separable Extension Is Nondegenerate]), and $u_i = 0$. This holds for every i, so $\overline{u} = 0$, and b is nondegenerate.

By Step 1, $p \nmid d_K$. Hence if $p \mid d_K$, then p is ramified, as we sought to show.

1.4.1 Example: Eisenstein Polynomials

We say a monic $f \in \mathbb{Z}[x]$ is *Eisenstein* with respect to p if p divides each coefficient other than the leading one but p^2 does not divide the constant coefficient. Then by Eisenstein's criterion ([3, Eisenstein's Criterion For $\mathbb{Z}[x]$]), such a polynomial is irreducible in $\mathbb{Q}[x]$.

Lemma 1.4.17: Eisenstein Polynomial Modular Roots

Let $K/\mathbb{Q}$ be a number field of degree n and assume $K = \mathbb{Q}(\alpha)$ for $\alpha \in \mathcal{O}_K$ with minimal polynomial $m_{\alpha,\mathbb{Q}}$ Eisenstein at p. Then then for some $f(x) \in \mathbb{Z}[x]$ of degree at most $n-1$ with coefficients $a_0, a_1, ..., a_{n-1}$, if:

$$f(\alpha) \equiv 0 \bmod p\mathcal{O}_K$$

Then:

$$a_i \equiv 0 \bmod p\mathbb{Z} \qquad 1 \leq i \leq n-1$$

***Proof*:**
we shall do an argument by induction. Certainly if $a_0 \equiv 0 \bmod p\mathcal{O}_K$ then $a_0 \equiv 0 \bmod p\mathbb{Z}$ (since $a_0 \in \mathbb{Z}$). Thus, assume for $i < j$, $j \in \{0, 1, ..., n-1\}$ that $a_i \equiv 0 \bmod p\mathbb{Z}$. Then for $i < j$:

$$a_j\alpha^j + a_{j+1}\alpha^{j+1} + \cdots + a_{n-1}\alpha^{n-1} \equiv \bmod p\mathcal{O}_K$$

Multiply the above by α^{n-1-j} to make all but the first term $a_j\alpha^{n-1}$ a multiple of α^n. Since α is a root of an Eisenstein polynomial at p, we have

$$\alpha^n \equiv 0 \bmod p\mathcal{O}_K$$

Hence:

$$a_j\alpha^{n-1} \equiv p\mathcal{O}_K$$

Then $a_j\alpha^{n-1} = p\gamma$, for $\gamma \in \mathcal{O}_K$. Taking the norm, we get:

$$a_j^n \mathrm{Nm}_{K/\mathbb{Q}}(\alpha)^{n-1} = p^n \mathrm{Nm}_{K/\mathbb{Q}}(\gamma) \in \mathbb{Z}$$

the left hand side is divisible by the constant term of the minimal polynomial (since that is the value of $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$). By Eisenstein, p divides $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$ exactly once, so since p^n must divide the left hand side gives that $p \big| a_j^n$, thus $p \big| a_j$, thus $a_i \equiv 0 \bmod p\mathbb{Z}$ for $i < j+1$, giving us the result.

Lemma 1.4.18: Eisenstein Polynomial divisibility by p

Let $K/\mathbb{Q}$ be a number field of degree n, $K = \mathbb{Q}(\alpha)$ where $\alpha \in \mathcal{O}_K$ with minimal polynomial Eisenstein at p. Then if

$$r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1} \in \mathcal{O}_K \qquad r_i \in \mathbb{Q}$$

then each r_i has no p in its denominator

***Proof*:**
For the sake of contradiction, assume some r_i has a p in its denominator. Let d be the lcm of the denominators so the r_i's so $p \big| d$ so that we may write $r_i = a_i/d$ for appropriate $a_i \in \mathbb{Z}$ where

$a_i \nmid p$. Then:

$$r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1} \in \mathcal{O}_K \implies \frac{a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1}}{d} \in \mathcal{O}_K$$

multiplying by d,w e get:

$$a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1} \in d\mathcal{O}_K \subseteq p\mathcal{O}_K$$

Then by lemma 50.4.17 we have that $a_i \in p\mathbb{Z}$ for each i, a contradiction!

Theorem 1.4.19: Eisenstein Polynomial Divisibility Test

Let $K = \mathbb{Q}(\alpha)$ where $\alpha \in \mathcal{O}_K$ a root of an Eisenstain polnomial at p of degree n. Then:

$$p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$$

***Proof*:**
For the sake of contradiction, suppose $p \mid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Then $\mathcal{O}_K/\mathbb{Z}[\alpha]$ has an element of order p, that is there is some $\gamma \in \mathcal{O}_K$ such that $\gamma \notin \mathbb{Z}[\alpha]$ but $p\gamma \in \mathbb{Z}[\alpha]$. Now, given basis $\{1, \alpha, ..., \alpha^{n-1}\}$ for $K/\mathbb{Q}$, then

$$\gamma = r_0 + r_1\alpha + \cdots + r_{n-1}\alpha^{n-1}$$

where $r_i \in \mathbb{Q}$. Since $\gamma \notin \mathbb{Z}[\alpha]<$ not all $r_i \in \mathbb{Z}>$ Since $p\gamma \in \mathbb{Z}[\alpha]$, $pr_i \in \mathbb{Z}$. Thus, r_i has p in its denominator, but that contradicts lemma 50.4.18, as we sought to show.

Example 1.4.20: Using Eisenstein Test

1. Let $K = \mathbb{Q}(\sqrt[3]{2})$. We shall quickly find $\mathcal{O}_K$. Recall that:

   $$-2^2 3^3 = -108 = \text{disc}(\mathbb{Z}[\sqrt[3]{2}]) = [\mathcal{O}_K : \mathbb{Z}[\sqrt[3]{2}]]^2 \text{disc}(\mathcal{O}_K)$$

   Since $\sqrt[3]{2}$ is a root of $x^3 - 2$, which is eisenstien at 2, we see that $2 \nmid [\mathcal{O}_K : \mathbb{Z}[\sqrt[3]{2}]]$. Furthermore, $1 + \sqrt[3]{2}$ is the root of

   $$(x-1)^3 - 2 = x^3 - 3x^2 + 3x - 3$$

   which is eisenstein at 3 s o

   $$3 \nmid [\mathcal{O}_K : \mathbb{Z}[1 + \sqrt[3]{2}]] = [\mathcal{O}_K : \mathbb{Z}[\sqrt[3]{2}]$$

   since $\mathbb{Z}[\sqrt[3]{2}] = \mathbb{Z}[1 + \sqrt[3]{2}]$. THus, the index must be 1, giving us:

   $$\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{2}]$$

2. Let $K = \mathbb{Q}(\sqrt[4]{2})$. Then:

   $$-2^{11} = \text{disc}(\mathbb{Z}[\sqrt[4]{2}]) = [\mathcal{O}_K : \mathbb{Z}[\sqrt[4]{2}]]^2 \text{disc}(\mathcal{O}_K)$$

   and $\sqrt[4]{2}$ is a root of $x^4 - 2$ which is eisenstein at 2, hence by the eisenstein divisibility test the idnex is 1 and

   $$\mathcal{O}_K = \mathbb{Z}[\sqrt[4]{2}]$$

Let $K = \mathbb{Q}(\sqrt[5]{2})$. Then

$$2^4 \cdot 5^5 = \text{disc}(\mathbb{Z}[\sqrt[5]{2}]) = [\mathcal{O}_K : \mathbb{Z}[\sqrt[5]{2}]]^2 \text{disc}(\mathcal{O}_K)$$

Certainly 2 does not divide the index. Next, the minimal polynomial of $\sqrt[5]{2} - 2$ is

$$(x+2)^5 - 2 = x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 30$$

which is Eisenstein at 5 and $\mathbb{Z}[\sqrt[5]{2} - 2] = \mathbb{Z}[\sqrt[5]{2}]$ Hence:

$$\mathcal{O}_K = \mathbb{Z}[\sqrt[5]{2}]$$

3. show that $\mathcal{O}_{\mathbb{Q}(\sqrt[3]{3})} = \mathbb{Z}[\sqrt[3]{3}]$ and $\mathcal{O}_{\mathbb{Q}(\sqrt[3]{5})} = \mathbb{Z}[\sqrt[3]{5}]$.

4. Recall the three cubic extensison K_1, K_2, K_3 in example 50.1.42 with the same discriminant, but it was claimed that they where not isomorphic as fields. The three polynomials in question where:

$$f_1(x) = x^3 - 18x - 6 \qquad f_2(x) = x^3 - 36x - 78 \qquad f_3(x) = x^3 - 54x - 150$$

each having discriminant $22345 = 5 \cdot 41 \cdot 109$. It can be verified that these three polynomials have real roots (by either graphing or doing some guess-work with derivative test and evaluating), and from the above methods we can check that

$$\mathcal{O}_{K_i} = \mathbb{Z}[\alpha_i]$$

for appropriate root α_i. Since the ring of integers is Monogenic, by Dedekind-Kummers theorem the factorization of primes p mirrors the factorization of $f_i(x)$ mod p. Then it can be verified that

   (a) f_1, f_2 are irreducible mod 5, but

$$f_3(x) \equiv x(x-2)(x-3) \bmod 5$$

   (b) f_2, f_3 are irreducible mod 11, but

$$f_1(x) \equiv (x-3)(x-9)(x-10) \bmod 11$$

   Thus, we see that these fields cannot be isomorphic, or else the prime factorization would have been equivalent (recall that ring homomorphisms map primes to primes)

It may be asked if $K = \mathbb{Q}(\sqrt[n]{2})$

$$\mathcal{O}_K = \mathbb{Z}[\sqrt[n]{2}]$$

This is true for $n \leq 1000$, but is *not always true*. For those interested, they may search for *Wieferich primes* (see Keith Conrad *The ring of integers in radical extensions*).

Eisenstein polynomials also allow us to identify when a polynomial totally ramifies:

Theorem 1.4.21: Eisenstein Polynomial And Total Ramification

Let K be a number field of degree n and p a prime number. Then p is totally ramified in K, that is $p\mathcal{O}_K = \mathfrak{B}^n$ for a prime ideal $\mathfrak{B}$ of $\mathcal{O}_K$, if and only if $K = \mathbb{Q}(\alpha)$ for some α that is a root of a polynomial in $\mathbb{Z}[x]$ which is Eisenstein at p.

Note how there is no reference to the ring of integers

***Proof*:**

Step 1: an Eisenstein generator makes p totally ramified. Suppose $K = \mathbb{Q}(\alpha)$ with $f(\alpha) = 0$ for $f = x^m + a_{m-1}x^{m-1} + \cdots + a_0$ in $\mathbb{Z}[x]$ Eisenstein at p. By [3, Eisenstein's Criterion For $\mathbb{Z}[x]$], f is irreducible over $\mathbb{Q}$, so it is the minimal polynomial of α and $m = [\mathbb{Q}(\alpha) : \mathbb{Q}] = n$. As f is monic with integer coefficients, $\alpha \in \mathcal{O}_K$ (definition 50.1.3), and

$$\alpha^n = -\left(a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0\right) \in p\mathcal{O}_K$$

because p divides every a_i. The ideal $p\mathcal{O}_K$ is proper, as $1/p \notin \mathcal{O}_K$ by theorem 50.1.9(1), so it has prime factors $\mathfrak{B}_1, \ldots, \mathfrak{B}_g$ with $g \geq 1$ (theorem 50.2.15). These are the primes $\mathfrak{q}$ of $\mathcal{O}_K$ with $\mathfrak{q} \cap \mathbb{Z} = p\mathbb{Z}$ (proposition 50.3.11). Each $\mathfrak{B}_i$ contains α^n, hence α, being prime.

By proposition 50.1.12 with $M = \mathbb{Q}(\alpha) = K$, $\mathrm{Nm}_{K/\mathbb{Q}}(\alpha)$ is the product of the roots of f, which is $(-1)^n a_0$. So $\mathfrak{N}(\alpha\mathcal{O}_K) = |a_0|$ by lemma 50.1.63, and p divides $|a_0|$ exactly once, since $p \big| a_0$ and $p^2 \nmid a_0$. Factor $\alpha\mathcal{O}_K = \prod_{\mathfrak{q}} \mathfrak{q}^{v_{\mathfrak{q}}}$ (theorem 50.2.15). By proposition 50.2.30 and lemma 50.2.29,

$$|a_0| = \prod_{\mathfrak{q}} \mathfrak{N}(\mathfrak{q})^{v_{\mathfrak{q}}} = \prod_{\mathfrak{q}} q_{\mathfrak{q}}^{f_{\mathfrak{q}} v_{\mathfrak{q}}}$$

where $\mathfrak{q} \cap \mathbb{Z} = q_{\mathfrak{q}}\mathbb{Z}$ and $f_{\mathfrak{q}} = [\mathcal{O}_K/\mathfrak{q} : \mathbb{F}_{q_{\mathfrak{q}}}] \geq 1$. Comparing the exponents of p on both sides gives

$$1 = \sum_{i=1}^{g} f_{\mathfrak{B}_i} v_{\mathfrak{B}_i}$$

Each $v_{\mathfrak{B}_i} \geq 1$, because $\alpha \in \mathfrak{B}_i$ means that $\mathfrak{B}_i$ divides $\alpha\mathcal{O}_K$ (corollary 50.2.18). So $g = 1$ and $f_{\mathfrak{B}_1} = 1$. Writing $p\mathcal{O}_K = \mathfrak{B}_1^e$, the fundamental identity (proposition 50.4.3) gives $e = n$, so $p\mathcal{O}_K = \mathfrak{B}_1^n$.

Step 2: a totally ramified prime gives an Eisenstein generator. Suppose $p\mathcal{O}_K = \mathfrak{B}^n$. By proposition 50.4.3, $n\, f(\mathfrak{B}/p) = n$, so $f(\mathfrak{B}/p) = 1$ and $\mathfrak{N}(\mathfrak{B}) = p$ (lemma 50.2.29). Moreover $\mathfrak{B}$ is the only prime of $\mathcal{O}_K$ containing p (proposition 50.3.11). Since $\mathfrak{B} \neq \mathfrak{B}^2$ (theorem 50.2.15), choose $\pi \in \mathfrak{B} \setminus \mathfrak{B}^2$. Then $\mathfrak{B}$ divides $\pi\mathcal{O}_K$ while $\mathfrak{B}^2$ does not, so $\pi\mathcal{O}_K = \mathfrak{B}\mathfrak{a}$ with $\mathfrak{B} \nmid \mathfrak{a}$. Every prime factor of $\mathfrak{a}$ differs from $\mathfrak{B}$, so it does not contain p, and its absolute norm is a power of a prime different from p (lemma 50.2.29). By lemma 50.1.63 and proposition 50.2.30,

$$\left|\mathrm{Nm}_{K/\mathbb{Q}}(\pi)\right| = \mathfrak{N}(\mathfrak{B})\mathfrak{N}(\mathfrak{a}) = p\,\mathfrak{N}(\mathfrak{a})$$

with $p \nmid \mathfrak{N}(\mathfrak{a})$, so p divides $\mathrm{Nm}_{K/\mathbb{Q}}(\pi)$ exactly once.

Let $\omega_1, \ldots, \omega_n$ be an integral basis (corollary 50.1.33), let $M \in M_n(\mathbb{Z})$ be the matrix of

multiplication by π in this basis, and let $\chi(x) = \det(xI - M) \in \mathbb{Z}[x]$. By [3, Properties Of The Characteristic Polynomial], χ is monic of degree n with constant term $(-1)^n \det(M) = (-1)^n \mathrm{Nm}_{K/\mathbb{Q}}(\pi)$ (definition 50.1.10). The classes of the ω_i form an $\mathbb{F}_p$-basis of $\mathcal{O}_K/p\mathcal{O}_K$, in which the reduction $\overline{M}$ of M modulo p is the matrix of multiplication by the class of π. The determinant is a polynomial in the entries, so the reduction $\overline{\chi}$ of χ is $\det(xI - \overline{M})$, computed in the commutative ring $\mathbb{F}_p[x]$. Since $\pi^n \in \mathfrak{B}^n = p\mathcal{O}_K$, we have $\overline{M}^n = 0$, and the matrices xI and $\overline{M}$ commute, so

$$(xI - \overline{M})\left(x^{n-1}I + x^{n-2}\overline{M} + \cdots + \overline{M}^{n-1}\right) = x^nI - \overline{M}^n = x^nI$$

Taking determinants ([3, Multiplicativity of the Determinant]) shows that $\overline{\chi}$ divides x^{n^2} in $\mathbb{F}_p[x]$. This ring is a unique factorization domain ([3, Field Then $\mathbb{F}[x]$ PID And UFD]) in which x is irreducible, so the monic divisor $\overline{\chi}$ of x^{n^2} is a power of x, and $\overline{\chi} = x^n$ as it has degree n. Hence every coefficient of χ other than the leading one is divisible by p, while the constant term $\pm\mathrm{Nm}_{K/\mathbb{Q}}(\pi)$ is not divisible by p^2. So χ is Eisenstein at p, and irreducible over $\mathbb{Q}$ by [3, Eisenstein's Criterion For $\mathbb{Z}[x]$].

Finally, by [3, Cayley-Hamilton Theorem] the minimal polynomial of M divides χ, so $\chi(M) = 0$. Sending $\beta \in K$ to the matrix of multiplication by β is a ring homomorphism $K \to M_n(\mathbb{Q})$, so $\chi(M)$ is the matrix of multiplication by $\chi(\pi)$, and $\chi(\pi) = \chi(\pi) \cdot 1 = 0$. Thus π is a root of the irreducible polynomial χ of degree n, so $[\mathbb{Q}(\pi) : \mathbb{Q}] = n = [K : \mathbb{Q}]$ and $K = \mathbb{Q}(\pi)$, completing the proof.

Example 1.4.22: Dececting Ramification Using Eisenstein

Since $\alpha = \sqrt[3]{10}$ is a root of $x^3 - 10$, which is eisenstein at 2 and 5, then in $K = \mathbb{Q}(\alpha)$, $(2) = \mathfrak{p}^3$ and $(5) = \mathfrak{q}^3$. Note however htat

$$\mathcal{O}_K \neq \mathbb{Z}[\sqrt[3]{10}]$$

It is in fact equal to

$$\mathcal{O}_K = \mathbb{Z}[\beta] \qquad \beta = \frac{1}{3}\left(1 + \sqrt[3]{1-} + \sqrt[3]{100}\right)$$

which is a root of $x^3 - x^2 - 3x - 3$.

1.4.2 Primes in Galois Extensions

The question of prime decomposition in extension becomes more interesting when L/K is also a Galois extension since $\mathrm{Gal}_K(L)$ can act on the set of primes.

Lemma 1.4.23: Galois Group Acts On Primes

Let L/K be a finite galois extension with galois group $G = \mathrm{Gal}_K(L)$, $\mathfrak{p} \subseteq \mathcal{O}_K$ a prime ideal of $\mathcal{O}_K$, and $\mathfrak{P} = \{\mathfrak{B}_1, \mathfrak{B}_2, ..., \mathfrak{B}_n\}$ the set of primes lying above $\mathfrak{p}$. Then:

$$G \curvearrowright \mathfrak{P} \qquad \text{via} \qquad \sigma \cdot \mathfrak{B}_i = \sigma(\mathfrak{B}_i)$$

***Proof*:**
let $G = \mathrm{Gal}_K(L)$. Then for each $a \in \mathcal{O}_L$, $\tau(a) \in \mathcal{O}_L$ for all $\tau \in G$ since K is invariant under τ and all conjugates have an integral minimal polynomial (i.e. the same minimal polynomial with coefficients in $\mathbb{Z} \subseteq K$), so there is a natural action of G on $\mathcal{O}_L$. Here $G \curvearrowright \mathcal{O}_L$. Next, given that τ is a field isomorphism, $\tau : \mathcal{O}_L \to \mathcal{O}_L$ is a ring isomorphism, importantly it sends primes to primes. If $\mathfrak{B}$ is a prime ideal of $\mathcal{O}_L$ above $\mathfrak{p}$, then $\tau(\mathfrak{B})$ is also a prime ideal above $\mathfrak{p}$ for each prime $\mathfrak{B}$ since

$$(\tau(\mathfrak{B})) \cap \mathcal{O}_K = \tau(\mathfrak{B} \cap \mathcal{O}_K) = \tau(\mathfrak{p}) = \mathfrak{p}$$

since τ is K-invariant, hence $\mathcal{O}_K \subseteq K$ is invariant.

Definition 1.4.24: Prime Ideal Conjugate

Let $\mathcal{O}_L/\mathcal{O}_K$ be an extension of rings of integers where L/K is Galois, and $\mathfrak{B}$ lie above $\mathfrak{p}$. An ideal $\tau(\mathfrak{B})$ is called a *prime ideal conjugate* to $\mathfrak{B}$.

As we saw, most primes split similarly to a minimal polynomial. The following is further parallel between prime numbers and polynomials: just like conjugate elements, the Galois group acts transitively:

Proposition 1.4.25: Galois Group Transitive Action On Ideals

Let $G = \mathrm{Gal}_K(L)$. Then G acts transitively on the set of all ideal $\mathfrak{B}$ of $\mathcal{O}$ lying above $\mathfrak{p}$, that is the prime ideals are all conjugates of each other.

***Proof*:**
Say $\mathfrak{B}, \mathfrak{B}'$ are two primes ideals lying above $\mathfrak{p}$ where, for the sake of contradiction, $\sigma(\mathfrak{B}) \neq \mathfrak{B}'$ for any $\sigma \in G$. By the CRT, there exists an $x \in \mathcal{O}$ such that

$$x \equiv 0 \bmod \mathfrak{B}' \qquad x \equiv 1 \bmod \sigma(\mathfrak{B}) \qquad \forall \sigma \in G$$

Then

$$\mathrm{Nm}_{L/K}(x) = \prod_{\sigma \in G} \sigma(x) \overset{!}{\in} \mathfrak{B}' \cap \mathcal{O}_K = \mathfrak{p}$$

On the other hand, $x \notin \sigma(\mathfrak{B})$ for all $\sigma \in G$, hence $\sigma(x) \notin \mathfrak{B}$ for all $\sigma \in G$ since if $\sigma(x) = b \in \mathfrak{B}$, then $x = \sigma^{-1}(b) = \tau(b)$ for some $\tau \in G$. Thus,

$$\prod_{\sigma \in G} \sigma(x) \overset{!}{\notin} \mathfrak{B} \cap \mathcal{O}_K = \mathfrak{p}$$

Hence, $\prod_{\sigma \in G} \sigma(x)$ is both in and not in $\mathfrak{p}$, a contradiction.

Using the Galois group, we may encode in group-theoretic terms the number of different primes ideals a prime ideal $\mathfrak{p} \subseteq \mathcal{O}_K$ decompose to in $\mathcal{O}_L$. One of the properties of an element being acted on by a [transitive] group is it *stabalizer*:

Definition 1.4.26: Decomposition Group And Field

Let $G = \mathrm{Gal}_K(L)$. Let $\mathfrak{B} \subseteq \mathcal{O}$ be a prime ideal. Then the subgroup

$$D_{\mathfrak{B}/\mathfrak{p}} = D_{\mathfrak{B}} = \{\tau \in G \mid \tau(\mathfrak{B}) = \mathfrak{B}\} = \mathrm{stab}_G(\mathfrak{B})$$

is called the *decomposition group* of $\mathfrak{B}$ over K. The fixed field

$$L^{D_{\mathfrak{B}}} = \{x \in L \mid \tau(x) = x \ \forall \tau \in G_{\mathfrak{B}}\}$$

is called the *decomposition field* of $\mathfrak{B}$ over K, and is often labeled as $L_{\mathfrak{B}}$ or $L_{\mathfrak{B}}$ if unambiguous.

Notice that the less primes that are fixed, the larger the decomposition field is, hence there is a contravariant relation between the size of the group and the extension of the field, just like in Galois theory. Due to transitivity and the introduction of Galois theory, we immediately get many counting arguments. If $\mathfrak{B}$ is a prime ideal lying over $\mathfrak{p}$, and we let σ vary over a set of representatives from the cosets in $G/D_{\mathfrak{B}}$, then by transitivity $\sigma(\mathfrak{B})$ varies over the different primes ideals above $\mathfrak{p}$, each occurring exactly once, hence the number of prime ideals is equal to the index $[G : D_{\mathfrak{B}}]$. For reference, we shall write:

Definition 1.4.27: Prime Counter

Let L/K Then the number of prime numbers is equal to:

$$g_{L/K}(\mathfrak{p}) = [G : D_{\mathfrak{B}}]$$

for any prime $\mathfrak{B}$ over $\mathfrak{p}$. If clear from context, we shall write $g(\mathfrak{p})$.

Thus, if $D_{\mathfrak{B}} = 1$, then no $\sigma \in G$ fixes $\mathfrak{B}$ unless it's the identity and $[G : D_{\mathfrak{B}}] = |G| = [L : K]$, and so by the fundamental identity (proposition 50.4.3):

$$D_{\mathfrak{B}} = 1 \iff L_D = L \iff \mathfrak{p} \text{ totally splits}$$

On the other hand, if $D_{\mathfrak{B}} = G$, then every $\mathfrak{B}$ lying over $\mathfrak{p}$ is fixed by σ, and so there must only be one prime ideal above $\mathfrak{p}$ by transitivity giving:

$$D_{\mathfrak{B}} = G \iff L_D = K \iff \mathfrak{p} \text{ is indecomposable}$$

If $1 \subsetneq D_{\mathfrak{B}} \subsetneq G$, there is ambiguity on how $\mathfrak{p}$ splits. For example, if $[L : K] = 50 = 2 \cdot 5^2$ and $|G/D_{\mathfrak{B}}| = 5$, then either

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^2\mathfrak{B}_2^2\mathfrak{B}_3^2\mathfrak{B}_4^2\mathfrak{B}_5^2 \qquad \text{or} \qquad \mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^5\mathfrak{B}_2^5\mathfrak{B}_3^5\mathfrak{B}_4^5\mathfrak{B}_5^5$$

depending on whether the inertia degree is 5 or 2 respectively. The distinction between the two shall be given with the inertia group in proposition 50.4.37. Choosing a different prime does change the resulting index: if we were to choose a different prime $\mathfrak{B}$ lying over $\mathfrak{p}$, then $D_{\mathfrak{B}}$ is conjugate to $D_{\sigma(\mathfrak{B})}$:

$$D_{\sigma(\mathfrak{B})} = \sigma D_{\mathfrak{B}} \sigma^{-1}$$

In particular:

$$\begin{aligned}\tau \in D_{\sigma(\mathfrak{B})} &\iff \tau\sigma\mathfrak{B} = \sigma\mathfrak{B} \\ &\iff \sigma^{-1}\tau\sigma\mathfrak{B} = \mathfrak{B} \\ &\iff \sigma^{-1}\tau\sigma \in D_{\mathfrak{B}} \\ &\iff \tau \in \sigma D_{\mathfrak{B}}\sigma^{-1}\end{aligned}$$

This should make sense, for by the proceeding theorem, the inertia and ramification of each prime lying ove $\mathfrak{p}$ in a galois extension is equal. At the end of this section, we shall show how to partially recover these results when working over non-Galois extensions. A key result when working with Galois extensions is that each prime has the same behavior in it's decomposition:

Proposition 1.4.28: Galois Extension, Inertia Degree And Ramification Index Independent

Let $\mathfrak{p} \subseteq K$ be a prime ideal, L/K a Galois extension, and

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}_1^{e_i} \cdots \mathfrak{B}_r^{e_r}$$

Then

$$f_1 = \cdots = f_r = f \qquad e_1 = \cdots = e_r = e$$

***Proof*:**
Let $\mathfrak{B} = \mathfrak{B}_1$ and $\mathfrak{B}_i = \sigma_i(\mathfrak{B})$ (which goes through all primes by transitivity). Then the isomorphism $\sigma_i : \mathcal{O}_L \to \mathcal{O}_L$ induces an isomorphism:

$$\mathcal{O}_L/\mathfrak{B} \xrightarrow{\sim} \mathcal{O}_L/\mathfrak{B}_i \qquad a \bmod \mathfrak{B} \mapsto \sigma_i(a) \bmod \sigma_i(\mathfrak{B})$$

But then:

$$f_i = [\mathcal{O}_L/\sigma_i(\mathfrak{B}) : \mathcal{O}_K/\mathfrak{p}] = [\mathcal{O}_L/\mathfrak{B} : \mathcal{O}_K/\mathfrak{p}]$$

Next, since $\sigma_i(\mathfrak{p}\mathcal{O}_L) = \mathfrak{p}\mathcal{O}_L$ (since $\mathfrak{p} \subseteq K$), we get

$$\mathfrak{B}^n \big| \mathfrak{p}\mathcal{O} \quad \iff \quad \sigma_i(\mathfrak{B}^n) \big| \sigma_i(\mathfrak{p}\mathcal{O}) \quad \iff \quad (\sigma_i(\mathfrak{B}))^n \big| \mathfrak{p}\mathcal{O}$$

showing that $e_1 = \cdots = e_r$.

Hence, the prime decomposition of $\mathfrak{p}$ can be written as :

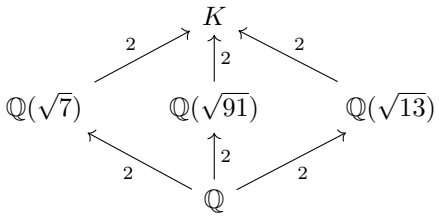
$$\mathfrak{p} = \left(\prod_{\substack{\sigma \in S \\ S \subseteq G/D_{\mathfrak{B}}}} \sigma(\mathfrak{B}) \right)^e$$

where $S \subseteq G/D_{\mathfrak{B}}$ represents a set of representatives of $G/D_{\mathfrak{B}}$ and $|S| = g_{L/K}(\mathfrak{p})$. This information is already very powerful, the following is a use-case of such information:

Example 1.4.29: Biquadratic Extension That Is Not Monogenic

Let $K = \mathbb{Q}(\sqrt{7}, \sqrt{13})$. Without computing $\mathcal{O}_K$, it can be shown that $\mathcal{O}_K$ is *not* monogenic, that is there does not exists $\alpha \in \mathcal{O}_K$ such that $1, \alpha, \alpha^2, \alpha^3$ generates $\mathcal{O}_K$, in other words $\mathcal{O}_K$ is not isomorphic to a quotient of $\mathbb{Z}[x]$. The key is to look at the behavior of factorizations of small primes (for they can only have so many rood $\bmod p$)

What we shall do is consider the factorization of (3) in $\mathcal{O}_K$, which will be our setup for a contradiction. First, notice that K has 3 intermediary fields[a]:



Since we know the ring of integers of quadratics, by direct computation, we see that (3) splits into two primes in each quadratic extension. For example, in $\mathbb{Q}(\sqrt{7})$, we have:

$$\mathcal{O}_{\mathbb{Q}(\sqrt{7})}/3\mathcal{O}_{\mathbb{Q}(\sqrt{7})} = \frac{\mathbb{F}_3[x]}{(x-1)(x+1)}$$

Hence:

$$(3) = (3, 1+\sqrt{7})(3, 1-\sqrt{7})$$

Doing a similar computation for the other two computation, we get:

$$(3) = (3, 1+\sqrt{13})(3, 1-\sqrt{13})$$
$$(3) = (3, 1+\sqrt{91})(3, 1-\sqrt{91})$$

Thus, over K, there must be at least 2 prime ideals, and since

$$g_{K/\mathbb{Q}}(3) = [G : D_{\mathfrak{B}}] \qquad [G : D_{\mathfrak{B}}] \mid |\mathrm{Gal}_{\mathbb{Q}}(K)| \qquad |\mathrm{Gal}_{\mathbb{Q}}(K)| = |(\mathbb{Z}/2\mathbb{Z})^2| = 4$$

for any prime ideal $\mathfrak{B}$ over (3), we get that $g_{K/\mathbb{Q}}(3) \big| 4$. I claim that $g_{K/\mathbb{Q}}(3) = 4$. For, if $g_{K/\mathbb{Q}}(3) = 2$, then for any of the two primes lying over (3) in $\mathcal{O}_K$, say $\mathfrak{B}$:

$$\{e\} \neq D_{\mathfrak{B}} = \{\sigma \in \mathrm{Gal}_{\mathbb{Q}}(K) \mid \sigma(\mathfrak{B}) = \mathfrak{B}\} \subsetneq \mathrm{Gal}_{\mathbb{Q}}(K)$$

Since $D_{\mathfrak{B}}$ non-trivial proper subgroup acting transitively on the two primes of K lying over (3), we have $|D_{\mathfrak{B}}| = 2$, implying there is one non-trivial automorphism $\rho \in \mathrm{Gal}_{\mathbb{Q}}(K)$ fixing the prime ideals (since there are two prime ideals, if $\rho(\mathfrak{B}) = \mathfrak{B}$, then $\rho(\mathfrak{B}') = \mathfrak{B}'$, or else ρ would not be bijective), and two non-trivial automorphisms σ, τ not fixing the primes. Consider now $\sigma \in \mathrm{Gal}_{\mathbb{Q}}(K)$ and take the fixed field $K^{\langle\sigma\rangle}$, which by some Galois theory we see must be one of the 3 quadratic extensions found above. Then by the 3rd fundamental theorem of Galois theory:

$$G := \mathrm{Gal}_{\mathbb{Q}}(K^{\langle\sigma\rangle}) \cong \frac{\mathrm{Gal}_{\mathbb{Q}}(K)}{\mathrm{Gal}_{K^{\langle\sigma\rangle}}(K)}$$

Importantly, since $G \cong \mathbb{Z}/2\mathbb{Z}$ and $[\sigma] = [\mathrm{id}]$, the map $\varphi : \mathrm{Gal}_{\mathbb{Q}}(K) \to \mathrm{Gal}_{\mathbb{Q}}(K^{\langle\sigma\rangle})$ maps $\rho, \tau \mapsto \rho|_{K^{\langle\sigma\rangle}}$. Since ρ is an automorphism and fixes both prime ideals in K over (3) (say $\mathfrak{B}, \mathfrak{B}'$), and all prime ideal of $K^{\langle\sigma\rangle}$ over (3) lie below $\mathfrak{B}, \mathfrak{B}'$:

$$\rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B} \cap K^{\langle\sigma\rangle}) = \rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B}) \cap K^{\langle\sigma\rangle} = \mathfrak{B} \cap K^{\langle\sigma\rangle}$$
$$\rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B}' \cap K^{\langle\sigma\rangle}) = \rho|_{K^{\langle\sigma\rangle}}(\mathfrak{B}') \cap K^{\langle\sigma\rangle} = \mathfrak{B}' \cap K^{\langle\sigma\rangle}$$

Hence, G fixes all primes. But we saw that all intermediate fields of K have two primes, thus this contradicts the transitivity of action of the Galois group on the prime ideals. Thus, it must be that $g_{K/\mathbb{Q}}(3) = 4$.

Now, for the sake of contradiction let's say

$$\mathcal{O}_K \cong \frac{\mathbb{Z}[x]}{m_\alpha(x)}$$

for some $\alpha \in \mathcal{O}_K$ and minimal polynomial $m_\alpha(x)$ of degree 4. Then

$$\mathcal{O}_K/3\mathcal{O}_K \cong \frac{\mathbb{F}_3[x]}{(m_\alpha(x) \bmod 3)} \tag{1.4}$$

By our previous calculations, we see that there must be 4 prime ideals in the right hand side. Decomposing $m_\alpha(x)$ mod 3 into irreducible, by the CRT we get that the right hand side is isomorphic to the direct sum of $\mathbb{F}_3[x]/(m_i(x) \bmod 3)$ for each irreducible $m_i(x)$ mod 3. Since each $m_i(x)$ mod 3 is irreducible, each direct summand is a field, and hence by some ring theory we see that the number of primes on the right hand side of equation 50.4 is equal to the number of direct summands, implying $m_\alpha(x)$ splits into linear factors giving us:

$$\frac{\mathbb{F}_3[x]}{(m_\alpha(x) \bmod 3)} \cong \bigoplus_i^4 \mathbb{F}_3$$

However, this is impossible: any degree 4 polynomial over $\mathbb{F}_3$ that breaks down into linear factors has the form:

$$(x-1)^\alpha (x+1)^\beta x^\gamma$$

where one of α, β, γ is 2 and the other two are 1. But then $\mathbb{F}_3[x]/((x-1)^\alpha(x+1)^\beta x^\gamma)$ is not isomorphic to $\bigoplus_i^4 \mathbb{F}_3$ (the quotient contains nilpotent elements), showing that our assumption that $\mathcal{O}_K$ is monogenic is not possible.

[a] this can be seen by taking the fixed fields $\sqrt{7}, \sqrt{13}$, and $\sqrt{91}$, corresponding to the subgroups of $\mathrm{Gal}_{\mathbb{Q}}(K) = \mathbb{Z}_2 \oplus \mathbb{Z}_2$

Our next goal is to break down extension L/K into intermediary extensions which will "capture" the inertial degree and ramification index, similarly to how any field extension F/k may be broken down into intermediary extensions into separable, inseparable, and transcendental extensions. In our case, we shall get the tower:

$$K \underset{1}{\overset{1}{\to}} L_D \underset{f}{\overset{1}{\to}} L_I \underset{1}{\overset{e}{\to}} L$$

where the ramification index of each field extension is on top, and the inertia degree is on the bottom.

Proposition 1.4.30: Prime Ideals in Decomposition Field

Let L/K be a finite field extension, $\mathfrak{p} \subseteq \mathcal{O}_K$ a prime ideal and $\mathfrak{B} \subseteq \mathcal{O}_L$ lying over $\mathfrak{p}$. Let $\mathfrak{B}_Z = \mathfrak{B} \cap L_D$ be the prime ideal of L_D below $\mathfrak{B}$. If $\mathfrak{B}$ has inertia degree f and ramification index e, then:

1. $\mathfrak{B}_Z$ is nonsplit in L, that is $\mathfrak{B}$ is the only prime ideal of L above $\mathfrak{B}_D$
2. $\mathfrak{B}$ over L_D has a ramification index e and inertia index f
3. The ramification index and inertia index of $\mathfrak{B}_D$ over K are both 1

Visually, we have:

$$\begin{array}{c} \mathfrak{B}^e J \\ {\scriptstyle e}\uparrow{\scriptstyle f} \\ \mathfrak{B}_D I \\ {\scriptstyle 1}\uparrow{\scriptstyle 1} \\ \mathfrak{p} \end{array}$$

where I and J are the remainder of the factorization. Note that the factorization of each prime factor of J lying above $\mathfrak{p}$ has the same e and f since L/K is Galois, but L_D/K is not necessarily Galois, and so the factorization of I may be quite different (for ex. another prime $\mathfrak{B}'_D$ in L_D lying ove $\mathfrak{B}$ may have ramification and inertia degree)

***Proof*:**

1. Notice that $\mathrm{Gal}_{L_\mathfrak{B}}(L) = D_\mathfrak{B}$: the automorphisms of L that fix $L_\mathfrak{B}$ are by definition the elements of $D_\mathfrak{B}$. Hence, the prime ideals above $\mathfrak{B}_D$ are the $\sigma(\mathfrak{B})$ for $\sigma \in \mathrm{Gal}_{L_\mathfrak{B}}(L)$, but $\sigma(\mathfrak{B}) = \mathfrak{B}$ by definition. Hence, $\mathfrak{B}_D$ is nonsplit in L.

2-3. The fundamental identity gives us:

$$n = efr$$

where $n = |\mathrm{Gal}_K(L)| = |G|$ and $r = [G : D_\mathfrak{B}]$. hence,

$$|D_\mathfrak{B}| = [L : L_\mathfrak{B}] = ef \tag{1.5}$$

Now, let e', e'' be the ramification index of $\mathfrak{B}$ over $L_\mathfrak{B}$ and $\mathfrak{B}_D$ over K respectively. Then $\mathfrak{p}$ is:

$$\mathfrak{p}\mathcal{O}_{L_\mathfrak{B}} = \mathfrak{B}_D^{e''} I$$

for some I and

$$\mathfrak{B}_D \mathcal{O}_L = \mathfrak{B}^{e'}$$

in L. Then

$$\mathfrak{p}\mathcal{O}_L = \mathfrak{B}^{e''e'} J$$

for some J. By the tower property, $e = e'e''$ and $f = f'f''$. Then the fundamental identity for the decomposition of $\mathfrak{B}_D$ in L gives us that the degree of L/L_D is:

$$[L : L_\mathfrak{B}] = e'f'$$

in particular, by equation (50.5), we get,

$$e'f' = ef = e'e''f'f''$$

Hence $e' = e$, $f' = f$, and $e'' = f'' = 1$, as we sought to show

This gives us:

$$K \xrightarrow[1]{1} L_{\mathfrak{B}} \xrightarrow[f]{e} L$$

Next, let us see how we can separate out an extension splitting the ramification and the inertia degree into different groups/fields. Since $\sigma(\mathcal{O}_L) = \mathcal{O}_L$ and $\sigma(\mathfrak{B}) = \mathfrak{B}$ for every $\sigma \in D_{\mathfrak{B}}$, there is an induced automorphism on the residue field $\mathcal{O}_L/\mathfrak{B}$:

$$\begin{aligned} \overline{\sigma} : \mathcal{O}_L/\mathfrak{B} &\to \mathcal{O}_L/\mathfrak{B} \\ (a \bmod \mathfrak{B}) &\mapsto (\sigma(a) \bmod \mathfrak{B}) \end{aligned}$$

Letting $\kappa(\mathfrak{B}) = \mathcal{O}_L/\mathfrak{B}$ and $\kappa(\mathfrak{p}) = \mathcal{O}_K/\mathfrak{p}$, we have that

$$f = [\kappa(\mathfrak{B}) : \kappa(\mathfrak{p})]$$

Recall that in Dedekind-Kummer's theorem, when $\mathfrak{p}$ is relatively prime to the conductor, the inertia degree of $\mathfrak{B}$ is given by the degree of $\overline{m_i}(x)$ which within the proof is derived by $\left[R/(\overline{m_i(x)}) : \overline{\mathcal{O}}_K\right]$. This shows that the above map has information about the inertia degree. The following links this quantity to $D_{\mathfrak{B}}$. To understand the result, the reader may see that $\mathrm{Gal}_K(L)$ would naturally surjects onto $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$ via restriction if the restriction was well-defined. The domain of this map can be restricted to the decomposition group to make it well-defined:

Proposition 1.4.31: Decomposition and Inertia Group Relation

The extension $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is normal and admits a surjective homomorphism:

$$D_{\mathfrak{B}} \twoheadrightarrow \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$

where $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) = \mathrm{Gal}_{\mathcal{O}_K/\mathfrak{p}}(\mathcal{O}_L/\mathfrak{B}) \cong \mathrm{Gal}_{\mathbb{F}_p}(\mathbb{F}_q)$.

***Proof*:**

As we are working over finite fields, we know that $\kappa(\mathfrak{B}) = \kappa(\mathfrak{p})(\overline{\theta})$ for some θ with minimal polynomial $\overline{g}(x)$ over $\kappa(\mathfrak{p})$ ($\overline{g}(x) \in \kappa(\mathfrak{p})[x]$). Let $\theta \in \mathcal{O}_L$ be a representative of $\overline{\theta}$, and let $f(x)$ be its minimal polynomial over $\mathcal{O}_K$. Then by definition $\overline{g}(\overline{\theta}) \equiv 0 \bmod \mathfrak{B}$. Since $f(\theta) = 0$, $\overline{f}(\overline{\theta}) \equiv 0 \bmod \mathfrak{B}$. Hence, taking $\overline{f}(x) \in \kappa(\mathfrak{p})[x]$, we have by minimality that:

$$\overline{g}(x) \big| \overline{f}(x)$$

Next, we know that finite field extensions are normal as we saw in [3, Finite Fields], however here is another way of seeing this worth pointing out: since L/K is normal, $f(x)$ splits over $\mathcal{O}_L$ into linear factors. Thus, $\overline{f}(x)$ splits into linear factors over $\kappa(\mathfrak{B})$, and similarly for $\overline{g}(x)$. But then, $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is a normal extension.

Now, consider:

$$\overline{\sigma} \in \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) = \mathrm{Gal}_{\kappa(\mathfrak{p})}\left(\kappa(\mathfrak{p})(\overline{\theta})\right)$$

By [3, Order of Fixed Automorphism Group], these homomorphisms are determined by where they send the root θ. Then $\overline{\sigma}(\overline{\theta})$ is a root of $\overline{g}(x)$ and hence of $\overline{f}(x)$, that is there exists a root θ' of $f(x)$ such that $\theta' \equiv \overline{\sigma}(\overline{\theta}) \bmod \mathfrak{B}$. θ' is a conjugate of θ, i.e. $\theta' = \sigma(\theta)$ for some $\sigma \in D_{\mathfrak{B}}$ (as the automorphism in $D_{\mathfrak{B}}$ fix $\mathfrak{B}$ and make the map well-defined). Since $\sigma(\theta) \equiv \overline{\sigma}(\overline{\theta}) \bmod \mathfrak{B}$, the automorphism σ is mapped by the homomorphism in question to $\overline{\sigma}$, proving surjectivity, as we sought to show.

Definition 1.4.32: Inertia Group And Field

The kernel $I_{\mathfrak{B}} \subseteq D_{\mathfrak{B}}$ of the homomorphism

$$D_{\mathfrak{B}} \to \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$

is called the *inertia group* of $\mathfrak{B}$ over K which is a normal subgroup of $D_{\mathfrak{B}}$ (as it is the kernel of a homomorphism). Equivalently:

$$I_{\mathfrak{B}} = \{\sigma \in \mathrm{Gal}_K(L) \mid \sigma(\alpha) \equiv \alpha \bmod \mathfrak{B}\}$$

The fixed field:

$$L_I = L^{I_{\mathfrak{B}}} = \{x \in L \mid \sigma(x) = x,\ \forall \sigma \in I_{\mathfrak{B}}\}$$

is called the *inertia field* of $\mathfrak{B}$ over K.

We thus get the following tower:

$$K \subseteq L_D \subseteq L_I \subseteq L$$

and short exact sequence:

$$1 \to I_{\mathfrak{B}} \to D_{\mathfrak{B}} \to \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \to 1$$

Corollary 1.4.33: Isomorphism Of Two Galois Groups

Let $L/K/\mathbb{Q}$ be a finite galois extension, and let $\mathfrak{B}$ be a prime lying over $\mathfrak{p} \subseteq \mathcal{O}_K$. Then:

$$\mathrm{Gal}_{L_D}(L_I) \cong \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$

***Proof*:**

Write $D = D_{\mathfrak{B}}$ and $I = I_{\mathfrak{B}}$, so that $L_D = L^D$ and $L_I = L^I$ (definition 50.4.26, definition 50.4.32). By [3, $H = \mathrm{Aut}_{F^H}(F)$], $\mathrm{Gal}_{L_D}(L) = D$ with $|D| = [L : L_D]$, and likewise $\mathrm{Gal}_{L_I}(L) = I$. In particular $|\mathrm{Gal}_{L_D}(L)| = [L : L_D]$, so L/L_D is a Galois extension by [3, Fundamental Theorem of Galois Theory I]. The subgroup $I = \mathrm{Gal}_{L_I}(L)$ is normal in $D = \mathrm{Gal}_{L_D}(L)$, being the kernel of the homomorphism of proposition 50.4.31 (definition 50.4.32). Hence [3, Fundamental Theorem Of Galois Theory III], applied to the Galois extension L/L_D with intermediate field L_I, shows that L_I/L_D is Galois with

$$\mathrm{Gal}_{L_D}(L_I) \cong D/I$$

On the other hand, the homomorphism $D \twoheadrightarrow \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$ of proposition 50.4.31 is surjective with kernel I, so [3, First Isomorphism Theorem] gives $D/I \cong \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$. Composing the two isomorphisms completes the proof.

Corollary 1.4.34: Quotient Group of Decomposition Group

Let L/K be a finite extension. Then the group $D_{\mathfrak{B}}/I_{\mathfrak{B}}$ is cyclic of order f (hence $[D_{\mathfrak{B}} : I_{\mathfrak{B}}] = f$), and the generator $[\varphi] \in D_{\mathfrak{B}}/I_{\mathfrak{B}}$ that generates the group (and hence $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \cong \mathbb{Z}/m\mathbb{Z}$) is called the *Frobenius Element*. The Frobenius element $\varphi \in D_{\mathfrak{B}}$ can be represented in the simple form

$$\varphi(x) = x^{\mathfrak{N}}(\mathfrak{p})$$

***Proof*:**
This is an immediate consequence of the surjectivity of $D_{\mathfrak{B}}$ onto the Galois group of finite field extensions, which is always cyclic.

The order of $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is f, giving the order of the quotient. As the galois group of a finite field is cyclic, it is generated by a Frobenius element. The Frobenius element can be explicitly described, namely

$$\varphi(x) = x^{\mathfrak{N}(\mathfrak{p})} = x^f$$

giving us the rest of the results.

Notice that this means that Frobenius elements and Cyclotomic extensions naturally appear when studying galois extensions. This motivates the study of Cyclotomic extensions in section 50.5.

Corollary 1.4.35: Equivalent Characterization Of Inertia Group

L/K be a finite Galois extension, $I_{\mathfrak{B}} \leqslant D_{\mathfrak{B}}$. Then:

$$I_{\mathfrak{B}} = \{\sigma \in \mathrm{Gal}_K(L) \mid \sigma(\mathfrak{B}) = \mathfrak{B}, \text{ and } \forall \alpha \in \mathcal{O}_L, \sigma(\alpha) - \alpha \in \mathfrak{B}\}$$

***Proof*:**
Let $\sigma \in I_{\mathfrak{B}}$ so that $\overline{\sigma\overline{\alpha}} = \overline{\alpha}$. Then $\sigma(\alpha) = \alpha + b$ for $b \in \mathfrak{B}$. But then:

$$\sigma(\alpha) - \alpha = b \in \mathfrak{B}$$

as we sought to show.

Corollary 1.4.36: Finding Element Of Galois Group

Let K be a number field, $f(x) \in \mathcal{O}_K[x]$ an irreducible polynomial over $\mathcal{O}_K$, L/K the splitting field extension of $f(x)$. Let $\mathfrak{p} \subseteq \mathcal{O}_K$ such that

$$f(x) \bmod \mathfrak{p} \equiv g_1(x) \cdots g_m(x)$$

for distinct irreducibles with degree $\deg(g_i(x)) = d_i$. Then in $\mathrm{Gal}_K(L) \lesssim S_n$, there exists an element α with cycle type

$$(d_1, d_2, ..., d_m)$$

***Proof*:**
Since L is the splitting field, let $\alpha_1, \alpha_2, ..., \alpha_m$ be the roots of f so that

$$f(\alpha_i) = 0$$

Let $\mathfrak{B}$ in L lie above $\mathfrak{p}$. Then:

$$f(\alpha_i) \bmod \mathfrak{p} \equiv 0$$

Since each each g_i are distinct, each $\alpha_i \bmod \mathfrak{p}$ are disticnt. Hence, $D_{\mathfrak{B}}$ acts on the roots of $f \bmod \mathfrak{p}$. Now, taking $g \in D_{\mathfrak{B}}$ that maps to the generator of $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$, we shall get the desired cycle type, completing the proof.

Proposition 1.4.37: Inertia Group Properties

The extension L_I/L_D is normal, and

$$\mathrm{Gal}_{L_D}(L_I) \cong \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B}))$$

and $\mathrm{Gal}_{L_I}(L) = I_{\mathfrak{B}}$. If the residue field extension $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is separable, then:

$$|I_{\mathfrak{B}}| = [L : L_I] = e \qquad [D_{\mathfrak{B}} : I_{\mathfrak{B}}] = [L_I : L_D] = f$$

and we can find the prime ideals $\mathfrak{B}_I$ of L_I bellow $\mathfrak{B}$:

1. The ramification index of $\mathfrak{B}$ over $\mathfrak{B}_I$ is e and the inertia degree is 1
2. The ramification index of $\mathfrak{B}_I$ over $\mathfrak{B}_D$ is 1, and the inertia degree if f

Hence:

$$\mathfrak{B}_Z\mathcal{O}_{L_I} = \mathfrak{B}_T I \qquad \mathfrak{B}_T\mathcal{O}_K = \mathfrak{B}^e J$$

for some appropriate ideals I, J.

***Proof*:**
The fact that:

$$|I_{\mathfrak{B}}| = [L : L_I] = e \qquad [D_{\mathfrak{B}} : I_{\mathfrak{B}}] = [L_I : L_D] = f$$

follow from $|D_{\mathfrak{B}}| = [L : L_D] = ef$ and Galois theory (namely $[L : L_I] = |\mathrm{Gal}_{L_I}(L)| = |I_{\mathfrak{B}}|$), hence we need to show $(1)-(2)$. Using the fundamental identity, it all follows from $\kappa(\mathfrak{B}_I) = \kappa(\mathfrak{B})$. Since the inertia group $I_{\mathfrak{B}}$ of $\mathfrak{B}$ over K is also the inertia group of $\mathfrak{B}$ over L_I, it follows from proposition 50.4.31 applied to L/L_I that $\mathrm{Gal}_{\kappa(\mathfrak{B}_I)}(\kappa(\mathfrak{B})) = 1$, thus

$$\kappa(\mathfrak{B}_I) = \kappa(\mathfrak{B})$$

but then the rest follows.

To sum up the results in a diagram

$$K \xrightarrow[1]{1} L_D \xrightarrow[f]{1} L_I \xrightarrow[1]{e} L$$

with a corresponding [contravariant] diagram of groups:

$$G \leftarrow D_{\mathfrak{B}} \leftarrow I_{\mathfrak{B}} \leftarrow \{e\}$$

where the ramification index of each field extension is on top, and the inertia degree is on the bottom. If the residue field extension $\kappa(\mathfrak{B})/\kappa(\mathfrak{p})$ is separable, then:

$$I_{\mathfrak{B}} = 1 \qquad \iff \qquad L_I = L \qquad \iff \qquad \mathfrak{p} \text{ is unramified in } L$$

and in this case the Galois group $\mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \cong D_{\mathfrak{B}}$, so the galois group of residue field extension can be viewed as a subgroup of $\mathrm{Gal}_K(L)$. This means that the inertia group can be thought of as the "ramification group". In fact, the generalizations of the inertia group (giving more nuanced information on the ramificaion) are called the *higher order ramification group* (which we shall cover soon). The reason the inertia group is not called the 1st ramification group is because the corresponding fixed field only effects the inertia of a prime. Ultimately this is due to the contravariant relation the galois correspondence gives, hence we must make a choice on naming convention depending if fields or groups are prioritised. In this case, fields were prioritised.

From the previous proposition, we may think that L_D and L_I are the *maximal* fields with these properties with respect to $\mathfrak{B}$, and this is indeed the case, making them "natural" with respect to these properties:

Proposition 1.4.38: Intermediary Fields Are Maximal

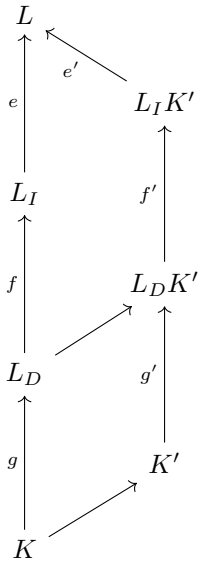
Let L/K be a finite extension of number fields, $\mathfrak{B}$ live over $\mathfrak{p}$. Then:

1. $L_{D_{\mathfrak{B}}}$ is the smallest intermediary field such that $\mathfrak{B}$ is the only prime of $\mathcal{O}_L$ lying over $\mathfrak{B}' = \mathfrak{B} \cap \mathcal{O}_{L_{D_{\mathfrak{B}}}}$
2. $L_{D_{\mathfrak{B}}}$ is the largest intermediate such that $e(\mathfrak{B}_D/\mathfrak{p}) = f(\mathfrak{B}_D/\mathfrak{p}) = 1$
3. $L_{I_{\mathfrak{B}}}$ is the smallest intermediary field such that $\mathfrak{B}$ is totally ramified over $\mathfrak{B}_I$ (that is $e(\mathfrak{B}/\mathfrak{B}_I) = [L : L_I]$)
4. $L_{I_{\mathfrak{B}}}$ is the largest intermediary field such that $e(\mathfrak{B}_I/\mathfrak{p}) = 1$

***Proof*:**
First, L_D and L_I have these properties as we have in proposition 50.4.30 and proposition 50.4.37, hence we prove they are maximal/minimal.

Let $K' = L^H$ for some $H \leqslant \mathrm{Gal}_K(L)$ where only $\mathfrak{B} \subseteq L$ is lying over $\mathfrak{p}' \subseteq K'$. Since H acts transitively on the primes of L over K' but there is only one prime $H \subseteq D_{\mathfrak{B}}$, so $L_D \subseteq K'$, giving (1).

Next, assume $e_{\mathfrak{p}'/\mathfrak{p}} = f_{\mathfrak{p}'/\mathfrak{p}} = 1$. Consider the following diagram:



Then by multiplicativity of towers, $e = e'$ and $f = f'$. Then by the above diagram, L_D and $L_D K'$ voth ahve the same index in L. Since one contains the other, this means they must be equal, hence

$$K' \subseteq L_D$$

giving (2). A similar arguments works for (3) and (4): if $e_{\mathfrak{p}'/\mathfrak{p}} = 1$, then $e = e'$, so $L_I = L_I K'$, so $K' \subseteq L_I$, and if $\mathfrak{B}$ is totally ramified over $\mathfrak{p}'$, then $[L : K'] = e'$, then by the above diagramwe get $K' = L_I K'$, so $L_I \subseteq K'$, as we sought to show.

Ramification

An importance consequence of this is that we may consider primes that are completely ramified/totally unramified in their galois closure. we first require the following result:

Theorem 1.4.39: Composition Field And Primes

Let L/K and M/K be finite extensions of number fields, and let $\mathfrak{p}$ be a prime of K. Then if $\mathfrak{p}$ is unramified in both L and M, then $\mathfrak{p}$ is unramified in LM. If $\mathfrak{p}$ splits completely in L and M, then $\mathfrak{p}$ splits completely in LM

***Proof*:**
Let $\mathfrak{p}$ be unramified in L and M, and take $\mathfrak{B}$ to be any prime in LM lying over $\mathfrak{p}$. We want to show $e(\mathfrak{B}/\mathfrak{p}) = 1$. Let F be the normal closure of K containing LM, and let $\mathfrak{A}$ be any prime in F lying over $\mathfrak{B}$, namely it lies over $\mathfrak{p}$. Let $I_{\mathfrak{A}/\mathfrak{p}}$ be the inertial group corresponding to the field F_I. Then we see that F_I contains both L and M since the primes $\mathfrak{A} \cap L$ and $\mathfrak{A} \cap M$ are necessarily unramified over $\mathfrak{p}$. Hence, F_I contains LM, implying that $\mathfrak{A} \cap LM$ is unramified over

$\mathfrak{p}$.

The argument for splitting follows exactly the same pattern, replacing $I_{\mathfrak{B}}$ with $D_{\mathfrak{B}}$.

Corollary 1.4.40: Primes In Normal Closure

Let L/K be a finite extension of number fields and $\mathfrak{p}$ a prime in K. Then:

1. If $\mathfrak{p}$ is unramified in L, then it is unramified in the normal closure of M of L over K
2. If $\mathfrak{p}$ splits completely in L, then it splits completely in the normal closure of M of L over K

***Proof*:**
If $\mathfrak{p}$ is unramified in L, then it is unramified in every $\sigma_i(L)$ where i runs over all the automorphism. But since $M = L\sigma_1(L)\cdots\sigma_n(L)$, we use induction on theorem 50.4.39 to get our desired result (similarly for splitting completely).

Example 1.4.41: More Decomposition Of Primes

Let $K = \mathbb{Q}(i, \sqrt{2}, \sqrt{5})$, which is abelian of degree 8. Take the prime $5 \in \mathbb{Z}$. Then it splits into tow primes in $\mathbb{Q}(i)$, stays inert in $\mathbb{Q}(\sqrt{2})$, and becomes a square in $\mathbb{Q}(\sqrt{5})$. Hence in L 5 must have ramification index at least 2, and split into at least 2 primes. Since the inertia field must be a field of degree 4 over $\mathbb{Q}$ in which 5 is unramified, the only choice is $\mathbb{Q}(i, \sqrt{2})$. Now, $(2 + i)$ and $(2 - i)$ remain prime in $\mathbb{Q}(\sqrt{i}, \sqrt{2})$, and become squares in L, hence 5 must have exactly 2 primes and ramification index 2.

Since normal subgroups over characteristic 0 fields are galois, it is meaningful to ask the behavior of normal subgroups of the galois subgroups:

Proposition 1.4.42: Decomposition Subgroup Is Normal

Suppose $D_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$. Then $\mathfrak{p}\mathcal{O}_L$ splits into r distinct primes in L_D. If $I_{\mathfrak{B}} \trianglelefteq G$, then each prime remains as primes in L_I (i.e. they are inert). Furthermore, each become an eth power in L.

In general, we can only hope for:

$$\mathfrak{B}_D\mathcal{O}_{L_I} = \mathfrak{B}_I I \qquad \mathfrak{B}_I\mathcal{O}_K = \mathfrak{B}^e J$$

for some ideals, while in the normal case of D and I, we get:

$$\begin{aligned}\mathfrak{p}\mathcal{O}_{L_D} &= \mathfrak{B}_{L_1}\cdots\mathfrak{B}_{L_e}\\ \mathfrak{p}\mathcal{O}_{L_I} &= \mathfrak{B}_{L_1}\cdots\mathfrak{B}_{L_e}\\ \mathfrak{p}\mathcal{O}_L &= \mathfrak{B}_1^e\cdots\mathfrak{B}_g^e\end{aligned}$$

which also shows better why the groups $D_{\mathfrak{B}}$ and $I_{\mathfrak{B}}$ are called the "decomposition" and "inertia" groups respectively.

***Proof*:**
This is taking advantage of the fact that $L^{D_{\mathfrak{B}}}/K$ would now be a Galois extension since $D_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$. Then we know that

$$f(\mathfrak{B}_D) = e(\mathfrak{B}_D) = 1$$

and that (since $L^{D_{\mathfrak{B}}}$ is Galois), each other prime must also have the same ramification and inertia degree. Thus, $\mathfrak{p}$ splits into exactly g distinct primes in $L^{D_{\mathfrak{B}}}$. Then by a simple counting argument, we see that each $\mathfrak{B}_D$ must lie under a unique $\mathfrak{B}_I$. It may be that some of these ramify. However, if $I_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$, then we have that each prime must have ramification index 1, showing they are all inert. Finally, a counting argument gives that each $\mathfrak{B}_I$ becomes an eth power in L

Corollary 1.4.43: Normal Subgroup And Completely Split Prime

Let $D_{\mathfrak{B}} \trianglelefteq \mathrm{Gal}_K(L)$ and let K' be an intermediate field of L/K. Then $\mathfrak{p}$ splits completely in K' if and only if $K' \subseteq L_D$

***Proof*:**
Suppose first that $\mathfrak{p}$ splits completely in K'. Then every prime of K' above $\mathfrak{p}$ has ramification index and inertia degree 1 over $\mathfrak{p}$, in particular the prime $\mathfrak{B} \cap K'$. By proposition 50.4.38(2), L_D is the largest intermediate field whose prime below $\mathfrak{B}$ has this property, so $K' \subseteq L_D$. This direction does not use the normality of $D_{\mathfrak{B}}$.

Conversely, suppose $K' \subseteq L_D$. We first check that $\mathfrak{p}$ splits completely in L_D. Since $L_D = L^{D_{\mathfrak{B}}}$, [3, $H = \mathrm{Aut}_{F^H}(F)$] gives $\mathrm{Gal}_{L_D}(L) = D_{\mathfrak{B}}$, which is normal in $\mathrm{Gal}_K(L)$ by hypothesis, so L_D/K is Galois by [3, Fundamental Theorem Of Galois Theory III]. By proposition 50.4.28, all primes in the factorization of $\mathfrak{p}\mathcal{O}_{L_D}$ have the same ramification index and the same inertia degree, and by proposition 50.4.30(3) the prime $\mathfrak{B}_D = \mathfrak{B} \cap L_D$ has both equal to 1. The fundamental identity (proposition 50.4.3) then forces the number of primes above $\mathfrak{p}$ in L_D to be $[L_D : K]$, so $\mathfrak{p}$ splits completely in L_D (definition 50.4.2).

Now let $\mathfrak{q}$ be any prime of K' above $\mathfrak{p}$. By proposition 50.3.10, $\mathfrak{q}\mathcal{O}_{L_D} \neq \mathcal{O}_{L_D}$, so some prime $\mathfrak{q}'$ of L_D divides $\mathfrak{q}\mathcal{O}_{L_D}$, and $\mathfrak{q}' \supseteq \mathfrak{q} \supseteq \mathfrak{p}$ shows that $\mathfrak{q}'$ lies above both $\mathfrak{q}$ and $\mathfrak{p}$ (proposition 50.3.11). Extending the factorization of $\mathfrak{p}\mathcal{O}_{K'}$ to $\mathcal{O}_{L_D}$ and factoring $\mathfrak{q}\mathcal{O}_{L_D}$ gives

$$e(\mathfrak{q}'/\mathfrak{p}) = e(\mathfrak{q}'/\mathfrak{q})\, e(\mathfrak{q}/\mathfrak{p})$$

since distinct primes of K' contribute disjoint sets of prime factors (corollary 50.3.12). Both factors are at least 1 and $e(\mathfrak{q}'/\mathfrak{p}) = 1$, so $e(\mathfrak{q}/\mathfrak{p}) = 1$. Similarly the residue field inclusions $\mathcal{O}_K/\mathfrak{p} \subseteq \mathcal{O}_{K'}/\mathfrak{q} \subseteq \mathcal{O}_{L_D}/\mathfrak{q}'$ give $f(\mathfrak{q}'/\mathfrak{p}) = f(\mathfrak{q}'/\mathfrak{q})f(\mathfrak{q}/\mathfrak{p})$ by [3, Multiplicity of Finite Extensions (Tower Law)], so $f(\mathfrak{q}/\mathfrak{p}) = 1$. Every prime of K' above $\mathfrak{p}$ therefore has ramification index and inertia degree 1, and the fundamental identity (proposition 50.4.3) makes their number $[K' : K]$. Thus $\mathfrak{p}$ splits completely in K', as we sought to show.

In the non-Galois case L/K, take $M/L/K$ where M/K is the Galois closure of L. Then M/L is

Galois. Let $G = \mathrm{Gal}_K(M)$ and $H = \mathrm{Gal}_L(M)$ so that $H \leqslant G$. Take the following tower of primes:

$$\begin{array}{ccc} M & \longrightarrow & \mathfrak{A} \\ \uparrow & & \uparrow \\ L & \longrightarrow & \mathfrak{B} \\ \uparrow & & \uparrow \\ K & \longrightarrow & \mathfrak{p} \end{array}$$

Then by definition:

$$D_{\mathfrak{A}/\mathfrak{p}} \cap H = D_{\mathfrak{A}/\mathfrak{B}}$$
$$I_{\mathfrak{A}/\mathfrak{p}} \cap H = I_{\mathfrak{A}/\mathfrak{B}}$$

Proposition 1.4.44: Prime Decomposition Over Finite Extension

Let

$$\mathfrak{p}_M = \prod_1^{g_{M/K}} \mathfrak{A}_i^{e_i}$$

Let $\mathfrak{O}_1, \mathfrak{O}_2, ..., \mathfrak{O}_{g_{L/K}}$ be the orbits of H acting on the primes $\{\mathfrak{A}_1, \mathfrak{A}_2, ..., \mathfrak{A}_{g_{M/K}}\}$. Then:

$$\mathfrak{p}_L = \prod_i^{g_{L/K}} \mathfrak{B}_i^{e_{\mathfrak{B}_i/\mathfrak{p}}}$$

where each $\mathfrak{B}_i$ comes from an $\mathfrak{O}_i$. Furthermore:

$$e_{\mathfrak{B}_i/\mathfrak{p}} = \frac{|I_{\mathfrak{A}_i/\mathfrak{p}}|}{|\mathrm{stab}_H(\mathfrak{A}_i) \cap I_{\mathfrak{A}_i/\mathfrak{p}}|} \qquad f_{\mathfrak{B}_i/\mathfrak{p}} = \frac{[D_{\mathfrak{A}_i/\mathfrak{p}} : I_{\mathfrak{A}_i/\mathfrak{p}}]}{[\mathrm{stab}_H(\mathfrak{A}_i) : \mathrm{stab}_H(\mathfrak{A}_i) \cap I_{\mathfrak{A}_i/\mathfrak{p}}]}$$

***Proof*:**

Throughout, the setup is the one fixed above: M/K is Galois with group G, $H = \mathrm{Gal}_L(M)$, and M/L is Galois with group H.

First, the primes of L above $\mathfrak{p}$ correspond to the orbits $\mathfrak{O}_1, \ldots, \mathfrak{O}_{g_{L/K}}$. Each $\mathfrak{A}_j$ contracts to the prime $\mathfrak{A}_j \cap L$ of L, which lies above $\mathfrak{p}$. If $\eta \in H$, then η fixes L pointwise, so $\eta(\mathfrak{A}_j) \cap L = \mathfrak{A}_j \cap L$: primes in the same orbit have the same contraction. Conversely, if $\mathfrak{A}_j \cap L = \mathfrak{A}_k \cap L = \mathfrak{B}$, then $\mathfrak{A}_j$ and $\mathfrak{A}_k$ both divide $\mathfrak{B}\mathcal{O}_M$ (proposition 50.3.11), so proposition 50.4.25, applied to the Galois extension M/L, provides $\eta \in H$ with $\eta(\mathfrak{A}_j) = \mathfrak{A}_k$: equal contractions force the same orbit. Finally, every prime $\mathfrak{B}$ of L above $\mathfrak{p}$ arises this way: $\mathfrak{B}\mathcal{O}_M \neq \mathcal{O}_M$ by proposition 50.3.10, and any prime factor of $\mathfrak{B}\mathcal{O}_M$ lies above $\mathfrak{p}$ and contracts to $\mathfrak{B}$ (proposition 50.3.11). Hence $\mathfrak{A}_j \mapsto \mathfrak{A}_j \cap L$ induces a bijection from the orbits onto the primes of L above $\mathfrak{p}$, and writing $\mathfrak{B}_i$ for the common contraction of the orbit $\mathfrak{O}_i$, the factorization

$$\mathfrak{p}_L = \prod_i^{g_{L/K}} \mathfrak{B}_i^{e_{\mathfrak{B}_i/\mathfrak{p}}}$$

holds by the definition of the ramification index.

For the formulas, fix i, write $\mathfrak{A} = \mathfrak{A}_i$ and $\mathfrak{B} = \mathfrak{B}_i$, and abbreviate $D = D_{\mathfrak{A}/\mathfrak{p}}$ and $I = I_{\mathfrak{A}/\mathfrak{p}}$. By definition 50.4.26 and definition 50.4.32 applied in M/L,

$$D_{\mathfrak{A}/\mathfrak{B}} = D \cap H = \mathrm{stab}_H(\mathfrak{A}) \qquad I_{\mathfrak{A}/\mathfrak{B}} = I \cap H = \mathrm{stab}_H(\mathfrak{A}) \cap I$$

where the last equality holds because $I \subseteq D$. The residue fields are finite, so the separability hypothesis of proposition 50.4.37 holds, and applying that proposition in M/K and in M/L gives

$$e_{\mathfrak{A}/\mathfrak{p}} = |I| \qquad f_{\mathfrak{A}/\mathfrak{p}} = [D : I] \qquad e_{\mathfrak{A}/\mathfrak{B}} = |\mathrm{stab}_H(\mathfrak{A}) \cap I| \qquad f_{\mathfrak{A}/\mathfrak{B}} = [\mathrm{stab}_H(\mathfrak{A}) : \mathrm{stab}_H(\mathfrak{A}) \cap I]$$

Extending the factorization of $\mathfrak{p}\mathcal{O}_L$ to $\mathcal{O}_M$ and factoring $\mathfrak{B}\mathcal{O}_M$ gives $e_{\mathfrak{A}/\mathfrak{p}} = e_{\mathfrak{A}/\mathfrak{B}}\, e_{\mathfrak{B}/\mathfrak{p}}$, since distinct primes of L contribute disjoint sets of prime factors (corollary 50.3.12). The residue field inclusions $\mathcal{O}_K/\mathfrak{p} \subseteq \mathcal{O}_L/\mathfrak{B} \subseteq \mathcal{O}_M/\mathfrak{A}$ give $f_{\mathfrak{A}/\mathfrak{p}} = f_{\mathfrak{A}/\mathfrak{B}}\, f_{\mathfrak{B}/\mathfrak{p}}$ by [3, Multiplicity of Finite Extensions (Tower Law)]. Dividing the two displayed computations of e and of f yields

$$e_{\mathfrak{B}/\mathfrak{p}} = \frac{|I|}{|\mathrm{stab}_H(\mathfrak{A}) \cap I|} \qquad f_{\mathfrak{B}/\mathfrak{p}} = \frac{[D : I]}{[\mathrm{stab}_H(\mathfrak{A}) : \mathrm{stab}_H(\mathfrak{A}) \cap I]}$$

as we sought to show.

It is also enlightening to have a second perspective here: for subgroup $U, V \leqslant G$, we may consider the equivalence class $\sigma \sim \sigma' \iff \sigma' = u\sigma v$ giving us the double-coset[26]:

$$U\sigma V = \{u\sigma v \mid u \in U, v \in V\}$$

This partition will give us the similar information in the non-Galois case. Let $N/L/K$ be two separable extensions where N/K is Galois with Galois group G. Let $H = \mathrm{Gal}_L(N) \leqslant \mathrm{Gal}_K(N)$. Let $\mathfrak{p}$ be a prime ideal of K and $P_{\mathfrak{p}}$ the set of all prime ideal of L above $\mathfrak{p}$. Then if $\mathfrak{B}$ is a prime ideal of N above $\mathfrak{p}$, then we get a well-defined bijection:

$$H \setminus G / D_{\mathfrak{B}} \to P_{\mathfrak{p}} \qquad H\sigma D_{\mathfrak{B}} \mapsto \sigma\mathfrak{B} \cap L$$

Exercise 1.4.1

1. Let L/K be a galois extension of Number fields, $G = \mathrm{Gal}_K(L)$, $\mathfrak{B}$ lie over $\mathfrak{p}$.
   a) If $\mathfrak{B}$ is inert, then G Is cyclic
   b) If $\mathfrak{p}$ is totally ramified in each intermediary field but not totally ramified in L, then there does not exist non-trivial proper intermediary field, hence G is cyclic (of prime order)
   c) Simiarly, Suppose every intermediate field contains a unique prime lying over $\mathfrak{p}$, but L does not. Show the same as the previous
   d) Let $\mathfrak{p}$ be unramified in every proper non-trivial intermediate field, but ramified in L. Show that G has unique smallest subgroup H and that $H \trianglelefteq G$. Use this to show that $|G| = p^k$ for some k and prime p, that $|H| = p^l$ for some l, and $H \leqslant Z(G)$.
   e) Similarly, show the same for when $\mathfrak{p}$ splits compeltely in every proper non-trivial intermediate field, but not in L.

[26] warning: double cosets need not all have the same size!

f) Let $\mathfrak{p}$ be inert in every proper non-trivial intermediate field but not in L. Prove that G is a cyclic where $|G| = p^k$.

2. (the non-normal case for Frobenius element by taking normal closure, p.78 Marcus)

1.4.3 Detecting Ramification

1.4.45 AKLB setup For generality, let A be a Dedekind domain, K the field of fraction of A, L a finite separable extension of K, and B the integral closure of A in L.

Furthermore, let κ be the residue field of A and λ be the residue field for L. Assume λ/κ and B/A are separable extensions.

Definition 1.4.46: Trace Dual of Fractional Ideal

Take AKLB and let $\mathfrak{P} \subseteq B$ be a fractional ideal. Then the *trace dual* of $\mathfrak{P}$ is:

$$^*\mathfrak{P} = \{x \in L \mid \mathrm{tr}(x\mathfrak{P}) \subseteq A\}$$

By proposition 50.1.32, $^*\mathfrak{P}$ is a finitely generated B-module, and hence by definition a fractional ideal. Note that as the trace-pairing is non-degenerate:

$$^*\mathfrak{P} \xrightarrow{\cong} \mathrm{Hom}_A(\mathfrak{P}, A) \qquad x \mapsto (y \mapsto \mathrm{tr}(xy))$$

which works even when $\mathfrak{P} = (1)_B$, that is:

$$^*B \cong \mathrm{Hom}_A(B, A)$$

This particular trace dual shall be very relevant in detecting ramification:

Definition 1.4.47: Different

Assume AKLB. The fractional ideal

$$\mathfrak{C}_{B/A} = \{x \in L \mid \mathrm{tr}(xB) \subseteq A\} = {}^*B$$

is called *Dedekind's complementary module*, or the *inverse different*. Its inverse

$$\mathfrak{D}_{B/A} := \mathfrak{C}_{B/A}^{-1}$$

is called the *Different* of B/A

Since $\mathfrak{C}_{B/A} \supseteq \mathcal{O}$, $\mathfrak{D}_{B/A} \subseteq \mathcal{O}$ is an integral ideal of L. Naturally, we will often denote the different as $\mathfrak{D}_{L/K}$ and $\mathfrak{C}_{B/A}$ as $\mathfrak{C}_{L/K}$ given the rings B an A are evident from context.

Proposition 1.4.48: Tower Law For Different

Assume $AKLB$ and let M/L be a finite separable extension. Then

1. If $M/L/K$, then $\mathfrak{D}_{M/K} = \mathfrak{D}_{M/L}\mathfrak{D}_{L/K}$
2. If $S \subseteq \mathfrak{o}$ is a multiplicative subset with $0 \notin S$, then $\mathfrak{D}_{S^{-1}\mathcal{O}/S^{-1}\mathfrak{o}} = S^{-1}\mathfrak{D}_{\mathcal{O}/\mathfrak{o}}$

***Proof*:**

1. Write $\mathcal{O}_K = A$, $\mathcal{O}_L = B$ and $\mathcal{O}_M$ for the integral closure of A in M, which is also the integral closure of $\mathcal{O}_L$ in M ([2, Transitivity of Integrality]). Write $\mathfrak{C}_{M/K}$, $\mathfrak{C}_{M/L}$ and $\mathfrak{C}_{L/K}$ for the complementary modules of $\mathcal{O}_M/\mathcal{O}_K$, $\mathcal{O}_M/\mathcal{O}_L$ and $\mathcal{O}_L/\mathcal{O}_K$, so that $\mathfrak{D}_{L/K}$ in the statement stands for its extension $\mathfrak{D}_{L/K}\mathcal{O}_M$. Let $x \in M$ and put $T = \mathrm{tr}_{M/L}(x\mathcal{O}_M)$, an $\mathcal{O}_L$-submodule of L because $\mathrm{tr}_{M/L}$ is L-linear. The extensions M/L and L/K are separable, by hypothesis and by the AKLB setup, so proposition 50.1.15 gives $\mathrm{tr}_{M/K} = \mathrm{tr}_{L/K} \circ \mathrm{tr}_{M/L}$, and
$$x \in \mathfrak{C}_{M/K} \iff \mathrm{tr}_{L/K}(T) \subseteq \mathcal{O}_K \iff T \subseteq \mathfrak{C}_{L/K}$$
where the second equivalence holds because $T\mathcal{O}_L = T$. Since $\mathfrak{C}_{L/K}\mathfrak{D}_{L/K} = \mathcal{O}_L$ (theorem 50.2.22), $T \subseteq \mathfrak{C}_{L/K}$ holds exactly when $\mathfrak{D}_{L/K}T \subseteq \mathcal{O}_L$: multiply by $\mathfrak{D}_{L/K}$ for one direction and by $\mathfrak{C}_{L/K}$ for the other. As $\mathrm{tr}_{M/L}$ is L-linear, $\mathfrak{D}_{L/K}T = \mathrm{tr}_{M/L}(x\mathfrak{D}_{L/K}\mathcal{O}_M)$, and since $x\mathfrak{D}_{L/K}\mathcal{O}_M$ is an $\mathcal{O}_M$-module, the condition $\mathrm{tr}_{M/L}(x\mathfrak{D}_{L/K}\mathcal{O}_M) \subseteq \mathcal{O}_L$ says that $x\mathfrak{D}_{L/K}\mathcal{O}_M \subseteq \mathfrak{C}_{M/L}$. Extension of ideals is multiplicative, both sides being generated by the same products, so $(\mathfrak{D}_{L/K}\mathcal{O}_M)(\mathfrak{C}_{L/K}\mathcal{O}_M) = (\mathfrak{D}_{L/K}\mathfrak{C}_{L/K})\mathcal{O}_M = \mathcal{O}_L\mathcal{O}_M = \mathcal{O}_M$. Multiplying by $\mathfrak{C}_{L/K}\mathcal{O}_M$ for one direction and by $\mathfrak{D}_{L/K}\mathcal{O}_M$ for the other,
$$x\mathfrak{D}_{L/K}\mathcal{O}_M \subseteq \mathfrak{C}_{M/L} \iff x \in \mathfrak{C}_{M/L}\mathfrak{C}_{L/K}\mathcal{O}_M$$
Altogether $\mathfrak{C}_{M/K} = \mathfrak{C}_{M/L}\mathfrak{C}_{L/K}\mathcal{O}_M$, and taking inverses in the group of fractional ideals of $\mathcal{O}_M$ (theorem 50.2.22) gives $\mathfrak{D}_{M/K} = \mathfrak{D}_{M/L}\mathfrak{D}_{L/K}\mathcal{O}_M$.

2. Let $S \subseteq \mathfrak{o}$ be multiplicative with $0 \notin S$. If $x \in L$ satisfies $x^d + (a_1/s_1)x^{d-1} + \cdots + a_d/s_d = 0$ with $a_i \in \mathfrak{o}$ and $s_i \in S$, then multiplying by s^d, where $s = s_1 \cdots s_d$, shows that sx is integral over $\mathfrak{o}$, so $sx \in \mathcal{O}$ and $x \in S^{-1}\mathcal{O}$. Conversely, if $b \in \mathcal{O}$ satisfies a monic equation of degree d over $\mathfrak{o}$, dividing it by s^d shows that b/s is integral over $S^{-1}\mathfrak{o}$ for every $s \in S$. So $S^{-1}\mathcal{O}$ is the integral closure of $S^{-1}\mathfrak{o}$ in L.

   Next, $\mathfrak{C}_{S^{-1}\mathcal{O}/S^{-1}\mathfrak{o}} = S^{-1}\mathfrak{C}_{\mathcal{O}/\mathfrak{o}}$. If $x \in \mathfrak{C}_{\mathcal{O}/\mathfrak{o}}$, then $\mathrm{tr}_{L/K}((x/s)(b/t)) = \mathrm{tr}_{L/K}(xb)/(st)$ lies in $S^{-1}\mathfrak{o}$ for all $s, t \in S$ and $b \in \mathcal{O}$, so x/s lies in the left-hand side. Conversely, let x lie in the left-hand side. The ring $\mathfrak{o}$ is Noetherian, so $\mathcal{O}$ is a finitely generated $\mathfrak{o}$-module (proposition 50.1.32), say by $b_1, \ldots, b_r$, and $\mathrm{tr}_{L/K}(xb_k) = c_k/t_k$ with $c_k \in \mathfrak{o}$ and $t_k \in S$. With $t = t_1 \cdots t_r$, the K-linearity of the trace gives $\mathrm{tr}_{L/K}(txb_k) \in \mathfrak{o}$ for every k, hence $\mathrm{tr}_{L/K}(tx\mathcal{O}) \subseteq \mathfrak{o}$. So $tx \in \mathfrak{C}_{\mathcal{O}/\mathfrak{o}}$ and $x \in S^{-1}\mathfrak{C}_{\mathcal{O}/\mathfrak{o}}$.

   Finally, $S^{-1}\mathcal{O}$ is a Dedekind domain (lemma 50.2.5), and putting a finite sum of products over a common denominator shows that localization carries products of $\mathcal{O}$-modules to products, so
$$\mathfrak{C}_{S^{-1}\mathcal{O}/S^{-1}\mathfrak{o}} \cdot S^{-1}\mathfrak{D}_{\mathcal{O}/\mathfrak{o}} = S^{-1}\left(\mathfrak{C}_{\mathcal{O}/\mathfrak{o}}\mathfrak{D}_{\mathcal{O}/\mathfrak{o}}\right) = S^{-1}\mathcal{O}$$

by theorem 50.2.22. Inverses in the group of fractional ideals of $S^{-1}\mathcal{O}$ are unique, so $S^{-1}\mathfrak{D}_{\mathcal{O}/o}$ is the inverse of $\mathfrak{C}_{S^{-1}\mathcal{O}/S^{-1}o}$, which is $\mathfrak{D}_{S^{-1}\mathcal{O}/S^{-1}o}$, completing the proof.

1.4.49 The Different and Completion The different is also compatible with completion: if $\mathfrak{B}$ is a prime of $\mathcal{O}$ above $\mathfrak{p}$ and $\mathcal{O}_{\mathfrak{B}}$, $o_{\mathfrak{p}}$ denote the completions, then $\mathfrak{D}_{\mathcal{O}/o}\mathcal{O}_{\mathfrak{B}} = \mathfrak{D}_{\mathcal{O}_{\mathfrak{B}}/o_{\mathfrak{p}}}$. This book does not construct completions and does not prove this statement. Its proof rests on the decomposition of $L \otimes_K K_{\mathfrak{p}}$ into the completions of L at the primes above $\mathfrak{p}$, *Local Field Theory* [5, Global To Local Correspondence], under which the trace of L/K becomes the sum of the local traces, [5, Global To Local Correspondence, Norm and Trace].

Example 1.4.50: Different

1. Let $K = \mathbb{Q}(\sqrt{5})$ so that $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{5}}{2}\right]$. Then:
$$\mathfrak{C}_{K/\mathbb{Q}} = \sqrt{5}^{-1}\mathcal{O}_K \implies \mathfrak{D}_{K/\mathbb{Q}} = \sqrt{5}\mathcal{O}_K$$
2. Let $K = \mathbb{Q}(\sqrt{3})$. So that $\mathcal{O}_K = \mathbb{Z}[\sqrt{3}]$. Then:
$$\mathfrak{C}_{K/\mathbb{Q}} = \langle\frac{1}{2}, \frac{1}{\sqrt{3}}\rangle_{\mathbb{Z}} = \frac{1}{2\sqrt{3}}\mathcal{O}_K \implies \mathfrak{D}_{K/\mathbb{Q}} = 2\sqrt{3}\mathcal{O}_K$$

Theorem 1.4.51: Norm Of Different

$$\mathfrak{N}(\mathfrak{D}_{K/\mathbb{Q}}) = |\mathrm{disc}(\mathcal{O}_K)|$$

***Proof*:**
Here $A = \mathbb{Z}$ and $B = \mathcal{O}_K$. Write $\mathfrak{C} = \mathfrak{C}_{K/\mathbb{Q}}$ and $\mathfrak{D} = \mathfrak{D}_{K/\mathbb{Q}} = \mathfrak{C}^{-1}$ (definition 50.4.47), let $n = [K : \mathbb{Q}]$, and let $\omega_1, \ldots, \omega_n$ be an integral basis of $\mathcal{O}_K$ (corollary 50.1.33).

Step 1: $\mathfrak{C}$ is free on the dual basis. The extension $K/\mathbb{Q}$ is separable, so by [3, The Trace Form of a Separable Extension Is Nondegenerate] there is a $\mathbb{Q}$-basis $\omega_1^*, \ldots, \omega_n^*$ of K with $\mathrm{tr}_{K/\mathbb{Q}}(\omega_i\omega_j^*) = \delta_{ij}$. Write $x \in K$ as $x = \sum_j c_j\omega_j^*$ with $c_j \in \mathbb{Q}$. Then $\mathrm{tr}_{K/\mathbb{Q}}(x\omega_i) = c_i$, and since $\mathcal{O}_K = \bigoplus_i \mathbb{Z}\omega_i$ and the trace is $\mathbb{Z}$-linear, $\mathrm{tr}_{K/\mathbb{Q}}(x\mathcal{O}_K) \subseteq \mathbb{Z}$ holds exactly when every $c_i \in \mathbb{Z}$. So $\mathfrak{C} = \bigoplus_j \mathbb{Z}\omega_j^*$.

Step 2: $[\mathfrak{C} : \mathcal{O}_K] = |d_K|$. Applying Step 1 to $x = \omega_i$ gives
$$\omega_i = \sum_{j=1}^{n} T_{ij}\,\omega_j^* \qquad T_{ij} = \mathrm{tr}_{K/\mathbb{Q}}(\omega_i\omega_j)$$
where $T_{ij} \in \mathbb{Z}$ by corollary 50.1.14, so that $\mathcal{O}_K \subseteq \mathfrak{C}$. Moreover $\det(T) = d_K$ by proposition 50.1.26 and definition 50.1.38, a nonzero integer (corollary 50.1.28). Let $\sigma_1, \ldots, \sigma_n$ be the embeddings of K into $\mathbb{C}$. Applying σ_k to the display shows that the matrix $(\sigma_k(\omega_i))_{k,i}$ equals $(\sigma_k(\omega_j^*))_{k,j}\,T^t$,

so definition 50.1.24 gives

$$d_K = \mathrm{disc}_{K/\mathbb{Q}}(\omega_1, \dots, \omega_n) = \det(T)^2 \mathrm{disc}_{K/\mathbb{Q}}(\omega_1^*, \dots, \omega_n^*)$$

Multiplying the first display by the adjugate of T, which has integer entries, gives $d_K \omega_j^* \in \bigoplus_i \mathbb{Z}\omega_i$ for every j. So $d_K \mathfrak{C} \subseteq \mathcal{O}_K \subseteq \mathfrak{C}$, and $\mathfrak{C}/\mathcal{O}_K$ is finite, being a quotient of $\mathfrak{C}/d_K\mathfrak{C} \cong (\mathbb{Z}/d_K\mathbb{Z})^n$. By proposition 50.1.39,

$$d_K = [\mathfrak{C} : \mathcal{O}_K]^2 \mathrm{disc}_{K/\mathbb{Q}}(\omega_1^*, \dots, \omega_n^*)$$

The discriminant of the $\mathbb{Q}$-basis $\omega_1^*, \dots, \omega_n^*$ is nonzero (corollary 50.1.28), so comparing the two displays gives $[\mathfrak{C} : \mathcal{O}_K]^2 = \det(T)^2 = d_K^2$, that is $[\mathfrak{C} : \mathcal{O}_K] = |d_K|$.

Step 3: $\mathfrak{N}(\mathfrak{D}) = [\mathfrak{C} : \mathcal{O}_K]$. Since $\mathcal{O}_K \subseteq \mathfrak{C}$ (Step 2) and $\mathfrak{C}\mathfrak{D} = \mathcal{O}_K$ (theorem 50.2.22), multiplying the inclusion by $\mathfrak{D}$ gives $\mathfrak{D} \subseteq \mathcal{O}_K$, so $\mathfrak{D}$ is a nonzero ideal of $\mathcal{O}_K$ and $\mathfrak{N}(\mathfrak{D})$ is defined (definition 50.1.61). Put $m = |d_K|$ and $\mathfrak{b} = m\mathfrak{C}$, a nonzero ideal of $\mathcal{O}_K$ by Step 2. Then $\mathfrak{b}\mathfrak{D} = m\mathcal{O}_K$. The absolute norm is multiplicative (proposition 50.2.30), so by lemma 50.1.63

$$\mathfrak{N}(\mathfrak{b})\mathfrak{N}(\mathfrak{D}) = \mathfrak{N}(m\mathcal{O}_K) = \left|\mathrm{Nm}_{K/\mathbb{Q}}(m)\right| = m^n$$

On the other hand $m\mathcal{O}_K \subseteq \mathfrak{b} \subseteq \mathcal{O}_K$, and multiplication by m is a group isomorphism $\mathfrak{C} \to \mathfrak{b}$ carrying $\mathcal{O}_K$ onto $m\mathcal{O}_K$, so

$$m^n = [\mathcal{O}_K : m\mathcal{O}_K] = [\mathcal{O}_K : \mathfrak{b}]\,[\mathfrak{b} : m\mathcal{O}_K] = \mathfrak{N}(\mathfrak{b})\,[\mathfrak{C} : \mathcal{O}_K]$$

Comparing the two displays and dividing by $\mathfrak{N}(\mathfrak{b}) > 0$ gives $\mathfrak{N}(\mathfrak{D}) = [\mathfrak{C} : \mathcal{O}_K] = |d_K|$, as claimed.

(This next proposition gives a special case of the different and the origin of it's name)

Proposition 1.4.52: Different In Monogenic Extension

Let $\mathcal{O} = \mathfrak{o}[\alpha]$ and let $f \in \mathfrak{o}[X]$ be the minimal polynomial of α over K. Then the different is equal to:

$$\mathfrak{D}_{L/K} = (\delta_{L/K}(\alpha))$$

where

$$\delta_{L/K}(\alpha) = \begin{cases} f'(\alpha) & L = K(\alpha) \\ 0 & L \neq K(\alpha) \end{cases}$$

***Proof*:**
The ring $\mathcal{O}$ is the integral closure of $\mathfrak{o}$ in L and has fraction field L, so $\mathcal{O} = \mathfrak{o}[\alpha]$ forces $L = K(\alpha)$ and $\delta_{L/K}(\alpha) = f'(\alpha)$. Write $n = [L : K]$ and $f(X) = a_0 + a_1 X + \cdots + a_n X^n$ with $a_n = 1$, let $\alpha_1, \dots, \alpha_n$ be the roots of f in a normal closure of L/K, and define $b_0, \dots, b_{n-1} \in \mathcal{O}$ by

$$\frac{f(X)}{X - \alpha} = b_0 + b_1 X + \cdots + b_{n-1} X^{n-1}$$

The first step is that $b_0/f'(\alpha), \dots, b_{n-1}/f'(\alpha)$ is the basis dual to $1, \alpha, \dots, \alpha^{n-1}$ for the trace

pairing. For $0 \le s \le n-1$ the polynomial identity

$$\sum_{i=1}^{n} \frac{f(X)}{X-\alpha_i} \cdot \frac{\alpha_i^s}{f'(\alpha_i)} = X^s$$

holds, because the difference of the two sides has degree at most $n-1$ and vanishes at each α_j: the jth summand contributes $f'(\alpha_j)\alpha_j^s/f'(\alpha_j) = \alpha_j^s$ and the others vanish, since $f(X)/(X-\alpha_i) = \prod_{k\neq i}(X-\alpha_k)$. The n summands on the left are the images of $\frac{f(X)}{X-\alpha}\cdot\frac{\alpha^s}{f'(\alpha)}$ under the n embeddings of L over K, so by corollary 50.1.13 the identity reads

$$\operatorname{tr}_{L/K}\left(\frac{f(X)}{X-\alpha}\cdot\frac{\alpha^s}{f'(\alpha)}\right) = X^s$$

and comparing the coefficient of X^i on each side gives that $\operatorname{tr}_{L/K}(b_i\alpha^s/f'(\alpha))$ is 1 when $i=s$ and 0 otherwise.

The elements $b_i/f'(\alpha)$ therefore form a K-basis of L, and for $x = \sum_i c_i b_i/f'(\alpha)$ with $c_i \in K$ one has $\operatorname{tr}_{L/K}(x\alpha^s) = c_s$. Since $\mathcal{O} = \mathfrak{o} + \mathfrak{o}\alpha + \cdots + \mathfrak{o}\alpha^{n-1}$, the condition $\operatorname{tr}_{L/K}(x\mathcal{O}) \subseteq \mathfrak{o}$ is exactly the condition that every c_i lie in $\mathfrak{o}$, so

$$\mathfrak{C}_{L/K} = {}^*\mathcal{O} = \frac{1}{f'(\alpha)}\left(\mathfrak{o}b_0 + \cdots + \mathfrak{o}b_{n-1}\right)$$

Expanding $f(X) = (X-\alpha)(b_0 + b_1X + \cdots + b_{n-1}X^{n-1})$ and comparing coefficients gives $b_{n-1} = 1$ and $b_{k-1} = a_k + \alpha b_k$ for $1 \le k \le n-1$, hence

$$b_{n-i} = \alpha^{i-1} + a_{n-1}\alpha^{i-2} + \cdots + a_{n-i+1} \qquad (1 \le i \le n)$$

Each b_{n-i} is monic of degree $i-1$ in α with coefficients in $\mathfrak{o}$, so the b_i are a second $\mathfrak{o}$-basis of $\mathfrak{o} + \mathfrak{o}\alpha + \cdots + \mathfrak{o}\alpha^{n-1} = \mathcal{O}$. Therefore $\mathfrak{C}_{L/K} = f'(\alpha)^{-1}\mathcal{O}$ and

$$\mathfrak{D}_{L/K} = \mathfrak{C}_{L/K}^{-1} = (f'(\alpha)) = (\delta_{L/K}(\alpha))$$

Only the $\mathfrak{o}$-basis $1, \alpha, \ldots, \alpha^{n-1}$ of $\mathfrak{o}[\alpha]$ entered this computation, never the equality $\mathcal{O} = \mathfrak{o}[\alpha]$. Since $\mathfrak{o}$ is integrally closed, the minimal polynomial of any $\alpha \in \mathcal{O}$ again has coefficients in $\mathfrak{o}$, so for *any* $\alpha \in \mathcal{O}$ with $L = K(\alpha)$ the same argument gives the dual of the $\mathfrak{o}$-module $\mathfrak{o}[\alpha]$,

$${}^*\mathfrak{o}[\alpha] := \left\{x \in L \;\middle|\; \operatorname{tr}_{L/K}(x\,\mathfrak{o}[\alpha]) \subseteq \mathfrak{o}\right\} = \frac{1}{f'(\alpha)}\mathfrak{o}[\alpha] \tag{1.6}$$

a fact used in the next proof.

Theorem 1.4.53: Generation Of Different

For every $\alpha \in \mathcal{O}$, the different $\mathfrak{D}_{L/K}$ divides the principal ideal $(\delta_{L/K}(\alpha))$. In other words, the ideal generated by the elements $\delta_{L/K}(\alpha)$ for $\alpha \in \mathcal{O}$ is contained in $\mathfrak{D}_{L/K}$.

***Proof*:**
Let $\mathfrak{a} \subseteq \mathcal{O}$ be the ideal generated by all $\delta_{L/K}(\alpha)$ for $\alpha \in \mathcal{O}$. Only those α with $L = K(\alpha)$ contribute, since $\delta_{L/K}(\alpha) = 0$ otherwise.

Fix such an α, let f be its minimal polynomial over K, and put $b = f'(\alpha)$, which is nonzero because L/K is separable. Let $\mathscr{F}_\alpha = \{x \in L \mid x\mathcal{O} \subseteq o[\alpha]\}$ be the conductor of $o[\alpha]$ in the sense of definition 50.4.6. For $x \in L$ the chain of equivalences

$$x\mathcal{O} \subseteq o[\alpha] \iff b^{-1}x\mathcal{O} \subseteq \frac{1}{b}o[\alpha] = {}^*o[\alpha] \iff \operatorname{tr}_{L/K}(b^{-1}x\mathcal{O}) \subseteq o \iff b^{-1}x \in {}^*\mathcal{O} = \mathfrak{C}_{L/K}$$

holds: the first equivalence is multiplication by b^{-1} together with equation (50.6), and the second holds because $o[\alpha]$ is a subring of $\mathcal{O}$ containing 1, so $\mathcal{O} \cdot o[\alpha] = \mathcal{O}$ turns the defining condition of ${}^*o[\alpha]$ on $b^{-1}x\mathcal{O}$ into $\operatorname{tr}_{L/K}(b^{-1}x\mathcal{O}) \subseteq o$, while the converse direction is the case $1 \in o[\alpha]$. The third is the definition of ${}^*\mathcal{O}$. Hence $\mathscr{F}_\alpha = b\,\mathfrak{C}_{L/K} = b\,\mathfrak{D}_{L/K}^{-1}$, that is

$$(\delta_{L/K}(\alpha)) = \mathscr{F}_\alpha \mathfrak{D}_{L/K} \tag{1.7}$$

Since $\mathscr{F}_\alpha \subseteq \mathcal{O}$ is an integral ideal, $\mathfrak{D}_{L/K}$ divides $(\delta_{L/K}(\alpha))$. This holds for every α, so $\mathfrak{a} \subseteq \mathfrak{D}_{L/K}$, as claimed.

Equality holds: $\mathfrak{D}_{L/K}$ is the ideal generated by all $\delta_{L/K}(\alpha)$. Summing (50.7) over all $\alpha \in \mathcal{O}$ with $L = K(\alpha)$ shows that the reverse inclusion amounts to $\sum_\alpha \mathscr{F}_\alpha = \mathcal{O}$, that is, to finding for every prime $\mathfrak{B}$ of $\mathcal{O}$ some α with $L = K(\alpha)$ and $\mathfrak{B} \nmid \mathscr{F}_\alpha$. This is proved by completing at $\mathfrak{B}$, where the valuation ring of the completion of L is generated by one element over that of $K_\mathfrak{p}$, *Local Field Theory* [5, Local Dedekind-Kummer Theorem], and passing back through the compatibility of the different with completion (box 50.4.49). Nothing in this book uses the reverse inclusion.

Theorem 1.4.54: Detecting Ramification Using Different

A prime ideal $\mathfrak{P}$ of L is ramified over K if and only if

$$\mathfrak{P} \mid \mathfrak{D}_{L/K}$$

***Proof*:**
Let $\mathfrak{p} = \mathfrak{P} \cap o$, let $e = e(\mathfrak{P}/\mathfrak{p})$, and let $\lambda = \mathcal{O}/\mathfrak{P}$ and $\kappa = o/\mathfrak{p}$ be the residue fields, with λ/κ separable by the AKLB setup. Factor

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}^e\mathfrak{q} \qquad \mathfrak{q} = \prod_{\mathfrak{P}' \neq \mathfrak{P}} \mathfrak{P}'^{\,e(\mathfrak{P}'/\mathfrak{p})}$$

by theorem 50.2.15, so $\mathfrak{q}$ is coprime to $\mathfrak{P}$. By definition 50.4.10 and the separability of λ/κ, the prime $\mathfrak{P}$ is ramified over K exactly when $e \geq 2$.

First, $\mathfrak{P} \mid \mathfrak{D}_{L/K}$ is equivalent to $\mathfrak{P}^{-1} \subseteq \mathfrak{D}_{L/K}^{-1} = \mathfrak{C}_{L/K}$, and since $\mathfrak{P}^{-1}\mathcal{O} = \mathfrak{P}^{-1}$ the defining condition of $\mathfrak{C}_{L/K}$ reduces to

$$\mathfrak{P} \mid \mathfrak{D}_{L/K} \iff \operatorname{tr}_{L/K}(\mathfrak{P}^{-1}) \subseteq o$$

Extension of ideals is multiplicative, so $\mathfrak{p}^{-1}\mathcal{O} = (\mathfrak{p}\mathcal{O})^{-1}$ and

$$\mathfrak{p}^{-1}\mathfrak{b} = (\mathfrak{P}^e\mathfrak{q})^{-1}\mathfrak{P}^{e-1}\mathfrak{q} = \mathfrak{P}^{-1} \qquad \mathfrak{b} := \mathfrak{P}^{e-1}\mathfrak{q}$$

The trace is $\mathfrak{o}$-linear and $\mathfrak{p}^{-1} \subseteq K$, so $\mathrm{tr}_{L/K}(\mathfrak{P}^{-1}) = \mathfrak{p}^{-1}\mathrm{tr}_{L/K}(\mathfrak{b})$, and therefore $\mathfrak{P} \big| \mathfrak{D}_{L/K}$ if and only if $\mathrm{tr}_{L/K}(\mathfrak{b}) \subseteq \mathfrak{p}$.

Next, pass to $\mathfrak{o}_\mathfrak{p}$, a discrete valuation ring. Localisation commutes with integral closure: if $x \in L$ satisfies a monic equation over $\mathfrak{o}_\mathfrak{p}$ then clearing denominators makes sx integral over $\mathfrak{o}$ for some $s \in \mathfrak{o} \setminus \mathfrak{p}$, so $x \in \mathcal{O}_\mathfrak{p}$, and conversely $\mathcal{O}_\mathfrak{p}$ is integral over $\mathfrak{o}_\mathfrak{p}$. Hence $\mathcal{O}_\mathfrak{p}$ is the integral closure of $\mathfrak{o}_\mathfrak{p}$ in L and is a free $\mathfrak{o}_\mathfrak{p}$-module of rank $n = [L : K]$ by proposition 50.1.32. A $\mathfrak{o}_\mathfrak{p}$-basis of $\mathcal{O}_\mathfrak{p}$ is a K-basis of L, so by definition 50.1.10 the element $\mathrm{tr}_{L/K}(x)$ is the trace of the matrix of multiplication by x in that basis, with entries in $\mathfrak{o}_\mathfrak{p}$. Every element of $\mathfrak{o} \setminus \mathfrak{p}$ acts invertibly on $\mathcal{O}/\mathfrak{p}\mathcal{O}$, so $\overline{B} := \mathcal{O}_\mathfrak{p}/\mathfrak{p}\mathcal{O}_\mathfrak{p} = \mathcal{O}/\mathfrak{p}\mathcal{O}$, a κ-algebra of dimension n, and reducing the matrix modulo $\mathfrak{p}$ gives

$$\mathrm{tr}_{L/K}(x) \bmod \mathfrak{p} = \mathrm{tr}_{\overline{B}/\kappa}(\overline{x}) \qquad (x \in \mathcal{O}_\mathfrak{p})$$

Since $\mathfrak{p}\mathcal{O} \subseteq \mathfrak{b}$, the condition $\mathrm{tr}_{L/K}(\mathfrak{b}) \subseteq \mathfrak{p}$ says that $\mathrm{tr}_{\overline{B}/\kappa}$ vanishes on the image $\overline{\mathfrak{b}} = \mathfrak{b}/\mathfrak{p}\mathcal{O}$.

The Chinese remainder theorem splits $\overline{B}$ as a product of κ-algebras

$$\overline{B} = \mathcal{O}/\mathfrak{p}\mathcal{O} \cong \mathcal{O}/\mathfrak{P}^e \times \prod_{\mathfrak{P}' \neq \mathfrak{P}} \mathcal{O}/\mathfrak{P}'^{\,e(\mathfrak{P}'/\mathfrak{p})}$$

The image of $\mathfrak{b} = \mathfrak{P}^{e-1}\mathfrak{q}$ is 0 in every factor beyond the first, because $\mathfrak{q} \subseteq \mathfrak{P}'^{\,e(\mathfrak{P}'/\mathfrak{p})}$, and it is $(\mathfrak{P}^{e-1}\mathfrak{q} + \mathfrak{P}^e)/\mathfrak{P}^e = \mathfrak{P}^{e-1}/\mathfrak{P}^e$ in the first, because $\mathfrak{q} + \mathfrak{P} = \mathcal{O}$. Multiplication by an element supported in one factor of a product of κ-algebras annihilates the other factors, so $\mathrm{tr}_{\overline{B}/\kappa}$ restricted to $\overline{\mathfrak{b}}$ is the trace map of $\mathcal{O}/\mathfrak{P}^e$ over κ restricted to $\mathfrak{P}^{e-1}/\mathfrak{P}^e$.

If $e \geq 2$ then $\mathfrak{P}^{2e-2} \subseteq \mathfrak{P}^e$, so every element of $\mathfrak{P}^{e-1}/\mathfrak{P}^e$ squares to zero in $\mathcal{O}/\mathfrak{P}^e$. Multiplication by such an element is a nilpotent endomorphism of a finite-dimensional κ-vector space, and a nilpotent endomorphism has trace 0, its characteristic polynomial being a power of X. So $\mathrm{tr}_{\overline{B}/\kappa}$ vanishes on $\overline{\mathfrak{b}}$ and $\mathfrak{P} \big| \mathfrak{D}_{L/K}$.

If $e = 1$ then $\mathcal{O}/\mathfrak{P}^e = \lambda$ and $\mathfrak{P}^{e-1}/\mathfrak{P}^e = \lambda$, so the map in question is $\mathrm{tr}_{\lambda/\kappa}$. The extension λ/κ is separable, so its trace form is nondegenerate by [3, The Trace Form of a Separable Extension Is Nondegenerate], and applying nondegeneracy to $x = 1$ produces $y \in \lambda$ with $\mathrm{tr}_{\lambda/\kappa}(y) \neq 0$. So $\mathrm{tr}_{\overline{B}/\kappa}$ does not vanish on $\overline{\mathfrak{b}}$ and $\mathfrak{P} \nmid \mathfrak{D}_{L/K}$.

Thus $\mathfrak{P} \big| \mathfrak{D}_{L/K}$ if and only if $e \geq 2$, which is ramification of $\mathfrak{P}$ over K.

(relation between different and discriminant)

Definition 1.4.55: Relative Discriminant

The *discriminant* $\mathfrak{d}_{\mathcal{O}/\mathfrak{o}}$ is the ideal of $\mathfrak{o}$ generated by the discriminants $d(\alpha_1, ..., \alpha_n)$ for all bases $\alpha_1, \alpha_2, ..., \alpha_n$ of L/K contained in $\mathcal{O}$:

$$\mathfrak{d}_{\mathcal{O}/\mathfrak{o}} = \langle \{ d(\alpha_1, \alpha_2, ..., \alpha_n) \mid \alpha_1, \alpha_2, ..., \alpha_n \text{ basis }, \alpha_i \in \mathcal{O} \} \rangle_\mathfrak{o}$$

In the case where $K = \mathbb{Q}$ and $L/\mathbb{Q}$ is a number field, then $\mathfrak{d} = (d_L)$ is the principal ideal of the usual discriminant.

Theorem 1.4.56: Relating Different And Relative Discriminant

$$\mathfrak{d}_{L/K} = \mathfrak{N}_{L/K}(\mathfrak{D}_{L/K})$$

***Proof*:**
Both sides are ideals of o, and an ideal of a Dedekind domain is determined by the exponents in its prime factorisation (corollary 50.2.23), which may be read off after localising at each prime. All three constructions survive localisation at a multiplicative set $S \subseteq o \setminus \{0\}$. For the different this is proposition 50.4.48(2). For the discriminant, a basis of L/K inside $S^{-1}\mathcal{O}$ may be scaled by a single $s \in S$ into $\mathcal{O}$, and $d(s\alpha_1, \dots, s\alpha_n) = s^{2n} d(\alpha_1, \dots, \alpha_n)$ by definition 50.1.24, so $S^{-1}\mathfrak{d}_{\mathcal{O}/o} = \mathfrak{d}_{S^{-1}\mathcal{O}/S^{-1}o}$. For the norm, definition 50.3.13 generates $\mathfrak{N}_{L/K}(I)$ by the $\mathrm{Nm}_{L/K}(\alpha)$ with $\alpha \in I$, and $\mathrm{Nm}_{L/K}(\alpha/s) = \mathrm{Nm}_{L/K}(\alpha)/s^n$, so $S^{-1}\mathfrak{N}_{L/K}(I) = \mathfrak{N}_{L/K}(S^{-1}I)$. We may therefore assume o is a discrete valuation ring with maximal ideal $\mathfrak{p}$.

Under that assumption $\mathcal{O}$ is a Dedekind domain (lemma 50.3.2) with finitely many primes $\mathfrak{P}_1, \dots, \mathfrak{P}_g$, namely the primes dividing $\mathfrak{p}\mathcal{O}$. A nonzero prime $\mathfrak{P}$ of $\mathcal{O}$ contains a nonzero x, whose minimal polynomial over K has coefficients in o and nonzero constant term, and that constant term lies in $\mathfrak{P} \cap o$. So $\mathfrak{P} \cap o$ is a nonzero prime of o, which is $\mathfrak{p}$, and $\mathfrak{P}$ divides $\mathfrak{p}\mathcal{O}$ by proposition 50.3.11. Such a ring is a principal ideal domain: given $\mathfrak{a} = \prod_i \mathfrak{P}_i^{a_i}$, the Chinese remainder theorem gives $x \in \mathcal{O}$ with $x \in \mathfrak{P}_i^{a_i} \setminus \mathfrak{P}_i^{a_i+1}$ for every i, and since there is no other prime, $(x) = \mathfrak{a}$. Also $\mathcal{O}$ is a free o-module of rank $n = [L : K]$ by proposition 50.1.32. Fix a o-basis $\alpha_1, \dots, \alpha_n$ of $\mathcal{O}$. Any K-basis of L contained in $\mathcal{O}$ is obtained from it by a matrix over o, so definition 50.1.24 gives $\mathfrak{d}_{L/K} = (d(\alpha_1, \dots, \alpha_n))$.

Let $\alpha_1', \dots, \alpha_n'$ be the dual basis for the trace pairing, which exists by [3, The Trace Form of a Separable Extension Is Nondegenerate]. Writing $x \in L$ as $x = \sum_j c_j \alpha_j'$ with $c_j \in K$, we get $\mathrm{tr}_{L/K}(x\alpha_i) = c_i$. As $\mathcal{O} = \bigoplus_i o\alpha_i$ and the trace is K-linear, $\mathrm{tr}_{L/K}(x\mathcal{O}) \subseteq o$ exactly when every c_i lies in o, so $\mathfrak{C}_{L/K}$ is o-free on $\alpha_1', \dots, \alpha_n'$. Since $\mathfrak{D}_{L/K} \subseteq \mathcal{O}$ is integral and $\mathcal{O}$ is a principal ideal domain, $\mathfrak{D}_{L/K} = (\gamma)$ for some $\gamma \in \mathcal{O}$, and $\mathfrak{C}_{L/K} = \gamma^{-1}\mathcal{O}$ is o-free on $\gamma^{-1}\alpha_1, \dots, \gamma^{-1}\alpha_n$. Two o-bases of the same free module differ by a matrix in $\mathrm{GL}_n(o)$, whose determinant is a unit, so

$$(d(\alpha_1', \dots, \alpha_n')) = (d(\gamma^{-1}\alpha_1, \dots, \gamma^{-1}\alpha_n))$$

Writing $P = (\sigma_i(\alpha_j))$ and $Q = (\sigma_i(\alpha_j'))$ for the K-embeddings σ_i of L, corollary 50.1.13 gives $(P^tQ)_{jk} = \mathrm{tr}_{L/K}(\alpha_j\alpha_k')$, which is the identity matrix, so $\det P \det Q = 1$ and

$$d(\alpha_1, \dots, \alpha_n)\, d(\alpha_1', \dots, \alpha_n') = 1$$

Likewise $(\sigma_i(\gamma^{-1}\alpha_j)) = \mathrm{diag}(\sigma_i(\gamma^{-1})) \cdot P$, so

$$d(\gamma^{-1}\alpha_1, \dots, \gamma^{-1}\alpha_n) = \mathrm{Nm}_{L/K}(\gamma)^{-2} d(\alpha_1, \dots, \alpha_n)$$

Finally $\mathfrak{N}_{L/K}(\mathfrak{D}_{L/K}) = (\mathrm{Nm}_{L/K}(\gamma))$: by definition 50.3.13 the left side is generated by the $\mathrm{Nm}_{L/K}(\gamma x) = \mathrm{Nm}_{L/K}(\gamma)\mathrm{Nm}_{L/K}(x)$ for $x \in \mathcal{O}$, and the $\mathrm{Nm}_{L/K}(x)$ generate o because they lie in o by corollary 50.1.14 and include $\mathrm{Nm}_{L/K}(1) = 1$. This is lemma 50.3.15(1) in the number field case. Combining,

$$\mathfrak{d}_{L/K}^{-1} = (d(\alpha_1, \dots, \alpha_n)^{-1}) = (d(\alpha_1', \dots, \alpha_n')) = (\mathrm{Nm}_{L/K}(\gamma)^{-2})\, \mathfrak{d}_{L/K} = \mathfrak{N}_{L/K}(\mathfrak{D}_{L/K})^{-2}\mathfrak{d}_{L/K}$$

so $\mathfrak{d}_{L/K}^2 = \mathfrak{N}_{L/K}(\mathfrak{D}_{L/K})^2$. Both are powers of $\mathfrak{p}$, so the exponents agree and $\mathfrak{d}_{L/K} = \mathfrak{N}_{L/K}(\mathfrak{D}_{L/K})$.

Corollary 1.4.57: Tower For Relative Discriminant

Let $M/L/K$. Then:

$$\mathfrak{d}_{M/K} = \mathfrak{d}_{L/K}^{[M:L]} \mathfrak{N}_{L/K}(\mathfrak{d}_{M/L})$$

***Proof*:**
Take $M/L/K$ to be number fields, the setting in which the ideal norm of definition 50.3.13 carries the properties collected in lemma 50.3.15. The tower law for the different, proposition 50.4.48(1), reads

$$\mathfrak{D}_{M/K} = \mathfrak{D}_{M/L} \left(\mathfrak{D}_{L/K} \mathcal{O}_M \right)$$

Apply $\mathfrak{N}_{M/K}$ and use theorem 50.4.56 on the left, transitivity $\mathfrak{N}_{M/K} = \mathfrak{N}_{L/K} \circ \mathfrak{N}_{M/L}$ from lemma 50.3.15(2), and multiplicativity from lemma 50.3.15(3):

$$\mathfrak{d}_{M/K} = \mathfrak{N}_{L/K} \left(\mathfrak{N}_{M/L}(\mathfrak{D}_{M/L}) \cdot \mathfrak{N}_{M/L} \left(\mathfrak{D}_{L/K} \mathcal{O}_M \right) \right)$$

The first inner factor is $\mathfrak{d}_{M/L}$ by theorem 50.4.56, and the second is $\mathfrak{D}_{L/K}^{[M:L]}$ by lemma 50.3.15(4). Multiplicativity of $\mathfrak{N}_{L/K}$ then gives

$$\mathfrak{d}_{M/K} = \mathfrak{N}_{L/K}(\mathfrak{d}_{M/L}) \cdot \mathfrak{N}_{L/K}(\mathfrak{D}_{L/K})^{[M:L]} = \mathfrak{N}_{L/K}(\mathfrak{d}_{M/L}) \, \mathfrak{d}_{L/K}^{[M:L]}$$

where the last step is theorem 50.4.56 once more.

Corollary 1.4.58: Discriminant And Ramified Primes

Let L/K be an extension of number fields and let $\mathfrak{p}$ be a nonzero prime of $\mathcal{O}_K$. Then $\mathfrak{p}$ is ramified in L if and only if

$$\mathfrak{p} \big| \mathfrak{d}_{L/K}$$

In particular, the extension L/K is unramified if and only if

$$\mathfrak{d}_{L/K} = (1)$$

***Proof*:**
Write $\mathfrak{D} = \mathfrak{D}_{L/K}$ and $\mathfrak{C} = \mathfrak{C}_{L/K}$. Traces of elements of $\mathcal{O}_L$ lie in $\mathcal{O}_K$ (corollary 50.1.14), so $\mathcal{O}_L \subseteq \mathfrak{C}$, and multiplying by $\mathfrak{D}$ gives $\mathfrak{D} \subseteq \mathfrak{C}\mathfrak{D} = \mathcal{O}_L$ (theorem 50.2.22). So $\mathfrak{D}$ is a nonzero integral ideal, and $\mathfrak{D} = \prod_{\mathfrak{P}} \mathfrak{P}^{d_{\mathfrak{P}}}$ with integers $d_{\mathfrak{P}} \geq 0$, almost all zero (theorem 50.2.15). By theorem 50.4.56, the multiplicativity of the ideal norm (lemma 50.3.15(3)) and corollary 50.3.16,

$$\mathfrak{d}_{L/K} = \mathfrak{N}_{L/K}(\mathfrak{D}) = \prod_{\mathfrak{P}} \mathfrak{N}_{L/K}(\mathfrak{P})^{d_{\mathfrak{P}}} = \prod_{\mathfrak{P}} (\mathfrak{P} \cap \mathcal{O}_K)^{f(\mathfrak{P}) d_{\mathfrak{P}}}$$

where $f(\mathfrak{P})$ is the inertia degree of $\mathfrak{P}$ over $\mathfrak{P} \cap \mathcal{O}_K$. By unique factorization in $\mathcal{O}_K$, the exponent of $\mathfrak{p}$ in $\mathfrak{d}_{L/K}$ is $\sum_{\mathfrak{P}} f(\mathfrak{P}) d_{\mathfrak{P}}$, the sum running over the primes $\mathfrak{P}$ of $\mathcal{O}_L$ above $\mathfrak{p}$. Every $f(\mathfrak{P}) \geq 1$, so $\mathfrak{p} \big| \mathfrak{d}_{L/K}$ if and only if $\mathfrak{P} \big| \mathfrak{D}$ for some $\mathfrak{P}$ above $\mathfrak{p}$. By theorem 50.4.54, this happens if and only if some prime above $\mathfrak{p}$ is ramified over K, that is, if and only if $\mathfrak{p}$ is ramified in L (definition 50.4.10).

For the second statement, $\mathfrak{d}_{L/K} = \mathfrak{N}_{L/K}(\mathfrak{D})$ is a nonzero integral ideal of $\mathcal{O}_K$, being generated by the norms of the elements of $\mathfrak{D} \subseteq \mathcal{O}_L$, which lie in $\mathcal{O}_K$ (corollary 50.1.14). By theorem 50.2.15, it equals (1) exactly when no prime divides it, that is, by the first statement, exactly when no prime of $\mathcal{O}_K$ is ramified in L, as claimed.

Proposition 1.4.59: Coprime Discriminants And Difference

Let K_1, K_2 be Galois over $\mathbb{Q}$ with coprime discriminants, of degrees n_1 and n_2, with $K_1 \cap K_2 = \mathbb{Q}$, and let $K = K_1K_2$. Then:

$$\mathfrak{d}_{K/\mathbb{Q}} = \mathfrak{d}_{K_1/\mathbb{Q}}^{n_2} \mathfrak{d}_{K_2/\mathbb{Q}}^{n_1}$$

***Proof*:**
This is theorem 50.1.47 applied over the base field $\mathbb{Q}$: its hypotheses are that $K_1/\mathbb{Q}$ and $K_2/\mathbb{Q}$ are Galois with coprime discriminants and intersection $\mathbb{Q}$, and over $\mathbb{Q}$ the integral bases it posits exist. Its conclusion gives the integral basis $\omega_i\omega_j'$ of $\mathcal{O}_K$ together with the discriminant $\mathfrak{d}_{K_1/\mathbb{Q}}^{n_2}\mathfrak{d}_{K_2/\mathbb{Q}}^{n_1}$, an identity of integers that reads as the stated identity of ideals.

(Neukirch goes on to prove some important results which are really cool! For example, if K is an algebraic number field and S a finite set of primes of K, then there exists only finitely many extensions L/K of degree n which are unramified outside of S!)

Exercise 1.4.2

1. Show that L_D is the maximal field in which the ramification and inertial degree is 1, and similarly for L_I.
2. Let K be a number fields. Show that if $\mathrm{disc}(K)$ is square free, then K has no intermediary number field

Wild and Tame Ramification

(see this! here)

(This section is essentially here for the proof of Kronecker-Weber Theorem. May choose to move it down to that section).

We end off by exploring some more about the inertia group, namely we shall like at *wild* and *tame* inertia.

Definition 1.4.60: Wild Inertia Group

Let L/K be an extension of number fields, $I_{\mathfrak{B}} \leqslant D_{\mathfrak{B}}$ be the inertia group. Then $W_{\mathfrak{B}} \trianglelefteq I_{\mathfrak{B}}$ is defined to be:

$$W_{\mathfrak{B}} := \left\{ g \in I_{\mathfrak{B}} \;\middle|\; \forall \overline{\alpha} \in \mathcal{O}_L/\mathfrak{B}^2,\ g(\overline{\alpha}) = \overline{\alpha} \right\}$$

the wild inertia group can be thought of as a parallel to "local coordinates". In particular if we have an extension $K/\mathbb{Q}$ of number fields, then we have an extension $K/\mathbb{Q}_p$ then $W_{\mathfrak{B}}$ is the unique sylow

p-subgroup of the inertia group. If you know some algebraic geometry, an intuitive reason to see why the quotient $\mathfrak{B}^2$ shows up is since the tangent space of a variety/scheme is given by $\mathfrak{m}/\mathfrak{m}^2$.

Theorem 1.4.61: Wild Inertia Group Is Sylow Subgroup

Let L/K be a Galois extension of number fields (for instance $K = \mathbb{Q}$), let Q be a nonzero prime of $\mathcal{O}_L$, $\kappa = \mathcal{O}_L/Q$, and let p be the prime number with $Q \cap \mathbb{Z} = p\mathbb{Z}$. For $\sigma \in I_Q$ there is a unique $\theta(\sigma) \in \kappa^\times$ with $\sigma(x) + Q^2 = \theta(\sigma)(x + Q^2)$ for every $x \in Q$, where Q/Q^2 is a κ-vector space through $(a + Q)(x + Q^2) = ax + Q^2$. The map $\theta\colon I_Q \to \kappa^\times$ is a group homomorphism with kernel W_Q, and W_Q is the unique Sylow p-subgroup of I_Q, hence $W_Q \trianglelefteq I_Q$.

***Proof*:**
Write $I = I_Q$, $W = W_Q$, $\kappa = \mathcal{O}_L/Q$ and $q = |\kappa|$. By lemma 50.2.29, κ is a finite field of characteristic p, so q is a power of p, and $p \in Q$. By definition 50.4.32 and corollary 50.4.35, every $\sigma \in I$ satisfies $\sigma(Q) = Q$ and $\sigma(\alpha) - \alpha \in Q$ for all $\alpha \in \mathcal{O}_L$, and by definition 50.4.60 such a σ lies in W exactly when $\sigma(\alpha) - \alpha \in Q^2$ for all $\alpha \in \mathcal{O}_L$. Each $\sigma \in I$ is a ring automorphism of $\mathcal{O}_L$ (lemma 50.4.23) with $\sigma(Q) = Q$, so $\sigma(Q^k) = Q^k$ for every $k \geq 1$.

Step 1: W is the kernel of a homomorphism $I \to \kappa^\times$. The group $V = Q/Q^2$ is a κ-vector space under $(a + Q)(x + Q^2) = ax + Q^2$, which is well defined since $Q \cdot Q = Q^2$. It has $[Q : Q^2] = \mathfrak{N}(Q^2)/\mathfrak{N}(Q) = \mathfrak{N}(Q) = q$ elements (definition 50.1.61, proposition 50.2.30), so $\dim_\kappa V = 1$. Each $\sigma \in I$ induces a bijection $\overline{\sigma}\colon V \to V$, $x + Q^2 \mapsto \sigma(x) + Q^2$, which is κ-linear: for $a \in \mathcal{O}_L$ and $x \in Q$,

$$\sigma(ax) - a\sigma(x) = (\sigma(a) - a)\sigma(x) \in Q \cdot Q = Q^2$$

A κ-linear bijection of the one-dimensional space V is multiplication by a unique scalar, so $\overline{\sigma}$ is multiplication by some $\theta(\sigma) \in \kappa^\times$. Since $\overline{\sigma\tau} = \overline{\sigma}\,\overline{\tau}$, the map $\theta\colon I \to \kappa^\times$ is a group homomorphism, and $W \subseteq \ker\theta$.

For the reverse inclusion let $\sigma \in \ker\theta$, so that $\sigma(x) - x \in Q^2$ for every $x \in Q$, and let $\alpha \in \mathcal{O}_L$. Every $y \in \kappa$ satisfies $y^q = y$, for $y \neq 0$ by [3, Lagrange Theorem] in $\kappa^\times$, which has order $q - 1$. So $x := \alpha - \alpha^q \in Q$. Write $\sigma(\alpha) = \alpha + z$ with $z \in Q$. By the binomial theorem,

$$\sigma(\alpha^q) = (\alpha + z)^q = \alpha^q + q\alpha^{q-1}z + \sum_{k=2}^{q} \binom{q}{k} \alpha^{q-k} z^k$$

where $q\alpha^{q-1}z \in Q^2$ because $q \in p\mathbb{Z} \subseteq Q$ and $z \in Q$, and each later term lies in Q^2. So $\sigma(\alpha^q) - \alpha^q \in Q^2$, and

$$\sigma(\alpha) - \alpha = (\sigma(\alpha^q) - \alpha^q) + (\sigma(x) - x) \in Q^2$$

Hence $\sigma \in W$ and $W = \ker\theta$. In particular $W \trianglelefteq I$, and I/W is isomorphic to a subgroup of $\kappa^\times$ ([3, First Isomorphism Theorem]), so $[I : W]$ divides $q - 1$ ([3, Lagrange Theorem]). As q is a power of p, $p \nmid [I : W]$.

Step 2: an element of W of order prime to p is trivial. Let $\tau \in W$ have order m with $p \nmid m$, and

put $\delta(\alpha) = \tau(\alpha) - \alpha$ for $\alpha \in \mathcal{O}_L$. We show by induction on $k \geq 2$ that $\delta(\mathcal{O}_L) \subseteq Q^k$. The case $k = 2$ is $\tau \in W$. Assume $\delta(\mathcal{O}_L) \subseteq Q^k$ for some $k \geq 2$. First, $\delta(Q^k) \subseteq Q^{k+1}$: the group Q^k is additively generated by products $y_1 \cdots y_k$ with $y_i \in Q$, the map δ is additive, and

$$\delta(y_1 \cdots y_k) = \sum_{i=1}^{k} \tau(y_1) \cdots \tau(y_{i-1})\, \delta(y_i)\, y_{i+1} \cdots y_k$$

where in each term $\delta(y_i) \in Q^k$ and the other $k-1$ factors lie in Q, so each term lies in $Q^{2k-1} \subseteq Q^{k+1}$. Next, by induction on $j \geq 1$,

$$\tau^j(\alpha) - \alpha - j\delta(\alpha) \in Q^{k+1} \qquad \alpha \in \mathcal{O}_L$$

The case $j = 1$ is the definition of δ. If it holds for j, write $\tau^j(\alpha) = \alpha + j\delta(\alpha) + z$ with $z \in Q^{k+1}$. Then $\tau^{j+1}(\alpha) = \alpha + \delta(\alpha) + j\delta(\alpha) + j\delta(\delta(\alpha)) + \tau(z)$, where $\delta(\delta(\alpha)) \in \delta(Q^k) \subseteq Q^{k+1}$ and $\tau(z) \in Q^{k+1}$, which is the case $j+1$. Taking $j = m$ gives $m\delta(\alpha) \in Q^{k+1}$. If $\delta(\alpha) \neq 0$, then the exponent of Q in the factorization of $m\delta(\alpha)\mathcal{O}_L = (m\mathcal{O}_L)(\delta(\alpha)\mathcal{O}_L)$ is at least $k+1$ (corollary 50.2.18, theorem 50.2.15). As $m \notin p\mathbb{Z} = Q \cap \mathbb{Z}$, the prime Q does not divide $m\mathcal{O}_L$, so the exponent of Q in $\delta(\alpha)\mathcal{O}_L$ is at least $k+1$, that is $\delta(\alpha) \in Q^{k+1}$. This completes the induction.

So $\delta(\alpha) \in Q^k$ for every k. If $\delta(\alpha) \neq 0$, then Q^k divides $\delta(\alpha)\mathcal{O}_L$ for every k, which is impossible because the exponent of Q in the factorization of $\delta(\alpha)\mathcal{O}_L$ is finite. Hence τ fixes $\mathcal{O}_L$ pointwise, and therefore fixes L pointwise, as every element of L is a quotient of elements of $\mathcal{O}_L$ (proposition 50.1.7). So $\tau = \mathrm{id}$.

Step 3: conclusion. Let $\sigma \in W$ have order $p^a m$ with $p \nmid m$. Then $\sigma^{p^a} \in W$ has order m, so $\sigma^{p^a} = \mathrm{id}$ by Step 2, and $m = 1$. If a prime $\ell \neq p$ divided $|W|$, then [3, Cauchy's Theorem] would give an element of W of order ℓ, which is impossible. So $|W|$ is a power of p. Since $|I| = |W|\,[I : W]$ with $p \nmid [I : W]$ by Step 1, W is a Sylow p-subgroup of I. It is normal in I by Step 1, so it is the unique Sylow p-subgroup of I ([3, Characterizations of a Unique Sylow Subgroup]), as we sought to show.

Lemma 1.4.62: Wild Inertia And Uniformizer

In the setting of theorem 50.4.61, let $\pi \in Q \setminus Q^2$ be a uniformizer for Q and let $\sigma \in I_Q$. Then:

$$\sigma \in W_Q \qquad \Leftrightarrow \qquad \sigma(\pi) - \pi \in Q^2$$

that is, $\sigma(\pi) = \pi \bmod Q^2$

***Proof*:**
By theorem 50.4.61, $\sigma \in W_Q$ exactly when $\theta(\sigma) = 1$, and $\sigma(\pi) + Q^2 = \theta(\sigma)(\pi + Q^2)$. So $\sigma(\pi) - \pi + Q^2 = (\theta(\sigma) - 1)(\pi + Q^2)$. As $\pi \notin Q^2$, the vector $\pi + Q^2$ of the κ-vector space Q/Q^2 is nonzero, so this is 0 exactly when $\theta(\sigma) = 1$, completing the proof.

Example 1.4.63: Example from Class for wild inertia

Let $\zeta = e^{2\pi i/8}$, a primitive 8th root of unity, and $K = \mathbb{Q}(\zeta)$, which contains $i = \zeta^2$ and $\sqrt{2} = \zeta + \zeta^{-1}$. By [3, Cyclotomic Polynomials] and [3, Irreducibility Of Cyclotomic Polynomials], the minimal polynomial of ζ is $\Phi_8(x) = \prod_{a\in\{1,3,5,7\}}(x - \zeta^a) = x^4 + 1$, so $[K : \mathbb{Q}] = 4$ and $G = \mathrm{Gal}_{\mathbb{Q}}(K) = \{\sigma_1, \sigma_3, \sigma_5, \sigma_7\}$ with $\sigma_a(\zeta) = \zeta^a$.

The prime above 2. Setting $x = 1$ gives $2 = \prod_a(1-\zeta^a)$. For odd a, $1-\zeta^a = (1-\zeta)(1+\zeta+\cdots+\zeta^{a-1})$, and the second factor is a unit, with inverse $(1-\zeta)/(1-\zeta^a) = 1+\zeta^a+\cdots+\zeta^{a(a'-1)}$ for $aa' \equiv 1 \bmod 8$. So $2\mathcal{O}_K = \mathfrak{B}^4$ with $\mathfrak{B} = (1-\zeta)$, which is a proper ideal as $1/2 \notin \mathcal{O}_K$ (theorem 50.1.9(1)). If $\mathfrak{B} = \mathfrak{q}_1^{a_1}\cdots\mathfrak{q}_t^{a_t}$, then proposition 50.4.3 gives $\sum_j 4a_jf_j = [K : \mathbb{Q}] = 4$, so $\mathfrak{B}$ is a prime of degree 1. So 2 is totally ramified, and $I_{\mathfrak{B}} = G$, as $|I_{\mathfrak{B}}| = e = 4 = |G|$ (proposition 50.4.37, the residue fields being finite).

The wild inertia group. Let $v(x)$ be the exponent of $\mathfrak{B}$ in $x\mathcal{O}_K$, so that $v(xy) = v(x) + v(y)$, $v(2) = 4$, and $\pi = 1-\zeta$ has $v(\pi) = 1$, a uniformizer. For $\sigma_a \neq \sigma_1$, $\sigma_a(\pi)-\pi = \zeta-\zeta^a = \zeta(1-\zeta^{a-1})$ with ζ a unit, so $v(\sigma_a(\pi) - \pi) = v(1 - \zeta^{a-1})$, and:

- for $a = 3$, $1 - \zeta^2 = 1 - i$ and $(1 - i)^2 = -2i$, so $2v(1 - i) = v(2) = 4$ and $v = 2$
- for $a = 7$, $1 - \zeta^6 = 1 + i = i(1 - i)$, so $v = 2$
- for $a = 5$, $1 - \zeta^4 = 2$, so $v = 4$

So $\sigma(\pi) - \pi \in \mathfrak{B}^2$ for every $\sigma \in G$, and lemma 50.4.62 gives $W_{\mathfrak{B}} = I_{\mathfrak{B}} = G$, a 2-group, as theorem 50.4.61 predicts. The subgroups $\{\sigma \in G \mid v(\sigma(\pi) - \pi) \geq j\}$, with $v(0) = \infty$, are G for $j \leq 2$, then $\{\sigma_1, \sigma_5\}$ for $j = 3, 4$, then $\{\sigma_1\}$ for $j \geq 5$.

Theorem 1.4.64: Abelian Extension Wild Map

Let L/K be a Galois extension of number fields, $G = \mathrm{Gal}_K(L)$, $P \subseteq \mathcal{O}_K$ a nonzero prime, Q a prime of $\mathcal{O}_L$ lying over P, W_Q the wild inertia group, and $\theta\colon I_Q \to \mathbb{F}_Q^\times$ the homomorphism of theorem 50.4.61, where $\mathbb{F}_Q = \mathcal{O}_L/Q$ and $\mathbb{F}_P = \mathcal{O}_K/P \subseteq \mathbb{F}_Q$. For a uniformizer π of Q, $\theta(\sigma)$ is the class of $\sigma(\pi)/\pi$ in the sense that $\sigma(\pi) \equiv \theta(\sigma)\pi \bmod Q^2$, and θ induces an injective homomorphism

$$\varphi : I_Q/W_Q \hookrightarrow \mathbb{F}_Q^\times \qquad [\sigma] \mapsto \theta(\sigma)$$

If G is abelian, then $\mathrm{im}\varphi \subseteq \mathbb{F}_P^\times$.

***Proof*:**

Step 1: the map. By theorem 50.4.61, θ is a homomorphism with kernel W_Q, so it induces the injective homomorphism φ ([3, First Isomorphism Theorem]), and $\sigma(\pi) + Q^2 = \theta(\sigma)(\pi + Q^2)$ by definition.

Step 2: conjugation. Let $\tau \in D_Q$ and $\sigma \in I_Q$, and let $\overline{\tau}$ be the automorphism $a + Q \mapsto \tau(a) + Q$ of $\mathbb{F}_Q$ fixing $\mathbb{F}_P$. Lift $\theta(\sigma)$ to $c \in \mathcal{O}_L$, so that $\sigma(y) - cy \in Q^2$ for $y \in Q$. For $x \in Q$ we have $\tau^{-1}(x) \in Q$, so $\sigma(\tau^{-1}(x)) - c\tau^{-1}(x) \in Q^2$, and applying τ, which preserves Q^2, gives

$\tau\sigma\tau^{-1}(x) - \tau(c)x \in Q^2$. So $\theta(\tau\sigma\tau^{-1}) = \overline{\tau}(\theta(\sigma))$.

Step 3: abelian groups. If G is abelian, then $\tau\sigma\tau^{-1} = \sigma$, so $\theta(\sigma)$ is fixed by $\overline{\tau}$ for every $\tau \in D_Q$. The homomorphism $D_Q \to \mathrm{Gal}_{\mathbb{F}_P}(\mathbb{F}_Q)$, $\tau \mapsto \overline{\tau}$, is surjective (proposition 50.4.31), so $\theta(\sigma)$ is fixed by $\mathrm{Gal}_{\mathbb{F}_P}(\mathbb{F}_Q)$. Let $q = |\mathbb{F}_P|$. Every $a \in \mathbb{F}_P$ satisfies $a^q = a$, by [3, Lagrange Theorem] in $\mathbb{F}_P^\times$, so the map $y \mapsto y^q$ is an injective ring endomorphism of the finite field $\mathbb{F}_Q$ fixing $\mathbb{F}_P$, hence an element of $\mathrm{Gal}_{\mathbb{F}_P}(\mathbb{F}_Q)$. So $\theta(\sigma)$ is a root of $X^q - X$, whose at most q roots in $\mathbb{F}_Q$ are the elements of $\mathbb{F}_P$, and $\theta(\sigma) \in \mathbb{F}_P^\times$, completing the proof.

Exercise 1.4.3

1. (Define higher-order ramification. Show that the inertia group is solvable, and when the residue fields are finite then the decomposition group is solvable).

1.4.4 Frobenius Automorphism

This is a short section to single out the Frobenius automorphisms from corollary 50.4.34 due to its importance in Class Field Theory (CFT). Namely the Frobenius automorphism is central to relating the structure of abelian galois groups to the structure of the class group. The full exposition is beyond the scope of this textbook, but the reader can still get an appreciation behind why the Frobenius automorphism is so important. For a full exposition, see [8].

Assume $\mathfrak{B}$ over $\mathfrak{p}$ is unramified in $L/K/\mathbb{Q}$, so that $I_{\mathfrak{B}}$ is trivial (which by proposition 50.4.12 and proposition 50.4.16 can be shown to be the case for all put finitely many primes). Then $D_{\mathfrak{B}} \cong \mathrm{Gal}_{\kappa(\mathfrak{p})}(\kappa(\mathfrak{B})) \cong (\mathbb{Z}/n\mathbb{Z})$. By finite field theory, we know that there exists the *Frobenius generator* sending $x \mapsto x^{\mathfrak{N}(\mathfrak{B})}$ since $\mathfrak{N}(\mathfrak{B}) = f$ and the order of this group is f. Thus, for each prime $\mathfrak{B}/\mathfrak{p}$, have a corresponding automorphism in the decomposition group, and hence the Galois group:

Definition 1.4.65: Frobenius Automorphism of a Prime

Let $\sigma \in D_{\mathfrak{B}}$. Then σ is called the *Frobenius* if $\sigma(\alpha) \equiv \alpha^f \mod \mathfrak{B}$ where f is the inertia degree of $\mathfrak{B}$ over $\mathfrak{p}$. The collection of element $\sigma \in D_{\mathfrak{B}}$ with this property is denoted $\mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}}$ so that is $\sigma \in \mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}}$:

$$\sigma(\alpha) \equiv \alpha^f \mod \mathfrak{B}$$

Note that if $\mathfrak{B}$ were ramified, then the Frobenius automorphism is defined up to the coset $I_{\mathfrak{B}}$: $\sigma + I_{\mathfrak{B}}$. If another $\mathfrak{B}' = \sigma(\mathfrak{B})$[27] is considered, the Frobenius element differ conjugation:

$$\mathrm{Frob}_{\sigma(\mathfrak{B}/\mathfrak{p})} = \sigma\mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}}\sigma^{-1}$$

If $D_{\mathfrak{B}}$ is abelian, then the Frobenius element is *only* determined by the prime $\mathfrak{p}$ as

$$\mathrm{Frob}_{\sigma(\mathfrak{B}/\mathfrak{p})} = \sigma\mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}}\sigma^{-1} = \mathrm{Frob}_{\mathfrak{B}/\mathfrak{p}}$$

Thus, we may denote it as $\mathrm{Frob}_{\mathfrak{p}}$. In this case, we have:

$$\sigma(\alpha) = \alpha^{\mathfrak{N}(\mathfrak{p})} \mod \mathfrak{p}\mathcal{O}_L$$

[27] by transitivity and definition of the decomposition group, any $\mathfrak{B}'$ is of this form

because $\mathfrak{p}\mathcal{O}_L$ is the product of all primes $\mathfrak{B}$ over $\mathfrak{p}$. Another immediate observation is that if $\mathfrak{p}$ spits completely, then $f = 1$, and the Frobenius element for $\mathfrak{p}$ shall be the identity, and hence the Frobenius element captures behaviour in the case where a prime does not completely split.

The important realization is that if $\mathrm{Gal}_k(F)$ is abelian, there exist $\sigma \in \mathrm{Gal}_k(F)$ that are associated to primes $\mathfrak{p}$ in the sense that the structure of σ mod $\mathfrak{B}/\mathfrak{p}$ is simply the power map (i.e. cyclic)! If each element $\sigma \in \mathrm{Gal}_k(F)$ can have such an association, then that means that primes have a strong link to the structure of an abelian galois group. Class field theory [8] makes this precise: for a finite abelian extension F/k of number fields, the Artin map, which sends an unramified prime $\mathfrak{p}$ to its Frobenius element, identifies $\mathrm{Gal}_k(F)$ with a quotient of a ray class group of k.

When $\mathrm{Gal}_k(F)$ is non-abelian, the situation become more tenuous. Robert Langland is credited for finding the right framework to extend this correspondence through the use of *representation theory*. This is the origins of the famous *Langland's Conjecture*.

1.4.5 Example: Cyclotomic Fields

By example 50.1.31, the discriminant of $\mathbb{Z}[\zeta_{p^k}]$, which is the ring of integers of $\mathbb{Q}(\zeta_{p^k})$ by proposition 50.1.50, is up to sign a power of p. So by corollary 50.4.58, a prime that ramifies in $\mathbb{Q}(\zeta_{p^k})$ divides this discriminant, hence equals p.

Lemma 1.4.66: Cyclotomic Fields Lemma

Let $n = p^k$ be a prime power, ζ_{p^k} a primitive p^kth root of unity, and $\lambda = 1 - \zeta_{p^k}$. Then the principal ideal $(\lambda) \subseteq \mathcal{O}$ in the ring of integers of $\mathbb{Q}(\zeta_{p^k})$ is a prime ideal of degree 1, and:

$$p\mathcal{O} = (\lambda)^d$$

where $d = \varphi(n) = [\mathbb{Q}(\zeta_{p^k}) : \mathbb{Q}]$. Furthermore, the basis $1, \zeta_{p^k}, ..., \zeta_{p^k}^{d-1}$ of $\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}$ has discriminant:

$$d_{\mathbb{Q}(\zeta_{p^k})/\mathbb{Q}}(1, \zeta_{p^k}, ..., \zeta_{p^k}^{d-1}) = \pm p^s$$

where $s = p^{k-1}(kp - k - 1)$

***Proof*:**
Write $\zeta = \zeta_{p^k}$, $K = \mathbb{Q}(\zeta)$ and $\Phi = \Phi_{p^k}$.

Step 1: $p = \prod_r (1 - \zeta^r)$. By [3, Cyclotomic Polynomials] and [3, Cyclotomic Polynomial divisor Representation],

$$\Phi(x) = \prod_{r \in (\mathbb{Z}/p^k\mathbb{Z})^\times} (x - \zeta^r) = \frac{x^{p^k} - 1}{x^{p^{k-1}} - 1} = x^{p^{k-1}(p-1)} + \cdots + x^{p^{k-1}} + 1$$

It is irreducible over $\mathbb{Q}$ ([3, Irreducibility Of Cyclotomic Polynomials]), so it is the minimal polynomial of ζ and $d = [K : \mathbb{Q}] = \varphi(p^k)$. Setting $x = 1$ gives

$$p = \Phi(1) = \prod_{r \in (\mathbb{Z}/p^k\mathbb{Z})^\times} (1 - \zeta^r)$$

Step 2: $p\mathcal{O} = (\lambda)^d$. For $r \in (\mathbb{Z}/p^k\mathbb{Z})^\times$, represented by a positive integer, $1-\zeta^r = (1-\zeta)\epsilon_r$ with $\epsilon_r = 1+\zeta+\cdots+\zeta^{r-1} \in \mathcal{O}$. If r' is a positive integer with $rr' \equiv 1 \bmod p^k$, then

$$\frac{1-\zeta}{1-\zeta^r} = \frac{1-(\zeta^r)^{r'}}{1-\zeta^r} = 1+\zeta^r+\cdots+(\zeta^r)^{r'-1} \in \mathcal{O}$$

so ϵ_r is a unit of $\mathcal{O}$. By Step 1, $p = \epsilon\lambda^d$ with the unit $\epsilon = \prod_r \epsilon_r$, so $p\mathcal{O} = (\lambda)^d$.

Step 3: (λ) *is prime of degree* 1. The element λ is not a unit, since otherwise $p\mathcal{O} = \mathcal{O}$ and $1/p \in \mathcal{O} \cap \mathbb{Q} = \mathbb{Z}$ (theorem 50.1.9(1)). Write $(\lambda) = \mathfrak{q}_1^{a_1}\cdots\mathfrak{q}_t^{a_t}$ with distinct primes and $a_j \geq 1$, so that $p\mathcal{O} = \mathfrak{q}_1^{da_1}\cdots\mathfrak{q}_t^{da_t}$. By proposition 50.4.3, with $\mathfrak{o} = \mathbb{Z}$ and $\mathcal{O} = \mathcal{O}_K$, $\sum_j da_j f_j = d$, so $t = 1$, $a_1 = 1$ and $f_1 = 1$: (λ) is a prime of degree 1.

Step 4: the discriminant. Let $\eta = \zeta^{p^{k-1}}$, a primitive pth root of unity. By proposition 50.1.29, $\operatorname{disc}(1,\zeta,\ldots,\zeta^{d-1}) = \pm\operatorname{Nm}_{K/\mathbb{Q}}(\Phi'(\zeta))$. Differentiating $(x^{p^{k-1}}-1)\Phi(x) = x^{p^k}-1$ and substituting $x = \zeta$, where $\Phi(\zeta) = 0$, gives

$$(\eta-1)\Phi'(\zeta) = p^k\zeta^{p^k-1} = p^k\zeta^{-1}$$

The norm is multiplicative, $\operatorname{Nm}_{K/\mathbb{Q}}(p^k) = p^{kd}$, and $\operatorname{Nm}_{K/\mathbb{Q}}(\zeta^{-1}) = \pm 1$ as ζ^{-1} is a unit (proposition 50.1.18). By Step 1 for $k = 1$ and corollary 50.1.13, $\operatorname{Nm}_{\mathbb{Q}(\eta)/\mathbb{Q}}(1-\eta) = \prod_{r\in(\mathbb{Z}/p\mathbb{Z})^\times}(1-\eta^r) = p$, so $\operatorname{Nm}_{\mathbb{Q}(\eta)/\mathbb{Q}}(\eta-1) = \pm p$. As $[K:\mathbb{Q}(\eta)] = \varphi(p^k)/\varphi(p) = p^{k-1}$, proposition 50.1.12 gives $\operatorname{Nm}_{K/\mathbb{Q}}(\eta-1) = \operatorname{Nm}_{\mathbb{Q}(\eta)/\mathbb{Q}}(\eta-1)^{p^{k-1}} = \pm p^{p^{k-1}}$. Hence

$$\operatorname{Nm}_{K/\mathbb{Q}}(\Phi'(\zeta)) = \pm\frac{\operatorname{Nm}_{K/\mathbb{Q}}(p^k\zeta^{-1})}{\operatorname{Nm}_{K/\mathbb{Q}}(\eta-1)} = \pm\frac{p^{k\varphi(p^k)}}{p^{p^{k-1}}} = \pm p^{p^{k-1}(kp-k-1)}$$

and $\operatorname{disc}(1,\zeta,\ldots,\zeta^{d-1}) = \pm p^s$, completing the proof.

For general n, the ring of integers of $\mathbb{Q}(\zeta_n)$ is $\mathbb{Z}[\zeta_n]$ (proposition 50.1.51), and its discriminant $\operatorname{disc}(\zeta_n)$ divides $n^{\varphi(n)}$ (example 50.1.31). So by corollary 50.4.58, only primes dividing n can ramify in $\mathbb{Q}(\zeta_n)$.

Proposition 1.4.67: Prime Decomposition In Cyclotomic Field

Let $p_1, p_2, \ldots$ enumerate the primes, and let $n = \prod_k p_k^{n_k}$, almost all $n_k = 0$, be the prime factorization of $n \in \mathbb{Z}_{>0}$. Let f_k be the smallest positive integer such that

$$p_k^{f_k} \equiv 1 \bmod (n/p_k^{n_k})$$

Then in $\mathbb{Q}(\zeta_n)$, we have:

$$p_k = (\mathfrak{p}_1\cdots\mathfrak{p}_r)^{\varphi(p_k^{n_k})}$$

where $\mathfrak{p}_1, ..., \mathfrak{p}_r$ are distinct prime ideals all of degree f_k.

Note that if $p_k \nmid n$, then $n_k = 0$, so the condition becomes

$$p_k^{f_k} \equiv 1 \bmod n$$

and $\varphi(p_k^{n_k}) = 1$, giving us:

$$p_k = \mathfrak{p}_1 \cdots \mathfrak{p}_r$$

***Proof*:**
Fix $p = p_k$, write $\nu = n_k$ and $m = n/p^\nu$, so $p \nmid m$, and let $f = f_k$, the least positive integer with $p^f \equiv 1 \bmod m$. Let $K = \mathbb{Q}(\zeta_n)$.

Step 1: reduction to Φ_m. By proposition 50.1.51, $\mathcal{O}_K = \mathbb{Z}[\zeta_n]$, so the conductor of $\mathbb{Z}[\zeta_n]$ is $\mathcal{O}_K$ and theorem 50.4.7 applies to every prime: $p\mathcal{O}_K$ factors as Φ_n, the minimal polynomial of ζ_n ([3, Irreducibility Of Cyclotomic Polynomials]), factors modulo p. As $\gcd(m, p^\nu) = 1$, the primitive nth roots of unity are the products $\xi\eta$ of a primitive mth root of unity ξ and a primitive p^νth root of unity η, so

$$\Phi_n(x) = \prod_{\xi,\eta}(x - \xi\eta)$$

Let $\mathfrak{p}$ be a prime of $\mathcal{O}_K$ over p. Each η satisfies $(\eta - 1)^{p^\nu} \equiv \eta^{p^\nu} - 1 = 0 \bmod p$, as the binomial coefficients $\binom{p}{i}$ with $0 < i < p$ are divisible by p, so $\eta - 1 \in \mathfrak{p}$. Hence $\Phi_n(x) \equiv \prod_{\xi,\eta}(x - \xi) = \Phi_m(x)^{\varphi(p^\nu)}$ coefficientwise modulo $\mathfrak{p}$, and as both sides lie in $\mathbb{Z}[x]$ and $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$, the congruence $\Phi_n \equiv \Phi_m^{\varphi(p^\nu)}$ holds modulo p.

Step 2: $\overline{\Phi}_m$ has distinct irreducible factors. Let $\overline{g}$ denote reduction modulo p. As $p \nmid m$, the derivative mx^{m-1} of $x^m - 1$ is coprime to it over $\mathbb{F}_p$, so $x^m - 1$ is separable over $\mathbb{F}_p$ ([3, Separable Polynomial Test]). By [3, Cyclotomic Polynomial divisor Representation], $x^m - 1 = \prod_{e \mid m} \overline{\Phi}_e$ over $\mathbb{F}_p$, so $\overline{\Phi}_m$ has no repeated irreducible factor.

Step 3: each factor has degree f. Let $h \in \mathbb{F}_p[x]$ be an irreducible factor of $\overline{\Phi}_m$. In the principal ideal domain $\mathbb{F}_p[x]$ ([3, Field Then $\mathbb{F}[x]$ PID And UFD]) the irreducible h is prime ([3, Prime If And Only If Irreducible In PID]), so (h) is maximal ([3, In PID, Prime Ideal iff Maximal Ideal]) and $\mathbb{F}_p[x]/(h)$ is a field. Let $\overline{\xi}$ be the class of x in it, a root of $\overline{\Phi}_m$. Then $\overline{\xi}^m = 1$. If $\overline{\xi}$ had order $e < m$, it would be a root of $x^e - 1 = \prod_{e' \mid e} \overline{\Phi}_{e'}$, hence a root of two different factors $\overline{\Phi}_{e'}$ and $\overline{\Phi}_m$ of $x^m - 1$, a repeated root, contrary to Step 2. So $\overline{\xi}$ has order m. Let $f' = \deg h$, so that $\mathbb{F}_p(\overline{\xi})$ has $p^{f'}$ elements. As $\overline{\xi}$ has order m in $\mathbb{F}_p(\overline{\xi})^\times$, [3, Lagrange Theorem] gives $m \mid p^{f'} - 1$, so $f \leq f'$. Conversely, $y \mapsto y^p$ is a ring endomorphism of $\mathbb{F}_p(\overline{\xi})$, and $y \mapsto y^{p^f}$ fixes $\overline{\xi}$, as $\overline{\xi}^{p^f - 1} = 1$, and fixes $\mathbb{F}_p$, so it fixes every element of $\mathbb{F}_p(\overline{\xi}) = \mathbb{F}_p[\overline{\xi}]$. So the $p^{f'}$ elements of $\mathbb{F}_p(\overline{\xi})$ are roots of $x^{p^f} - x$, and $p^{f'} \leq p^f$. Hence $f' = f$.

Step 4: conclusion. By Steps 2 and 3, $\overline{\Phi}_m = h_1 \cdots h_r$ with distinct irreducible h_i of degree f, and by Step 1, $\overline{\Phi}_n = (h_1 \cdots h_r)^{\varphi(p^\nu)}$. Theorem 50.4.7 gives $p\mathcal{O}_K = (\mathfrak{p}_1 \cdots \mathfrak{p}_r)^{\varphi(p^\nu)}$ with distinct primes $\mathfrak{p}_i$ of inertia degree f, completing the proof.

Corollary 1.4.68: Special Case Of Factorization

Let $n \geq 1$ and let p be a prime number.

1. If $p \nmid n$, then p is unramified in $\mathbb{Q}(\zeta_n)$, and it is totally split if and only if $p \equiv 1 \bmod n$.
2. If $p \mid n$, then p is ramified in $\mathbb{Q}(\zeta_n)$, except when $p = 2$ and $n \equiv 2 \bmod 4$, in which case 2 is unramified.

***Proof*:**

Write $n = p^\nu m$ with $p \nmid m$. By proposition 50.4.67, $p\mathcal{O} = (\mathfrak{p}_1 \cdots \mathfrak{p}_r)^{\varphi(p^\nu)}$ with distinct primes $\mathfrak{p}_i$, all of inertia degree f, the least positive integer with $p^f \equiv 1 \bmod m$. So every ramification index equals $\varphi(p^\nu)$.

1. If $p \nmid n$, then $\nu = 0$ and $\varphi(1) = 1$, so p is unramified. It is totally split exactly when also $f = 1$, that is, when $p \equiv 1 \bmod m = n$.
2. If $p \mid n$, then $\nu \geq 1$ and $\varphi(p^\nu) = p^{\nu-1}(p-1)$, which equals 1 only for $p = 2$ and $\nu = 1$, that is, $n \equiv 2 \bmod 4$. Otherwise the ramification index exceeds 1 and p ramifies.

This completes the proof.

Corollary 1.4.69: Subfield Of Cyclotomic Field

Let $K \subseteq \mathbb{Q}(\zeta_n) = L$ with corresponding subgroup $H = \mathrm{Gal}_K(L) \leqslant \mathrm{Gal}_\mathbb{Q}(L)$, considered as a subgroup of $(\mathbb{Z}/n\mathbb{Z})^\times$ through the isomorphism $\mathrm{Gal}_\mathbb{Q}(L) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ matching the automorphism σ_a with $\sigma_a(\zeta_n) = \zeta_n^a$ to the class of a. For prime p where $p \nmid n$, let f be the least positive integer such that $\overline{p^f} \in H$. Then for prime $\mathfrak{p}$ lying above p in K:

$$f = f_{\mathfrak{p}/p}$$

***Proof*:**

Let $G = \mathrm{Gal}_\mathbb{Q}(L)$. The field L is the splitting field of $x^n - 1$ over $\mathbb{Q}$, so $L/\mathbb{Q}$ is Galois ([3, Finite Normal Extension Equivalences], [3, Separability Over Characteristic 0]), and so is L/K, with group H ([3, Tower Law On Normal Extensions]). By [3, Cyclotomic Field Automorphism], $G \cong (\mathbb{Z}/n\mathbb{Z})^\times$, the class of a corresponding to the automorphism σ_a with $\sigma_a(\zeta_n) = \zeta_n^a$. Then $\sigma_a\sigma_b = \sigma_{ab}$, so $\sigma_p^j = \sigma_{p^j}$ for every $j \geq 1$. Let $\mathfrak{B}$ be a prime of $\mathcal{O}_L$ above $\mathfrak{p}$, which exists because $\mathfrak{p}\mathcal{O}_L \neq \mathcal{O}_L$ (proposition 50.3.10).

Step 1: the inertia degree as an index. Since $p \nmid n$, proposition 50.4.67 gives $p\mathcal{O}_L = \mathfrak{p}_1 \cdots \mathfrak{p}_r$ with distinct primes, so $e(\mathfrak{B}/p) = 1$, and $e(\mathfrak{B}/\mathfrak{p}) = 1$ by corollary 50.4.4. The residue fields are finite, hence perfect ([3, Finite Fields Are Perfect]), so proposition 50.4.37 applies to $\mathfrak{B}$ in $L/\mathbb{Q}$ and in L/K. With $D = D_{\mathfrak{B}/p} \leqslant G$ the decomposition group in $L/\mathbb{Q}$, it gives $|D| = f(\mathfrak{B}/p)$ and $|I_{\mathfrak{B}/p}| = 1$. The decomposition group of $\mathfrak{B}$ in L/K is $\{\tau \in H \mid \tau(\mathfrak{B}) = \mathfrak{B}\} = D \cap H$

(definition 50.4.26), so $|D \cap H| = f(\mathfrak{B}/\mathfrak{p})$. By corollary 50.4.4,

$$f(\mathfrak{p}/p) = \frac{f(\mathfrak{B}/p)}{f(\mathfrak{B}/\mathfrak{p})} = \frac{|D|}{|D \cap H|}$$

Step 2: D is generated by σ_p. By proposition 50.1.51, $\mathcal{O}_L = \mathbb{Z}[\zeta_n]$. For $u, v \in \mathcal{O}_L$, the binomial theorem gives $(u+v)^p - u^p - v^p \in p\mathcal{O}_L$, since p divides $\binom{p}{k}$ for $0 < k < p$, and $c^p \equiv c \bmod p$ for $c \in \mathbb{Z}$ by [3, Lagrange Theorem] in $(\mathbb{Z}/p\mathbb{Z})^\times$. Applying the first fact repeatedly, for $x = \sum_i c_i \zeta_n^i$ with $c_i \in \mathbb{Z}$,

$$x^p \equiv \sum_i c_i^p \zeta_n^{ip} \equiv \sum_i c_i \zeta_n^{ip} = \sigma_p(x) \bmod p\mathcal{O}_L$$

and $\sigma_p(x) - x^p \in p\mathcal{O}_L \subseteq \mathfrak{B}$ for every $x \in \mathcal{O}_L$. In particular $\sigma_p(\mathfrak{B}) \subseteq \mathfrak{B}$, and $\sigma_p(\mathfrak{B})$ is a nonzero prime of $\mathcal{O}_L$ (lemma 50.4.23), hence maximal (theorem 50.2.6), so $\sigma_p(\mathfrak{B}) = \mathfrak{B}$ and $\sigma_p \in D$. The homomorphism $D \to \mathrm{Gal}_{\mathbb{F}_p}(\mathcal{O}_L/\mathfrak{B})$ of proposition 50.4.31 is surjective, with kernel $I_{\mathfrak{B}/p}$ (definition 50.4.32), which is trivial by Step 1. So it is an isomorphism. It sends σ_p to the map $y \mapsto y^p$ of $\mathcal{O}_L/\mathfrak{B}$, which generates $\mathrm{Gal}_{\mathbb{F}_p}(\mathcal{O}_L/\mathfrak{B})$ by [3, Structure of Aut Of Finite Fields]. Hence $D = \langle \sigma_p \rangle$.

Step 3: conclusion. Let f_0 be the least positive integer with $\sigma_p^{f_0} \in H$, which exists because σ_p has finite order N. If $\sigma_p^j \in H$, write $j = af_0 + b$ with $0 \le b < f_0$. Then $\sigma_p^b = \sigma_p^j(\sigma_p^{f_0})^{-a} \in H$, so $b = 0$. Hence $\langle \sigma_p \rangle \cap H = \langle \sigma_p^{f_0} \rangle$, and f_0 divides N, as $\sigma_p^N = 1 \in H$. The subgroup $\langle \sigma_p^{f_0} \rangle$ has order N/f_0, so Steps 1 and 2 give

$$f(\mathfrak{p}/p) = \frac{|\langle \sigma_p \rangle|}{|\langle \sigma_p \rangle \cap H|} = \frac{N}{N/f_0} = f_0$$

Under $G \cong (\mathbb{Z}/n\mathbb{Z})^\times$, $\sigma_p^j = \sigma_{p^j}$ corresponds to $\overline{p^j}$, so f_0 is the least positive integer with $\overline{p^{f_0}} \in H$, that is $f_0 = f$, completing the proof.

In summary, the ring of integers of $\mathbb{Q}(\zeta_n)$ is $\mathbb{Z}[\zeta_n]$ (proposition 50.1.51), so it is monogenic, and proposition 50.4.67 gives the factorization of every prime in it. By corollary 50.4.68, a prime p ramifies in $\mathbb{Q}(\zeta_n)$ if and only if $p \big| n$, except that 2 is unramified when $n \equiv 2 \bmod 4$.

1.4.6 Application: Quadratic Extensions and Quadratic Reciprocity

We now take a moment to solve a classical number theory problems with the developed tools. Let $p \in \mathbb{Z}$ be a prime number. In [3, Application: Sums of Primes and Gaussian Integers], it was shown that p can be written as the sum of two squares if and only if $p \equiv 1 \bmod 4$. Let us now ask another question that may shed further light on the nature of primes: the square-root of primes is always an irrational number. However, when modding by $n \in \mathbb{N}$, it is possible that there is an answer. By the Chinese Remainder Theorem, it suffices to check when $n = q$ is a prime. Thus:

what primes p have the property that there exists an n such that $n^2 \equiv p \bmod q$

Example 1.4.70: Concrete Example
Let $p = 3$. Then:

mod	1^2	2^2	3^2	4^2	5^2	6^2
2	1					
3	1	1				
5	1	4	4	1		
7	1	4	2	2	4	1
11	1	4	9	5	***3***	***3***

Note the table is mirrored as $(-n)^2 = n^2$. Hence, for a reason that is yet to be explained in this book, the prime 3 is somehow tied to 11 when looking for square-roots of 3. The reader can verify that 13, 23, 37, and so forth also have a square-root of 3.

The reader should play around with different pairs (p, q) and flip them around to consider (q, p), there shall appear some symmetry that at first is not obvious.

This relation between primes and square-root in modular arithmetics was first documented to be discovered by Euler, however it took a generation before a complete proof by Gauss appeared. As it turns out, the condition translates into a condition on rings of integers of quadratic extensions!

Definition 1.4.71: Quadratic Residue

Let p be a prime and $n \in \mathbb{Z}$. If there exists an $m \in \mathbb{Z}$ such that $m^2 \equiv n \bmod p$, then n is called a *quadratic residue* mod p.

Whether p has a square root modulo q is the question whether $x^2 - p \equiv 0 \bmod q$ has a solution. Let $K = \mathbb{Q}(\sqrt{p})$ and consider $\mathcal{O}_K/q\mathcal{O}_K$, which is isomorphic to $(\mathbb{Z}/q\mathbb{Z})^2$ as an abelian group. The factorization of $q\mathcal{O}_K$ detects the answer: for odd $q \nmid p$, theorem 50.4.72(3) below shows that $q\mathcal{O}_K$ is a product of two distinct primes exactly when p is a square modulo q, and the same holds for $K = \mathbb{Q}(\sqrt{m})$ with m square-free and $q \nmid m$. By example 50.1.17, $\mathcal{O}_K$ has the integral basis

$$\begin{cases} \{1, \sqrt{m}\} & m \not\equiv 1 \bmod 4 \\ \left\{1, \dfrac{1+\sqrt{m}}{2}\right\} & m \equiv 1 \bmod 4 \end{cases}$$

and $\operatorname{disc}(\mathcal{O}_K) = 4m$ in the first case and $\operatorname{disc}(\mathcal{O}_K) = m$ in the second. Now, take $(p) \subseteq \mathbb{Z}$ a prime ideal. Then we know that either p can split as:

$$p\mathcal{O}_K = \mathfrak{B} \qquad p\mathcal{O}_K = \mathfrak{B}^2 \qquad p\mathcal{O}_K = \mathfrak{B}_1\mathfrak{B}_2$$

and that only m and possibly 2 ramifies.

Theorem 1.4.72: Primes In Quadratic Extensions

Let p be a prime number and $K = \mathbb{Q}(\sqrt{m})$ a quadratic extension where $m \neq 0, 1$ is square-free. Then:

1. If $p \mid m$:
$$p\mathcal{O}_K = (p, \sqrt{m})^2$$
with $(p, \sqrt{m})$ a prime ideal.

2. If m is odd, then 2 splits as:
$$2\mathcal{O}_K = \begin{cases} (2, 1+\sqrt{m})^2 & m \equiv 3 \bmod 4 \\ \left(2, \frac{1+\sqrt{m}}{2}\right)\left(2, \frac{1-\sqrt{m}}{2}\right) & m \equiv 1 \bmod 8 \\ (2) & m \equiv 5 \bmod 8 \end{cases}$$

3. If p is odd and $p \nmid m$:
$$p\mathcal{O}_K = \begin{cases} (p, n+\sqrt{m})(p, n-\sqrt{m}) & m \equiv n^2 \bmod p \\ (p) & m \text{ not a square modulo } p \end{cases}$$

In each case the ideals displayed on the right are distinct prime ideals.

***Proof*:**
All cases come from corollary 50.4.8 with $K = \mathbb{Q}$ and $\mathfrak{p} = p\mathbb{Z}$, applied to a generator θ of K whose minimal polynomial lies in $\mathbb{Z}[x]$. By example 50.1.17, $\mathcal{O}_K = \mathbb{Z}[\sqrt{m}]$ if $m \equiv 2, 3 \bmod 4$, and $\mathcal{O}_K = \mathbb{Z}[\omega] = \mathbb{Z} + \mathbb{Z}\omega$ with $\omega = \frac{1+\sqrt{m}}{2}$ if $m \equiv 1 \bmod 4$. In the second case $\mathbb{Z}[\sqrt{m}] = \mathbb{Z} + 2\mathbb{Z}\omega$, as $\sqrt{m} = 2\omega - 1$, so $[\mathcal{O}_K : \mathbb{Z}[\sqrt{m}]] = 2$. So for $\theta = \sqrt{m}$, with minimal polynomial $x^2 - m$, the index $[\mathcal{O}_K : \mathbb{Z}[\theta]]$ is 1 or 2, and for $\theta = \omega$ when $m \equiv 1 \bmod 4$, with minimal polynomial $x^2 - x + \frac{1-m}{4}$, the index is 1.

Step 1: proof of (1). Take $\theta = \sqrt{m}$. If p is odd, p does not divide the index. If $p = 2$, then m is even and square-free, so $m \equiv 2 \bmod 4$ and the index is 1. As $p \mid m$, $x^2 - m \equiv x^2 \bmod p$, so corollary 50.4.8 gives the prime $(p, \sqrt{m})$ of inertia degree 1 with $p\mathcal{O}_K = (p, \sqrt{m})^2$.

Step 2: proof of (2). If $m \equiv 3 \bmod 4$, take $\theta = \sqrt{m}$, of index 1. Modulo 2, $x^2 - m \equiv x^2 + 1 \equiv (x+1)^2$, so $2\mathcal{O}_K = (2, 1+\sqrt{m})^2$. If $m \equiv 1 \bmod 4$, take $\theta = \omega$. If $m \equiv 1 \bmod 8$, then $\frac{1-m}{4}$ is even and $x^2 - x + \frac{1-m}{4} \equiv x(x-1) \bmod 2$, so $2\mathcal{O}_K = (2, \omega)(2, \omega - 1)$, and $(2, \omega - 1) = (2, \frac{1-\sqrt{m}}{2})$ because $\frac{1-\sqrt{m}}{2} = 1 - \omega$. If $m \equiv 5 \bmod 8$, then $\frac{1-m}{4}$ is odd and $x^2 - x + \frac{1-m}{4} \equiv x^2 + x + 1 \bmod 2$, which has no root in $\mathbb{F}_2$ and so is irreducible, being of degree 2. Then $2\mathcal{O}_K$ is prime, of inertia degree 2.

Step 3: proof of (3). Take $\theta = \sqrt{m}$, whose index 1 or 2 is prime to the odd p. If $m \equiv n^2 \bmod p$, then $n \not\equiv 0$ and $n \not\equiv -n \bmod p$, as $p \nmid m$ and p is odd, so $x^2 - m \equiv (x-n)(x+n) \bmod p$ with distinct factors, and $p\mathcal{O}_K = (p, \sqrt{m} - n)(p, \sqrt{m} + n)$. If m is not a square modulo p, then $x^2 - m$

has no root in $\mathbb{F}_p$ and is irreducible, so $p\mathcal{O}_K$ is prime, completing the proof.

Definition 1.4.73: Legendre Symbol

Let p be an odd prime in $\mathbb{Z}$. Then for $n \in \mathbb{Z}$, $p \nmid n$, the *Legendre symbol* is defined to be:

$$\left(\frac{n}{p}\right) = \begin{cases} 1 & \exists m,\ m^2 \equiv n \bmod p \\ -1 & \text{otherwise} \end{cases}$$

The symbol depends only on the class of n modulo p, since the congruence $m^2 \equiv n \bmod p$ does.

Lemma 1.4.74: Multiplicativity Of The Legendre Symbol

Let p be an odd prime. The squares in $(\mathbb{Z}/p\mathbb{Z})^\times$ form a subgroup S of index 2, and for $a, b \in \mathbb{Z}$ prime to p

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$$

***Proof*:**
The map $(\mathbb{Z}/p\mathbb{Z})^\times \to (\mathbb{Z}/p\mathbb{Z})^\times$, $x \mapsto x^2$, is a homomorphism, as the group is abelian, and its image is S. Its kernel consists of the x with $(x-1)(x+1) = x^2 - 1 = 0$ in the field $\mathbb{Z}/p\mathbb{Z}$, namely $x = 1$ and $x = -1$, which are distinct as p is odd. So $|S| = (p-1)/2$ ([3, First Isomorphism Theorem], [3, Lagrange Theorem]), and S has index 2.

By definition 50.4.73, $\left(\frac{a}{p}\right) = 1$ exactly when the class $\bar{a}$ lies in S. The quotient $(\mathbb{Z}/p\mathbb{Z})^\times/S$ has order 2, so the bijection from it to the group $\{\pm 1\}$ sending S to 1 and the other coset to -1 is an isomorphism. Composing it with the quotient map gives $\bar{a} \mapsto \left(\frac{a}{p}\right)$, which is therefore a homomorphism, as claimed.

Proposition 1.4.75: First Supplement To Quadratic Reciprocity

Let p be an odd prime. Then

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$$

that is, -1 is a quadratic residue modulo p if and only if $p \equiv 1 \bmod 4$.

***Proof*:**
As $i = \zeta_4$, the field $K = \mathbb{Q}(\zeta_4)$ is the quadratic field $\mathbb{Q}(\sqrt{-1})$. Apply corollary 50.4.69 with $n = 4$ and $K = L = \mathbb{Q}(\zeta_4)$, so that $H = \{\bar{1}\}$, to the prime $p \nmid 4$. A prime $\mathfrak{p}$ of K above p has inertia degree the least $f \geq 1$ with $\overline{p^f} = \bar{1}$ in $(\mathbb{Z}/4\mathbb{Z})^\times$, so $f(\mathfrak{p}/p) = 1$ if and only if $p \equiv 1 \bmod 4$.

On the other hand, p is odd and does not divide $m = -1$, so by theorem 50.4.72(3), $p\mathcal{O}_K$ is a product of two distinct primes if -1 is a square modulo p, and is prime otherwise. As $\sum_i e_i f_i = 2$

by proposition 50.4.3, the primes above p have inertia degree 1 in the first case and 2 in the second. So -1 is a square modulo p if and only if $p \equiv 1 \bmod 4$, that is, if and only if $\frac{p-1}{2}$ is even, as we sought to show.

Theorem 1.4.76: Quadratic Reciprocity

Let p, q be distinct odd primes, and let $p^* = (-1)^{\frac{p-1}{2}} p$, so that $p^* = p$ if $p \equiv 1 \bmod 4$ and $p^* = -p$ if $p \equiv 3 \bmod 4$. Then:

1. q is a quadratic residue modulo p if and only if p^* is a quadratic residue modulo q, that is, $\left(\frac{q}{p}\right) = \left(\frac{p^*}{q}\right)$.
2. The two symbols satisfy
$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}$$
3. $\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right)$ if $p \equiv 1 \bmod 4$ or $q \equiv 1 \bmod 4$, and $\left(\frac{q}{p}\right) = -\left(\frac{p}{q}\right)$ if $p \equiv q \equiv 3 \bmod 4$.

***Proof*:**
Let $\zeta = \zeta_p$ and $L = \mathbb{Q}(\zeta)$, and for $b \in \mathbb{Z}$ prime to p let $\sigma_b \in \mathrm{Gal}_{\mathbb{Q}}(L)$ be the automorphism with $\sigma_b(\zeta) = \zeta^b$, which corresponds to $\bar{b} \in (\mathbb{Z}/p\mathbb{Z})^\times$ as in corollary 50.4.69. Write $\chi(a) = \left(\frac{a}{p}\right)$ for a prime to p. It depends only on a modulo p, it satisfies $\chi(a)^2 = 1$, and it is multiplicative by lemma 50.4.74.

A square root of p^* in L. Let $g = \sum_{a=1}^{p-1} \chi(a)\zeta^a \in L$. If $p \nmid d$, then $\eta = \zeta^d$ is a pth root of unity other than 1, and $(\eta - 1)\sum_{a=0}^{p-1} \eta^a = \eta^p - 1 = 0$ gives $\sum_{a=1}^{p-1} \zeta^{ad} = -1$. For fixed a, as b runs through $1, \dots, p-1$, the integer $c \in \{1, \dots, p-1\}$ with $b \equiv ac \bmod p$ runs through $1, \dots, p-1$, and $\chi(a)\chi(b) = \chi(a)^2\chi(c) = \chi(c)$. Hence

$$g^2 = \sum_{a=1}^{p-1}\sum_{b=1}^{p-1} \chi(a)\chi(b)\zeta^{a+b} = \sum_{c=1}^{p-1} \chi(c) \sum_{a=1}^{p-1} \zeta^{a(1+c)}$$

The inner sum is $p-1$ for $c = p-1$, and -1 for the other c by the case $d = 1 + c$ of the sum above. By lemma 50.4.74, χ takes the value 1 at exactly half of $1, \dots, p-1$, so $\sum_{c=1}^{p-1} \chi(c) = 0$ and

$$g^2 = (p-1)\chi(-1) - \sum_{c=1}^{p-2} \chi(c) = p\chi(-1) - \sum_{c=1}^{p-1} \chi(c) = p\chi(-1)$$

By proposition 50.4.75, $\chi(-1) = (-1)^{\frac{p-1}{2}}$, so $g^2 = p^*$. As $p^* = \pm p$ is square-free and differs from 0 and 1, the field $K = \mathbb{Q}(g) \subseteq L$ is the quadratic field $\mathbb{Q}(\sqrt{p^*})$, with $\sqrt{p^*} = g$.

The subgroup of K. Let b be prime to p. As a runs through $1, \dots, p-1$, the integer $c \in \{1, \dots, p-1\}$ with $c \equiv ab \bmod p$ runs through $1, \dots, p-1$, and $\chi(a) = \chi(a)\chi(b)^2 = \chi(c)\chi(b)$.

So

$$\sigma_b(g) = \sum_{a=1}^{p-1} \chi(a)\zeta^{ab} = \chi(b)\sum_{c=1}^{p-1} \chi(c)\zeta^c = \chi(b)g$$

As $g^2 = p^* \neq 0$, σ_b fixes g, equivalently fixes $K = \mathbb{Q}(g)$ pointwise, if and only if $\chi(b) = 1$. So the subgroup $H = \mathrm{Gal}_K(L)$ of corollary 50.4.69, viewed in $(\mathbb{Z}/p\mathbb{Z})^\times$, is the subgroup of squares.

1. Let $\mathfrak{q}$ be a prime of K above q. As $q \nmid p$, corollary 50.4.69 gives $f(\mathfrak{q}/q) = 1$ if and only if $\bar{q} \in H$, that is, if and only if q is a quadratic residue modulo p. On the other hand, q is odd and does not divide p^*, so by theorem 50.4.72(3) with $m = p^*$, $q\mathcal{O}_K$ is a product of two distinct primes if p^* is a square modulo q, and is prime otherwise. As $\sum_i e_i f_i = 2$ by proposition 50.4.3, $f(\mathfrak{q}/q) = 1$ in the first case and $f(\mathfrak{q}/q) = 2$ in the second. So $\left(\frac{q}{p}\right) = 1$ if and only if $\left(\frac{p^*}{q}\right) = 1$, and the two symbols, each equal to ± 1, coincide.
2. By lemma 50.4.74 and proposition 50.4.75 for the prime q,

$$\left(\frac{p^*}{q}\right) = \left(\frac{-1}{q}\right)^{\frac{p-1}{2}} \left(\frac{p}{q}\right) = (-1)^{\frac{q-1}{2}\frac{p-1}{2}} \left(\frac{p}{q}\right)$$

   Multiplying (1) by $\left(\frac{p}{q}\right)$ and using $\left(\frac{p}{q}\right)^2 = 1$ gives the product formula.
3. The exponent $\frac{p-1}{2}\frac{q-1}{2}$ is odd exactly when $\frac{p-1}{2}$ and $\frac{q-1}{2}$ are both odd, that is, when $p \equiv q \equiv 3 \bmod 4$. Multiplying (2) by $\left(\frac{p}{q}\right)$ gives (3).

This completes the proof.

Proposition 1.4.77: Second Supplement To Quadratic Reciprocity

Let p be an odd prime. Then

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$$

that is, 2 is a quadratic residue modulo p if and only if $p \equiv \pm 1 \bmod 8$.

***Proof*:**

Let $\zeta = \zeta_8$, $L = \mathbb{Q}(\zeta)$ and $\theta = \zeta + \zeta^{-1}$. As $\zeta^2 = i$ and $\zeta^{-2} = -i$, $\theta^2 = \zeta^2 + 2 + \zeta^{-2} = 2$, so $K = \mathbb{Q}(\theta) \subseteq L$ is the quadratic field $\mathbb{Q}(\sqrt{2})$, with $\sqrt{2} = \theta \neq 0$. For odd b let $\sigma_b \in \mathrm{Gal}_{\mathbb{Q}}(L)$ be the automorphism with $\sigma_b(\zeta) = \zeta^b$. Then $\sigma_{-1}(\theta) = \zeta^{-1} + \zeta = \theta$. As $\zeta^4 = -1$, $\zeta^3 = -\zeta^{-1}$ and $\zeta^{-3} = -\zeta$, so $\sigma_3(\theta) = \sigma_{-3}(\theta) = \zeta^3 + \zeta^{-3} = -\theta \neq \theta$. So the subgroup $H = \mathrm{Gal}_K(L)$ of corollary 50.4.69 is $\{\pm\bar{1}\}$ inside $(\mathbb{Z}/8\mathbb{Z})^\times = \{\pm\bar{1}, \pm\bar{3}\}$.

As $p \nmid 8$, corollary 50.4.69 shows that a prime of K above p has inertia degree 1 if and only if $\bar{p} \in H$, that is, $p \equiv \pm 1 \bmod 8$. As p is odd and does not divide $m = 2$, theorem 50.4.72(3) shows that $p\mathcal{O}_K$ is a product of two distinct primes if 2 is a square modulo p, and is prime otherwise, so by proposition 50.4.3 the primes above p have inertia degree 1 exactly when 2 is a square modulo p. So 2 is a square modulo p if and only if $p \equiv \pm 1 \bmod 8$. Finally, writing $p = 8k + r$

with $k \in \mathbb{Z}$ and $r \in \{\pm 1, \pm 3\}$,

$$\frac{p^2-1}{8} = 8k^2 + 2kr + \frac{r^2-1}{8}$$

where $\frac{r^2-1}{8}$ is 0 for $r = \pm 1$ and 1 for $r = \pm 3$. So $(-1)^{\frac{p^2-1}{8}} = 1$ exactly when $p \equiv \pm 1 \bmod 8$, as we sought to show.

1.5 Application: Kronecker Weber Theorem

We now apply our finding to Field Theory and find some important consequences' on *Abelian Extension*. This shall also lead to proving one of the central theorems in field theory, The *Kronecker-Weber Theorem*, which states that all finite abelian extensions show up as sub-extensions of Cyclotomic extensions over $\mathbb{Q}$! Since Cyclotomic extensions are generated by roots of unity, every algebraic integers whose Galois group is abelian can be written as a $\mathbb{Q}$-linear combination of roots of unity! For example:

$$\sqrt{5} = e^{2\pi i/5} - e^{4\pi i/5} - e^{6\pi i/5} + e^{8\pi i/5}$$

namely, all square roots of integers may be written as $\mathbb{Q}$-linear combinations of roots of unity! We first give the setting in which we work in:

Definition 1.5.1: Universal Cyclotomic Field and Abelian Closure

1. Let $K^{\mathbf{cycl}} = \bigcup_{n\in\mathbb{N}} K(\zeta_n)$. If $K = \mathbb{Q}$, then:

   $$\mathrm{Gal}(\mathbb{Q}^{\mathbf{cycl}}) = \varprojlim \mathrm{Gal}(\mathbb{Q}(\zeta_n)) = \varprojlim (\mathbb{Z}/n\mathbb{Z})^\times = \hat{\mathbb{Z}}^\times$$

   is the pro-finite group given by the limit of units of cyclic groups.

2. Let K be a field. Then

   $$K^{\mathbf{ab}} \subseteq \overline{K}$$

   is the maximal extension for which $\mathrm{Gal}_K(K^{\mathbf{ab}})$ is abelian.

Proposition 1.5.2: Properties of Abelian Extensions

1. If $L_1, L_2 \subseteq K^{\mathrm{ab}}$ have finite abelian groups, then $\mathrm{Gal}_K(L_1L_2)$ is a finite abelian group.
2. If $G_K = \mathrm{Gal}_K(\overline{K})$ and $H = [G, G]$ (the commutator [closed] subgroup), then

   $$K^{\mathrm{ab}} = \overline{K}^H$$

   with Galois group G_K/H.

***Proof*:**

1. See [3, Galois Composite Fields II]
2. This comes from the universal property of abelian quotients, see [3, Commutator Subgroup Properties]

Note that in general, $K^{\mathbf{cycl}} \subsetneq K^{\text{ab}}$:

Example 1.5.3: Abelian, But Not Cyclotomic

Let $k = \mathbb{Q}(i)$ and $F = \mathbb{Q}(\sqrt{1+2i})/\mathbb{Q}(i)$. Since every quadratic extension is abelian, F is an abelian extension. However, it is not a cyclotomic extension: for if it was a sub-cyclotomic extension, then since $\mathbb{Q}(i)$ is a cyclotomic extension over $\mathbb{Q}$, then $\mathbb{Q}(\sqrt{1+2i})$ would be a cyclotomic extension, but that is not possible since $\mathbb{Q}(\sqrt{1+2i})/\mathbb{Q}$ is not Galois (but all sub-extensions of a cyclotomic extension are necessarily Galois).

It is thus quite interesting that if $K = \mathbb{Q}$, we have the following:

Theorem 1.5.4: Kronecker Weber's Theorem

Let $K = \mathbb{Q}$. Then:

$$\mathbb{Q}^{\mathbf{cycl}} = \mathbb{Q}^{\text{ab}}$$

In other words, every abelian extension of $\mathbb{Q}$ is contained in a cyclotomic extension.

We will first show we can assume $\text{Gal}_{\mathbb{Q}}(K)$ is a p-group for some prime p. For if $\text{Gal}_{\mathbb{Q}}(K)$ is abelian, we have

$$\text{Gal}_{\mathbb{Q}}(K) \cong \bigoplus_p G_p$$

where G_p is a p-group.

Proof. **Reduction #1**:

Define $K_{p'} = K^{(\bigoplus_{p \neq p'} G_p)}$. Then $K = K_2 K_3 \cdots$ is the join of all these fields. Hence, if all such fields is inside a Cyclotomic extension, so is K. This can naturally be extended to reduce to cyclic groups in the same manner. □

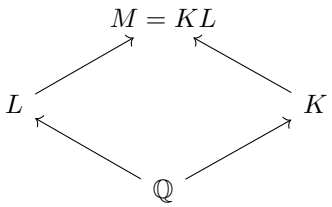
Next, we show that p is the *only* prime which ramifies in K

Proof. **Reduction #2**:

Assume not. Assume ℓ ramifies in K as well. Let I be the inertia group (does not matter what prime we pick). Then the wield inertia group W is trivial (by group theory). Then by theorem 50.4.64:

$$I \hookrightarrow \mathbb{F}_\ell^\times \qquad I \cong \mathbb{Z}/\ell\mathbb{Z}, \quad e \Big| \ell - 1$$

Let $L \in \mathbb{Q}(\mu_\ell)$ be the (unique) extension such that $[L:\mathbb{Q}] = e$. So ℓ totally ramifies in L. Then take:



K and L are abelian, thus so is M. Let I_M be the inertia group of a prime over ℓ. Then

$$I_M \hookrightarrow \mathbb{Z}/e\mathbb{Z} \oplus \mathbb{Z}/\ell\mathbb{Z}$$

Hence

$$I_M \cong \mathbb{Z}/e\mathbb{Z}$$

Let $K' = M^{I_M}$. Then $K' \cap L = \mathbb{Q}$ since nothing ramifies in $K' \cap L$.

Now, $K'L/K'$ has degree at least e, in particular$\geq e$, since ℓ has to ramify (ℓ does not ramify in K', and ramifies in L by e). But then,

$$K'L = M = KL$$

Note that we have not introduced any new ramifications, since no other primes ramify that ramify in L, K. We can afterward inductively remove (if it ramifies, it has to have some inertia group, but it will be trivial in the intermediate extensions).

With this, let's now prove the $p = 2$ case.

Assume $K/\mathbb{Q}$ is abelian $[K:\mathbb{Q}] = 2^n$ and 2 is the only prime which ramifies. I claim that K is cyclotomic.

Without loss of generality, let's assume $\mathbb{Q}(i) \subseteq K$ (just adjoin it if need be). Then $K_0 = K \cap \mathbb{R}$ (fixed field under conjugation). This is still abelian since K is abelian, and $[K:K_0] = 2$, so $K = K_0(i)$. Now, let $K_n/\mathbb{Q}$ be the real cyclotomic field where $\mathrm{Gal}(K_n/\mathbb{Q}) \cong \mathbb{Z}/2^n\mathbb{Z}$ (done it in comments). Consider now

$$M = KK_n$$

Then $\mathrm{Gal}_{\mathbb{Q}}(M) \subseteq \mathbb{Z}/2^n\mathbb{Z} \oplus \mathbb{Z}/2^n\mathbb{Z}$. If $K_n \neq K$, then this Galois group is *not* cyclic (too large order). Therefore, M has ≥ 3 real quadratic subfields (real, since everything is real). But there is only 2 with ramified primes, a contradiction ($\mathbb{Q}(\sqrt{2})$ is the only real quadratic extension in which 2 ramifies). $\square$

Now, for the other primes, we have the following key proposition:

Proposition 1.5.5: Ramification Of Primes in $\mathbb{Z}/n\mathbb{Z}$-Extensions

Let p be an odd prime. Then there exists a unique field $K/\mathbb{Q}$ which is Galois with Galois group $\mathbb{Z}/p\mathbb{Z}$ in which p is the only ramified prime.

***Proof*:**
Let ζ be a pth primitive root of 1, let $L = \mathbb{Q}(\zeta)$. $\pi = 1 - \zeta$. So that $(\pi)^{p-1} = (p)$. Let $\gamma \in \mathcal{O}_L$ be a non pth power. Assume $\gamma \equiv 1 \bmod \pi^p$. Take the field $M = L(\sqrt[p]{\gamma})/L$. Then π is unramified.

Suppose not. Then $\pi\mathcal{O}_M = Q^p$ for some prime Q where $\mathbb{F}_Q \cong \mathbb{F}_{(\pi)} \cong \mathbb{F}_p$. Let δ be such that $\delta^p = \gamma$. Let $r = \frac{\delta-1}{\pi}$. Claim that $r \in \mathcal{O}_M$. Take the conjugates of r over L, i.e. they are

$$\frac{\zeta^i\delta - 1}{\pi} \qquad i \in \mathbb{Z}/p\mathbb{Z}$$

So the minimal polynomial is the product:

$$\prod\left(x - \frac{\zeta^i\delta - 1}{\pi}\right) = \frac{1}{\pi^p}\prod_{i=1}^{p-1}((\pi x - 1) - \zeta^i\delta) = \frac{(\pi x - 1)^p - \gamma}{\pi^p}$$

So this is inside $\mathcal{O}_L[x]$, so r is inside of integers. Thus, there exists some $m \in \mathbb{Z}$ such that $r - m \in \mathbb{Q}$ (because $\mathbb{F}_Q$ is just $\mathbb{F}_p$). But then for all $\sigma \in \mathrm{Gal}_K(M)$, $\sigma(r-m) \in \mathbb{Q}$ since it's the inerta group. But

$$\frac{\zeta\gamma - 1}{\pi} - m \in \mathbb{Q}$$

and so is $\frac{\gamma-1}{\pi} \in \mathbb{Q}$. Subtracting we get

$$\frac{1-\gamma}{\pi}\gamma = \frac{\pi}{\pi}\gamma = \gamma \in \mathbb{Q}$$

But this is a contradiction of our $\gamma \equiv 1 \bmod Q$, as we sought to show.

With this, we can continue the second reduction

Proof. **Reduction #2 cont.**:

By reduction # 2, we only need to work with $K/\mathbb{Q}$ abelian of prime power order p^n, and without loss of generality $K/\mathbb{Q}$ may be assumed to be cyclic since every field is a compositum which have cyclic subgroups.

Now consider $K_n \subseteq \mathbb{Q}(\zeta_{n+1})$ be the field such that

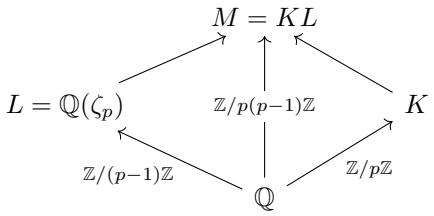
$$[K_n : \mathbb{Q}] = p^n \qquad \mathrm{Gal}(K_n/\mathbb{Q}) \cong \mathbb{Z}/p^n\mathbb{Z}$$

Let $M + KK_n$. In general $\mathrm{Gal}(M/\mathbb{Q}) \hookrightarrow \mathbb{Z}/p^n\mathbb{Z} \oplus \mathbb{Z}/p^n\mathbb{Z}$. If $K \neq K_n$, then like for the $p = 2$ case, this Galois group is not cyclic. Thus, M contains at least $p+1$ subfields which have Galois group $\mathbb{Z}/p\mathbb{Z}$ over $\mathbb{Q}$ in which only p ramifies, a contradiction □

***Proof*:**
Of Kronecker-Weber's Theorem

We shall show that if p is an odd prime, $K/\mathbb{Q}$ is a $\mathbb{Z}/p\mathbb{Z}$ extension, and p is the only prime that ramified in K, then $K \subseteq \mathbb{Q}(\zeta_{p^2})$, which by our earlier reduction shall complete the proof.

Let $\mathrm{Gal}_{\mathbb{Q}}(M) = \langle \sigma \mid \sigma^{p(p-1)} = \mathrm{id} \rangle$, $K = M^{\langle \sigma^p \rangle}$ and $L = M^{\langle \sigma^{p-1} \rangle}$. Consider:



We hat that p is the only prime that ramifies in M. Thus, by Kummer theory, we know that $M = L(\alpha)$ where $\alpha^p = \beta \in L^\times$. Next, we know that

$$\frac{\sigma(\alpha)}{\alpha}$$

is a non-trivial rot of unity. Without loss of generality say that $\frac{\sigma^{p-1}(\alpha)}{\alpha} = \zeta_p$ (so $\langle \sigma^{p-1} \rangle = \mathrm{Gal}_L(M)$). Note that $\sigma(\zeta_p) = \zeta_p^n$, where $\overline{n} \in (\mathbb{Z}/p\mathbb{Z})^\times$ (i.e. is a primitive root).

I claim that

$$M = L\left(\frac{\sigma(\alpha)}{\alpha}\right)$$

For suppose not. Then $\frac{\sigma(a)}{\alpha} \in L$, and

$$\frac{\sigma(\alpha)}{\alpha} = \sigma^{p-1}\left(\frac{\sigma(\alpha)}{\alpha}\right) = \frac{\sigma^p(\alpha)}{\sigma^{p-1}(\alpha)} = \frac{\sigma(\zeta_p\alpha)}{\zeta_p\alpha} \implies 1 = \frac{\sigma(\zeta_p)}{\zeta_p}$$

but that is a contradiction, hence $M = L\left(\frac{\sigma(\alpha)}{\alpha}\right)$.

For the rest of this proof, we write

$$\alpha \mapsto \frac{\sigma(\alpha)}{\alpha} = \alpha' \qquad \beta \mapsto \frac{\sigma(\beta)}{\beta} = \beta'$$

Let P_k, P_L, P_M be the primes ideals above $p\mathbb{Z}$. Take the prime decomposition:

$$\beta\mathcal{O}_L = \prod Q^{n_Q} = P_L^r \cdot \prod_{Q \neq P} Q^{n_Q} \qquad n_Q \in \mathbb{Z}$$

Then:

$$\sigma(\beta)\mathcal{O}_L = P_L^r \prod_{Q \neq P_L} \sigma(Q)^{n_Q}$$

$$\frac{\sigma(\beta)\mathcal{O}_L}{\beta} = \prod_{Q \notin P_L} Q^{m_Q}$$

Hence, P_L does not occur in $\beta\mathcal{O}_L$. Take $\beta'' = \beta' N^P$ and $\alpha'' = \alpha' N$ for $N \in \mathbb{Z}$ such that $\beta'' \in \mathcal{O}_L$ Then:

$$M = L(\alpha'') \qquad (\alpha'')^P = \beta''$$

Now let $\pi = 1 - \zeta_p \in \mathcal{O}_L$ so that $P_L = (\pi)$. Then $\mathbb{F}_{P_L} \cong \mathbb{F}_P$ implies $\beta'' \bmod P_L = m \bmod P_L$ for $m \leq N$. Let $m_0 = m^{-1} \bmod p$, so $\beta'' \to m_0\beta''$ and $\alpha'' \to m_0\alpha''$. Without loss of generality say $\beta'' = 1 \bmod P_L$. Then for some $c \in \mathbb{Z}/p\mathbb{Z}$

$$\beta'' = 1 + c\pi \bmod P_L^2 \qquad \zeta_p = 1 - \pi \bmod P_L^2 \qquad \zeta_p^r = 1 - r\pi \bmod P_L^2 \implies \beta'' = \zeta_p^{-c}\nu$$

where $\nu \equiv 1 \bmod P_L^2$. I claim that $\nu \in (L^\times)^P$, which will imply $M = \mathbb{Q}(\zeta_{p^2})$ and complete the proof. First:

$$\begin{aligned}
(\alpha'')^p &= \beta'' = \zeta_p^{-c}\nu \\
\sigma^{p-1}(\alpha'') &= \alpha''\zeta_p \\
\implies \sigma(\zeta_p) &= \zeta_p^n && n \in (\mathbb{Z}/p\mathbb{Z})^\times \\
\implies \sigma^p(\alpha'') &= \sigma(\sigma^{p-1}(\alpha'')) = \sigma(\alpha''\zeta_p) = \sigma(\alpha'') \cdot \zeta_p^n \\
\implies \sigma^{p-1}((\alpha'')^n) &= (\alpha'')^n\zeta_p^n \\
\implies \sigma^{p-1}\left(\frac{\sigma(\alpha'')}{(\alpha'')^n}\right) &= \frac{\sigma(\alpha'')}{(\alpha'')^n} \\
\implies \frac{\sigma(\alpha'')}{(\alpha'')^n} &\in M^{\langle\sigma^{p-1}\rangle} = L \\
\implies \frac{\sigma(\beta'')}{(\beta'')^n} &\in (L^\times)^P \\
\implies \frac{\sigma(\nu\zeta_p^{-c})}{\nu^n\zeta_p^{-nc}} &\in (L^\times)^p
\end{aligned}$$

Thus $\frac{\sigma(\nu)}{\nu^n} \in (L^\times)^p$.

Next, we claim that $\nu = 1 \bmod P_L^p$. Suppose not. Then let $p^M \| \nu - 1$ where $1 \leq m \leq p-1$, and $\nu = 1 + c\pi^m \bmod P_L^{m+1}$ (where $c \in (\mathbb{Z}/p\mathbb{Z})^\times$). Then by this:

$$\nu^n = 1 + nc\pi^m \bmod P_L^{m+1}$$

hence,

$$\sigma(\pi) = \sigma(1 - \zeta_p) = 1 - \zeta_p^n = (1 - \zeta_p)(1 + \zeta_p + \cdots + \zeta_p^{n-1}) = \pi n \bmod P_L^2$$

implying

$$\sigma(\pi^m) = \pi^m n^m \bmod P_L^{m+1} \implies \sigma(\nu) = 1 + cn^m\pi^m \bmod P_L^{m+1}$$

Note that $n^m \not\equiv n \bmod p$ for $n \in (\mathbb{Z}/p\mathbb{Z})^\times$. Now as a corollary:

$$\frac{\sigma(\nu)}{\nu^m} \equiv 1 + d\pi^m \bmod P_L^{m+1} \qquad d \in \mathbb{Z}/p\mathbb{Z}^\times$$

However,

$$\frac{\sigma(\nu)}{n\nu} = S^p \qquad S \in L$$

where P_L does not occur in S, and:

$$S^p \equiv 1 \bmod P_L \implies S = 1 \bmod P_L$$

Giving us $S = 1 + t$, $P_L \big| t$, so

$$S^P = 1 + pt + \cdots + t^p \equiv 1 \bmod P_L^P$$

a contradiction!

Finally, $L(\sqrt[p]{\nu})/L$ is unramified above P_L. Let

$$F = L(\sqrt[p]{\nu}),\ F \subseteq M(\zeta_{p^2}) \qquad F/Q \text{ abelian}$$

and $I \leqslant \mathrm{Gal}_{\mathbb{Q}}(\mathbb{F})$ where I is the inertia group above P, and $|I| = p-1$, so $F^I/\mathbb{Q}$ is unramified everywhere, namely $F^I = \mathbb{Q}$, so $F = L$. This implies

$$\nu \in (L^\times)^P$$

But then

$$M = \mathbb{Q}(\zeta_{p^2})$$

as we sought to show.

Let us use it in an actual example:

Example 1.5.6: Using Kronecker-Weber

Let $K/\mathbb{Q}$ be a galois extension with Galois group $\mathbb{Z}/5\mathbb{Z}$ whose discriminant is only divisible by $5, 11$, and 31. How many such number fields are there?

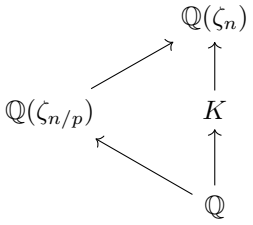
Proof. By Kronecker Weber, we know that K is an intermediary field of a Cyclotomic extension:

$$K \subseteq \mathbb{Q}(\zeta_n)$$

We may without loss of generailty assume n is minnimal for which this containment makes sense. Since only $5, 11, 31$, divide the discriminant, we know the discriminant of K is of the form:

$$\mathrm{disc}(\mathcal{O}_K) = 5^a 11^b 31^c$$

which also mean only $5, 11, 31$ ramify in K. Now, I claim that only $5, 11, 31$ divide n and no other number those. For the sake of contradiction, let's say $p \big| n$ but $p \notin \{5, 11, 31\}$. By minimality, we have:



Let $I_{\mathfrak{B}}$ be the inertia group (for any prime $\mathfrak{B}$ lying above p). Since $p \nmid \mathrm{disc}(\mathcal{O}_K)$, p is unramified. If p is unramified in $\mathbb{Q}(\zeta_{n/p})$, then since $\mathbb{Q}(\zeta_{n/p}) \subseteq \mathbb{Q}(\zeta_n)$, we get that $\mathbb{Q}(\zeta_{n/p}) \subseteq \mathbb{Q}(\zeta_n)^{I_p} \subseteq \mathbb{Q}(\zeta_n)$

$$K \subseteq \mathbb{Q}(\zeta_n)^{I_{\mathfrak{B}}} \subseteq \mathbb{Q}(\zeta_{n/p})$$

contradicting minimality of n. Hence, we get $n = 5^a 11^b 31^c$. Thus:

$$\mathrm{Gal}_{\mathbb{Q}}(\mathbb{Q}(\zeta_n)) \cong (\mathbb{Z}/n\mathbb{Z})^\times \cong \mathbb{Z}/4\mathbb{Z} \times (\mathbb{Z}/5^{a-1}\mathbb{Z}) \times \mathbb{Z}/10\mathbb{Z} \times (\mathbb{Z}/11^{b-1}\mathbb{Z}) \times \mathbb{Z}/30\mathbb{Z} \times (\mathbb{Z}/31^{c-1}\mathbb{Z})$$

which now become a counting argument of how many subgroups of index 5 there are, counting the number of intermediary subfields, thus completing the proof. □

1.6 *Modules Over Dedekind Domains

(Useful for alg-geo, theory itself is also really nice. Also $\mathcal{O}_K$-modules appear in the next section)

Lemma 1.6.1: all Ideal co-maximal up to Module Isomorphism

Let I, J be fractional ideals of a Dedekind domain R. Then $I \oplus J \cong_R R \oplus IJ$.

***Proof*:**
Step 1: comaximal integral ideals. Let $D, E \subseteq R$ be ideals with $D+E = R$, and write $1 = d_0 + e_0$ with $d_0 \in D$ and $e_0 \in E$. By [3, Chinese Remainder Theorem], $D \cap E = DE$. The R-linear maps

$$D \oplus E \to DE \oplus R \quad (d, e) \mapsto (de_0 - ed_0, d+e) \qquad DE \oplus R \to D \oplus E \quad (c, r) \mapsto (c + rd_0, -c + re_0)$$

are well defined, as $de_0, ed_0 \in DE$ and $c \in DE \subseteq D \cap E$, and they are mutually inverse: using $d_0 + e_0 = 1$, the first composite sends (d, e) to $(d(e_0 + d_0), e(d_0 + e_0)) = (d, e)$, and the second sends (c, r) to $(c(e_0 + d_0), r(d_0 + e_0)) = (c, r)$. So $D \oplus E \cong DE \oplus R$.

Step 2: the general case. Choose $y \in K^\times$ with $yJ \subseteq R$ (corollary 50.2.12). By corollary 50.2.21 there is $x \in K^\times$ such that xI is an ideal with $xI + yJ = R$. Multiplication by a nonzero element of K is an R-module isomorphism, so Step 1 gives

$$I \oplus J \cong xI \oplus yJ \cong R \oplus xyIJ \cong R \oplus IJ$$

completing the proof.

Proposition 1.6.2: Ideals Are Projective

Let R be a Dedekind domain and $I \subseteq R$ an ideal. Then I is a projective R-module

***Proof*:**
The zero module is projective. If $I \neq 0$, corollary 50.2.19 gives a nonzero ideal J and $a \neq 0$ with $IJ = aR$, and $aR \cong R$ by multiplication by a. By lemma 50.6.1,

$$I \oplus J \cong R \oplus IJ = R \oplus aR \cong R^2$$

so I is isomorphic to a direct summand of a free module, and it is projective by [3, Equivalencies for Projective Modules], completing the proof.

Proposition 1.6.3: IBN equivalent for Dedekind Domains

Let $I_1, ..., I_n$ be fractional ideals of a Dedekind domain R. Then

$$I_1 \oplus I_2 \oplus \cdots \oplus I_n \cong_R \underbrace{R \oplus R \oplus \cdots \oplus R}_{n-1 \text{ times}} \oplus (I_1 I_2 \cdots I_n)$$

Consequently, if $J_1, \ldots, J_n$ are fractional ideals with $I_1 I_2 \cdots I_n = (a) J_1 J_2 \cdots J_n$ for some $a \in K(R)^\times$, that is, if the products lie in the same class of the class group, then

$$I_1 \oplus I_2 \oplus \cdots \oplus I_n \cong_R J_1 \oplus J_2 \oplus \cdots \oplus J_n$$

***Proof*:**
We induct on n, the case $n = 1$ being trivial. For $n \geq 2$, the induction hypothesis and lemma 50.6.1 give

$$I_1 \oplus \cdots \oplus I_n \cong R^{n-2} \oplus (I_1 \cdots I_{n-1}) \oplus I_n \cong R^{n-2} \oplus R \oplus I_1 \cdots I_n$$

For the consequence, multiplication by a is an isomorphism $J_1 \cdots J_n \cong (a) J_1 \cdots J_n = I_1 \cdots I_n$, so both sums are isomorphic to $R^{n-1} \oplus I_1 \cdots I_n$, completing the proof.

The converse also holds, a theorem of Steinitz: if $I_1 \oplus \cdots \oplus I_n \cong J_1 \oplus \cdots \oplus J_m$, then $n = m$ and the products $I_1 \cdots I_n$ and $J_1 \cdots J_m$ differ by a principal ideal. In particular $R^n \oplus I \cong R^n \oplus J$ exactly when $I = (a)J$ for some $a \in K(R)^\times$. This book does not prove the converse.

Theorem 1.6.4: Classifying Torsion-Free Modules Over Dedekind Domain

Let M be a finitely generated torsion-free R-module over a Dedekind domain R of rank $n \geq 1$, in the sense of [3, [Free] Rank]. Then

$$M \cong R^{n-1} \oplus I$$

for a nonzero ideal I of R, where I may be replaced with any ideal J such that $I = (a)J$ (they belong to the same element in the class group). As a consequence, any finitely generated torsion-free module over a Dedekind domain is projective.

***Proof*:**
Let K be the fraction field of R. Elements of K^n are linearly independent over R exactly when they are linearly independent over K, by clearing denominators, so the rank of a submodule $N \subseteq K^n$ is $\dim_K(KN)$, where KN is its K-span.

Step 1: M lies in R^n. Let $y_1, \ldots, y_n \in M$ be linearly independent and let $m_1, \ldots, m_k$ generate M. By maximality of n, each m_j satisfies $c_j m_j = \sum_i r_{ji} y_i$ with $c_j, r_{ji} \in R$ not all zero, and $c_j \neq 0$ as the y_i are independent. For $c = c_1 \cdots c_k \neq 0$, cM lies in the free module $Ry_1 + \cdots + Ry_n \cong R^n$, and multiplication by c is injective on M, as M is torsion-free. So M is isomorphic to a submodule of R^n, and we may assume $M \subseteq R^n \subseteq K^n$, with $KM = K^n$ as M has rank n.

Step 2: splitting off an ideal. We induct on n. If $n = 1$, then $M \subseteq R$ is a nonzero ideal, and $M \cong R^0 \oplus M$. Let $n \geq 2$ and let $\pi\colon R^n \to R$ be the last coordinate. If $\pi(M) = 0$, then $KM \subseteq K^{n-1} \times 0$, contradicting $KM = K^n$. So $I = \pi(M)$ is a nonzero ideal, projective by proposition 50.6.2, and the sequence $0 \to M' \to M \to I \to 0$ with $M' = \ker(\pi|_M)$ splits ([3, Equivalencies for Projective Modules]), so $M \cong M' \oplus I$. Here M' is torsion-free, and finitely generated, being a quotient of M through this splitting.

Step 3: the rank of M'. Choose $z_0 \in M$ with $a = \pi(z_0) \neq 0$. For $z \in M$, the element $az - \pi(z)z_0$ lies in M', so $z = a^{-1}(az - \pi(z)z_0) + a^{-1}\pi(z)z_0 \in KM' + Kz_0$. Hence $K^n = KM \subseteq KM' + Kz_0$, so $\dim_K KM' \geq n - 1$, and $KM' \subseteq K^{n-1} \times 0$ gives $\dim_K KM' = n - 1$. So M' has rank $n - 1 \geq 1$.

Step 4: conclusion. By the induction hypothesis, $M' \cong R^{n-2} \oplus I'$ for a nonzero ideal I', and by lemma 50.6.1,

$$M \cong R^{n-2} \oplus I' \oplus I \cong R^{n-2} \oplus R \oplus I'I = R^{n-1} \oplus I'I$$

with $I'I$ a nonzero ideal. So for every $n \geq 1$, $M \cong R^{n-1} \oplus I_0$ for a nonzero ideal I_0, namely $I_0 = M$ if $n = 1$ and $I_0 = I'I$ if $n \geq 2$. If $I_0 = (a)J$, then multiplication by $a \neq 0$ is an isomorphism $J \cong I_0$, so I_0 may be replaced by J. Finally, corollary 50.2.19 gives a nonzero ideal J' and $b \neq 0$ with $I_0J' = bR \cong R$, and lemma 50.6.1 gives $I_0 \oplus J' \cong R \oplus I_0J' \cong R^2$, so $R^{n-1} \oplus I_0$ is a direct summand of R^{n+1} and is projective ([3, Equivalencies for Projective Modules]). A torsion-free module of rank 0 is 0, as any nonzero element is linearly independent, completing the proof.

1.7 *Geometric Idea: 1-Dimensional Schemes

See Neukirch, and put orders here too for their geometric interpretation. I think it is important to include (especially once this chapter becomes its own book) since (to me) this explains why ramification happens. We naturally cannot prove the Riemann-Hurwitz Formula here, but this still is very interesting.

1.7.1 Localization of Dedekind Domains

For the motivation to CFT

1.7.2 Orders: Curves with Singularities

This entire section is completely inspired by Neukirch p.72

should also include how if finite L/K then Spec $(\mathcal{O}_L)$ is the normalization of Spec $(\mathcal{O}_K[\alpha])$ where $L = K(\alpha)$. And I would really love a discussion linking Spec $(\mathcal{O}_L)$ to the "generic point" Spec (L) and to $y - m_{\alpha,K}(x)$ where $m_{\alpha,K}(x)$ is the minimal polynomial of L/K

All smooth 1-dimensional curves correspond to Dedekind domains. However, there are some curves with singularities. These curves are associated to sub-rings of Dedekind domains called *orders*

Definition 1.7.1: Order

Let $K/\mathbb{Q}$ be a number field of degree n. Then an order of K is a subring of $\mathcal{O}_K$ which contains an integral basis of length n. The ring $\mathcal{O}_K$ is called a *maximal order*.

By the above definition, an order is of the form:

$$\mathbb{Z}[\alpha_1, ..., \alpha_n]$$

where $K = \mathbb{Q}(\alpha_1, ..., \alpha_n)$, which motivates why $\mathcal{O}_K$ is the maximal order. It is the maximal ring with this property. This also implies that orders need not be normal, hence an order is a one-dimensional Noetherian integral domain (prime remain maximal as if $a \in \mathfrak{p} \cap \mathbb{Z}$ then $a\mathcal{O} \subseteq \mathfrak{p} \subseteq \mathcal{O}$ that is $\mathfrak{p}$ and $\mathcal{O}$ have the same rank n, and thus $\mathcal{O}/\mathfrak{p}$ is a finite integral domain, showing $\mathfrak{p}$ is maximal) . An example of an order is $\mathbb{Z} = \mathbb{Z}(\sqrt{5})$. Many order's naturally are as "rings of multipliers. For example if $\alpha_1, .., \alpha_n$ is a basis for $K/\mathbb{Q}$ and $M = \sum_i \mathbb{Z}\alpha_i$, then

$$\mathcal{O}_M = \{\alpha \in K \mid \alpha M \subseteq M\}$$

is an order.

It is important that we re-prove the Chinese remainder

Here

As orders are not Dedekind domain in general, *not* all ideals are invertible. This also means that the collection of fractional ideals does not form a group.

Proposition 1.7.2: Invertible Fracitonal Ideals

Let I be a fractional ideal of a Noetherian domain R. Then I is invertible iff its localization at every maximal ideal of R is invertible, equivalently, iff its localization at every prime ideal of R is invertible.

***Proof*:**
Write $K = K(R)$, and for a multiplicative set $S \subseteq R \setminus \{0\}$ and an R-submodule $M \subseteq K$ let $S^{-1}M = M \cdot S^{-1}R \subseteq K$. Write $M_{\mathfrak{p}}$ when $S = R \setminus \mathfrak{p}$. Recall from definition 50.2.9 that $(R : I) = \{x \in K \mid xI \subseteq R\}$, and note $I(R : I) \subseteq R$ always, since $x \in (R : I)$ and $y \in I$ give $xy \in R$. The product $I(R : I)$ is an R-submodule of R, that is an ideal.

First, $(IJ)_S = I_S J_S$ for fractional ideals I and J: an element of $I_S J_S$ is a finite sum $\sum_k (i_k/s_k)(j_k/t_k)$, and putting the terms over the common denominator $\prod_k s_k t_k$ exhibits it in $(IJ)_S$, while the reverse inclusion is immediate. Second, $(R : J)_S = (R_S : J_S)$ for a nonzero fractional ideal J. For $\subseteq$, if $xJ \subseteq R$ then $(x/s)J_S \subseteq R_S$. For $\supseteq$, let $x \in K$ satisfy $xJ_S \subseteq R_S$. Since R is Noetherian, $J = (j_1, \ldots, j_r)$ is finitely generated, and each $xj_k \in R_S$ gives $s_k \in S$ with $s_k x j_k \in R$. With $s = s_1 \cdots s_r$ we get $sxJ \subseteq R$, so $sx \in (R : J)$ and $x \in (R : J)_S$.

Suppose I is invertible, say $IJ = R$ for a fractional ideal J. Then $jI \subseteq R$ for every $j \in J$, so $J \subseteq (R : I)$ and

$$R = IJ \subseteq I(R : I) \subseteq R$$

forcing $I(R : I) = R$. Localising at any prime $\mathfrak{p}$ and applying the two identities gives $I_{\mathfrak{p}}(R_{\mathfrak{p}} : I_{\mathfrak{p}}) = (I(R : I))_{\mathfrak{p}} = R_{\mathfrak{p}}$, so $I_{\mathfrak{p}}$ is invertible. This covers every prime, hence every maximal ideal.

Conversely, suppose $I_{\mathfrak{m}}$ is invertible for every maximal ideal $\mathfrak{m}$. By the paragraph proved above,

$I_{\mathfrak{m}}(R_{\mathfrak{m}} : I_{\mathfrak{m}}) = R_{\mathfrak{m}}$ for every such $\mathfrak{m}$. If the ideal $I(R : I)$ were proper it would lie in some maximal ideal $\mathfrak{m}$, and then

$$R_{\mathfrak{m}} = I_{\mathfrak{m}}(R_{\mathfrak{m}} : I_{\mathfrak{m}}) = (I(R : I))_{\mathfrak{m}} \subseteq \mathfrak{m}R_{\mathfrak{m}} \neq R_{\mathfrak{m}}$$

a contradiction. So $I(R : I) = R$ and I is invertible. Finally, if $I_{\mathfrak{p}}$ is invertible at every prime $\mathfrak{p}$ then in particular at every maximal ideal, so the prime version of the criterion follows from the maximal one.

Example 1.7.3: non-invertible ideal of an order
put this example here

(I want this build up here)

Definition 1.7.4: Picard Group Of An Order

Let $J(\mathcal{O})$ be the collection of invertible ideals and $P(\mathcal{O})$ the fractional principal ideals. Then

$$\text{Pic}\,(\mathcal{O}) = \frac{J(\mathcal{O})}{P(\mathcal{O})}$$

A lot more to be said.

The Picard group ignore s ideals that are not invertible. Another approach that gives some useful results is to define the following for A Dedekind domain:

$$\text{Div}(\mathcal{O}) = \mathcal{O}_{\mathfrak{p}}\mathbb{Z}\mathfrak{p}$$

Then $\text{Div}(\mathcal{O})$ is isomorphic to the ideal group. For an order, the divisor group shall be a quotient of this group. We would then need to define the divisor elements, which gives the divisor class group/chow group.

1.8 *Profinite Groups

(original title: infinite galois extensions)

I moved this from the galois theory chapter. I actually want to make this its own chapter when I extract these notes into their own book so that the many books that use profinite groups can reference this namely: the class field theory book, the p-adic rep book, and the galois cohomology book.

In this section, we shall explore the case of infinite separable normal extensions L/K. This section uses topological groups [6, Topological Group] and the following facts of point-set topology from *Topology* [6]: Tychonoff's theorem [6, Tychonoff's Theorem], that closed subsets of compact spaces are compact [6, closed subsets of compact spaces are compact], that compact Hausdorff spaces are normal [6, Compact Hausdorff Spaces Are Normal], that continuous images and closures of connected sets are connected [6, Continuous Images of Connected Spaces Are Connected] [6, Connectedness Survives Adding Limit Points], and that a continuous bijection from a compact space onto a Hausdorff space is a homeomorphism [6, making continuous function homeomorphisms]. Most of this material will be important for a course in *Class Field Theory* [8].

In this section, a Galois extension is a normal separable extension, where the finiteness condition is dropped. We may want to keep the galois correspondence between the fixed fields and the subgroups, however there are simply way too many subgroups. Consider

$$\overline{\mathbb{Q}}/\mathbb{Q} \qquad \text{and} \qquad \mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$$

Then the reader should check that there are only countably many intermediary extension of degree 2 corresponding to the extensions of the form $\mathbb{Q}(\sqrt{\alpha})/\mathbb{Q}$ where α is either a prime p, ip, or a finite compositum of such elements. However, there are uncountably many index two subgroups. To see this, consider $\varphi_S : \mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}}) \to \mathbb{Z}/2\mathbb{Z}$ where S is a finite collection of primes where if σ fixes each $\sqrt{p}$. Then $\varphi_S(s) = 0$, and it is 1 otherwise. This is a group-homomorphism, as the composition of two maps that fix each $\sqrt{p}$ for $p \in S$ shall still fix those primes. Furthermore, There are certainly always morphisms τ such that $\varphi_S(\tau) = 1$ for each finite subset S. Thus, each φ_S is a surjective homomorphism, whose kernel is index 2. But then, there are uncountably distinct $\ker(\varphi_S)$. In fact, we do not even need S to be finite (exercise). In this case, we shall even have index two subgroups whose fixed field corresponds to $\mathbb{Q}$!

The reader may feel that there is some simple solutions that can be found, as it seems there are many galois subgroups corresponding to an intermediary. Indeed, we may still recover the galois correspondence if we appropriately limit which subgroups of $\mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ we consider (and more generally which subgroups of $\mathrm{Gal}_K(L)$ we consider). This is done by taking the *Krull topology* on the galois group, which we shall naturally derive by first taking the *profinite topology*. The profinite topology shall make us limit our attention to normal subgroups of finite index (namely, such subgroups shall be open in the profinite topology), while in the Krull topology we shall limit our attention to normal subgroups of finite index which also have a [finite] galois group (namely, such subgroups shall be open in the Krull topology). Both of these topologies are achieved by showing that the infinite galois group, as a topological group, is a profinite group. The main idea is that all of the information of infinite galois extension is in fact fully captured by the finite normal extension, due to the fact galois groups represent the relationship of polynomials (where an infinite galois groups stores the information of infinitely many polynomials).

1.8.1 Profinite Groups

In this section, a brief introduction to profinite groups is given. This is not an in-depth coverage, but rather it gives enough information to describe infinite galois theory.

Definition 1.8.1: Profinite Group

Take an inverse system of finite groups with the discrete topology. Then a *profinite group* is a topological group that is the limit of the system. Given any topological group G, we can take its *profinite completion* by taking:

$$\widehat{G} = \varprojlim_N G/N$$

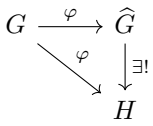
where N ranges over all finite index open normal subgroups ordered by containment.

Any group G can be given a topology by given it the profinite topology given by taking the cosets of finite-index normal subgroups as a basis. The reader should check that if G is a topological group and $\widehat{G}$ is its profinite completion, then the continuous homomorphism $\varphi : G \to \widehat{G}$ can be shown that G is dense in $\widehat{G}$.

Example 1.8.2: Profinite Groups

1. Let G be finite. Then all normal subgroups have finite index and there are finitely many. It follows that $G \cong \widehat{G}$.
2. Let $G = \mathbb{Z}$. Then
$$\widehat{Z} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$
where $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is the p-adic completion of $\mathbb{Z}$.
3. Take $G = \mathbb{Q}$. Then $\mathbb{Q}$ has only one finite-index normal subgroup, namely itself. Hence $\widehat{\mathbb{Q}} = \{1\}$.

The above examples show that the map $\varphi : G \to \widehat{G}$ could be injective, surjective, both. The map will be injective if the intersections of all finite-index open normal subgroups of G is trivial (where we would then say G is residually finite). The completion comes equipped with a natural universal property where a continuous homomorphism $\varphi : G \to H$ with H a profinite group has a unique continuous homomorphism such that the following diagram commutes:



A word on strongly complete? MIT notes p.275 Remark 26.15

Theorem 1.8.3: Characterizing Profinite Groups

A topological group is profinite if and only if it is a totally disconnected compact group

***Proof*:**
Step 1: profinite groups are compact and totally disconnected. An isomorphism of topological groups preserves both properties, so by definition 50.8.1 we may take $G = \varprojlim_{i\in I} G_i \subseteq P = \prod_{i\in I} G_i$ for an inverse system of finite discrete groups G_i over a directed set I, with transition maps $\varphi_{ji}\colon G_j \to G_i$ for $i \le j$ and projections $p_i\colon P \to G_i$. Each G_i is finite, hence compact, so P is compact by Tychonoff's theorem. The space P is Hausdorff: two distinct points differ in some coordinate i, and the preimages under p_i of the two values are disjoint open sets. For $i \le j$ the set
$$C_{ij} = \{g \in P \mid \varphi_{ji}(p_j(g)) = p_i(g)\}$$
is closed, since its complement is the union, over the pairs $(a, b) \in G_j \times G_i$ with $\varphi_{ji}(a) \neq b$, of the open sets $p_j^{-1}(a) \cap p_i^{-1}(b)$. So $G = \bigcap_{i\le j} C_{ij}$ is closed in P, hence compact, as a closed subset of a compact space. If $C \subseteq G$ is connected, then each $p_i(C)$ is a connected subset of the discrete space G_i, being a continuous image of a connected set, so it is a single point. Points of C therefore agree in every coordinate, and C has at most one point. So G is totally disconnected.

Step 2: a compact totally disconnected group is Hausdorff. Let G be a compact totally disconnected group. The closure of the connected set $\{1\}$ is connected, hence equal to $\{1\}$, so

$\{1\}$ is closed, and so is every $\{g\}$, because left translation is a homeomorphism. Let $x \neq y$ in G. Then $G \setminus \{x^{-1}y\}$ is an open neighbourhood of 1, and continuity of $(a,b) \mapsto ab^{-1}$ at $(1,1)$ gives an open neighbourhood V of 1 with $VV^{-1} \subseteq G \setminus \{x^{-1}y\}$. If $xv = yw$ with $v, w \in V$, then $x^{-1}y = vw^{-1} \in VV^{-1}$, which is impossible. So xV and yV are disjoint open sets containing x and y.

Step 3: every open neighbourhood of 1 *contains a clopen neighbourhood of* 1. Let Y be the intersection of all clopen subsets of G containing 1, a closed set. Suppose $Y = A \cup B$ with A and B disjoint and closed in Y, hence closed in G, and $1 \in A$. A compact Hausdorff space is normal, so there are disjoint open sets $U_A \supseteq A$ and $U_B \supseteq B$. The closed sets $C \setminus (U_A \cup U_B)$, for C clopen with $1 \in C$, have empty intersection, because $Y \subseteq U_A \cup U_B$. By compactness finitely many of them already have empty intersection, and the intersection C of the corresponding clopen sets is clopen, contains 1 and lies in $U_A \cup U_B$. Then $C \cap U_A$ is open, and it equals $C \setminus U_B$, which is closed. So $C \cap U_A$ is a clopen set containing 1, hence $Y \subseteq C \cap U_A$, and $B \subseteq Y \subseteq U_A$. As $B \subseteq U_B$ and $U_A \cap U_B = \emptyset$, B is empty. So Y is connected, and $Y = \{1\}$ because G is totally disconnected. Now let U be an open neighbourhood of 1. The closed sets $C \setminus U$, for C clopen with $1 \in C$, have intersection $Y \setminus U = \emptyset$, so by compactness finitely many of them do, and the intersection of the corresponding clopen sets is a clopen neighbourhood of 1 inside U.

Step 4: every clopen neighbourhood C of 1 *contains an open normal subgroup.* The set C is closed in the compact space G, hence compact. For each $c \in C$, the set C is an open neighbourhood of $c \cdot 1$, so continuity of multiplication at $(c, 1)$ gives open sets $W_c \ni c$ and $V_c \ni 1$ with $W_c V_c \subseteq C$. Finitely many $W_{c_1}, \ldots, W_{c_k}$ cover C, and $V = \bigcap_i \left(V_{c_i} \cap V_{c_i}^{-1}\right)$ is an open neighbourhood of 1 with $V = V^{-1}$ and $CV \subseteq C$. By induction $CV^r \subseteq C$ for every $r \geq 1$, where V^r is the set of products of r elements of V. The union $H = \bigcup_{r \geq 1} V^r$ is a subgroup, being closed under products and, as $V = V^{-1}$, under inverses. It is open, each $V^r = \bigcup_{g \in V^{r-1}} gV$ being a union of translates of V. Since $1 \in C$, $H \subseteq CH \subseteq C$. The left cosets of H are disjoint open sets covering G, so by compactness there are finitely many of them, and H has finite index. The conjugate gHg^{-1} depends only on the coset gH, so $N = \bigcap_{g \in G} gHg^{-1}$ is a finite intersection of open subgroups. It is an open normal subgroup with $N \subseteq H \subseteq C$.

Step 5: G is the limit of its finite quotients. Let $\mathcal{N}$ be the set of open normal subgroups of G, directed by reverse inclusion, as $N_1 \cap N_2 \in \mathcal{N}$ for $N_1, N_2 \in \mathcal{N}$. Each G/N with $N \in \mathcal{N}$ is finite, by the coset argument of Step 4, and carries the discrete topology. The map

$$\Phi \colon G \to \varprojlim_{N \in \mathcal{N}} G/N \qquad g \mapsto (gN)_N$$

is a group homomorphism. It is continuous, because each $G \to G/N$ is: the preimage of any subset of G/N is a union of cosets of the open subgroup N. It is injective: for $g \neq 1$, the set $G \setminus \{g\}$ is an open neighbourhood of 1 by Step 2, it contains a clopen neighbourhood of 1 by Step 3, and that contains some $N \in \mathcal{N}$ by Step 4, so $g \notin N$. It is surjective: let $(x_N N)_N$ be a compatible family. Each coset $x_N N$ is closed, its complement being a union of cosets of the open subgroup N. For $N_1, \ldots, N_k \in \mathcal{N}$ and $N = N_1 \cap \cdots \cap N_k$, compatibility gives $x_N N_i = x_{N_i} N_i$, so $x_N \in x_{N_1} N_1 \cap \cdots \cap x_{N_k} N_k$. By compactness the intersection of all the cosets $x_N N$ contains some g, and then $\Phi(g) = (x_N N)_N$. Finally, $\varprojlim_N G/N$ is Hausdorff, as a subspace of a product of discrete spaces (Step 1), and a continuous bijection from a compact space onto a Hausdorff space is a homeomorphism. So Φ is an isomorphism of topological groups, and G is profinite,

completing the proof.

Corollary 1.8.4: Distinguishing Profinite Completion

Let G be a profinite group. Then G is naturally isomorphic to its profinite completion, or even stronger:

$$G \cong \varprojlim G/U$$

where U ranges over all open normal subgroups ordered by containment.
Moreover G is *strongly complete*, meaning that the natural map from G to the profinite completion of its underlying abstract group is an isomorphism, if and only if every finite index subgroup of G is open

***Proof*:**
By definition 50.8.1 write $G = \varprojlim_{i\in I} G_i \subseteq \prod_i G_i$ for an inverse system of finite discrete groups over a directed index set I. Let $p_i : G \to G_i$ be the projections and $N_i = \ker p_i$, an open normal subgroup since p_i is continuous and G_i is discrete. If $j \geq i$ then the transition map carries $p_j(g)$ to $p_i(g)$, so $N_j \subseteq N_i$. A basic open neighbourhood of 1 in the subspace topology of $\prod_i G_i$ is a finite intersection $\bigcap_{i\in F} N_i$, and by directedness there is j with $j \geq i$ for all $i \in F$, so the N_i form a neighbourhood basis at 1.

By theorem 50.8.3 the group G is compact, so every open subgroup U has finite index: its cosets form an open cover of G by disjoint sets, and a finite subcover must consist of all of them. Any open normal subgroup U is a neighbourhood of 1, hence contains some N_i, so the family $\{N_i\}$ is cofinal among open normal subgroups and $\varprojlim_U G/U = \varprojlim_i G/N_i$. The natural map $G \to \varprojlim_i G/N_i$ is injective because $\bigcap_i N_i = \{1\}$, and it is surjective because a compatible family in $\prod_i G/N_i \cong \prod_i p_i(G)$ is a compatible family in $\prod_i G_i$, hence a point of $\varprojlim_i G_i = G$. It is continuous, since each $G \to G/N_i$ is continuous for N_i open, and a continuous bijection from a compact space to a Hausdorff space is a homeomorphism. So $G \cong \varprojlim_U G/U$ through the natural map, and since every open normal subgroup has finite index, this limit is the profinite completion of the topological group G.

For the second statement, let $\widehat{G}_{\text{abs}} = \varprojlim_N G/N$ where N ranges over *all* finite index normal subgroups of G, and let $\varphi : G \to \widehat{G}_{\text{abs}}$ be the natural map. Every open normal subgroup has finite index, so forgetting the coordinates indexed by the non-open N gives a continuous homomorphism $\psi : \widehat{G}_{\text{abs}} \to \varprojlim_U G/U = G$ with $\psi \circ \varphi = \mathrm{id}_G$.

If every finite index subgroup of G is open, the two index families coincide, $\widehat{G}_{\text{abs}} = \varprojlim_U G/U$, and φ is the isomorphism of the first part. Conversely, suppose φ is an isomorphism. Then $\psi = \varphi^{-1}$ is a continuous bijection, and $\widehat{G}_{\text{abs}}$ is compact, being a closed subgroup of the product of the finite discrete groups G/N, while G is Hausdorff. So ψ, and hence φ, is an isomorphism of topological groups. Let $H \leqslant G$ have finite index and set $N = \bigcap_{g\in G} gHg^{-1}$. The map $g \mapsto gHg^{-1}$ factors through the cosets of the normaliser of H, which contains H and therefore has finite index, so there are finitely many conjugates, each of finite index. Their intersection N again has finite index, because G embeds into the product of the finitely many finite coset spaces, and N is normal by construction. The subgroup $\ker\left(\widehat{G}_{\text{abs}} \to G/N\right)$ is open by construction and its preimage under φ is N, because $G \to \widehat{G}_{\text{abs}} \to G/N$ is the quotient map. Hence N is open, and H, a union of cosets of N, is open as well.

1.8.2 Infinite Galois Theory

Let us now find which topology to put on galois groups. Let us first make sure that the notion of there being a correspondence between subfields and fixed fixed fields of subgroups of galois group is still well-defined

Lemma 1.8.5: Fixed Field Infinite Extension

Let L/K be a galois extension with galois group $G = \mathrm{Gal}_K(L)$ and let F be an intermediate field. Then $H = \mathrm{Gal}_F(L)$ has fixed field F. If moreover F/K is normal, then $H \trianglelefteq G$ is a normal subgroup of G, and there is a short exact sequence:

$$1 \to \mathrm{Gal}_F(L) \to \mathrm{Gal}_K(L) \to \mathrm{Gal}_K(F) \to 1$$

which implies:

$$G/H \cong \mathrm{Gal}_K(F)$$

***Proof*:**
Let $\overline{L}$ be an algebraically closed field containing L, which exists by [3, Union Of The Root-Adjoining Chain Is Algebraically Closed] and the chain construction before it.

Step 1: extending embeddings. Let E be an intermediate field of L/K and $\rho\colon E \to L$ a field homomorphism fixing K. The extension L/E is algebraic and $\overline{L}$ is algebraically closed, so, regarding $\overline{L}$ as an extension of E through ρ, [3, Embedding Into Algebraic Extension] gives a field homomorphism $\sigma\colon L \to \overline{L}$ restricting to ρ on E. For $\beta \in L$ with minimal polynomial f over K, $\sigma(\beta)$ is a root of f, because σ fixes K. Since L/K is normal and f has the root β in L, f splits over L ([3, Normal Extension]), so $\sigma(\beta) \in L$. Moreover σ maps the finite set of roots of f in L injectively into itself, hence onto itself, so $\beta \in \sigma(L)$. Thus σ is an automorphism of L fixing K, that is $\sigma \in G$, and it extends ρ.

Step 2: the restriction map, for normal F/K. Suppose F/K is normal, and let $\sigma \in G$ and $\beta \in F$ with minimal polynomial f over K. As F/K is normal, f splits over F. Since σ fixes K, it maps the finite set of roots of f in F injectively, hence bijectively, to itself. So $\sigma(F) = F$ and $\sigma|_F \in \mathrm{Gal}_K(F)$. The map $G \to \mathrm{Gal}_K(F)$, $\sigma \mapsto \sigma|_F$, is a group homomorphism with kernel $\mathrm{Gal}_F(L) = H$, which is therefore normal in G. It is surjective: given $\tau \in \mathrm{Gal}_K(F)$, Step 1 with $E = F$ and $\rho = \tau$ gives $\sigma \in G$ with $\sigma|_F = \tau$. This is the short exact sequence of the statement, and $G/H \cong \mathrm{Gal}_K(F)$ by [3, First Isomorphism Theorem].

Step 3: the fixed field of H is F. Every element of H fixes F, so $F \subseteq L^H$. Let $\alpha \in L \setminus F$ and let g be its minimal polynomial over F, which has degree at least 2. It divides the minimal polynomial of α over K, which has distinct roots, as L/K is separable, and splits over L, as L/K is normal. So g has a root $\alpha' \in L$ with $\alpha' \neq \alpha$. By [3, Extending Isomorphisms to Adjoint Fields] there is an isomorphism $F(\alpha) \to F(\alpha')$ fixing F and sending α to α', and Step 1 with $E = F(\alpha)$ extends it to some $\sigma \in G$. Then σ fixes F, so $\sigma \in H$, while $\sigma(\alpha) = \alpha' \neq \alpha$. So $\alpha \notin L^H$, and $L^H = F$, completing the proof.

Hence, every normal subextension has (at least one) corresponding normal subgroup. However, there

are many normal subgroups whose fixed field F is *not* equal to $\mathrm{Gal}_F(L)$, but is contained in $\mathrm{Gal}_F(L)$ The index two example of $\mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ is the prototypical example, where there are uncountably many index 2 subgroups $H \leqslant \mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ whose fixed field is $\mathbb{Q}$ but naturally H is not the galois subgroup of $\overline{\mathbb{Q}}/\mathbb{Q}$, nor is it the galois group of $\overline{\mathbb{Q}}/K$ for any subextension $K/\mathbb{Q}$. Hence, the following topology limits our attention to the right groups:

Definition 1.8.6: Krull Topology On Galois Group

Let L/K be a Galois (i.e. normal separable) extension with $G = \mathrm{Gal}_K(L)$. Then the *Krull Topology* on G is the basis given by the basis consisting of cosets of the subgroups

$$H_F = \mathrm{Gal}_F(L)$$

where F ranges over finite normal extensions of K in L.

In this topology, every open normal subgroups has finite index, but not every normal subgroup of finite index is open (think $\mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$ again). Hence, this topology is (strictly) coarser than the porfinite topology, however it is still a profinite group:

Proposition 1.8.7: Galois Group is Profinite Group

Let L/K be a galois extension. Then under the Krull topology, the restriction maps induce a natural isomorphism of topological groups:

$$\mathrm{Gal}_K(L) \cong \varprojlim_F \mathrm{Gal}_K(F)$$

where F ranges over finite galois extension of K in L. Hence, $\mathrm{Gal}_K(L)$ is a profinite group whose open normal subgroups are those of the form $\mathrm{Gal}_F(L)$ for some finite normal extension F/K.

***Proof*:**
Let's first show the map is bijective. For injectivity, for every $\alpha \in L$ it is algebraic over K, and hence liens ins some finite normal subextensions F/K, and so every automorphism in $\mathrm{Gal}_K(L)($ is uniquely determined by all its restrictions to finite normal F/K, showing φ is injective. For surjectivity, given any $(\sigma_F) \in \varprojlim_F \mathrm{Gal}_K(F)$, we may define a corresponding $\sigma \in \mathrm{Gal}_K(L)$ that maps to it by point-wise defining:

$$\sigma(\alpha) := \sigma_F(\vec{\alpha})$$

where F is the normal closure of $K(\alpha)$. This map is well-defined by construction of the inverse limit, giving surjectivity.

For continuity, Letting $G = \mathrm{Gal}_K(L)$ and $H_F = \mathrm{Gal}_F(L)$, we can view φ as the natural continuous map:

$$\varphi : G \to \varprojlim G/H_F$$

Finally, to show the inverse is continuous, we will show the equivalent of φ being an open map, which translates to showing that φ maps open subgroups $H \subseteq G$ (i.e. the basis) to open subsets in $\varprojlim G/H_F$. Taking $H = \mathrm{Gal}_F(L)$, we get:

$$\varphi(H) = \{(\sigma_E) \mid \sigma_{E_{E\cap F}} = \mathrm{id}_{E\cap F}\} = \pi_F^{-1}(\mathrm{id}|_F)$$

as E/K ranges over all finite normal subextension of L/K and π_F is the projection from the limit to $\mathrm{Gal}_K(F)$. Then the singleton set $\{\mathrm{id}|_F\}$ is open in the discrete group $\mathrm{Gal}_K(F)$, and so its inverse is open, completing the proof.

Theorem 1.8.8: Fundamental Theorem of Galois Theory, Infinite Extensions

Let L/K be a galois extension with galois group $G = \mathrm{Gal}_K(L)$ endowed with the Krull topology. Then the maps

$$F \mapsto \mathrm{Gal}_F(L) \qquad H \mapsto L^H$$

define inclusion reversing bijections between subextensions F/K of L/K and closed subgroups H of G. Thus, subextensions of finite degree n correspond to subgroups of finite index n, and normal subextensions F/K correspond to normal subgroups $H \leqslant G$ such that as topological groups:

$$\mathrm{Gal}_K(F) \cong G/H$$

***Proof*:**
First off, recall that every open subgroup is clopen, as the compliment is the union of non-trivial cosets, and so closed subgroups of finite index are open. Next, the correspondence between finite galois subextensions F/K and finite index closed normal subgroups H is given by proposition 50.8.7, and furthermore we have $[F : K] = [G : H]$ as $G/H \cong \mathrm{Gal}_K(F)$ by lemma 50.8.5.

For the correspondence maps, if F/K is any finite subextension with normal closure E, then $H = \mathrm{Gal}(L/F)$ contains the normal subgroup $N = \mathrm{Gal}(L/E)$ with finite index. The subgroup N is open and therefore closed, thus H is closed since it is a finite union of cosets of N. The fixed field of H is F by lemma 50.8.5, thus finite subextensions correspond to closed subgroups of finite index. Conversely, every closed subgroup H of finite index has a fixed field F of finite degree, since the intersection of its conjugates is a normal closed subgroup $N = \mathrm{Gal}(L/E)$ of finite index whose fixed field E contains F and has finite degree. The degrees and indices match because $[G : N] = [G : H][H : N]$ and $[E : K] = [F : K][E : F]$. By the previous argument for finite normal subextensions, $[E : K] = [G : N]$ and $[E : F] = [H : N]$ (for the second equality, replace L/K with L/F and G with H).]

Next, for the closed part, any subextension F/K is the union of its finite subextensions E/K. The intersection of the corresponding closed finite index subgroups $\mathrm{Gal}(L/E)$ is equal to $\mathrm{Gal}(L/F)$, which is therefore closed. Conversely, every closed subgroup H of G is an intersection of basic closed subgroups, all of which have the form $\mathrm{Gal}(L/E)$ for some finite subextension E/K, thus $H = \mathrm{Gal}(L/F)$, where F is the union of the E, completing the proof.

Finally, let F/K be normal and $H = \mathrm{Gal}_F(L)$. By lemma 50.8.5, restriction $\rho\colon G \to \mathrm{Gal}_K(F)$, $\sigma \mapsto \sigma|_F$, is a surjective homomorphism with kernel H, so it induces a group isomorphism $\overline{\rho}\colon G/H \to \mathrm{Gal}_K(F)$. The map ρ is continuous: a basic open set of $\mathrm{Gal}_K(F)$ is a coset $\tau\mathrm{Gal}_E(F)$ with E/K finite normal inside F (definition 50.8.6), and if $\rho(\sigma_0) = \tau$ then $\rho^{-1}(\tau\mathrm{Gal}_E(F)) = \sigma_0\mathrm{Gal}_E(L)$, a basic open set of G, since an element of G fixes E exactly when its restriction to F does. So $\overline{\rho}$ is continuous for the quotient topology on G/H. The group G is compact (proposition 50.8.7 and theorem 50.8.3), so G/H is compact: an open cover of G/H pulls back to an open cover of G, a finite subcover of which maps onto a finite subcover. The group $\mathrm{Gal}_K(F)$ is Hausdorff: if $\tau \neq \tau'$, choose $x \in F$ with $\tau(x) \neq \tau'(x)$ and a finite normal E/K

inside F containing x, the normal closure of $K(x)$. Then the cosets $\tau\mathrm{Gal}_E(F)$ and $\tau'\mathrm{Gal}_E(F)$ are disjoint, their elements sending x to $\tau(x)$ and to $\tau'(x)$ respectively. A continuous bijection from a compact space onto a Hausdorff space is a homeomorphism, as recalled at the start of this section, so $\overline{\rho}$ is an isomorphism of topological groups, completing the proof.

Corollary 1.8.9: Closure in Krull Topology

Let L/K be a galois extension and $H \leqslant \mathrm{Gal}_K(L)$ with fixed field F. Then the closure $\overline{H}$ in the Krull Topology is $\mathrm{Gal}_F(L)$:

$$\overline{H} = \mathrm{Gal}_F(L)$$

***Proof*:**
First, $H \subseteq \mathrm{Gal}_F(L)$ since by definition $\mathrm{Gal}_F(L)$ contains every $\sigma \in \mathrm{Gal}_K(L)$ that fixes F . Next, $\mathrm{Gal}_F(L)$ is closed and is the group characterized by $L^{\mathrm{Gal}_F(L)} = F$ by theorem 50.8.8. Thus

$$H \subseteq \overline{H} \subseteq \mathrm{Gal}_F(L)$$

Then in the bijection presented in theorem theorem 50.8.8, we have that they are equal, completing the proof.

Let us conclude with a result proven by Waterhouse in 1973 that generalizes [3, Finite Group Represented by Galois Group]:

Theorem 1.8.10: Profinite Group Representable By Galois Group

Let G be a profinite group. Then G is isomorphic to the galois group of some galois extension L/K

***Proof*:**
Let $\mathcal{N}$ be the set of open normal subgroups of G and let

$$X = \coprod_{N \in \mathcal{N}} G/N$$

with G acting on each summand by left translation. Each $N \in \mathcal{N}$ has finite index, since G is compact by theorem 50.8.3 and the cosets of N form a cover of G by disjoint open sets. So every G-orbit in X is finite. The action is faithful: if g fixes every point then $g \in N$ for all $N \in \mathcal{N}$, and $\bigcap_N N = \{1\}$ because the natural map $G \to \varprojlim_N G/N$ is injective by corollary 50.8.4. The stabiliser of a point gN is gNg^{-1}, which is open.

Let k be any field and let $L = k(X)$ be purely transcendental over k with one indeterminate x_ξ for each $\xi \in X$. Each $\sigma \in G$ permutes X, hence induces the k-automorphism $x_\xi \mapsto x_{\sigma\xi}$ of L, and faithfulness makes $G \to \mathrm{Aut}_k(L)$ injective. Identify G with its image and put $K = L^G$.

The extension L/K is Galois. Since $k \subseteq K$, we have $L = k(X) \subseteq K(X) \subseteq L$, so $L = K(X)$. For each $\xi \in X$ the orbit $G\xi$ is finite, and the polynomial $\prod_{\eta \in G\xi}(T - x_\eta)$ has coefficients fixed by G, hence in K. Its roots are distinct indeterminates and all lie in L, so x_ξ is separable over K with minimal polynomial splitting in L. Therefore $L = K(X)$ is algebraic, separable and normal over K.

By construction $G \leqslant \mathrm{Gal}_K(L)$ with fixed field K, so G is dense in $\mathrm{Gal}_K(L)$ for the Krull topology by corollary 50.8.9. The inclusion $\iota : G \to \mathrm{Gal}_K(L)$ is continuous: by definition 50.8.6 it is enough that $\iota^{-1}(\mathrm{Gal}_F(L))$ be open in G for every finite normal subextension F/K, since the preimage of a coset is empty or a coset of that subgroup. Write $F = K(\alpha_1, \ldots, \alpha_r)$. Each α_i is a rational function in finitely many of the x_ξ, so there is a finite $Y \subseteq X$ such that any $\sigma \in G$ fixing Y pointwise fixes every α_i, hence fixes F pointwise. The pointwise stabiliser of Y is a finite intersection of open stabilisers, so it is open, and $\iota^{-1}(\mathrm{Gal}_F(L))$ is a union of its cosets.

Finally, $\iota(G)$ is compact as the continuous image of a compact group, and $\mathrm{Gal}_K(L)$ is Hausdorff because proposition 50.8.7 realises it as a subspace of a product of discrete groups. So $\iota(G)$ is closed, and being dense it is all of $\mathrm{Gal}_K(L)$. Thus ι is a continuous bijection from a compact group to a Hausdorff group, hence an isomorphism of topological groups $G \cong \mathrm{Gal}_K(L)$.

Just like for finite groups where $K = \mathbb{Q}$ is a difficult (as of writing this book unsolved) problem called the *inverse Galois Problem*, there is little control on what base field K we can choose.

Exercise 1.8.1

1. Find $G = \mathrm{Gal}_{\mathbb{F}_p}(\overline{\mathbb{F}}_p)$ recalling that for each degree n, there is only one extension of $\mathbb{F}_p$. Show that there are too many subgroups to correspond to the fixed fields. Define the Krull topology on G.